NUS-H2: First-Principles Derivations

Nuclear-Upgraded Steam Methane Reforming with Carbon Capture
for Low-Carbon Hydrogen in Singapore

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Every number on these pages comes from the tools script. Of its 619 numbers, 360 are read from the results files, 109 are computed from results values, 139 are computed here from published inputs, 3 are published and 2 are assumed. Five values of the Case A sizing walk are read from the design decisions table and the low reactor corner of 163.0 is taken from the screening table, because no results row holds them. The 83 validation status lines are written from the validation results file.

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1 Reference Frame

This section is the language of the whole document. It lists every symbol, every numbered equation and the assumptions and ranges needed to read them. Sections A to L then take each equation in turn.

Basis. Case B is the headline plant: 100,000 $Nm^3\ h^{-1}$ of hydrogen, or 66,925 t y^{-1}, with 120.8 MW_{th} of process heat, 23.2 MMSCFD of natural gas, a reactor rated 210 MW_{th} against 178.5 MW_{th} required, and 578,052 t y^{-1} of direct abatement, all on the Headline preset of the final results. Case A, at 130 MMSCFD of hydrogen, is the INL reference. PSA tail gas is recycled across the plant boundary. SMR means steam methane reforming throughout this document. The unabated SMR is a steam methane reforming plant with no capture and the same hydrogen output, the comparison case.

How to read an equation card. Each card gives the number, the statement, a physical interpretation, the working that fits its tier, the variables with SI units, the assumptions, the source, the implementation reference and the validation method. Implementation references name the engine item that computes the equation, by crate and module path. A card whose item carries a doc line "eq" with its number is linked, and a card with no engine item says that it is not implemented. A card that the engine does not implement may name the item planned for it.

Tier R rigorous, meaning first principles or a published model used inside its stated range. O reduced-order. E empirical correlation. F fixed design assumption. X external boundary. A published model used outside its range is tier O and its card says so.

Evidence [**P**] published, with the page given. [**T**] textbook result, chapter only, page not checked because the book is not in the library. [**C**] calculated here. [**A**] assumed. [**Mn**] source missing, listed in Section M.

Assumption codes SS steady state. 1D axial only. PF plug flow. IG ideal gas. AD adiabatic. CP constant heat capacity. NKE kinetic and potential energy neglected. IN inert carrier gases in the solvent.

Validation codes ID analytic identity in a unit test. GOLD agreement with the screening reference values, which is verification of a port and never validation. PUB comparison with a published calculated value. EXP comparison with measured data. NONE no benchmark exists, and the card says so.

Page numbers "PDF p." is the page in the file. "Printed p." is the number on the page.

Status of this version. Every number on these pages comes from the tools script. Of its 619 numbers, 360 are read from the results files, 109 are computed from results values, 139 are computed here from published inputs, 3 are published and 2 are assumed. Five values of the Case A sizing walk are read from the design decisions table and the low reactor corner of 163.0 is taken from the screening table, because no results row holds them. The 83 validation status lines are written from the validation results file. The engine test suite ran on the engine at commit 0405ff0: 368 tests passed, 0 failed and 0 were ignored. Of these, 109 carry an equation name and 109 of them passed. Of the 299 cards, 265 are linked to the engine: a doc line names the card on an engine item and the implementation line names that item. Of the rest, 27 are stated as not implemented, 4 are theory and 3 are listed only. Of the 281 cards with a benchmark code other than NONE, 101 have an engine test named for them that passed in the recorded run and 180 have none. A validation line that names rows of the validation results prints the outcome those rows record, so the text follows the results on every build. A benchmark stated on a card without a results outcome is the target of its named test. Where this document has computed the comparison itself, the card says "computed here" and gives the result.

Symbols that collide in the sources. R is the gas constant, and reactor rating is $\dot{Q}_{rated}$. A is area, f_{av} availability and $\mathcal{A}$ abatement. Every efficiency carries a subscript: η_{pc} power conversion, η_{pp} per-pass capture, η_{pl} plant capture, η_j^{cat} catalyst effectiveness, η_{is} isentropic, η_m motor. ε_b is bed porosity and ε_{HX} exchanger effectiveness. α is CO_2 loading and h a heat transfer coefficient, although JAERI writes α for the latter. ϕ_c is a carbon split and φ_i a fugacity coefficient. γ_i is an activity coefficient and $\kappa = c_p/c_v$.

Units. SI at every interface. Kinetics are coded in the published units (bar, kmol, kg of catalyst, hour) and converted at the function boundary, because the Xu and Froment notation mixes them. Conversions used: 1 MMSCFD = 13.83 mol s^{-1} at 60 °F, 1 st = 907.185 kg, 1 MMBtu h^{-1} = 0.29307 MW.

1.1 Table of variables

Symbol	Definition	SI units	Case B	Case A	Class
s	Scale on INL product mass	1	0.6833	1.0000	calculated
$\dot{m}_{H_2}$	Hydrogen product, external PSA boundary	kg s^{-1}	2.497	3.654	calculated, published
M_{H_2}	Annual hydrogen	t y^{-1}	66,925	97,946	calculated
f_{av}	Availability	1	0.85	0.85	assumed
t_h	Annual operating hours	h y^{-1}	7,446	7,446	calculated
$\dot{V}_{NG}$	Natural gas	MMSCFD	23.2	34.0	calculated, published
$\dot{Q}_{rated}$	Selected reactor rating	MW_{th}	210	none	assumed
$\dot{Q}_R$	Reactor heat required	MW_{th}	178.5	260.7	calculated
$\dot{Q}_{spare}$	Rating less requirement, no revenue	MW_{th}	31.5	-50.7	calculated
T_{ROT}	Reactor outlet temperature	K	1198.15	1198.15	published
$T_{s,sup}$	Secondary helium supply temperature	K	1173.15	1173.15	published
$T_{s,ret}$	Secondary helium return temperature	K	739.15	739.15	published
$T_{ref,out}$	Reformer exit temperature	K	1144.15	1144.15	published
$\dot{m}_s$	Secondary helium flow	kg s^{-1}	53.6	78.49	calculated, published
$\dot{m}_p$	Primary helium flow	kg s^{-1}	79.2	116	calculated
$\dot{Q}_{proc}$	Process heat to secondary helium	MW_{th}	120.8	176.8	calculated, published
$\dot{Q}_{rec}$	Sensible heat of recycled CO_2 slip	MW_{th}	0.88	1.28	calculated
$\dot{W}_s$	Secondary circulator shaft power	MW	3.13	4.59	calculated
$\dot{W}_p$	Primary circulator shaft power	MW	3.57	5.21	calculated
$\dot{Q}_{ves}$	Vessel heat loss	MW	2.66	3.42	calculated
$\dot{Q}_{ihx}$	IHX heat loss	MW	0.97	1.42	calculated
$\dot{Q}_{duct}$	Duct heat loss	MW	0.31	0.45	calculated
η_{pc}	Power conversion efficiency	1	0.397	0.397	published
r_w	Back-pressure work ratio	$MW_e\ MW_{th}^{-1}$	0.169	0.169	calculated
$\dot{Q}_{reb}$	Reboiler duty	MW	51.0	74.6	calculated
$\dot{Q}_{stack}$	Plant steam credit	MW	37.9	55.4	calculated
$\dot{Q}_{gap}$	Reboiler gap carried by the reactor	MW	13.1	19.2	calculated
E_{proc}	Process electricity	MW_e	12.33	18.05	calculated
E_{tot}	Total electricity	MW_e	19.78	28.94	calculated
η_{pp}	Per-pass capture	1	0.90	0.90	assumed

Symbol	Definition	SI units	Case B	Case A	Class
η_{pl}	Plant capture, captured over feed carbon	1	0.9005	0.9005	calculated
ϕ_c	CO_2 share of carbon after shift	1	0.775	0.775	calculated
f_r	Recycled fraction of tail-gas carbon	1	0.752	0.752	calculated
SRD	Specific reboiler duty	GJ t^{-1}	3.8 (3.4 to 4.2)	same	assumed
α_{rich}	Rich loading	mol mol^{-1}	0.50	0.50	assumed
α_{lean}	Lean loading	mol mol^{-1}	0.25	0.25	assumed
$\dot{m}_{CO_2}$	Captured CO_2	t h^{-1}	48.3	70.7	calculated
$\dot{m}_{solv}$	Lean solvent	t h^{-1}	894	1,309	calculated
p_{abs}	Absorber pressure	Pa	26.8 bar	26.8 bar	published
T_{abs}	Absorber temperature	K	313.15	313.15	published
e_{base}	Baseline intensity	kg kg^{-1}	9.21(9.00 to 9.35)	same	calculated, published
E_{base}	Baseline CO_2	t y^{-1}	616,361	902,060	calculated
E_{atm}	Residual CO_2 to atmosphere	t y^{-1}	38,309	56,066	calculated
$\mathcal{A}$	Direct abatement	t y^{-1}	578,052	845,994	calculated
i	Discount rate	1	0.08	0.08	assumed
t_ℓ	Life	y	25 and 60	same	assumed
n_s	Reactor size exponent	1	1.0, with 0.6 and 0.31 at equal prominence	same	assumed
c_{kWt}	Overnight capital cost, no turbine	US$$_{2019}$ kWt^{-1}	2,184	same	calculated from published
p_{gas}	Gas price	S$ GJ^{-1}	15	15	published
p_{TS}	Transport and storage price	S$ t^{-1}	60	60	assumed
AC	Abatement cost, L3, BOAK, $n_s = 1$, 210 MW_{th}, 25 y and 60 y	S$ t^{-1}	137.7, 117.2	see K and L	calculated
ρ	density	kg m^{-3}	varies	varies	variable
ψ	conserved quantity per unit mass	per kg	varies	varies	variable
$\mathbf{u}$	velocity	m s^{-1}	varies	varies	variable
$\mathbf{n}$	outward unit normal	1	varies	varies	variable
$\mathbf{j}_\psi$	non-convective flux of ψ	per m^2 s	varies	varies	variable
$\dot{s}_\psi$	volumetric source of ψ	per m^3 s	varies	varies	variable
V	control volume	m^3	varies	varies	variable
S	control surface	m^2	varies	varies	variable
M	mass held in the control volume	kg	varies	varies	variable
$\dot{m}$	mass flow of a stream	kg s^{-1}	varies	varies	variable
n_i	moles of species i held	mol	varies	varies	variable
F_i	molar flow of species i	mol s^{-1}	varies	varies	variable
R_i	volumetric production rate of i	mol m^{-3} s^{-1}	varies	varies	variable
ν_{ij}	stoichiometric coefficient of i in reaction j	1	varies	varies	variable
r_j	rate of reaction j	mol m^{-3} s^{-1}, or mol kg^{-1} s^{-1} per mass of catalyst	varies	varies	variable
ρ_B	catalyst bed density	kg m^{-3}	varies	varies	variable
η_j^{cat}	effectiveness factor of reaction j	1	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
a_{ki}	atoms of element k per molecule of i	1	varies	varies	variable
$\mathbf{a}$	element by species matrix	1	varies	varies	variable
$\boldsymbol{\nu}$	species by reaction matrix	1	varies	varies	variable
N_r	number of independent reactions	1	varies	varies	variable
N_s	number of species	1	varies	varies	variable
E	energy held in the control volume	J	varies	varies	variable
$\dot{Q}$	heat into the control volume	W	varies	varies	variable
$\dot{W}_{sh}$	shaft work out of the control volume	W	varies	varies	variable
h	specific enthalpy	J kg^{-1}	varies	varies	variable
u	velocity magnitude	m s^{-1}	varies	varies	variable
g	gravity	m s^{-2}	varies	varies	variable
z	elevation	m	varies	varies	variable
h_i	molar enthalpy of species i, formation basis	J mol^{-1}	varies	varies	variable
$\Delta_f h_i^\circ$	standard enthalpy of formation	J mol^{-1}	varies	varies	variable
$c_{p,i}$	molar heat capacity of species i	J mol^{-1} K^{-1}	varies	varies	variable
T	temperature	K	varies	varies	variable
T_{ref}	reference temperature, 298.15	K	varies	varies	variable
t	temperature over 1000	1	varies	varies	variable
A, B, C, D, E_S, F, H	Shomate coefficients	mixed, per NIST	varies	varies	variable
$\Delta_r h_j$	enthalpy of reaction j	J mol^{-1}	varies	varies	variable
p	pressure	Pa	varies	varies	variable
f_D	Darcy friction factor	1	varies	varies	variable
D	hydraulic diameter	m	varies	varies	variable
θ	channel inclination	rad	varies	varies	variable
G	superficial mass velocity	kg m^{-2} s^{-1}	varies	varies	variable
D_p	particle diameter	m	varies	varies	variable
ε_b	bed porosity	1	varies	varies	variable
μ	gas viscosity	Pa s	varies	varies	variable
A_c	flow cross section	m^2	varies	varies	variable
$\dot{q}'$	heat into the gas per unit length	W m^{-1}	varies	varies	variable
p_i	partial pressure of i	Pa	varies	varies	variable
y_i	mole fraction of i	1	varies	varies	variable
c	molar concentration	mol m^{-3}	varies	varies	variable
Z	compressibility factor	1	varies	varies	variable
R	gas constant, 8.314462	J mol^{-1} K^{-1}	varies	varies	variable
ϵ_m	relative mass residual	1	varies	varies	variable
ϵ_k	relative residual of element k	1	varies	varies	variable
ϵ_E	relative energy residual	1	varies	varies	variable
P	helium pressure in bar, in correlations only	bar	varies	varies	variable
c_p	isobaric heat capacity	J kg^{-1} K^{-1}	varies	varies	variable
c_v	isochoric heat capacity	J kg^{-1} K^{-1}	varies	varies	variable
κ	heat capacity ratio	1	varies	varies	variable
k	thermal conductivity	W m^{-1} K^{-1}	varies	varies	variable
T_{RI}	reactor inlet temperature	K	varies	varies	variable
$T_{h,out}$	hot-side helium leaving the IHX	K	varies	varies	variable
$\dot{Q}_p$	heat lost by primary helium	W	varies	varies	variable
ΔT_{app}	IHX approach temperature	K	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
$\dot{Q}_s$	heat carried by secondary helium	W	varies	varies	variable
$\dot{W}$	shaft power	W	varies	varies	variable
T_1	gas temperature at circulator inlet	K	varies	varies	variable
Δp	loop pressure drop	Pa	varies	varies	variable
η_{is}	isentropic efficiency, 0.75	1	varies	varies	variable
Δp_s	secondary loop pressure drop	Pa	varies	varies	variable
Δp_p	primary loop pressure drop	Pa	varies	varies	variable
E_{circ}	circulator electricity	W	varies	varies	variable
η_m	motor and drive efficiency	1	varies	varies	variable
f_{ihx}	IHX loss fraction	1	varies	varies	variable
$\dot{Q}_{av}$	heat available at the process interface	MW	varies	varies	variable
T_c	cycle cold temperature	K	varies	varies	variable
T_h	cycle hot temperature	K	varies	varies	variable
$\dot{Q}_{pc}$	heat into the power cycle	W	varies	varies	variable
w	work from the back-pressure stage	MW	varies	varies	variable
$\dot{Q}_{br}$	heat to the power and steam branch	MW	varies	varies	variable
$\dot{Q}_{ex}$	excess heat for power	W	varies	varies	variable
$T_{s,ideal}$	ideal helium return temperature, 461 C	K	varies	varies	variable
$\dot{Q}_h$	heat lost by the hot stream	W	varies	varies	variable
$T_{h,in}$	hot stream inlet	K	varies	varies	variable
$\dot{Q}_c$	heat gained by the cold stream	W	varies	varies	variable
$T_{c,in}$	cold stream inlet	K	varies	varies	variable
$T_{c,out}$	cold stream outlet	K	varies	varies	variable
UA	overall conductance	W K^{-1}	varies	varies	variable
h_{fi}	film coefficient, tube side	W m^{-2} K^{-1}	varies	varies	variable
h_{fo}	film coefficient, shell side	W m^{-2} K^{-1}	varies	varies	variable
A_i	inner surface area	m^2	varies	varies	variable
A_o	outer surface area	m^2	varies	varies	variable
$R_{f,i}$	inner fouling resistance	m^2 K W^{-1}	varies	varies	variable
$R_{f,o}$	outer fouling resistance	m^2 K W^{-1}	varies	varies	variable
d_i	tube inner diameter	m	varies	varies	variable
d_o	tube outer diameter	m	varies	varies	variable
k_w	wall conductivity	W m^{-1} K^{-1}	varies	varies	variable
L	tube length	m	varies	varies	variable
N_t	number of tubes	1	varies	varies	variable
U_o	overall coefficient on outer area	W m^{-2} K^{-1}	varies	varies	variable
Nu_{out}	shell-side Nusselt number	1	varies	varies	variable
Nu_{in}	tube-side Nusselt number	1	varies	varies	variable
C_0	shell-side correction factor, value not printed	1	varies	varies	variable
Re	Reynolds number	1	varies	varies	variable
Pr	Prandtl number	1	varies	varies	variable
d_{in}	tube inner diameter in the source	m	varies	varies	variable
D_h	helix diameter in the source	m	varies	varies	variable
ΔT_{lm}	log-mean temperature difference	K	varies	varies	variable
ΔT_1	terminal difference at end 1	K	varies	varies	variable
ΔT_2	terminal difference at end 2	K	varies	varies	variable
U	overall heat transfer coefficient	W m^{-2} K^{-1}	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
A	heat transfer area	m^2	varies	varies	variable
C_h	hot stream capacity rate	W K^{-1}	varies	varies	variable
C_c	cold stream capacity rate	W K^{-1}	varies	varies	variable
ΔT	local temperature difference	K	varies	varies	variable
ε_{HX}	exchanger effectiveness	1	varies	varies	variable
C_{min}	smaller capacity rate	W K^{-1}	varies	varies	variable
NTU	number of transfer units	1	varies	varies	variable
C^*	capacity ratio	1	varies	varies	variable
C_{max}	larger capacity rate	W K^{-1}	varies	varies	variable
UA_{mod}	conductance per module	W K^{-1}	varies	varies	variable
UA_{HTTR}	measured HTTR conductance	W K^{-1}	varies	varies	variable
N_{mod}	number of modules	1	varies	varies	variable
ΔT_{cold}	cold-end temperature difference	K	varies	varies	variable
ΔT_{hot}	hot-end temperature difference	K	varies	varies	variable
ΔT_{duct}	duct temperature drop	K	varies	varies	variable
T_{tube}	IHX tube temperature	K, stated in C	varies	varies	variable
Δp_{tube}	differential pressure across the tube	Pa	varies	varies	variable
p_s	secondary helium pressure	Pa	varies	varies	variable
p_p	primary helium pressure	Pa	varies	varies	variable
$\Delta_r h^\circ_{298}$	standard enthalpy of reaction at 298 K	J mol^{-1}	varies	varies	variable
n	carbon atoms in the hydrocarbon	1	varies	varies	variable
m	hydrogen atoms in the hydrocarbon	1	varies	varies	variable
F_{H_2O}	steam molar feed	mol s^{-1}	varies	varies	variable
F_C	carbon molar feed	mol s^{-1}	varies	varies	variable
x_1	extent of reaction I, which is total methane conversion	1	varies	varies	variable
x_2	extent of reaction II	1	varies	varies	variable
ξ_{CO_2}	CO_2 fed per mole of CH_4	1	varies	varies	variable
ξ_{H_2}	H_2 fed per mole of CH_4	1	varies	varies	variable
S_C	steam fed per mole of CH_4	1	varies	varies	variable
Σ_0	total moles fed per mole of CH_4	1	varies	varies	variable
K_1	equilibrium constant of reaction I	bar^2	varies	varies	variable
K_2	equilibrium constant of reaction II	1	varies	varies	variable
K_3	equilibrium constant of reaction III	bar^2	varies	varies	variable
Q_1	quotient of reaction I	bar^2	varies	varies	variable
Q_2	quotient of reaction II	1	varies	varies	variable
Q_3	quotient of reaction III	bar^2	varies	varies	variable
n_{tot}	total moles per mole of CH_4 fed	1	varies	varies	variable
K	equilibrium constant	mixed	varies	varies	variable
k_i	rate coefficient of reaction i	XF units, see working	varies	varies	variable
$A(k_i)$	pre-exponential factor of k_i	XF units	varies	varies	variable
$E_{a,i}$	activation energy of reaction i	J mol^{-1}	varies	varies	variable
K_j	adsorption constant of species j	bar^{-1}, or 1 for H_2O	varies	varies	variable
$A(K_j)$	pre-exponential factor of K_j	as K_j	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
ΔH_j	adsorption enthalpy of species j	J mol^{-1}	varies	varies	variable
DEN	adsorption denominator	1	varies	varies	variable
p_{CO}	partial pressure of CO, in bar in this form	bar	varies	varies	variable
r_1	rate of reaction I	kmol kg^{-1} h^{-1}	varies	varies	variable
r_2	rate of reaction II	kmol kg^{-1} h^{-1}	varies	varies	variable
p_{CO_2}	partial pressure of CO_2, in bar in this form	bar	varies	varies	variable
r_3	rate of reaction III	kmol kg^{-1} h^{-1}	varies	varies	variable
p_{H_2}	partial pressure of H_2, in bar in this form	bar	varies	varies	variable
r_{CO}	rate of formation of CO	kmol kg^{-1} h^{-1}	varies	varies	variable
r_{CO_2}	rate of formation of CO_2	kmol kg^{-1} h^{-1}	varies	varies	variable
r_{CH_4}	rate of disappearance of CH_4	kmol kg^{-1} h^{-1}	varies	varies	variable
ρ_s	catalyst solid density	kg m^{-3}	varies	varies	variable
$\mathbf{p}$	partial pressures inside the pellet	bar	varies	varies	variable
$\mathbf{p}^s$	partial pressures at the pellet surface	bar	varies	varies	variable
V_p	pellet volume	m^3	varies	varies	variable
a_{cat}	lumped catalyst activity	1	varies	varies	variable
x_{CH_4}	total methane conversion	1	varies	varies	variable
Ω	tube cross section	m^2	varies	varies	variable
$F^0_{CH_4}$	methane feed rate per tube	kmol h^{-1}	varies	varies	variable
x_{CO_2}	conversion of methane into CO_2	1	varies	varies	variable
ρ_g	process gas density	kg m^{-3}	varies	varies	variable
u_s	superficial velocity	m s^{-1}	varies	varies	variable
T_w	tube wall temperature	K	varies	varies	variable
R_f	fouling resistance	m^2 K W^{-1}	varies	varies	variable
t_w	wall thickness	m	varies	varies	variable
$\dot{m}_{He}$	helium flow on the shell side	kg s^{-1}	varies	varies	variable
T_{He}	helium temperature	K	varies	varies	variable
Nu	Nusselt number	1	varies	varies	variable
S_f	fin spacing	m	varies	varies	variable
H_f	fin height	m	varies	varies	variable
D_e	equivalent diameter of the annulus	m	varies	varies	variable
h_w	wall to bed film coefficient	W m^{-2} K^{-1}	varies	varies	variable
k_g	gas conductivity	W m^{-1} K^{-1}	varies	varies	variable
q	radiative heat per unit shell volume	W m^{-3}	varies	varies	variable
σ	Stefan-Boltzmann constant	W m^{-2} K^{-4}	varies	varies	variable
$\varepsilon_{r,a}$	emissivity of surface a	1	varies	varies	variable
$\varepsilon_{r,b}$	emissivity of surface b	1	varies	varies	variable
T_a	temperature of surface a	K	varies	varies	variable
T_b	temperature of surface b	K	varies	varies	variable
A_{sh}	shell cross section	m^2	varies	varies	variable
X_{CH_4}	methane conversion	1	varies	varies	variable
Y_{H_2}	hydrogen yield	1	varies	varies	variable
$\dot{Q}_{tube}$	reformer tube duty	W	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
q''	average heat flux on outer area	W m^{-2}	varies	varies	variable
X_{CO}	CO conversion	1	varies	varies	variable
$F_{CO,in}$	CO molar flow into the bed	mol s^{-1}	varies	varies	variable
$F_{CO,out}$	CO molar flow out of the bed	mol s^{-1}	varies	varies	variable
ξ	extent of shift across a bed	mol s^{-1}	varies	varies	variable
T_{out}	gas temperature at the bed exit	K	varies	varies	variable
ΔT_{eq}	equilibrium approach temperature	K	varies	varies	variable
$\sum_i F_i h_i$	enthalpy flow of the stream	W	varies	varies	variable
$\dot{Q}_{WGS}$	heat released by shift	W	varies	varies	variable
k_{HTS}	pseudo-first-order shift constant, empty bed basis	s^{-1}	varies	varies	variable
$p_{CO,eq}$	CO partial pressure at equilibrium	Pa	varies	varies	variable
t_r	contact time on the empty bed	s	varies	varies	variable
S_v	space velocity at reaction conditions, empty bed	s^{-1}	varies	varies	variable
$X_{CO,e}$	CO conversion at equilibrium	1	varies	varies	variable
V_{cat}	catalyst volume, empty bed basis	m^3	varies	varies	variable
$\dot{V}$	gas volume flow at reaction conditions	m^3 s^{-1}	varies	varies	variable
$dT/d\xi$	temperature rise per unit extent	K mol^{-1} s	varies	varies	variable
Δp_{bed}	pressure drop across one shift bed	Pa	varies	varies	variable
S/DG	steam to dry gas mole ratio	1	varies	varies	variable
$T_{HTS,in}$	high-temperature bed inlet	K	varies	varies	variable
$T_{LTS,in}$	low-temperature bed inlet	K	varies	varies	variable
T_{dp}	dew point	K	varies	varies	variable
$T_{LTS,out}$	low-temperature bed exit	K	varies	varies	variable
r	shift rate per unit mass of catalyst	mol g^{-1} s^{-1}	varies	varies	variable
k_0	pre-exponential factor	mol g^{-1} Pa$^{-1.71}$ s^{-1}	varies	varies	variable
$E_{a,K}$	activation energy of this law	J mol^{-1}	varies	varies	variable
T_{in}	cooler inlet temperature	K	varies	varies	variable
$\dot{Q}_{train}$	total duty of the cooling train	W	varies	varies	variable
y_{H_2O}	water mole fraction in the gas	1	varies	varies	variable
p_{sat}	water saturation pressure	Pa	varies	varies	variable
$p_{sat}(T)$	saturation pressure at T	Pa	varies	varies	variable
y_w	water mole fraction in the gas	1	varies	varies	variable
x_w	water mole fraction in the liquid	1	varies	varies	variable
γ_w	water activity coefficient	1	varies	varies	variable
x_i	liquid mole fraction of i	1	varies	varies	variable
H_x	Henry constant, mole fraction basis	Pa	varies	varies	variable
$H^w_{CO_2}$	Henry constant of CO_2 in water	Pa m^3 mol^{-1}	varies	varies	variable
c_w	molar density of liquid water	mol m^{-3}	varies	varies	variable
z_i	feed mole fraction of i	1	varies	varies	variable
K_i	vapour to liquid ratio of i	1	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
β_V	vapour fraction of the feed	1	varies	varies	variable
F_f	flash feed	mol s^{-1}	varies	varies	variable
V_f	flash vapour	mol s^{-1}	varies	varies	variable
L_f	flash liquid	mol s^{-1}	varies	varies	variable
h_V	molar enthalpy of the vapour	J mol^{-1}	varies	varies	variable
h_L	molar enthalpy of the liquid	J mol^{-1}	varies	varies	variable
h_F	molar enthalpy of the feed	J mol^{-1}	varies	varies	variable
L_{cond}	total condensate	kg s^{-1}	varies	varies	variable
L_{ret}	condensate returned to the reformer	kg s^{-1}	varies	varies	variable
L_{treat}	condensate sent to treatment	kg s^{-1}	varies	varies	variable
T_{min}	lowest useful gas temperature	K	varies	varies	variable
$\dot{C}$	gas capacity rate	W K^{-1}	varies	varies	variable
$\dot{m}_{cond}$	condensate formed over the interval	kg s^{-1}	varies	varies	variable
Δh_{vap}	latent heat of water	J kg^{-1}	varies	varies	variable
$\dot{m}_{CO_2,cond}$	CO_2 leaving in condensate	kg s^{-1}	varies	varies	variable
M_w	molar mass of water	kg mol^{-1}	varies	varies	variable
M_{CO_2}	molar mass of CO_2	kg mol^{-1}	varies	varies	variable
$y_{i,dry}$	dry mole fraction of i	1	varies	varies	variable
MEA	monoethanolamine, C_2H_7NO	species	varies	varies	variable
$K_{x,j}$	mole fraction equilibrium constant of reaction j	1	varies	varies	variable
C_1, C_2, C_3, C_4	coefficients of the fit	1, K, 1, K^{-1}	varies	varies	variable
μ_i	chemical potential of i	J mol^{-1}	varies	varies	variable
μ_i°	standard chemical potential of i	J mol^{-1}	varies	varies	variable
γ_i	activity coefficient of i	1	varies	varies	variable
γ_i^*	unsymmetric activity coefficient of i	1	varies	varies	variable
γ_i^∞	activity coefficient of i at infinite dilution in water	1	varies	varies	variable
$K_{x,MEA,eff}$	MEA constant on the pure liquid basis	1	varies	varies	variable
n^0	total MEA charged	mol	varies	varies	variable
α	CO_2 loading	mol mol^{-1}	varies	varies	variable
$n^0_{H_2O}$	apparent water charged	mol	varies	varies	variable
γ_i^C	combinatorial activity coefficient	1	varies	varies	variable
γ_i^R	residual activity coefficient	1	varies	varies	variable
γ_i^{DH}	Debye-Hückel activity coefficient	1	varies	varies	variable
ϕ_i	volume fraction of i	1	varies	varies	variable
θ_i	surface fraction of i	1	varies	varies	variable
r_i	UNIQUAC volume parameter	1	varies	varies	variable
q_i	UNIQUAC surface parameter	1	varies	varies	variable
z_c	coordination number, 10	1	varies	varies	variable
ψ_{ki}	interaction factor between k and i	1	varies	varies	variable
u_{ki}	interaction energy between k and i	K	varies	varies	variable
u^0_{ki}	interaction energy at 298.15 K	K	varies	varies	variable
u^T_{ki}	temperature slope of the interaction energy	1	varies	varies	variable
Z_i	ion charge of i	1	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
A_{DH}	Debye-Hückel constant	$kg^{1/2}$ $mol^{-1/2}$	varies	varies	variable
b_{DH}	Debye-Hückel size parameter, 1.5	$kg^{1/2}$ $mol^{-1/2}$	varies	varies	variable
σ_{DH}	Debye-Hückel water function	1	varies	varies	variable
t_C	temperature in Celsius, here only	C	varies	varies	variable
γ_w^{DH}	Debye-Hückel coefficient of water	1	varies	varies	variable
I	ionic strength	mol kg^{-1}	varies	varies	variable
m_i	molality of i	mol kg^{-1}	varies	varies	variable
φ_i	fugacity coefficient of i	1	varies	varies	variable
$v_{CO_2}^{\infty}$	partial molar volume of CO_2 at infinite dilution	m^3 mol^{-1}	varies	varies	variable
$p_{H_2O}^s$	saturation pressure of water	Pa	varies	varies	variable
$p_{CO_2}^*$	equilibrium partial pressure of CO_2	Pa	varies	varies	variable
p_i^s	saturation pressure of i	Pa	varies	varies	variable
φ_i^s	fugacity coefficient of saturated i	1	varies	varies	variable
v_i	partial molar volume of i	m^3 mol^{-1}	varies	varies	variable
A_S	SRK attraction group	1	varies	varies	variable
B_S	SRK volume group	1	varies	varies	variable
a_m	SRK mixture attraction	Pa m^6 mol^{-2}	varies	varies	variable
b_m	SRK mixture covolume	m^3 mol^{-1}	varies	varies	variable
H_{app}	apparent Henry constant of CO_2 in the solvent	Pa m^3 mol^{-1}	varies	varies	variable
H_{N_2O}	Henry constant of N_2O in the solvent	Pa m^3 mol^{-1}	varies	varies	variable
$H_{N_2O}^w$	Henry constant of N_2O in water	Pa m^3 mol^{-1}	varies	varies	variable
Δh_{abs}	heat of absorption of CO_2, positive when exothermic	J mol^{-1}	varies	varies	variable
$c_{p,s}$	solvent mass heat capacity	J kg^{-1} K^{-1}	varies	varies	variable
V_{sln}	molar volume of the solution	m^3 mol^{-1}	varies	varies	variable
V_{MEA}	molar volume of MEA	m^3 mol^{-1}	varies	varies	variable
V_{H_2O}	molar volume of water	m^3 mol^{-1}	varies	varies	variable
V_{CO_2}	dissolved volume of CO_2	m^3 mol^{-1}	varies	varies	variable
V^*	interaction volume of MEA and water	m^3 mol^{-1}	varies	varies	variable
ρ_{sln}	solution density	kg m^{-3}	varies	varies	variable
a_ρ, b_ρ, c_ρ	MEA density coefficients	g mL^{-1} with T in K	varies	varies	variable
μ_{sln}	solution viscosity	Pa s	varies	varies	variable
μ_w	water viscosity	Pa s	varies	varies	variable
Ω_w	MEA mass percent	1	varies	varies	variable
D_w	CO_2 diffusivity in water	m^2 s^{-1}	varies	varies	variable
D_l	CO_2 diffusivity in solution	m^2 s^{-1}	varies	varies	variable
f_{free}	free fraction of MEA	1	varies	varies	variable
p_{Xu}^*	equilibrium CO_2 pressure by the Xu fit	Pa	varies	varies	variable
α^*	equilibrium loading	mol mol^{-1}	varies	varies	variable
w_r	four-point Lagrange weight	1	varies	varies	variable
Λ_{rs}	stored value of $\ln p^* - 2\ln\alpha$	1	varies	varies	variable
N_{CO_2}	CO_2 flux from gas to liquid	mol m^{-2} s^{-1}	varies	varies	variable
k_G	gas film coefficient	mol m^{-2} s^{-1} Pa^{-1}	varies	varies	variable
p_{int}	CO_2 partial pressure at the interface	Pa	varies	varies	variable
$\mathcal{E}$	enhancement factor	1	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
k_L°	physical liquid film coefficient	m s^{-1}	varies	varies	variable
H	Henry constant of CO_2 on the rate side	Pa m^3 mol^{-1}	varies	varies	variable
C_{int}	CO_2 concentration at the interface	mol m^{-3}	varies	varies	variable
K_G	overall gas side coefficient	mol m^{-2} s^{-1} Pa^{-1}	varies	varies	variable
k_2	second-order rate constant	m^3 $kmol^{-1}$ s^{-1}	varies	varies	variable
$\mathcal{M}$	Hatta group, the Hatta number squared	1	varies	varies	variable
$\mathcal{E}_1$	pseudo-first-order enhancement	1	varies	varies	variable
C_B	free MEA in the bulk liquid	kmol m^{-3}	varies	varies	variable
$\mathcal{E}_\infty$	enhancement for an instantaneous reaction	1	varies	varies	variable
D_B	MEA diffusivity in solution	m^2 s^{-1}	varies	varies	variable
b	MEA consumed per CO_2, 2	1	varies	varies	variable
C_{MEA}	total MEA concentration	kmol m^{-3}	varies	varies	variable
C_E	surface renewal correction, 0.9	1	varies	varies	variable
U_L	superficial liquid velocity	m s^{-1}	varies	varies	variable
S_{ch}	corrugation channel side	m	varies	varies	variable
L_m	liquid mass velocity	kg m^{-2} s^{-1}	varies	varies	variable
a_w	wetted area per unit volume, Onda	m^2 m^{-3}	varies	varies	variable
a_p	packing area per unit volume	m^2 m^{-3}	varies	varies	variable
d_p	nominal packing size	m	varies	varies	variable
ρ_L	liquid density	kg m^{-3}	varies	varies	variable
μ_L	liquid viscosity	Pa s	varies	varies	variable
$k_{G,v}$	gas film coefficient as a velocity	m s^{-1}	varies	varies	variable
D_G	gas diffusivity of CO_2	m^2 s^{-1}	varies	varies	variable
U_G	superficial gas velocity	m s^{-1}	varies	varies	variable
ρ_G	gas density	kg m^{-3}	varies	varies	variable
μ_G	gas viscosity	Pa s	varies	varies	variable
G_m	gas mass velocity	kg m^{-2} s^{-1}	varies	varies	variable
C_1	Onda packing constant	1	varies	varies	variable
a_e	effective area per unit volume	m^2 m^{-3}	varies	varies	variable
$C_{h,BS}$	Billet and Schultes packing constant	1	varies	varies	variable
Re_L	liquid Reynolds number, $U_L\rho_L/a_p\mu_L$	1	varies	varies	variable
F_{SE}	surface enhancement factor	1	varies	varies	variable
We_L	liquid Weber number	1	varies	varies	variable
Fr_L	liquid Froude number	1	varies	varies	variable
ε_p	packing void fraction	1	varies	varies	variable
γ_c	liquid contact angle	rad	varies	varies	variable
G_n	total gas molar flow	mol s^{-1}	varies	varies	variable
N_{H_2O}	water flux from gas to liquid	mol m^{-2} s^{-1}	varies	varies	variable
y_{CO_2}	CO_2 mole fraction in the gas	1	varies	varies	variable
L_n	liquid molar flow, CO_2-free basis	mol s^{-1}	varies	varies	variable
X_{CO_2}	CO_2 in the liquid per mole of CO_2-free liquid	1	varies	varies	variable
X_{H_2O}	water fraction of the CO_2-free liquid	1	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
T_G	gas temperature	K	varies	varies	variable
q_{GL}	gas to liquid heat flux	W m^{-2}	varies	varies	variable
$c_{p,G}$	gas molar heat capacity	J mol^{-1} K^{-1}	varies	varies	variable
T_L	liquid temperature	K	varies	varies	variable
$c_{p,L}$	liquid molar heat capacity	J mol^{-1} K^{-1}	varies	varies	variable
Δh_{CO_2}	enthalpy change of CO_2 absorption	J mol^{-1}	varies	varies	variable
Δh_{H_2O}	enthalpy change of water condensation	J mol^{-1}	varies	varies	variable
h_{GL}	gas side heat transfer coefficient	W m^{-2} K^{-1}	varies	varies	variable
c_G	gas molar density	mol m^{-3}	varies	varies	variable
Sc	gas Schmidt number	1	varies	varies	variable
k_{G,H_2O}	gas film coefficient of water	mol m^{-2} s^{-1} Pa^{-1}	varies	varies	variable
p_{H_2O}	water partial pressure in the bulk gas	Pa	varies	varies	variable
$p^*_{H_2O}$	water pressure at equilibrium with the solvent	Pa	varies	varies	variable
Z_c	packed height	m	varies	varies	variable
F_s	gas load factor	Pa$^{1/2}$	varies	varies	variable
N_z	number of slices	1	varies	varies	variable
Δz	slice height	m	varies	varies	variable
$\mathbf{w}$	all unknowns on the grid	mixed	varies	varies	variable
$\mathbf{f}$	residual vector	mixed	varies	varies	variable
$\mathbf{J}$	Jacobian of the residual	mixed	varies	varies	variable
λ	damping factor	1	varies	varies	variable
y_{in}	CO_2 fraction in the feed gas	1	varies	varies	variable
y_{out}	CO_2 fraction in the treated gas	1	varies	varies	variable
n_{in}	gas flow in	mol s^{-1}	varies	varies	variable
n_{out}	gas flow out	mol s^{-1}	varies	varies	variable
$\dot{n}_{CO_2}$	CO_2 absorbed	mol s^{-1}	varies	varies	variable
M_{MEA}	molar mass of MEA	kg mol^{-1}	varies	varies	variable
w_{MEA}	MEA mass fraction, CO_2-free basis	1	varies	varies	variable
D_c	column diameter	m	varies	varies	variable
$\dot{V}_L$	liquid volume flow	m^3 s^{-1}	varies	varies	variable
ℓ_L	liquid load	m s^{-1}	varies	varies	variable
$\dot{n}_{H_2,abs}$	H_2 absorbed	mol s^{-1}	varies	varies	variable
$\dot{n}_{CH_4,abs}$	CH_4 absorbed	mol s^{-1}	varies	varies	variable
$\dot{n}_{CO,abs}$	CO absorbed	mol s^{-1}	varies	varies	variable
f	pressure modification index	1	varies	varies	variable
L_v	liquid flow in Isa's balance	m^3 s^{-1}	varies	varies	variable
C^0_{MEA}	MEA concentration fed	mol m^{-3}	varies	varies	variable
k_{MEA}	reaction rate constant of Isa Eq 43	m^3 mol^{-1} s^{-1}	varies	varies	variable
$\dot{n}_{MEA}$	MEA circulation	mol s^{-1}	varies	varies	variable
p^*	equilibrium CO_2 pressure	Pa	varies	varies	variable
$y_{CO_2,k}$	CO_2 mole fraction on stage k	1	varies	varies	variable
T_k	stage temperature	K	varies	varies	variable
α_k	stage loading	mol mol^{-1}	varies	varies	variable
N	number of equilibrium stages	1	varies	varies	variable
$\dot{n}_{H_2O}$	water in the overhead	mol s^{-1}	varies	varies	variable
T_{reb}	reboiler temperature	K	varies	varies	variable
p^*_{MEA}	equilibrium MEA pressure over the solvent	Pa	varies	varies	variable
p_{strip}	stripper pressure, 2.0 bar	Pa	varies	varies	variable
$\dot{n}$	molar flow of a stream	mol s^{-1}	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
$\dot{Q}_{cond}$	condenser duty	W	varies	varies	variable
$\dot{Q}_{H_2O}$	cooling of condensed water	W	varies	varies	variable
$\dot{Q}_{steam}$	cooling and condensing of steam	W	varies	varies	variable
$\dot{Q}_{CO_2}$	cooling of CO_2	W	varies	varies	variable
$\dot{n}_{ref}$	condensate returned	mol s^{-1}	varies	varies	variable
T_{cond}	separator temperature	K	varies	varies	variable
$\dot{m}_r$	rich solvent flow	kg s^{-1}	varies	varies	variable
$\dot{m}_l$	lean solvent flow	kg s^{-1}	varies	varies	variable
$T_{r,out}$	rich solvent leaving the exchanger	K	varies	varies	variable
$T_{l,in}$	lean solvent entering the exchanger	K	varies	varies	variable
ΔT_{hx}	hot-end approach, 10 K	K	varies	varies	variable
$\dot{m}_{st}$	heating steam flow	kg s^{-1}	varies	varies	variable
Δh_{cond}	latent heat of the heating steam	J kg^{-1}	varies	varies	variable
q_{des}	desorption term	J kg^{-1}	varies	varies	variable
q_{sens}	sensible term	J kg^{-1}	varies	varies	variable
q_{strip}	stripping steam term	J kg^{-1}	varies	varies	variable
I_{str}	stripping intensity	J kg^{-1}	varies	varies	variable
$\dot{m}_{rich}$	rich solvent flow	kg s^{-1}	varies	varies	variable
U_r	reboiler overall coefficient	W m^{-2} K^{-1}	varies	varies	variable
A_r	reboiler area	m^2	varies	varies	variable
ΔT_{LMTD}	reboiler log-mean difference	K	varies	varies	variable
T_{source}	heating medium temperature	K	varies	varies	variable
ΔT_{min}	minimum approach	K	varies	varies	variable
SRD_{tot}	total stripping heat per unit CO_2	J kg^{-1}	varies	varies	variable
SRD_{reb}	reboiler share	J kg^{-1}	varies	varies	variable
q_{IH}	interheater share	J kg^{-1}	varies	varies	variable
$\dot{m}_{H_2,A}$	Case A hydrogen product	kg s^{-1}	varies	varies	variable
$\dot{m}_{H_2,B}$	Case B hydrogen product	kg s^{-1}	varies	varies	variable
ρ_N	hydrogen normal density	kg m^{-3}	varies	varies	variable
$\dot{V}_N$	normal volumetric flow, 100,000	Nm^3 h^{-1}	varies	varies	variable
$\dot{n}_{NG}$	natural gas molar flow	mol s^{-1}	varies	varies	variable
$F_{CO_2,abs}$	CO_2 at the absorber inlet	mol s^{-1}	varies	varies	variable
T_C	tube carbon	mol s^{-1}	varies	varies	variable
T_6	tube carbon of INL Case 6	mol s^{-1}	varies	varies	variable
$F_{cap,6}$	carbon captured in INL Case 6	mol s^{-1}	varies	varies	variable
G_6	tail-gas carbon of INL Case 6	mol s^{-1}	varies	varies	variable
η_{Sel}	Selexol capture, 0.99	1	varies	varies	variable
C	carbon captured	mol s^{-1}	varies	varies	variable
$\dot{n}_{rec}$	recycled carbon	mol s^{-1}	varies	varies	variable
$\dot{n}_{burn}$	carbon burned in the nitrogen-rich cut	mol s^{-1}	varies	varies	variable
$F_{H_2,prod}$	hydrogen in the product	mol s^{-1}	varies	varies	variable
$F_{H_2,in}$	hydrogen to the PSA	mol s^{-1}	varies	varies	variable
ξ_{PSA}	PSA hydrogen recovery	1	varies	varies	variable
$F_{H_2,rej}$	hydrogen in the PSA reject	mol s^{-1}	varies	varies	variable
ΔF_{slip}	extra CO_2 slip over INL	mol s^{-1}	varies	varies	variable
c_{p,CO_2}	CO_2 heat capacity, 50	J mol^{-1} K^{-1}	varies	varies	variable
$\dot{Q}_{Sel}$	freed Selexol steam	W	varies	varies	variable
$\dot{Q}_{exp}$	export steam	W	varies	varies	variable
$\dot{Q}_{syn}$	syngas heat above 130 C	W	varies	varies	variable
E_i	electric row i	W	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
$E_{i,6}$	row i in INL Case 6	W	varies	varies	variable
E_{pump}	lean pump electricity	W	varies	varies	variable
η_{pump}	pump efficiency, 0.75	1	varies	varies	variable
E_{fans}	cooling electricity	W	varies	varies	variable
k_{cool}	electric load per unit reboiler heat	1	varies	varies	variable
E_{CO_2}	CO_2 compression electricity	W	varies	varies	variable
C_6	carbon captured in INL Case 6	mol s^{-1}	varies	varies	variable
E_{pur}	recycle and nitrogen removal electricity	W	varies	varies	variable
$\dot{Q}_{R,c}$	reactor heat at corner c	W	varies	varies	variable
E_{imp}	imported electricity	W	varies	varies	variable
E_{nuc}	electricity asked of the reactor	W	varies	varies	variable
$\dot{Q}_{heat}$	heat duties at fixed rating	W	varies	varies	variable
$T_{stack,j}$	temperature of stack stream j	K	varies	varies	variable
$T_{source,min}$	lowest usable source	K	varies	varies	variable
$\dot{Q}_{cool}$	total cooling duty	W	varies	varies	variable
$\dot{Q}_{KO}$	knock-out drum cooler	W	varies	varies	variable
$\dot{Q}_{HW}$	hot water trim cooler	W	varies	varies	variable
$\dot{Q}_{lean}$	lean solvent cooler	W	varies	varies	variable
η_{th}	thermal efficiency	1	varies	varies	variable
LHV_{H_2}	hydrogen lower heating value	J kg^{-1}	varies	varies	variable
LHV_{NG}	gas lower heating value	J mol^{-1}	varies	varies	variable
Y_m	gas mass per unit hydrogen	kg kg^{-1}	varies	varies	variable
Y_v	hydrogen volume per unit gas	1	varies	varies	variable
$\dot{m}_{NG}$	gas mass flow	kg s^{-1}	varies	varies	variable
$\dot{V}_{H_2}$	hydrogen standard volume flow	m^3 s^{-1}	varies	varies	variable
$\dot{Q}_{R,0}$	first estimate of reactor heat	W	varies	varies	variable
$\dot{E}_{base,A}$	CO_2 of the INL conventional plant	kg s^{-1}	varies	varies	variable
$\dot{E}_{furn}$	furnace CO_2	kg s^{-1}	varies	varies	variable
f_{fire}	share of gas fired	1	varies	varies	variable
F_C^{conv}	feed carbon of the conventional plant	mol s^{-1}	varies	varies	variable
$\dot{n}_{rej}^{conv}$	reject carbon burned	mol s^{-1}	varies	varies	variable
e_{st}	stoichiometric CO_2 per unit H_2	kg kg^{-1}	varies	varies	variable
e_{feed}	feed carbon intensity	kg kg^{-1}	varies	varies	variable
E_{cap}	CO_2 captured	t y^{-1}	varies	varies	variable
E_{el}	CO_2 from imported power	t y^{-1}	varies	varies	variable
EF_{grid}	grid emission factor, 0.402	t MWh^{-1}	varies	varies	variable
CI	direct carbon intensity	kg kg^{-1}	varies	varies	variable
$\mathcal{A}_{fate}$	abatement by tail-gas fate	t y^{-1}	varies	varies	variable
$\dot{n}_{exp}$	carbon leaving the boundary as fuel gas	mol s^{-1}	varies	varies	variable
$\mathcal{A}_{sys}$	system abatement	t y^{-1}	varies	varies	variable
$E_{lost\,steam}$	CO_2 to replace lost steam	t y^{-1}	varies	varies	variable
$E_{base\,power}$	CO_2 of the avoided baseline grid power	t y^{-1}	varies	varies	variable
$\mathcal{A}_{LC}$	lifecycle abatement	t y^{-1}	varies	varies	variable
E_{up}	saved upstream gas emissions	t y^{-1}	varies	varies	variable
E_{TS}	transport and storage burden	t y^{-1}	varies	varies	variable
e_{bk}	intensity in backup hours	kg kg^{-1}	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
e_{site}	site intensity	kg kg^{-1}	varies	varies	variable
EF	combustion emission factor	kg J^{-1}	varies	varies	variable
η_{bk}	backup firing efficiency, 0.85	1	varies	varies	variable
η_{cmp}	comparator capture	1	varies	varies	variable
$\dot{E}_{cap}^{cmp}$	comparator CO_2 captured	kg s^{-1}	varies	varies	variable
$\dot{E}_{em}^{cmp}$	comparator CO_2 emitted	kg s^{-1}	varies	varies	variable
CRF	capital recovery factor	y^{-1}	varies	varies	variable
y	year index	y	varies	varies	variable
C_{ann}	annualised capital	S\$ y^{-1}	varies	varies	variable
SFF	sinking fund factor	y^{-1}	varies	varies	variable
f_{IDC}	interest during construction factor	1	varies	varies	variable
t_c	construction time, 5.5	y	varies	varies	variable
c_{kWe}	BOAK overnight capital cost	US\$$_{2019}$ kWe^{-1}	varies	varies	variable
C_R	reactor capital	S\$	varies	varies	variable
F_{19}	US\$2019 to S\$2025 factor, 1.6195	S\$ US\$$^{-1}$	varies	varies	variable
m_{stage}	stage multiplier	1	varies	varies	variable
$\dot{Q}_{ref}$	reference size, 600	MW$_{\mathrm{th}}$	varies	varies	variable
C_{OM}	reactor O&M and fuel	S\$ y^{-1}	varies	varies	variable
c_{OM}	O&M rate, 10	US\$$_{2019}$ MWth^{-1} h^{-1}	varies	varies	variable
f_{fuel}	fuel share of OPEX, 0.4438	1	varies	varies	variable
C_{cap}	capture capital	S\$	varies	varies	variable
K_{cap}	capture capital per unit rate	S\$ t^{-1} h	varies	varies	variable
C_{FOM}	capture fixed O&M	S\$ y^{-1}	varies	varies	variable
C_{VOM}	capture variable O&M	S\$ y^{-1}	varies	varies	variable
F_{14}	EUR2014 to S\$2025 factor, 2.322	S\$ EUR^{-1}	varies	varies	variable
C_{TS}	transport and storage cost	S\$ y^{-1}	varies	varies	variable
C_{gas}	gas cost difference	S\$ y^{-1}	varies	varies	variable
HHV	gas higher heating value	J m^{-3}	varies	varies	variable
d	operating days, 310.25	d y^{-1}	varies	varies	variable
C_{el}	imported electricity cost	S\$ y^{-1}	varies	varies	variable
p_{el}	electricity price, 0.20	S\$ kWh^{-1}	varies	varies	variable
C_{ihx}	IHX and loop capital	S\$	varies	varies	variable
C_a	loop cost anchor	S\$	varies	varies	variable
C_{lic}	annualised licensing cost	S\$ y^{-1}	varies	varies	variable
C_{NRC}	design certification fee proxy	US\$	varies	varies	variable
FX	S\$ per US\$, 1.3065	S\$ US\$$^{-1}$	varies	varies	variable
C_{dec}	decommissioning fund payment	S\$ y^{-1}	varies	varies	variable
c_{dec}	decommissioning cost	US\$$_{2022}$ kWe^{-1}	varies	varies	variable
C_{IDC}	annualised construction interest	S\$ y^{-1}	varies	varies	variable
C_{pi}	power island capital	S\$	varies	varies	variable
c_{pi}	turbine share, 540	US\$$_{2019}$ kWe^{-1}	varies	varies	variable
C_{rec}	recycle train capital	S\$	varies	varies	variable
$C_{rec,A}$	recycle train capital at Case A	S\$	varies	varies	variable
$C_{ann,rec}$	recycle train annual line	S\$ y^{-1}	varies	varies	variable
C_{st}	lost steam line	S\$ y^{-1}	varies	varies	variable
C_{pw}	baseline power line	S\$ y^{-1}	varies	varies	variable
$\dot{Q}_{exp}^{conv}$	export steam of the unabated SMR	W	varies	varies	variable
$\dot{Q}_{exp}^{kept}$	export steam the plant keeps	W	varies	varies	variable

Symbol	Definition	SI units	Case B	Case A	Class
E_{conv}	electricity of the unabated SMR	W	varies	varies	variable
ΔC_k	cumulative annual cost difference at rung k	S\$ y^{-1}	varies	varies	variable
C_j	ledger line j	S\$ y^{-1}	varies	varies	variable
ΔC	annual cost difference	S\$ y^{-1}	varies	varies	variable
LAC	levelised abatement cost	S\$ t^{-1}	varies	varies	variable
NPV	net present value	S\$	varies	varies	variable
q^*	value of an input that meets the target	various	varies	varies	variable
AC_{cmp}	comparator cost	S\$ t^{-1}	varies	varies	variable
CAC_3	IEAGHG Case 3 avoidance cost	EUR t^{-1}	varies	varies	variable
$p_{TS,3}$	T&S price in the IEAGHG case, 23.2	S\$ t^{-1}	varies	varies	variable
$e_{cap,3}$	CO_2 captured per unit H_2	kg Nm^{-3}	varies	varies	variable
$e_{av,3}$	CO_2 avoided per unit H_2	kg Nm^{-3}	varies	varies	variable
NPV_x	present cost of plant x	S\$	varies	varies	variable
$c_{j,x}$	cash flow of line j on plant x	S\$ y^{-1}	varies	varies	variable
c_j^{rep}	purchase or credit of a replaced item	S\$	varies	varies	variable
$LCOH_x$	levelised cost of hydrogen of plant x	S\$ kg^{-1}	varies	varies	variable
NPV_{NUS}	present cost of NUS-H2	S\$	varies	varies	variable
NPV_{SMR}	present cost of the unabated SMR	S\$	varies	varies	variable
AC_{L5}	ledger abatement cost at L5	S\$ t^{-1}	varies	varies	variable
f_s	spend factor of the discrete schedule	1	varies	varies	variable
Θ	present worth of replacements per unit capital	1	varies	varies	variable
CRF_{25}	capital recovery factor at 25 y	y^{-1}	varies	varies	variable

1.2 Numbered equation list

Every equation in the document, grouped by section, in compact form. The full card for each one follows in its section.

Section A

A.1 General balance: $\frac{d}{dt}\int_V \rho\psi\,dV + \oint_S \rho\psi\,\mathbf{u}\cdot\mathbf{n}\,dS = -\oint_S \mathbf{j}_\psi\cdot\mathbf{n}\,dS + \int_V \dot{s}_\psi\,dV$ [R]

A.2 Total mass: $dM/dt = \sum_{in}\dot{m} - \sum_{out}\dot{m}$, SS gives $\sum_{in}\dot{m} = \sum_{out}\dot{m}$ [R]

A.3 Component: $dn_i/dt = \sum_{in}F_i - \sum_{out}F_i + \int_V R_i\,dV$ [R]

A.4 Source term: $R_i = \sum_j \nu_{ij} r_j$; catalytic form $R_i = \rho_B \sum_j \nu_{ij}\eta_j^{cat} r_j$ [R]

A.5 Elements: $\sum_i a_{ki}(F_{i,out} - F_{i,in}) = 0$, requires $\mathbf{a}\boldsymbol{\nu} = 0$ [R]

A.6 Independent reactions: $N_r = N_s - \mathrm{rank}(\mathbf{a})$. Reformer $5-3=2$, so $K_3 = K_1K_2$. MEA $9-4=5$ with charge as a row [R]

A.7 First law: $dE/dt = \dot{Q} - \dot{W}_{sh} + \sum_{in}\dot{m}(h+u^2/2+gz) - \sum_{out}\dot{m}(h+u^2/2+gz)$ [R]

A.8 Enthalpy balance: $\sum_{out}F_i h_i - \sum_{in}F_i h_i = \dot{Q} - \dot{W}_{sh}$ [R]

A.9 Species enthalpy: $h_i(T) = \Delta_f h_i^\circ + \int_{T_{ref}}^{T} c_{p,i}dT$, Shomate $c_{p,i} = A + Bt + Ct^2 + Dt^3 + E_S/t^2$, $t = T/1000$ [E]

A.10 Hess: $\Delta_r h_j(T) = \sum_i \nu_{ij} h_i(T)$ [R]

A.11 Momentum, channel: $dp/dz = -\rho u\,du/dz - (f_D/D)\rho u^2/2 - \rho g\sin\theta$ [R]

A.12 Packed bed, Ergun: $-dp/dz = \frac{G}{\rho D_p}\frac{1-\varepsilon_b}{\varepsilon_b^3}\left[\frac{150(1-\varepsilon_b)\mu}{D_p} + 1.75G\right]$ [E]

A.13 Plug-flow reduction: $dF_i/dz = A_c R_i$; $\sum_i F_i c_{p,i}\,dT/dz = \dot{q}' + A_c\sum_j(-\Delta_r h_j)r_j$ [R]

A.14 Gas closure: $p_i = y_i p$, $c = p/(ZRT)$ [R]
A.15 Residuals: ϵ_m, ϵ_k, ϵ_E relative to inlet flow. Limit 10^{-8} per unit, 10^{-6} flowsheet [F]

Section B

B.1 Helium density: $\rho = 48.14(P/T)/(1 + 0.4446P/T^{1.2})$ [E]
B.2 $c_p = 5195$, $c_v = 3117$ J/kg/K, $\kappa = 1.6667$ [E]
B.3 Viscosity: $\mu = 3.674 \times 10^{-7} T^{0.7}$ [E]
B.4 Conductivity: $k = 2.682 \times 10^{-3}(1 + 1.123 \times 10^{-3} P)T^{0.71(1-2\times10^{-4}P)}$ [E]
B.5 Reactor balance: $\dot{Q}_R - \dot{Q}_{ves} = \dot{m}_p c_p (T_{ROT} - T_{RI})$ [O]
B.6 Circulator rise: $T_{RI} = T_{h,out} + \dot{W}_p/(\dot{m}_p c_p)$ [O]
B.7 Primary: $\dot{Q}_p = \dot{m}_p c_p (T_{ROT} - T_{h,out})$, $T_{h,out} \geq T_{s,ret} + \Delta T_{app}$ [R]
B.8 Secondary: $\dot{Q}_s = \dot{m}_s c_p (T_{s,sup} - T_{s,ret})$ [R]
B.9 Flow need: $\dot{m}_s = \dot{Q}_{proc}/(c_p \Delta T_s)$, $\dot{m}_p = \dot{Q}_R/[c_p(T_{ROT} - T_{s,ret} - \Delta T_{app})]$ [R]
B.10 Compression: $\dot{W} = \dot{m} c_p T_1[(1 + \Delta p/p)^{(\kappa-1)/\kappa} - 1]/\eta_{is}$, exponent 0.4 [R]
B.11 Budget: 7.0 MPa, $\Delta p_s = 200$ kPa, $\Delta p_p = 150$ kPa. Low, high and 5.15 MPa sets [F]
B.12 Electric draw: $E_{circ} = (\dot{W}_s + \dot{W}_p)/\eta_m$, $\eta_m = 0.90$ [F]
B.13 Vessel loss: $\dot{Q}_{ves} = 0.81(\dot{Q}_R/30)^{2/3}$ MW; high case $0.027\dot{Q}_R$ [E]
B.14 Duct loss: $\dot{Q}_{duct} = 0.45(\dot{m}_s/78.49)^{1/2}$ MW at 90 m, 3.12 MW at 500 m [E]
B.15 IHX loss: $\dot{Q}_{ihx} = f_{ihx}(\dot{Q}_{proc} + \dot{Q}_{rec})$, $f_{ihx} = 0.008$, range 0 to 0.0458 [E]
B.16 Available heat: $\dot{Q}_{av} = \dot{Q}_R + \dot{W}_s + \dot{W}_p - \dot{Q}_{ves} - \dot{Q}_{ihx} - \dot{Q}_{duct}$ [O]
B.17 Efficiency: $\eta_{pc} = 0.397$; alternative $0.475 = 0.66(1 - T_c/T_h)$ [E]
B.18 Power branch: $E = \eta_{pc}\dot{Q}_{pc}$ [O]
B.19 Extraction: $w = \min(E, r_w \dot{Q}_{gap})$, $\dot{Q}_{br} = \dot{Q}_{gap} + w + (E - w)/\eta_{pc}$ [O]
B.20 Allocation: $\dot{Q}_R = \dot{Q}_{proc} + \dot{Q}_{rec} + \dot{Q}_{ihx} + \dot{Q}_{duct} + \dot{Q}_{ves} + \dot{Q}_{br} - \dot{W}_s - \dot{W}_p$ [O]
B.21 Margin: $\dot{Q}_{spare} = \dot{Q}_{rated} - \dot{Q}_R$. No revenue credited [F]
B.22 INL inlet fit: $T_{RI} = 0.8205 T_{ROT} - 242.68$ (C) [E]
B.23 INL excess heat: $\dot{Q}_{ex} = \dot{m}_s c_p (T_{s,ret} - T_{s,ideal})$ [R]

Section C

C.1 Hot side: $\dot{Q}_h = \dot{m}_p c_p (T_{h,in} - T_{h,out})$ [R]
C.2 Cold side: $\dot{Q}_c = \dot{m}_s c_p (T_{c,out} - T_{c,in})$ [R]
C.3 Closure: $\dot{Q}_h = \dot{Q}_c + \dot{Q}_{ihx}$ [R]
C.4 Network: $\frac{1}{UA} = \frac{1}{h_{fi} A_i} + \frac{R_{f,i}}{A_i} + \frac{\ln(d_o/d_i)}{2\pi k_w L N_t} + \frac{R_{f,o}}{A_o} + \frac{1}{h_{fo} A_o}$ [R]
C.5 Overall coefficient: $U_o = (UA)/A_o$ [R]
C.6 Nusselt: tube side Tochio Eq 8, shell side Eq 5 and 6 with C_0 [E]
C.7 LMTD: $\Delta T_{lm} = (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$, derived from $d\dot{Q} = U \Delta T \, dA$ [R]
C.8 Balanced limit: $\Delta T_{lm} \to \Delta T$ as $\Delta T_1 \to \Delta T_2$ [R]
C.9 Effectiveness: $\varepsilon_{HX} = \dot{Q}/[C_{min}(T_{h,in} - T_{c,in})]$ [R]
C.10 $NTU = UA/C_{min}$, $C^* = C_{min}/C_{max}$ [R]
C.11 Counterflow: $\varepsilon_{HX} = \frac{1 - e^{-NTU(1-C^*)}}{1 - C^* e^{-NTU(1-C^*)}}$ [R]
C.12 Balanced: $\varepsilon_{HX} = NTU/(1 + NTU)$ [R]
C.13 Area: $A = \dot{Q}/(U \Delta T_{lm})$ [R]
C.14 Module anchor: $UA_{mod} = UA/N_{mod}$, HTTR measured 85 kW/K [E]
C.15 Unbalanced option: choose $T_{h,out}$, get $\dot{m}_p$, cold end, LMTD, UA [O]
C.16 Tube friction: $\Delta p = f_D(L/D)\rho u^2/2$, $f_D = 0.3164 Re^{-1/4}$ [E]
C.17 Cascade: $T_{s,sup} = T_{ROT} - \Delta T_{app} - \Delta T_{duct}$ [F]
C.18 Constraints: $T_{s,sup} - T_{ref,out} \geq 10$ K, tube ≤ 955 C, tube $\Delta p \leq 0.3$ MPa [F]
C.19 Pressure sign: $p_s \geq p_p + 0.1$ MPa in HTTR. Project sign left to the safety case [F]

Section D

D.1 Reaction I: $CH_4 + H_2O \rightleftharpoons CO + 3H_2$, +206.1 kJ/mol [R]
D.2 Reaction II, shift: $CO + H_2O \rightleftharpoons CO_2 + H_2$, −41.15 kJ/mol [R]
D.3 Reaction III: $CH_4 + 2H_2O \rightleftharpoons CO_2 + 4H_2$, +164.9 kJ/mol, equal to I + II [R]
D.4 Heavier gas: $C_nH_m + nH_2O \to nCO + (n + m/2)H_2$ taken to completion at tube inlet [O]
D.5 Feed ratio: $S/C = F_{H_2O}/F_C$, 3.0 [F]

D.6 Extents, per mole CH4 fed: CH4 $1 - x_1$; CO $x_1 - x_2$; CO2 $\xi_{CO_2} + x_2$; H2 $\xi_{H_2} + 3x_1 + x_2$; H2O $S_C - x_1 - x_2$; total $\Sigma_0 + 2x_1$ [R]
D.7 $K_1(T) = 1.198 \times 10^{13} \exp(-26830/T)$ [E]
D.8 $K_2(T) = 1.767 \times 10^{-2} \exp(4400/T)$ [E]
D.9 $K_3 = K_1 K_2$ [R]
D.10 Quotients: $Q_1 = p_{H_2}^3 p_{CO}/(p_{CH_4} p_{H_2O})$, $Q_2 = p_{H_2} p_{CO_2}/(p_{CO} p_{H_2O})$, $Q_3 = Q_1 Q_2$ [R]
D.11 Equilibrium: solve $Q_1(x) = K_1(T)$, $Q_2(x) = K_2(T)$ at exit T, p [R]
D.12 Consistency: $d \ln K/d(1/T) = -\Delta_r h/R$ [R]
D.13 Rates: $k_i = A(k_i) \exp(-E_{a,i}/RT)$; $A = 4.225 \times 10^{15}$, 1.955×10^6, 1.020×10^{15}; $E_a = 240.1$, 67.13, 243.9 kJ/mol [E]
D.14 Adsorption: $K_j = A_j \exp(-\Delta H_j/RT)$; K_{CO} 8.23e-5, −70.65; K_{H_2} 6.12e-9, −82.90; K_{CH_4} 6.65e-4, −38.28; K_{H_2O} 1.77e5, +88.68 kJ/mol [E]
D.15 $DEN = 1 + K_{CO} p_{CO} + K_{H_2} p_{H_2} + K_{CH_4} p_{CH_4} + K_{H_2O} p_{H_2O}/p_{H_2}$ [E]
D.16 $r_1 = (k_1/p_{H_2}^{2.5})(p_{CH_4} p_{H_2O} - p_{H_2}^3 p_{CO}/K_1)/DEN^2$ [E]
D.17 $r_2 = (k_2/p_{H_2})(p_{CO} p_{H_2O} - p_{H_2} p_{CO_2}/K_2)/DEN^2$ [E]
D.18 $r_3 = (k_3/p_{H_2}^{3.5})(p_{CH_4} p_{H_2O}^2 - p_{H_2}^4 p_{CO_2}/K_3)/DEN^2$ [E]
D.19 Species: $r_{CO} = r_1 - r_2$, $r_{CO_2} = r_2 + r_3$, $r_{CH_4} = r_1 + r_3$ [R]
D.20 Effectiveness: $\eta_j^{cat} = \int \rho_s r_j \, dV/(\rho_s r_j^s V_p)$ [O]
D.21 Lumped activity: $\eta_1^{cat} = \eta_2^{cat} = \eta_3^{cat} = a_{cat}$, fitted on JAERI A-2 only [O]
D.22 Continuity: $dx_{CH_4}/dz = (\Omega \rho_B / F_{CH_4}^0)(\eta_1 r_1 + \eta_3 r_3)$ [O]
D.23 Continuity: $dx_{CO_2}/dz = (\Omega \rho_B / F_{CH_4}^0)(\eta_2 r_2 + \eta_3 r_3)$ [O]
D.24 Partial pressures from D.6: $p_i = p y_i$ [R]
D.25 Energy: $c_p \rho_g u_s \, dT/dz = \rho_B \sum_j (-\Delta_r h_j) \eta_j r_j + (4U/d_i)(T_w - T)$ [O]
D.26 Momentum: A.12 [E]
D.27 Wall: $1/U = 1/h_{fi} + R_f + t_w/k_w + (A_i/A_o)/h_{fo}$ [O]
D.28 Helium side: $\dot{m}_{He} c_p \, dT_{He}/dz = \pm \pi d_i N_t U (T_{He} - T)$ [O]
D.29 Finned annulus: $Nu = 0.039 Re^{0.87} Pr^{0.4} (S_f/D_e)^{0.4} (H_f/D_e)^{-0.19}$ [E]
D.30 Plain annulus: $Nu = 0.205 Re^{0.603} Pr^{0.33}$ [E]
D.31 Process side: $h_w = 0.4(k_g/D_p)(2.58 Re^{1/3} Pr^{1/3} + 0.094 Re^{0.8} Pr^{0.4})$ [E]
D.32 Radiation: $q = \sigma \varepsilon_{r,a} \varepsilon_{r,b} \pi d_o (T_a^4 - T_b^4)/A_{sh}$ [E]
D.33 Performance: $X_{CH_4} = x_1$ at exit; $Y_{H_2} = F_{H_2,out}/F_{CH_4,in}$ [R]
D.34 Tube duty: $\dot{Q}_{tube} = \sum F_i h_i|_{out} - \sum F_i h_i|_{in}$ [O]
D.35 Tube count: $N_t = \dot{Q}_{tube}/(\pi d_o L \, q'')$, $q'' = 40$ to 60 kW/m2 [O]
D.36 Hot end: $\Delta T_{hot} = T_{s,sup} - T_{ref,out} = 29$ K. Sensitivity 825 C, 800 C [F]

Section E

E.1 CO conversion: $X_{CO} = (F_{CO,in} - F_{CO,out})/F_{CO,in}$ [R]
E.2 Species: $F_i = F_{i,in} + \nu_i \xi$ with ν for D.2 [R]
E.3 Constraint: $Q_2(T_{out}) \le K_2(T_{out})$ [R]
E.4 Approach: $Q_2(T_{out}) = K_2(T_{out} + \Delta T_{eq})$, $\Delta T_{eq} = 10$ K [O]
E.5 Adiabatic bed: $\sum F_i h_i|_{out} = \sum F_i h_i|_{in}$ [R]
E.6 Heat release: $\dot{Q}_{WGS} = \xi(-\Delta_r h_2)$ [R]
E.7 HTS rate: $dp_{CO}/dt = -k(p_{CO} - p_{CO,eq})$ [E]
E.8 Integrated: $k = S_v \ln[1/(1 - C/C_e)]$ [E]
E.9 $k(T)$: interpolate Table 1, 310 to 450 C, 16 to 31 bar [E]
E.10 Catalyst volume: $V_{cat} = (\dot{V}/k) \ln[1/(1 - C/C_e)]$ [O]
E.11 Profile: $\sum F_i c_{p,i} \, dT/d\xi = -\Delta_r h_2$ [O]
E.12 Drop: fixed 17 psi across each bed, Ergun as check [F]
E.13 Steam ratio: $S/DG = F_{H_2O}/\sum_{dry} F_i$ [F]
E.14 Window: HTS inlet ≥ 600 F; LTS inlet ≥ 375 F and 27 F above dew [F]
E.15 LTS model: E.4 only. No rate law [M4] [O]
E.16 Listed, not used: Kraussler power law, Table 3 orders [E]

Section F

F.1 Sensible: $\dot{Q} = \sum_i F_i [h_i(T_1) - h_i(T_2)]$ [R]
F.2 Train: sum over HPS-GEN1, EX1, EX2, EX3, EX4, KO-DRM-1 [R]
F.3 Dew point: $y_{H_2O} p = p_{sat}(T_{dp})$ [R]
F.4 Saturation: IAPWS-IF97 via seuif97 [E]

F.5 Water VLE: $y_w p = x_w \gamma_w p_{sat}(T)$, $x_w \approx 1$ [R]
F.6 Dissolved gas: $x_{CO_2} = y_{CO_2} p / H_{CO_2}$, Aronu Eq 17 [E]
F.7 Flash: $\sum_i z_i(K_i - 1)/[1 + V(K_i - 1)] = 0$ [R]
F.8 Mass: $F = V + L$, $F z_i = V y_i + L x_i$ [R]
F.9 Energy: $\dot{Q} = V h_V + L h_L - F h_F$ [R]
F.10 Condensate split: $L = L_{ret} + L_{treat}$ [R]
F.11 Heat above T_{min}: $\dot{Q}(T_{min}) = \int_{T_{min}}^{149\,C} \dot{C}\, dT + \dot{m}_{cond} \Delta h_{vap}$ [O]
F.12 CO2 lost in condensate, kept outside the ledger [O]
F.13 Absorber feed: dry composition after KO, y_{CO_2}, y_{CO}, y_{CH_4}, y_{H_2} [R]

Section G

G.1 Reactions: $2H_2O \rightleftharpoons H_3O^+ + OH^-$; $2H_2O + CO_2 \rightleftharpoons H_3O^+ + HCO_3^-$; $H_2O + HCO_3^- \rightleftharpoons H_3O^+ + CO_3^{2-}$; $H_2O + MEAH^+ \rightleftharpoons H_3O^+ + MEA$; $H_2O + MEACOO^- \rightleftharpoons MEA + HCO_3^-$ [R]
G.2 $\ln K$ or $\ln H = C_1 + C_2/T + C_3 \ln T + C_4 T$ [E]
G.3 Chemical potential: $\mu_i = \mu_i^\circ(T) + RT \ln(\gamma_i x_i)$ [R]
G.4 $RT \ln K_j = -\sum_i \nu_{ij} \mu_i^\circ$ [R]
G.5 Equilibrium: $\ln K_j = \sum_i \nu_{ij} \ln(\gamma_i x_i)$ [R]
G.6 Standard states: H2O and MEA pure; solutes $\gamma_i^* = \gamma_i / \gamma_i^\infty$ in water [R]
G.7 Hidden convention: $\ln K_{MEA,eff} = \ln K_{MEA} + \ln \gamma_{MEA}^\infty$ [O]
G.8 MEA balance: $n_{MEA} + n_{MEAH^+} + n_{MEACOO^-} = n^0$ [R]
G.9 Carbon: $n_{CO_2} + n_{HCO_3^-} + n_{CO_3^{2-}} + n_{MEACOO^-} = \alpha n^0$ [R]
G.10 Hydrogen element balance over all nine species [R]
G.11 Electroneutrality: $n_{H_3O^+} + n_{MEAH^+} = n_{OH^-} + n_{HCO_3^-} + 2n_{CO_3^{2-}} + n_{MEACOO^-}$ [R]
G.12 Loading: α = total CO2 / total MEA [R]
G.13 $\ln \gamma_i = \ln \gamma_i^C + \ln \gamma_i^R + \ln \gamma_i^{DH}$ [R]
G.14 Combinatorial: $\ln \gamma_i^C = \ln(\phi_i/x_i) + 1 - \phi_i/x_i - (z/2) q_i [\ln(\phi_i/\theta_i) + 1 - \phi_i/\theta_i]$, $z = 10$ [R]
G.15 Residual: $\ln \gamma_i^R = q_i [1 - \ln \sum_k \theta_k \psi_{ki} - \sum_k \theta_k \psi_{ik} / \sum_l \theta_l \psi_{lk}]$, $\psi_{ki} = \exp[-(u_{ki} - u_{ii})/T]$ [R]
G.16 $u_{ki} = u_{ki}^0 + u_{ki}^T (T - 298.15)$ [E]
G.17 Debye-Hueckel: $\ln \gamma_i^{DH} = -z_i^2 A \sqrt{I} / (1 + b\sqrt{I})$, water term $(2/3) M_w A I^{3/2} \sigma(b\sqrt{I})$, $b = 1.5$, $A(T) = 1.131 + 1.335 \times 10^{-3} t + 1.164 \times 10^{-5} t^2$ [R]
G.18 Ionic strength: $I = \frac{1}{2} \sum_i m_i z_i^2$ [R]
G.19 CO2 VLE: $\varphi_{CO_2} y_{CO_2} p = \gamma_{CO_2}^* x_{CO_2} H \exp[v^\infty (p - p_s)/RT]$ [R]
G.20 H2O and MEA VLE: $\varphi_i y_i p = \gamma_i x_i p_i^s \varphi_i^s \exp[v_i (p - p_i^s)/RT]$ [R]
G.21 Vapour: SRK fugacity [R]
G.22 Henry in water: $H_w = 2.82 \times 10^6 \exp(-2044/T)$ [E]
G.23 N2O analogy: $H_{app} = H_{N_2O} H_{w,CO_2} / H_{w,N_2O}$ [E]
G.24 Heat of absorption: $\Delta h_{abs} = -R(\partial \ln p_{CO_2}^* / \partial(1/T))_x$ [O]
G.25 Solvent heat capacity: $c_{p,s} = 3.6$ kJ/kg/K [F]
G.26 Density (Weiland): $V = \sum_i x_i V_i + x_M x_W V^*$, $V_M = 61.08/(aT^2 + bT + c)$, $V_{CO_2} = 0.04747$ mL/mol [E]
G.27 Viscosity (Weiland): $\mu = \mu_w \exp\{(21.186\Omega + 2373)[\alpha(0.01015\Omega + 0.0093T - 2.2589) + 1]\Omega/T^2\}$ [E]
G.28 Diffusivity: $D_w = 2.35 \times 10^{-6} \exp(-2119/T)$, $D_l = D_w (\mu_w/\mu)^{0.8}$ [E]
G.29 Free MEA: n_{MEA}/n^0 from G.5 to G.11 [O]
G.30 Comparator: Dugas Eq 5.26 [E]
G.31 Plant point: $\alpha^*(p_{CO_2}, T)$, 0.741 at 378 kPa, 40 C [O]
G.32 Guards: T 313 to 393 K, $\alpha \le 0.50$ in columns, $p_{CO_2} \le 10$ bar [F]
G.33 Table $p^*(T, \alpha)$ for column calls, bilinear in $\ln p$ [O]

Section H

H.1 Gas film: $N_{CO_2} = k_G (p_{CO_2} - p_i)$ [O]
H.2 Liquid film: $N_{CO_2} = E k_L^\circ (p_i - p^*)/H$ [O]
H.3 Interface: $p_i = H C_i$ [R]
H.4 Overall: $N_{CO_2} = K_G (p_{CO_2} - p^*)$, $1/K_G = 1/k_G + H/(E k_L^\circ)$ [O]
H.5 $\log_{10} k_2 = 10.99 - 2152/T$ [E]
H.6 $M = D_l k_2 C_B / (k_L^\circ)^2$, $E_1 = \sqrt{M} / \tanh \sqrt{M}$ [O]
H.7 $E_\infty = 1 + C_B D_B / (b D_l C_i)$ [O]
H.8 Wellek: $E = 1 + 1/[(1/(E_\infty - 1))^{1.35} + (1/(E_1 - 1))^{1.35}]^{1/1.35}$ [O]
H.9 $H = 10^{5.3 - 0.035 C_{MEA} - 1140/T}$ [E]

H.10 $k_L = 2(D_L C_E U_L/\pi S)^{1/2}$, $C_E = 0.9$ [E]
H.11 $k_G = 0.054(D_G/S)[(U_L + U_G)\rho_G S/\mu_G]^{0.8}(\mu_G/D_G\rho_G)^{0.33}$ [E]
H.12 Area, central: Billet and Schultes $a_e/a = C_h(U_L\rho_L/a\mu_L)^{0.25}(U_L^2 a/g)^{0.1} \times 0.85$, $Re_L \geq 5$, $C_h = 0.554$ [E]
H.13 Area, alternatives: Rocha with $F_{SE} = 0.25$ or 0.35, Wang, Gualito [E]
H.14 Gas: $dG/dz = -(N_{CO_2} + N_{H_2O})aA_c$ [O]
H.15 $dy_{CO_2}/dz = [N_{CO_2} aA_c(y_{CO_2} - 1) + N_{H_2O} y_{CO_2} aA_c]/G$ [O]
H.16 $dy_{H_2O}/dz = [N_{H_2O} aA_c(y_{H_2O} - 1) + N_{CO_2} y_{H_2O} aA_c]/G$ [O]
H.17 $dL/dz = -N_{H_2O} aA_c$ [O]
H.18 $dX_{CO_2}/dz = (N_{H_2O} X_{CO_2} - N_{CO_2})aA_c/L$; $dX_{H_2O}/dz = N_{H_2O}(X_{H_2O} - 1)aA_c/L$ [O]
H.19 $dT_G/dz = -qaA_c/(Gc_{p,G})$ [O]
H.20 dT_L/dz: terms in q, $N_{CO_2} H_{CO_2} + N_{H_2O} H_{H_2O}$ and $(N_{CO_2} c_{p,CO_2} + N_{H_2O} c_{p,H_2O})(T_L - T_G)$, all over $Lc_{p,L}$ [O]
H.21 Heat flux: $q = h(T_G - T_L)$, $h = k_G c_G\, c_{p,G}(Sc/Pr)^{2/3}$ [O]
H.22 Water flux: $N_{H_2O} = k_{G,H_2O}(p_{H_2O} - p^*_{H_2O})$ [O]
H.23 Drop: $\Delta p/Z = (\Delta p/Z)_{ref}(F/F_{ref})^2$, $F = u_G\sqrt{\rho_G}$ [F]
H.24 Grid: N finite volumes, upwind in each phase [O]
H.25 Boundary: gas in at bottom, liquid in at top [O]
H.26 Solve: damped Newton on the banded system, $\|res\|_\infty < 10^{-8}$ [O]
H.27 Per-pass: $\eta_{pp} = (y_{in} n_{in} - y_{out} n_{out})/(y_{in} n_{in})$ [R]
H.28 Solvent: $\dot{m}_{solv} = \dot{n}_{CO_2} M_{MEA}/[w(\alpha_{rich} - \alpha_{lean})]$ [R]
H.29 Diameter from liquid load, 40 to 100 m3/m2/h [O]
H.30 Co-absorption: zero central. Reference: aMDEA 0.056% H2, 0.137% CH4 [F]
H.31 Listed, not used: Isa index f, Eq 42 [E]

Section I

I.1 CO2 balance: $\dot{n}_{CO_2} = \dot{n}_{MEA}(\alpha_{rich} - \alpha_{lean})$ [R]
I.2 Desorption driving force: H.4 with sign reversed [O]
I.3 Equilibrium stage: $y_{CO_2} p = p^*(T, \alpha)$ on each of N stages [O]
I.4 Rich-end pinch: $n_{H_2O}/n_{CO_2} = (p - p^*_{CO_2})/p^*_{CO_2}$ at top [O]
I.5 Lean-end: T_{reb} solves $p^*_{CO_2} + p^*_{H_2O} + p^*_{MEA} = p_{strip}$ at α_{lean} [R]
I.6 Reboiler: $\dot{Q}_{reb} = \sum_{out} \dot{n}h - \sum_{in} \dot{n}h$ [R]
I.7 Condenser: $\dot{Q}_{cond} = \dot{Q}_{H_2O} + \dot{Q}_{steam} + \dot{Q}_{CO_2}$ [R]
I.8 Reflux: condensate returned equals water in overhead at T_{cond} less vent water [R]
I.9 Circulation: H.28 [R]
I.10 Rich-lean exchanger: $\dot{m}_r c_p(T_{r,out} - T_{r,in}) = \dot{m}_l c_p(T_{l,in} - T_{l,out})$, approach 10 K [R]
I.11 Steam: $\dot{m}_{st} = \dot{Q}_{reb}/\Delta h_{cond}$ [R]
I.12 SRD: $SRD = \dot{Q}_{reb}/\dot{m}_{CO_2}$ [R]
I.13 Split: $SRD = q_{des} + q_{sens} + q_{strip}$ [O]
I.14 Carried: 3.8, range 3.4 to 4.2 [F]
I.15 Stripping intensity: $RD \cdot 3600/\dot{m}_{rich}$ [R]
I.16 Reboiler area: $RD = UA_r \Delta T_{LMTD}$ [R]
I.17 Source temperature: $T_{source} \geq T_{reb} + \Delta T_{min}$, steam near 145 to 153 C [F]
I.18 Listed, not carried: interheat, total heat unchanged, 3.77 GJ/t [F]

Section J

J.1 Scale: $s = \dot{m}_{H_2,B}/\dot{m}_{H_2,A} = 8988/13154$ [F]
J.2 $\dot{m} = \rho_N \dot{V}_N$, $\rho_N = 0.08988$ kg/Nm3 [R]
J.3 Annual: $M_{H_2} = \dot{m} t_h$, $t_h = 8760 f_{av}$ [R]
J.4 Gas: 1 MMSCFD $= 10^6/(379.49 \cdot 24)$ lbmol/h [R]
J.5 Feed carbon: $F_C = 1.05379\, \dot{n}_{NG}$ [R]
J.6 Split: $\phi_c = F_{CO_2,abs}/T_C = 0.775$, range 0.76 to 0.79 [O]
J.7 INL fraction: $T_6 = F_{cap,6}/(\eta_{Sel}\phi_c)$, $G_6 = T_6 - F_{cap,6} - P$, $f_r = (T_6 - F_C)/G_6$ [O]
J.8 Tube carbon: $T_C = (F_C - f_r P)/[1 - f_r(1 - \eta_{pp}\phi_c)]$, from $T_C = F_C + f_r G$ [O]
J.9 Captured: $C = \eta_{pp}\phi_c T_C$ [O]
J.10 Tail gas: $G = T_C - C - P$; recycled $f_r G$; burned $(1 - f_r)G$ [O]
J.11 Closure: $F_C = C + (1 - f_r)G + P$ [R]

J.12 PSA block, unsimulated: $F_{H_2,prod} = \xi_{PSA} F_{H_2,in}$, $\xi_{PSA} = 0.88$; reject crosses the boundary (project decision) [X]
J.13 Stream ledger: reformer syngas, H2-rich gas after MEA, product after PSA, recovery, annual H2, each a separate row [R]
J.14 Process heat: $\dot{Q}_{proc} = 176.8\,s$ [O]
J.15 Slip heat: $\dot{Q}_{rec} = \Delta F_{slip} c_{p,CO_2} \Delta T$, 50 J/mol/K, 571 K [O]
J.16 Reboiler: $\dot{Q}_{reb} = \dot{m}_{CO_2} SRD$ [R]
J.17 Stack: freed Selexol $\dot{Q}_{Sel}$, export $\dot{Q}_{exp}$, syngas above 130 C, in that order [O]
J.18 Gap: $\dot{Q}_{gap} = \max(0, \dot{Q}_{reb} - \dot{Q}_{stack})$. Unmet thermal demand if the reactor cannot cover it [O]
J.19 Rows: reforming, PSA, steam, cooling, water scaled from Table 2. Selexol row removed [O]
J.20 Lean pump: $E = \dot{m}_{solv}(p_{abs} - p_{strip})/(\rho \eta_{pump} \eta_m)$ [O]
J.21 Fans: $E = k_{cool} \dot{Q}_{reb}$, 0.015 [O]
J.22 Compression: $E = 6.4\,C/C_6$ MWe to 2,015 psia [O]
J.23 Recycle row: $E_{pur} = 5.7\,s(1 + \Delta F_{slip}/1473)$ [O]
J.24 Total: $E_{tot} = E_{proc} + E_{circ}$ [R]
J.25 Primary flow depends on $\dot{Q}_R$, so B.9, B.10, B.13 and B.20 are coupled [O]
J.26 Fixed point: iterate B.20 to 10^{-9} MW [O]
J.27 Minimum capacity with corners [O]
J.28 Heat basis sensitivity: replace J.14 by 143.4 MWth [O]
J.29 Fixed rating: $E_{imp} = E_{nuc} - \max(0, \dot{Q}_{rated} - \dot{Q}_{heat})\eta_{pc}$. Unmet electrical demand [O]
J.30 Pinch: stack grades against I.17, hot water loop excluded [F]
J.31 Cooling: $\dot{Q}_{cool} = \dot{Q}_{KO} + \dot{Q}_{HW} + \dot{Q}_{lean} + \dot{Q}_{cond} + (1 - \eta_{pc})\dot{Q}_{pc}$ [O]
J.32 Thermal efficiency: $\eta_{th} = \dot{m}_{H_2} LHV_{H_2}/(\dot{n}_{NG} LHV_{NG} + \dot{Q}_R)$ [O]
J.33 Capture: print η_{pp} and $\eta_{pl} = C/F_C$ together [R]
J.34 Yield: $\dot{m}_{H_2}/\dot{m}_{NG}$, and scf H2 per scf NG [R]
J.35 Walk: 206.0, then purge, losses, circulators, MEA swap, reboiler gap [O]
J.36 Comparison: rated 210 MWth against feasible 179.0 MWth, cost effect L.26 [O]

Section K

K.1 Baseline: $e_{base} = \dot{E}_{base,A}/\dot{m}_{H_2,A}$ from 3,205 st/d [R]
K.2 Bounds: 9.35 Cho, 9.00 IEAGHG [F]
K.3 Annual: $E_{base} = e_{base} M_{H_2}$ [R]
K.4 Furnace CO2: fired feed plus reject carbon times M_{CO_2} [R]
K.5 Process minimum: $e_{st} = M_{CO_2}/(4M_{H_2}) = 5.46$ [R]
K.6 Carbon bound: $e_{feed} = F_C M_{CO_2}/\dot{m}_{H_2} = 5.97$ [R]
K.7 Captured: $E_{cap} = C M_{CO_2} t_h$ [R]
K.8 Residual: $E_{atm} = (1 - f_r) G M_{CO_2} t_h$ [R]
K.9 Electricity: $E_{el} = E_{imp} t_h EF_{grid}$, 0.402 t/MWh [O]
K.10 Hours: $t_h = 8760 f_{av}$ [R]
K.11 Net abatement: $\mathcal{A} = E_{base} - E_{atm}$ [R]
K.12 Bounds: replace K.1 by K.2 [R]
K.13 Intensity: $CI = E_{atm}/M_{H_2}$ [R]
K.14 Pair: K.11 printed with η_{pp} and η_{pl} [R]
K.15 Capture sensitivity: $\mathcal{A}(\eta_{pp})$ through J.8 [O]
K.16 Availability: $\mathcal{A} \propto f_{av}$ [R]
K.17 Tail-gas fates: recycle, burn on site, export, each with atmosphere view. Exclusion by boundary never removes carbon [O]
K.18 System view: $\mathcal{A}_{sys} = \mathcal{A} - E_{lost\,steam} + E_{base\,power}$ [O]
K.19 Lifecycle: add upstream gas 11.5 kg/GJ, less nuclear 5.5 kg/MWh, less T&S 2.5% [O]
K.20 Backup firing: extra H2 in $(1 - f_{av})$ hours at $e_{bk} = e_{site} + \dot{Q}_{proc} EF/(\eta_{bk} \dot{m}_{H_2})$ [O]
K.21 Comparator: SMR plus CCS, new-build, same H2, one labelled row [O]
K.22 Target test: $\mathcal{A} > 0.25$ Mt/y [R]

Section L

L.1 $CRF = i/[1 - (1 + i)^{-t_\ell}]$, from an annuity sum [R]
L.2 Annualised capital: $C_{ann} = C\,CRF$ [R]
L.3 Sinking fund: $SFF = i/[(1 + i)^{t_\ell} - 1]$ [R]
L.4 IDC: $f_{IDC} = [(1 + i)^{t_c} - 1]/(it_c)$ [R]

L.5 OCC: $c_{kWt} = c_{kWe}(1 - 0.09) \cdot 0.40$ [R]
L.6 Reactor: $C_R = c_{kWt}\dot{Q}_{ref}F_{19}m_{stage}(\dot{Q}_{rated}/\dot{Q}_{ref})^{n_s}$ [O]
L.7 Stage: BOAK 1.0, FOAK 1.6 (1.3 to 2.1) [O]
L.8 Learning, route p46: $m = 0.9^{\log_2 10} = 0.7047$ from BOAK [O]
L.9 Learning, route p47: $m = 1.6 \cdot 0.7047 = 1.128$ from FOAK [O]
L.10 Size exponent: 1.0; 0.6 [A]; $0.31 = 1 + \ln(26553/18313)/\ln(350/600)$ [C] [E]
L.11 Reactor O&M: $C_{OM} = c_{OM}F_{19}t_h[f_{fuel}\dot{Q} + (1 - f_{fuel})\dot{Q}_{ref}(\dot{Q}/\dot{Q}_{ref})^{n_s}]$ [O]
L.12 Capture capital: $C_{cap} = K_{cap}\dot{m}_{CO_2}$, $K = 2.04$ S$M per t/h (1.74 to 5.09) [O]
L.13 Capture fixed O&M: $0.027C_{cap}$, 6% high [F]
L.14 Variable O&M: MEA make-up unpriced central; reference chemicals EUR 0.1M/y [F]
L.15 T&S: $C_{TS} = p_{TS}E_{cap}$, 60 (23.2 to 112) [F]
L.16 Gas saving: $C_{gas} = -(52.5s - 34.0s)\,HHV\,d\,p_{gas}$ [R]
L.17 Electricity: self-supplied, $C_{el} = E_{imp}t_h p_{el}$ only when importing [O]
L.18 IHX: duty over 170 MW times S$ anchor on CRF 25 [E]
L.19 Licensing: US$50M (0 to 137.5) on CRF [E]
L.20 Decommissioning: $1000(0.40\dot{Q})FX\,SFF$ [E]
L.21 IDC term: $C_R(f_{IDC} - 1)CRF$ [O]
L.22 Power island: US$540/kWe on E_{tot} [O]
L.23 Recycle train: S$71.5M at Case A, scaled by s [E]
L.24 Lost steam and baseline power, L5 [O]
L.25 Incremental cost: $\Delta C = \sum$ ladder rung L0 to L5 [R]
L.26 $AC = \Delta C/\mathcal{A}$ [R]
L.27 Table: n_s at 1.0, 0.6, 0.31 side by side, BOAK and FOAK, 25 y and 60 y [R]
L.28 Levelised abatement cost: $LAC = \text{NPV}(\Delta C)/\text{NPV}(\mathcal{A})$ [O]
L.29 Sensitivity: one-at-a-time over gas, T&S, n_s, stage, capital, availability [O]
L.30 Discount sweep: 5, 8, 10% through L.1, L.3, L.4 [O]
L.31 Levers to S$100/t: solve for single input [O]
L.32 Comparator: SMR plus CCS row, same ladder [O]
L.33 Guards: no double count of capture energy; FX 1.6195 and 2.322 factors; carbon tax printed apart [F]
L.34 Discounted cash flow: $\text{NPV}_x = \sum_j \sum_y c_{j,x}(y)(1 + i)^{-y}$, 25 y items bought again with a straight-line credit at the horizon [R]
L.35 Levelised cost of hydrogen: $LCOH_x = \text{NPV}_x/\text{NPV}(M_{H_2})$ [R]
L.36 Levelised abatement cost: $LAC = (\text{NPV}_{NUS} - \text{NPV}_{SMR})/(\mathcal{A}\sum_y(1 + i)^{-y})$ [R]
L.37 Gap test: $g = (LAC - AC_{L5})/AC_{L5}$, $|g| \leq 0.10$ [R]

1.3 Assumptions and parameter ranges

Parameter	Central	Range	Basis and status
Plant basis	Case B	Case A reference	Project decision
Availability	0.85	0.65 to 0.90	Assumed. Published studies assume 90 to 92%, and no measured HTGR value was found. Hydrogen runs only in those hours; backup firing is a sensitivity
Process heat basis	120.8 MW_{th}	143.4 MW_{th} (Ahn Table 6)	Surplus against the rating runs about 8 to 31 MW_{th}, and both ends are printed
IHX approach	25 K	50 K	INL assumption (TEV-961 printed p. 10). No measurement below 77 K
Secondary pressure	7.0 MPa	5.15 MPa	Assumed, because INL sources conflict (9.1 and 7 MPa)
Reformer exit	871 C, S/C 3.0, 30.3 bar	825 and 800 C	Case 6 S/C and pressure are not in the local file, so Case 5 values are used and disclosed
Shift	HTS plus LTS, 10 K approach	HTS only	INL and IEAGHG anchors
Solvent	30 wt% MEA	Selexol, aMDEA	Project decision. The literature on solvent choice for syngas is mixed
SRD	3.8 GJ t^{-1}	3.4 to 4.2	Judgement from flue gas pilots. No syngas measurement exists, and the direction of any bias is unknown

Parameter	Central	Range	Basis and status
Rich loading	0.50	0.55 upper case	Cap: no kinetics or properties above 0.50
Absorber	40 C, 26.8 bar	none	Case 5, TEV-961 A-53
Steam order	Freed Selexol steam, export steam, syngas above 130 C, then reactor steam	ZLD steam freed	Project decision
Baseline	9.21 kg kg^{-1}	9.00 to 9.35	Project decision
Reactor stage	BOAK	FOAK, 10-OAK by both INL routes	The route is stated on every result
Size exponent	1.0	0.6, 0.31	Project decision. 0.6 is an earlier project assumption. 0.31 is a ratio of two FOAK estimates and so a weak adverse bound
Gas price	S$15/GJ	10 to 20	Project basis
T and S	S$60/t	23.2 to 112	Project decision
Discount rate	8%	5% and 10%	IEAGHG 2017-02 PDF p. 12 basis

2 Variables

Each block below expands one row of the seed table: what the quantity means, its typical value, its units, the job it does in the model and the caveat that goes with it. Symbols that appear in a single card are defined on that card and listed in the Reference Frame table.

Plant scale and product

Variable: Scale factor (s)

Ratio of the Case B hydrogen mass flow to the INL Case A hydrogen mass flow. *Typical value:* 0.6833 at Case B and 1.0000 at Case A. *Units:* Dimensionless. *Role:* Scales INL stream data and flowsheet duties to the headline plant size. *Caveat:* Linear scaling assumes the INL flowsheet keeps its ratios at the smaller size. Every scaled row says so on its card.

Variable: Hydrogen product rate ($\dot{m}_{H_2}$)

Hydrogen leaving the external PSA boundary. *Typical value:* 2.497 at Case B and 3.654 at Case A. *Units:* kg s^{-1}. *Role:* Sets the size of every downstream ledger. *Caveat:* Stated at the external PSA boundary, because the PSA is outside the simulated plant.

Variable: Annual hydrogen (M_{H_2})

Hydrogen sold in a year at the assumed availability. *Typical value:* 66,925 at Case B and 97,946 at Case A. *Units:* t y^{-1}. *Role:* Denominator of intensity and cost per kilogram. *Caveat:* Hydrogen runs only in the availability hours. Backup firing is a sensitivity, so these tonnes carry no firing emissions.

Variable: Availability (f_{av})

Fraction of the year at rated output. *Typical value:* 0.85, with a range of 0.65 to 0.90. *Units:* Dimensionless. *Role:* Converts rates to annual totals through t_h. *Caveat:* Assumed. Published studies assume 90 to 92%, and no measured HTGR availability was found.

Variable: Annual hours (t_h)

Operating hours in a year, $8760 f_{av}$. *Typical value:* 7,446. *Units:* h y^{-1}. *Role:* Links hourly flows to annual ledgers. *Caveat:* Moves one for one with f_{av}.

Variable: Natural gas rate ($\dot{V}_{NG}$)

Natural gas drawn across the plant boundary. *Typical value:* 23.2 at Case B and 34.0 at Case A. *Units:* MMSCFD, with 1 MMSCFD = 13.83 mol s^{-1} at 60 °F. *Role:* Sets feed carbon and the gas bill. *Caveat:* The Case A value rests on the INL recycle of tail gas. Without recycle the demand rises.

Reactor and helium loops

Variable: Selected rating ($\dot{Q}_{rated}$)

Reactor thermal rating chosen for the plant. *Typical value:* 210 at Case B. Case A has no selected rating. *Units:* MW_{th}. *Role:* Sets reactor capital in the cost ladder. *Caveat:* A project choice. The rating earns no revenue above what the plant draws.

Variable: Reactor heat required ($\dot{Q}_R$)

Thermal power the plant actually draws from the core, from Eq. B.20. *Typical value:* 178.5 at Case B and 260.7 at Case A. *Units:* MW_{th}. *Role:* Closes the nuclear island balance. *Caveat:* Depends on η_{pc}, whose net or gross status is open.

Variable: Spare margin ($\dot{Q}_{spare}$)

Rating less requirement, Eq. B.21. *Typical value:* 31.5 at Case B. At Case A the value is -50.7, so 210 MW_{th} cannot carry that plant. *Units:* MW_{th}. *Role:* Shows how much of the rating sits idle. *Caveat:* Credited with no revenue.

Variable: Reactor outlet temperature (T_{ROT})

Helium temperature leaving the core. *Typical value:* 1198.15 (925 C). *Units:* K. *Role:* Top of the temperature cascade in Eq. C.17. *Caveat:* Taken from the INL Case 6 basis. No HTGR has run a long campaign at this value with a process heat load.

Variable: Secondary supply temperature ($T_{s,sup}$)

Secondary helium leaving the IHX for the reformer. *Typical value:* 1173.15 (900 C). *Units:* K. *Role:* Hot end of the reformer heat source. *Caveat:* Assumes a 25 K IHX approach and no duct drop.

Variable: Secondary return temperature ($T_{s,ret}$)

Secondary helium returning to the IHX. *Typical value:* 739.15 (466 C). *Units:* K. *Role:* Sets the secondary flow through Eq. B.8. *Caveat:* Published for INL Case 6 at its own preheat train. A different train moves it.

Variable: Reformer exit temperature ($T_{ref,out}$)

Process gas temperature at the catalyst exit. *Typical value:* 1144.15 (871 C). *Units:* K. *Role:* Sets methane conversion through Eq. D.11. *Caveat:* An INL Gibbs result with no tube model behind it. Sensitivities at 825 and 800 C are carried.

Variable: Secondary helium flow ($\dot{m}_s$)

Mass flow in the intermediate loop. *Typical value:* 53.6 at Case B on process heat alone, 54.0 with the slip heat, and 78.49 published at Case A. *Units:* kg s^{-1}. *Role:* Sets the secondary circulator and IHX. *Caveat:* Published by INL at Case A only.

Variable: Primary helium flow ($\dot{m}_p$)

Mass flow through the core at $\dot{Q}_R$. *Typical value:* 79.2 at Case B and 116 at Case A. *Units:* kg s^{-1}. *Role:* Sets the primary circulator and core rise. *Caveat:* Calculated. INL publishes no primary flow.

Variable: Process heat ($\dot{Q}_{proc}$)

Heat handed to the process by secondary helium. *Typical value:* 120.8 at Case B and 176.8 at Case A. *Units:* MW_{th}. *Role:* Largest term in the reactor balance. *Caveat:* Ahn and Lee print 143.4 MW_{th} on a comparable duty, so this basis has a stated upper case.

Variable: Slip recycle heat ($\dot{Q}_{rec}$)

Sensible heat to bring recycled CO_2 slip back to reformer conditions. *Typical value:* 0.88 at Case B and 1.28 at Case A. *Units:* MW_{th}. *Role:* Added to the secondary helium load. *Caveat:* Calculated. INL has no equivalent stream because Selexol slip differs.

Variable: Circulator shaft powers ($\dot{W}_s$, $\dot{W}_p$)

Shaft work into the secondary and primary helium. *Typical value:* 3.13 and 3.57 at Case B. *Units:* MW. *Role:* Heat the gas and draw electricity through Eq. B.12. *Caveat:* No measured circulator power exists in the library.

Variable: Heat losses ($\dot{Q}_{ves}$, $\dot{Q}_{ihx}$, $\dot{Q}_{duct}$)

Loss from the reactor vessel, the IHX and the hot duct. *Typical value:* 2.66, 0.97 and 0.31 at Case B. *Units:* MW. *Role:* Raise $\dot{Q}_R$ in Eq. B.20. *Caveat:* Each one rests on a single small anchor. The table prints the screening duct value and Eq. B.14 scales it differently.

Variable: Power conversion efficiency (η_{pc})

Electricity per unit of heat in the power branch. *Typical value:* 0.397, with 0.475 as an alternative. *Units:* Dimensionless. *Role:* Turns electricity demand into reactor heat through Eq. B.19. *Caveat:* Interpolated by INL from TEV-981, which is not in the library, so the net or gross status of the value is unresolved.

Variable: Work ratio (r_w)

Electricity from back-pressure work per unit of heat extracted for the reboiler. *Typical value:* 0.169, with 0.30 as a high case. *Units:* $\mathrm{MW_e\ MW_{th}^{-1}}$. *Role:* Lets extraction steam make power on its way to the reboiler. *Caveat:* Calculated from one IEAGHG case.

Heat recovery and electricity

Variable: Reboiler duty ($\dot{Q}_{reb}$)

Heat to regenerate the MEA solvent, SRD times captured CO_2. *Typical value:* 51.0 at Case B and 74.6 at Case A. *Units:* MW. *Role:* Largest heat load after reforming. *Caveat:* Inherits the uncertainty of SRD.

Variable: Plant steam credit ($\dot{Q}_{stack}$)

Heat recovered from the plant in the approved order for the reboiler. *Typical value:* 37.9 at Case B and 55.4 at Case A. *Units:* MW. *Role:* Reduces what the reactor must supply. *Caveat:* The whole stack is used at central cases. Order matters only when the reboiler is smaller than the stack.

Variable: Reboiler gap ($\dot{Q}_{gap}$)

Reboiler duty that the plant steam cannot cover. *Typical value:* 13.1 at Case B and 19.2 at Case A. *Units:* MW. *Role:* Passed to the reactor through Eq. B.19. *Caveat:* Falls to zero at low SRD when freed steam is counted.

Variable: Electricity (E_{proc}, E_{tot})

Process electricity, and the total with circulators. *Typical value:* 12.33 and 19.78 at Case B. *Units:* $\mathrm{MW_e}$. *Role:* Feed the power branch. *Caveat:* Row splits inherit INL scaling and the restored fuel-gas recycle line.

Capture

Variable: Per-pass capture (η_{pp})

CO_2 removed per pass through the absorber. *Typical value:* 0.90. *Units:* Dimensionless. *Role:* Fixed target for the absorber. *Caveat:* Assumed. The rate-based absorber tests it.

Variable: Plant capture (η_{pl})

Captured carbon over feed carbon. *Typical value:* 0.9005. *Units:* Dimensionless. *Role:* Reports capture on the whole plant, including recycle. *Caveat:* Its closeness to η_{pp} is a numerical result of this plant and is no general rule.

Variable: Carbon split (ϕ_c)

Share of tube carbon that reaches the absorber as CO_2 after shift. *Typical value:* 0.775. *Units:* Dimensionless. *Role:* Sets CO_2 to capture. *Caveat:* Follows from the INL anchors, so a different shift train changes it.

Variable: Recycle fraction (f_r)

Share of tail-gas carbon returned to the reformer. *Typical value:* 0.752. *Units:* Dimensionless. *Role:* Sets gas demand and residual emissions. *Caveat:* Held from INL, whose PSA and cryogenic block is outside this model.

Variable: Specific reboiler duty (SRD)

Reboiler heat per tonne of CO_2. *Typical value:* 3.8, with a range of 3.4 to 4.2. *Units:* GJ t^{-1}. *Role:* Sets $\dot{Q}_{reb}$. *Caveat:* A judgement from flue gas pilots. No syngas measurement exists, and the direction of any bias is unknown.

Variable: Solvent loadings (α_{rich}, α_{lean})

Moles of CO_2 per mole of MEA in rich and lean solvent. *Typical value:* 0.50 and 0.25. *Units:* mol mol^{-1}. *Role:* Set the solvent rate. *Caveat:* Rich loading is capped at 0.50 because kinetics and properties stop there.

Variable: Captured CO_2 ($\dot{m}_{CO_2}$)

CO_2 leaving the stripper. *Typical value:* 48.3 at Case B and 70.7 at Case A. *Units:* t h^{-1}. *Role:* Scales the reboiler, compression and storage. *Caveat:* Follows ϕ_c and η_{pp}.

Variable: Lean solvent rate ($\dot{m}_{solv}$)

Lean solvent circulated to the absorber. *Typical value:* 894 at Case B and 1,309 at Case A. *Units:* t h^{-1}. *Role:* Sets pumps and column size. *Caveat:* Depends on the assumed loading swing.

Variable: Absorber conditions (p_{abs}, T_{abs})

Absorber pressure and temperature. *Typical value:* 26.8 bar and 313.15 K. *Units:* Pa, K. *Role:* Fix the driving force. *Caveat:* Taken from INL Case 5 for Selexol, then held for MEA.

Emissions

Variable: Baseline intensity (e_{base})

Direct CO_2 per kilogram of hydrogen from an unabated SMR. *Typical value:* 9.21, with a range of 9.00 to 9.35. *Units:* kg kg^{-1}. *Role:* Reference for abatement. *Caveat:* The bounds come from the Cho and IEAGHG studies.

Variable: Baseline CO_2 (E_{base})

Annual CO_2 from the unabated SMR at the same output. *Typical value:* 616,361 at Case B and 902,060 at Case A. *Units:* t y^{-1}. *Role:* Reference term in abatement. *Caveat:* Direct emissions only.

Variable: Residual CO_2 (E_{atm})

CO_2 reaching the atmosphere from the plant. *Typical value:* 38,309 at Case B and 56,066 at Case A. *Units:* t y^{-1}. *Role:* Second term in abatement. *Caveat:* Excludes lifecycle emissions, which are a sensitivity.

Variable: Direct abatement ($\mathcal{A}$)

Baseline less residual. *Typical value:* 578,052 at Case B and 845,994 at Case A. *Units:* t y^{-1}. *Role:* Denominator of abatement cost. *Caveat:* Direct, at the atmosphere boundary.

Economics

Variable: Discount rate (i)

Real rate used for annuities and the cash flow. *Typical value:* 0.08, with 0.05 and 0.10. *Units:* Dimensionless. *Role:* Annualises capital. *Caveat:* Taken from the IEAGHG 2017-02 basis.

Variable: Life (t_ℓ)

Reactor life. *Typical value:* 25 and 60. *Units:* y. *Role:* Sets annuity length and the printed pair of results. *Caveat:* Items with shorter lives are replaced in the cash flow.

Variable: Size exponent (n_s)

Exponent of reactor cost on rating. *Typical value:* 1.0, with 0.6 and 0.31 at equal prominence. *Units:* Dimensionless. *Role:* Scales the INL reactor cost to 210 MW_{th}. *Caveat:* 0.6 is an earlier project assumption, and 0.31 is a ratio of two FOAK estimates, so it is a weak adverse bound.

Variable: Overnight capital cost (c_{kWt})

Reactor overnight cost per kilowatt thermal with no turbine. *Typical value:* 2,184. *Units:* $US\$_{2019}$ kWt^{-1}. *Role:* Base of reactor capital. *Caveat:* Calculated from published INL cost tables.

Variable: Gas and storage prices (p_{gas}, p_{TS})

Delivered gas price, and the transport and storage tariff. *Typical value:* 15 and 60. *Units:* S\$ GJ^{-1}, S\$ t^{-1}. *Role:* Main operating cost and credit. *Caveat:* Gas runs from 10 to 20. The tariff runs from 23.2 to 112.

Variable: Abatement cost (AC)

Incremental annual cost over direct abatement, at ladder step L3. *Typical value:* 137.7 at 25 y and 117.2 at 60 y, on the BOAK basis with $n_s = 1$ at 210 MW_{th}. *Units:* S\$ t^{-1}. *Role:* A named step in the cost ladder. *Caveat:* The headline is the rebuilt full cost, which includes every plant cost. Sections K and L state it.

3 A. Conservation laws

Every balance in this document is one statement applied to a chosen quantity: what accumulates in a control volume equals what crosses its boundary plus what is made inside. This section sets that statement down once, reduces it to the forms the engine solves, and fixes the closure tests that every unit must pass.

Equation A.1: General balance

Tier R

$$\frac{d}{dt}\int_V \rho\psi \, dV + \oint_S \rho\psi \, \mathbf{u}\cdot\mathbf{n} \, dS = -\oint_S \mathbf{j}_\psi \cdot \mathbf{n} \, dS + \int_V \dot{s}_\psi \, dV \tag{A.1}$$

Interpretation. The rate of change of ψ held in a fixed volume, plus what the flow carries out, equals what diffusion brings in plus what is produced inside. Picture a tank with pipes: the level changes because of the pipes and because of anything made in the tank.

Derivation. Start from a material volume and apply the Reynolds transport theorem to a fixed control volume, which gives the first two terms. Split the boundary transport into a convective part and a non-convective flux $\mathbf{j}_\psi$. Choosing ψ gives each balance used here: $\psi = 1$ with $\mathbf{j}_\psi = 0$ and $\dot{s}_\psi = 0$ gives total mass, Eq. A.2. ψ equal to a species mass fraction gives Eq. A.3. $\psi = \mathbf{u}$ gives momentum, Eq. A.11. $\psi = e + u^2/2 + gz$ gives energy, Eq. A.7.

Variables. ρ, density, kg m^{-3}; ψ, conserved quantity per unit mass, per kg; $\mathbf{u}$, velocity, m s^{-1}; $\mathbf{n}$, outward unit normal, 1; $\mathbf{j}_\psi$, non-convective flux of ψ, per m^2 s; $\dot{s}_\psi$, volumetric source of ψ, per m^3 s; V, control volume, m^3; S, control surface, m^2.

Assumptions. Continuum. Fixed control volume.

Source. [**T**] Bird, Stewart and Lightfoot [3], ch 3, 11, 19.

Implementation. Theory, no code. Parent of Eq. A.2, Eq. A.3, Eq. A.7 and Eq. A.11.

Validation. ID, test `eq_a_01`. The reduction in the working above is the identity. Each child equation carries its own test.

Equation A.2: Total mass

Tier R

$$\frac{dM}{dt} = \sum_{in} \dot{m} - \sum_{out} \dot{m}, \qquad \text{steady state: } \sum_{in} \dot{m} = \sum_{out} \dot{m} \tag{A.2}$$

Interpretation. Mass in equals mass out once nothing accumulates. It is the first test a stream table must pass.

Derivation. Set $\psi = 1$, $\mathbf{j}_\psi = 0$ and $\dot{s}_\psi = 0$ in Eq. A.1. The volume integral becomes dM/dt. The surface integral becomes a sum over the openings of $\rho\mathbf{u}\cdot\mathbf{n}\,A$, which is the mass flow with its sign.

Variables. M, mass held in the control volume, kg; $\dot{m}$, mass flow of a stream, kg s^{-1}.

Assumptions. None beyond Eq. A.1.

Source. [**T**] As Eq. A.1.

Implementation. `plant::flowsheet::check_mass`

Validation. ID, PUB, test `eq_a_02`. Computed here on INL Case 5, TEV-961 [36] A-56, PDF p. 80: 223,172 lb/h into the Selexol block equals 58,787 plus 164,385 lb/h out, exact to the printed digit.

Equation A.3: Component balance

Tier R

$$\frac{dn_i}{dt} = \sum_{in} F_i - \sum_{out} F_i + \int_V R_i \, dV \tag{A.3}$$

Interpretation. Moles of a species change through flow and through reaction. Reaction is what separates this from Eq. A.2: a species can appear or vanish while atoms stay fixed.

Derivation. Put ψ equal to the mass fraction of species i in Eq. A.1, with $\dot{s}_\psi$ the mass rate of production. Divide by the molar mass and let the control surface sit where diffusive flux is small against convection, so the flux term drops at the openings.

Variables. n_i, moles of species i held, mol; F_i, molar flow of species i, mol s^{-1}; R_i, volumetric production rate of i, mol m^{-3} s^{-1}.

Assumptions. A well defined control volume. Diffusion across the openings is negligible.

Source. [**T**] As Eq. A.1.

Implementation. `plant::flowsheet::check_species`

Validation. ID, test `eq_a_03`. Unit test on species residuals of each reactor.

Equation A.4: Source term

Tier R

$$R_i = \sum_j \nu_{ij} r_j, \qquad \text{catalytic: } R_i = \rho_B \sum_j \nu_{ij} \eta_j^{cat} r_j \tag{A.4}$$

Interpretation. Each reaction contributes to a species in proportion to its stoichiometric coefficient. In a packed bed the rate is carried per kilogram of catalyst, so the bed density turns it into a rate per unit of reactor volume, and the effectiveness factor corrects it for diffusion inside the pellet.

Derivation. One event of reaction j changes species i by ν_{ij} moles, so summing over reactions gives the first form with r_j per unit volume. For a bed, $\rho_B\, dV$ is the catalyst mass in dV, and the observed rate is $\eta_j^{cat} r_j$ evaluated at bulk conditions, which gives the second form.

Variables. ν_{ij}, stoichiometric coefficient of i in reaction j, 1; r_j, rate of reaction j, mol m^{-3} s^{-1}, or mol kg^{-1} s^{-1} per mass of catalyst; ρ_B, catalyst bed density, kg m^{-3}; η_j^{cat}, effectiveness factor of reaction j, 1.

Assumptions. Rates do not depend on the size of the control volume.

Source. [**T**] Fogler [7]. [**P**] Xu and Froment Part II [38], Eq 10a, 10b, PDF p. 3.

Implementation. `units::xf::Rates::ch4` and `units::xf::Rates::co2` sum the reaction rates into species rates, and `units::smr_tube::Tube::eval` multiplies them by the bed mass per metre, which carries ρ_B and the lumped activity of Eq. D.21

Validation. ID, test `eq_a_04`. Species rates from Eq. D.19 equal the stoichiometric sum.

Equation A.5: Element balance

Tier R

$$\sum_i a_{ki} \left(F_{i,out} - F_{i,in}\right) = 0, \qquad \mathbf{a}\boldsymbol{\nu} = 0 \tag{A.5}$$

Interpretation. Atoms survive every chemical reaction. A stream table that loses carbon has an error, whatever the kinetics say.

Derivation. Multiply the steady form of Eq. A.3 by a_{ki} and sum over species. The source becomes $\int_V \sum_j r_j \sum_i a_{ki}\nu_{ij}\, dV$. Each reaction conserves atoms, so $\sum_i a_{ki}\nu_{ij} = 0$ for every j, which is the matrix condition, and the source vanishes.

Variables. a_{ki}, atoms of element k per molecule of i, 1; $\mathbf{a}$, element by species matrix, 1; $\boldsymbol{\nu}$, species by reaction matrix, 1.

Assumptions. No nuclear change in the process streams.

Source. [**T**] As Eq. A.1.

Implementation. `plant::flowsheet::check_elements`

Validation. ID, PUB, test `eq_a_05`. INL prints a carbon closure of 99.4 to 99.9% across its eight cases. The worked example below closes the Case B carbon balance. The flowsheet envelope rows: Validation results, 6 rows (FS_ENVELOPE_ELEMENTS_FS_A_INL_SEL to FS_ENVELOPE_ELEMENTS_FS_B_REC_F_T825): 6 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation A.6: Independent reactions

Tier R

$$N_r = N_s - \operatorname{rank}(\mathbf{a}) \tag{A.6}$$

Interpretation. Only this many reactions can be set independently. The rest are sums of those, so their equilibrium constants are products and carry no new information.

Derivation. Allowed changes of composition form the null space of $\mathbf{a}$, and rank plus nullity equals the number of species. For the reformer the species are CH_4, H_2O, CO, CO_2 and H_2 with elements C, H and O, so $N_s = 5$, the rank computed here is 3 and $N_r = 2$. That is why $K_3 = K_1 K_2$ in Eq. D.9. For aqueous MEA, with MEA, $MEAH^+$, $MEACOO^-$, CO_2, HCO_3^-, CO_3^{2-}, H_2O, H_3O^+ and OH^-, charge joins the rows of $\mathbf{a}$. The rank computed here is 4, because four moieties span the set, so $N_r = 5$.

Variables. N_r, number of independent reactions, 1; N_s, number of species, 1.

Assumptions. A closed list of species.

Source. [**T**] As Eq. A.1. Aronu [2], PDF p. 4 prints "four" and lists five reactions. Section M records this.

Implementation. `plant::flowsheet::reaction_rank`

Validation. ID, test `eq_a_06`. Rank by exact arithmetic on the two matrices above, computed here and repeated by the test named for this card.

Equation A.7: First law

Tier R

$$\frac{dE}{dt} = \dot{Q} - \dot{W}_{sh} + \sum_{in} \dot{m} \left(h + \frac{u^2}{2} + gz \right) - \sum_{out} \dot{m} \left(h + \frac{u^2}{2} + gz \right) \tag{A.7}$$

Interpretation. Stored energy changes through heat, shaft work and the energy each stream carries. Enthalpy appears in place of internal energy because a stream also does flow work at its boundary.

Derivation. Set $\psi = e + u^2/2 + gz$ in Eq. A.1. The stress work at the openings is $\dot{m}\, p/\rho$, and adding it to the internal energy of the stream gives $h = e + p/\rho$. What remains on the boundary is $\dot{Q}$, and the moving parts inside supply $\dot{W}_{sh}$.

Variables. E, energy held in the control volume, J; $\dot{Q}$, heat into the control volume, W; $\dot{W}_{sh}$, shaft work out of the control volume, W; h, specific enthalpy, J kg^{-1}; u, velocity magnitude, m s^{-1}; g, gravity, m s^{-2}; z, elevation, m.

Assumptions. None beyond Eq. A.1.

Source. [**T**] As Eq. A.1.

Implementation. Theory, no code. Parent of Eq. A.8.

Validation. ID, test `eq_a_07`. Reduction to Eq. A.8 shown on that card.

Equation A.8: Enthalpy balance

Tier R

$$\sum_{out} F_i h_i - \sum_{in} F_i h_i = \dot{Q} - \dot{W}_{sh} \tag{A.8}$$

Interpretation. Each unit in the plant closes on this line. Because h_i is on a formation basis, the heat of reaction needs no separate term: it shows up as the difference between what comes out and what goes in.

Derivation. Set $dE/dt = 0$ in Eq. A.7, drop the kinetic and potential terms, and write the mass flows as molar flows over each species.

Variables. h_i, molar enthalpy of species i, formation basis, J mol^{-1}.

Assumptions. SS, NKE.

Source. [**T**] As Eq. A.1.

Implementation. `plant::flowsheet::envelope_enthalpy` and `plant::flowsheet::check_energy` on the flowsheet envelope, and `units::columns::island_energy_residual_w` on the capture island

Validation. ID, test `eq_a_08`. Residual inside the limit of Eq. A.15. The flowsheet envelope rows: Validation results, 6 rows (FS_ENVELOPE_ENERGY_FS_A_INL_SEL to FS_ENVELOPE_ENERGY_FS_B_REC_F_T825): 6 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation A.9: Species enthalpy

Tier E

$$h_i(T) = \Delta_f h_i^\circ + \int_{T_{ref}}^{T} c_{p,i}\, dT, \qquad c_{p,i} = A + Bt + Ct^2 + Dt^3 + \frac{E_S}{t^2}, \quad t = \frac{T}{1000} \tag{A.9}$$

Interpretation. A species holds its formation enthalpy at the reference state plus the sensible heat taken up on the way to T.

Fit range and origin. The form is Hoseinzade and Adams Eq 28, 29, PDF p. 6. The Shomate polynomial integrates to $h_i - h_i(T_{ref}) = At + Bt^2/2 + Ct^3/3 + Dt^4/4 - E_S/t + F - H$ in kJ mol^{-1}. Coefficients come from the NIST Chemistry WebBook gas phase tables [20], read on 2 November 2026. The ranges in K are CH_4 298 to 1300, CO 298 to 1300, CO_2 298 to 1200 and 1200 to 6000, H_2 298 to 1000 and 1000 to 2500, H_2O 500 to 1700 and N_2 500 to 2000. The H_2O and N_2 rows used here do not reach below 500 K.

Variables. $\Delta_f h_i^\circ$, standard enthalpy of formation, J mol^{-1}; $c_{p,i}$, molar heat capacity of species i, J mol^{-1} K^{-1}; T, temperature, K; T_{ref}, reference temperature, 298.15, K; t, temperature over 1000, 1; A, B, C, D, E_S, F, H, Shomate coefficients, mixed, per NIST.

Assumptions. Ideal gas mixture, so mixing adds no enthalpy.

Source. [**P**] Form from Hoseinzade and Adams [10], Eq 28, 29, PDF p. 6. Coefficients [**M6**]: absent from the library, read from NIST [20], listed there as Chase 1998.

Implementation. `props::shomate::Shomate::cp_j_mol_k` and `props::shomate::Shomate::dh_j_mol` evaluate the Shomate form, and `props::species::h_j_mol` adds the formation enthalpy

Validation. PUB, test `eq_a_09`. Shomate against the JANAF tables, one row per quantity at its worst point. Validation results, 3 rows (PROPS-SHOMATE-CP to PROPS-SHOMATE-S): 2 pass, 1 fail. Failed: PROPS-SHOMATE-CP gives 29.3 against 29.34 J/mol/K, limit 0.1% relative.

Equation A.10: Hess law

Tier R

$$\Delta_r h_j(T) = \sum_i \nu_{ij} h_i(T) \tag{A.10}$$

Interpretation. The heat of a reaction at any temperature is the enthalpy of its products less that of its reactants, on a common basis. It shifts with T because products and reactants have different heat capacities.

Derivation. Apply Eq. A.8 to one event of reaction j at constant temperature, with no shaft work. Substitute Eq. A.9 to carry the result to any T.

Variables. $\Delta_r h_j$, enthalpy of reaction j, J mol^{-1}.

Assumptions. As Eq. A.9.

Source. [**T**] Textbook result. [**P**] Values in Xu and Froment Part I [37], Table 7, PDF p. 8.

Implementation. Every enthalpy in the engine sits on the formation basis of `props::species::h_j_mol`, so the reaction heat enters each balance with no separate term. `plant::heatrec::wgs_release_w` evaluates the sum for reaction II

Validation. PUB, test `eq_a_10`. XF1 Table 7 prints 206.1, -41.15 and 164.9 kJ mol^{-1} at 298 K, and 224.0, -37.3 and 187.5 at 948 K, for reactions I, II and III. Computed here at 298 K from formation values: 206.17, -41.17 and 165.00. Computed here at 948 K from Shomate: 225.1, -35.3 and 189.8. The largest gap at 948 K is 2.3 kJ mol^{-1}, and the Shomate values are the ones the plant uses. The engine rows: Validation results, 6 rows (XF-3-I-298K to XF-3-III-948K): 5 pass, 1 fail. Failed: XF-3-III-948K gives 189.8 against 187.5 kJ/mol, limit 2 kJ/mol.

Equation A.11: Momentum in a channel

Tier R

$$\frac{dp}{dz} = -\rho u \frac{du}{dz} - \frac{f_D}{D}\frac{\rho u^2}{2} - \rho g \sin\theta \tag{A.11}$$

Interpretation. Wall drag usually dominates the pressure loss along a channel. Acceleration of the gas and lift against gravity make up the rest, and both are small in these loops.

Derivation. Write Eq. A.1 with $\psi = u$ on a slice $A\,dz$ at steady state. The pressure force per unit volume is $-dp/dz$, the convective term is $\rho u\,du/dz$, gravity gives $\rho g \sin\theta$, and the wall shear $\tau_w\,\pi D/A = 4\tau_w/D$ is written with the Darcy factor as $f_D \rho u^2/(2D)$.

Variables. p, pressure, Pa; f_D, Darcy friction factor, 1; D, hydraulic diameter, m; θ, channel inclination, rad.

Assumptions. 1D, SS.

Source. [**T**] As Eq. A.1.

Implementation. `units::circulator::dp_blasius`, on the level channel at constant speed of Eq. C.16

Validation. ID, test `eq_a_11`. The acceleration and gravity terms vanish for a level channel at constant speed, leaving the Darcy form used in Eq. C.16.

Equation A.12: Packed bed pressure drop

Tier E

$$-\frac{dp}{dz} = \frac{G}{\rho D_p}\,\frac{1-\varepsilon_b}{\varepsilon_b^3}\left[\frac{150(1-\varepsilon_b)\mu}{D_p} + 1.75G\right] \tag{A.12}$$

Interpretation. Gas forcing through catalyst pays a viscous term, which grows with flow, and an inertial term, which grows with flow squared. Porosity enters strongly, so a small change in packing moves the pressure drop a long way.

Fit range and origin. Ergun 1952, the friction law named in Xu and Froment Part II, PDF p. 4, for their Eq 10d. In factor form it is $f = \frac{1-\varepsilon_b}{\varepsilon_b^3}[150(1-\varepsilon_b)/Re + 1.75]$ with $Re = D_p G/\mu$ and $-dp/dz = fG^2/(\rho D_p)$, which rearranges to the statement. Hoseinzade and Adams write it with a positive sign because their z runs against the flow. Do not copy Yin Eq 15, which defines the diameter as the tube diameter and drops $(1-\varepsilon_b)$ from the viscous term.

Variables. G, superficial mass velocity, kg m^{-2} s^{-1}; D_p, particle diameter, m; ε_b, bed porosity, 1; μ, gas viscosity, Pa s.

Assumptions. Friction dominates the momentum balance, so acceleration is dropped.

Source. [**P**] Hoseinzade and Adams [10], Eq 26, PDF p. 5. Xu and Froment Part II [38], Eq 10d, PDF p. 4.

Implementation. Inside `units::smr_tube::Tube::eval` for the reformer bed. The shift beds take their drop from the printed pressures, Eq. E.12

Validation. EXP, test `eq_a_12`. Compared on Eq. D.26, the measured process drop of the JAERI tube.

Equation A.13: Plug-flow reduction

Tier R

$$\frac{dF_i}{dz} = A_c R_i, \qquad \sum_i F_i c_{p,i} \frac{dT}{dz} = \dot{q}' + A_c \sum_j \left(-\Delta_r h_j\right) r_j \tag{A.13}$$

Interpretation. Move along a tube in slices. Each slice changes composition through reaction, and its temperature changes with wall heat plus the heat released by reaction, shared across the heat capacity of the flow.

Derivation. Apply Eq. A.3 at steady state to the slice $A_c\,dz$, which gives $dF_i/dz = A_c R_i$. Apply Eq. A.8 to the same slice with heat $\dot{q}'dz$: $\frac{d}{dz}\sum_i F_i h_i = \dot{q}'$. Expand the derivative as $\sum_i h_i dF_i/dz + \sum_i F_i c_{p,i} dT/dz$. Substitute the species result and Eq. A.4, so $\sum_i h_i A_c \sum_j \nu_{ij} r_j = A_c \sum_j r_j \Delta_r h_j$ by Eq. A.10. Moving it to the right gives the statement. In a bed, r_j becomes $\rho_B \eta_j^{cat} r_j$.

Variables. A_c, flow cross section, m^2; $\dot{q}'$, heat into the gas per unit length, $\mathrm{W\ m^{-1}}$.

Assumptions. SS, PF, NKE.

Source. [**T**] Fogler [7].

Implementation. `units::smr_tube::Tube::residual` holds the discrete species and energy balances of the reformer tube, and `units::wgs::hts_extent` integrates the extent along the high-temperature bed

Validation. ID, test `eq_a_13`. A long tube limit returns the equilibrium of Eq. D.11. Validation results, 1 row (XF-4): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation A.14: Gas closure

Tier R

$$p_i = y_i p, \qquad c = \frac{p}{ZRT} \tag{A.14}$$

Interpretation. A partial pressure is a share of the total. Concentration follows from the gas law, with Z holding any departure from ideal behaviour.

Derivation. Dalton's law and the definition of Z. Syngas at reformer conditions is taken as ideal. Helium uses the correction in Eq. B.1.

Variables. p_i, partial pressure of i, Pa; y_i, mole fraction of i, 1; c, molar concentration, $\mathrm{mol\ m^{-3}}$; Z, compressibility factor, 1; R, gas constant, 8.314462, $\mathrm{J\ mol^{-1}\ K^{-1}}$.

Assumptions. $Z = 1$ for syngas. Eq. B.1 for helium.

Source. [**T**] As Eq. A.1.

Implementation. `props::mixture::mole_fractions` and `props::mixture::density_ideal_kg_m3`, with $Z = 1$ for the process gas. Helium carries its own Z in Eq. B.1

Validation. ID, test `eq_a_14`. Round trip between p_i, y_i and c.

Equation A.15: Closure residuals

Tier F

$$\epsilon_m = \frac{\left|\sum_{in} \dot{m} - \sum_{out} \dot{m}\right|}{\sum_{in} \dot{m}}, \quad \epsilon_k = \frac{\left|\sum_i a_{ki}(F_{i,out} - F_{i,in})\right|}{\sum_i a_{ki} F_{i,in}}, \quad \epsilon_E = \frac{\left|\sum_{out} F_i h_i - \sum_{in} F_i h_i - \dot{Q} + \dot{W}_{sh}\right|}{\sum_{in} |F_i h_i|} \tag{A.15}$$

Interpretation. These numbers say how far a result sits from obeying the laws above. Limits are 10^{-8} per unit and 10^{-6} on the flowsheet.

Justification. Fixed by the project. Solver tolerances sit below these limits, so a failure points first to a modelling error. Each residual is relative to its inlet flow, so the same limit serves every plant size.

Variables. ϵ_m, relative mass residual, 1; ϵ_k, relative residual of element k, 1; ϵ_E, relative energy residual, 1.

Assumptions. None.

Source. [**A**] Project limits.

Implementation. `units::gas::closure`, which every gas-side unit reports

Validation. ID, test `eq_a_15`. Evaluated on every run.

Worked example: carbon closure at Case B

Use Eq. A.5 for carbon around two control volumes, in kmol h^{-1}. Around the whole plant, 1219.23 enters with the gas. Leaving are 1097.90 captured, 116.90 in the part of the tail gas that is burned and 4.42 in the hydrogen, which sum to the feed within rounding. Around the reformer, tube carbon is feed plus recycle: 1219.23 plus 354.82 gives 1574.06. Plant capture is then 1097.90 over 1219.23, which is 0.9005, the value of η_{pl} in the Reference Frame. Recycle explains why capture relative to feed stays close to capture per pass: unconverted carbon gets a second chance at the absorber.

Link to the sources

The closure of INL Case 5 at the Selexol block in Eq. A.2, to the printed digit, shows that the INL stream tables close as mass balances, so their stream values can serve as anchors. The Hess comparison in Eq. A.10 shows the Shomate enthalpies sit within 2.3 kJ mol^{-1} of the values Xu and Froment used.

4 B. HTGR thermal model

This section follows heat from the core to the IHX and asks how much reactor power the plant must draw. The module is reduced order. It holds no neutronics and no core flow or fuel model: the reactor appears only as a heat source with an outlet temperature, so nothing here says the core can be built or licensed at these conditions. Helium is treated with published correlations first, because every balance after them depends on c_p.

Equation B.1: Helium density

Tier E

$$\rho = 48.14\,\frac{P}{T}\left[1 + 0.4446\,\frac{P}{T^{1.2}}\right]^{-1} \tag{B.1}$$

Interpretation. Helium is almost an ideal gas at reactor conditions. The bracket is the compressibility factor, which stays within about one percent of unity across the loop.

Fit range and origin. Petersen writes $P\cdot 10^5/(2077.3\,\rho T) = Z$ (Eq 3-2, 3-3) and fits $Z = 1+0.4446P/T^{1.2}$ (Eq 3-18). Since $10^5/2077.3 = 48.14$, the density follows (Eq 3-19). Petersen states validity from 1 to 100 bar and room temperature to about 1800 K, with a standard deviation of about 0.03% at 1 bar and 0.3% at 100 bar. The correlation is in bar and K, and the code converts at the function boundary.

Variables. P, helium pressure in bar, in correlations only, bar.

Assumptions. 1 to 100 bar, up to about 1800 K.

Source. [**P**] Petersen [23], Eq 3-19, printed p. 6, read as an image. [**M5**]: not in the library.

Implementation. `props::helium::density_kg_m3`, with Z from `props::helium::compressibility`

Validation. PUB, test `eq_b_01`. Computed here at 925 C and 7 MPa: $Z = 1.0063$ and $\rho = 2.795$ kg m^{-3}. The OUTRAM reference model gives 1.00643 from its helium equation of state, a project run that is unpublished, so it serves only as a comparison. Petersen's own tables: Validation results,

1 row (PROPS-HE-PETERSEN): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation B.2: Helium heat capacities

Tier E

$$c_p = 5195 \text{ J kg}^{-1} \text{ K}^{-1}, \quad c_v = 3117 \text{ J kg}^{-1} \text{ K}^{-1}, \quad \kappa = c_p/c_v = 1.6667 \tag{B.2}$$

Interpretation. A monatomic gas stores heat only in translation, so its heat capacity barely moves with temperature or pressure. This is why every helium balance below can use one number.

Fit range and origin. Petersen Eq 4-4 to 4-6. The ideal value is $5R/2M = 5193.2$ J kg^{-1} K^{-1}, computed here with $M = 4.002602$ g mol^{-1}, so the printed 5195 sits 1.8 J kg^{-1} K^{-1} above it. Petersen gives a standard deviation of about 0.05% at 273 K and 1 bar, 0.5% at 100 bar, and 0.05% at high temperature at all pressures. INL stream data imply 5190.

Variables. c_p, isobaric heat capacity, J kg^{-1} K^{-1}; c_v, isochoric heat capacity, J kg^{-1} K^{-1}; κ, heat capacity ratio, 1.

Assumptions. CP.

Source. [**P**] Petersen [23], Eq 4-4 to 4-6, printed p. 6. [**C**] Ideal value computed here.

Implementation. `props::helium::CP_J_KG_K`, `props::helium::CV_J_KG_K` and `props::helium::GAMMA`

Validation. ID, test `eq_b_02`. Target: the JAERI mock-up run sheet implies 5.193 to 5.195 kJ kg^{-1} K^{-1}.

Equation B.3: Helium viscosity

Tier E

$$\mu = 3.674 \times 10^{-7} T^{0.7} \tag{B.3}$$

Interpretation. Gas viscosity rises with temperature because molecules cross shear layers faster. Pressure plays no part in this form.

Fit range and origin. Petersen Eq 6-1, with a stated standard deviation of about 0.4% at 273 K and 2.7% at 1800 K. Viscosity enters Reynolds numbers only, and these feed the friction and film terms.

Variables. μ, gas viscosity, Pa s.

Assumptions. As Eq. B.1.

Source. [**P**] Petersen [23], Eq 6-1, printed p. 6.

Implementation. `props::helium::viscosity_pa_s`

Validation. PUB, test `eq_b_03`. Computed here at 900 C: 5.17 $\times 10^{-5}$ Pa s. Validation results, 2 rows (HE-2-MU to PROPS-HE-PETERSEN): 1 pass, 1 not run. The results mark 1 of these rows as using the same data as the model, and 1 as independent.

Equation B.4: Helium conductivity

Tier E

$$k = 2.682 \times 10^{-3} \left(1 + 1.123 \times 10^{-3} P\right) T^{0.71\left(1 - 2\times 10^{-4} P\right)} \tag{B.4}$$

Interpretation. Conductivity is high for a gas, which makes helium a good coolant, and it climbs with temperature almost as viscosity does. The Prandtl number therefore stays nearly flat.

Fit range and origin. Petersen Eq 7-1, with a stated standard deviation of about 1% at 273 K and 6% at 1800 K. That spread is the largest of the four helium properties, and it reaches the film coefficients directly.

Variables. k, thermal conductivity, W m^{-1} K^{-1}.

Assumptions. As Eq. B.1.

Source. [**P**] Petersen [23], Eq 7-1, printed p. 7.

Implementation. `props::helium::conductivity_w_m_k`

Validation. PUB, test `eq_b_04`. Computed here at 900 C and 51.5 bar: 0.407 W m^{-1} K^{-1}, and $Pr = 0.66$ from $c_p\mu/k$. Petersen Eq 8-1 gives 0.660 at the same point. Validation results, 2 rows (HE-2-K to PROPS-HE-PETERSEN): 1 pass, 1 not run. The results mark 1 of these rows as using the same data as the model, and 1 as independent.

Equation B.5: Reactor balance

Tier O

$$\dot{Q}_R - \dot{Q}_{ves} = \dot{m}_p c_p \left(T_{ROT} - T_{RI}\right) \tag{B.5}$$

Interpretation. Fission heat that does not leave through the vessel wall goes into the helium. Flow times heat capacity times temperature rise must match it. This is the only reactor physics in the model.

Derivation. Apply Eq. A.8 to the core with no shaft work and a constant c_p. The heat term is the core power less the vessel loss. No neutronics and no core flow or fuel model enter, so the equation cannot say whether a given peak fuel temperature is acceptable.

Variables. T_{RI}, reactor inlet temperature, K.

Assumptions. SS, CP. No core model.

Source. [**T**] Consequence of Eq. A.8.

Implementation. Not implemented in the engine at this build: the primary flow comes from Eq. B.9, which leaves out the vessel loss and the circulator rise, so this balance is computed in this document only. Planned name `units::htgr::core_balance`.

Validation. PUB, test `eq_b_05`. Target: thirteen published designs within 1.5%, HTTR test data within 6%. Computed here at Case B: left side 175.83 MW, right side 174.93 MW with the flow of Eq. B.9, a gap of 0.5%. The gap exists because Eq. B.9 ignores the vessel loss and the circulator rise. A primary flow of 79.6 kg s^{-1} closes the equation exactly.

Equation B.6: Circulator rise

Tier O

$$T_{RI} = T_{h,out} + \frac{\dot{W}_p}{\dot{m}_p c_p} \tag{B.6}$$

Interpretation. The circulator does work on the gas, and in a closed loop that work appears as heat. So the helium enters the core a little warmer than it left the IHX.

Derivation. Eq. A.8 on the circulator with $\dot{Q} = 0$ and shaft work into the gas gives $\dot{m}_p c_p(T_{RI} - T_{h,out}) = \dot{W}_p$.

Variables. $T_{h,out}$, hot-side helium leaving the IHX, K.

Assumptions. All shaft work enters the gas. Motor loss stays outside the gas.

Source. [**T**] Consequence of Eq. A.8.

Implementation. `units::circulator::rise`

Validation. PUB, test `eq_b_06`. Target: Sato Fig 12 shows 133 to 150 C across the secondary circulator at 2.519 kg s^{-1}, which gives 0.222 MW computed here. Case B gives a rise of 8.7 K and an inlet of 499.7 C.

Equation B.7: Primary side

Tier R

$$\dot{Q}_p = \dot{m}_p c_p \left(T_{ROT} - T_{h,out}\right), \qquad T_{h,out} \geq T_{s,ret} + \Delta T_{app} \tag{B.7}$$

Interpretation. Primary helium gives up heat in the IHX. Its outlet cannot fall below the secondary return plus the approach, so the return temperature sets a floor on reactor inlet temperature.

Derivation. Eq. A.8 on the hot side of the IHX. The inequality is the second law at the cold end of a counterflow exchanger.

Variables. $\dot{Q}_p$, heat lost by primary helium, W; ΔT_{app}, IHX approach temperature, K.

Assumptions. SS, CP.

Source. [**T**] Consequence of Eq. A.8.

Implementation. `units::ihx::hot_side`

Validation. ID, test `eq_b_07`. Sign and limit tests.

Equation B.8: Secondary side

Tier R

$$\dot{Q}_s = \dot{m}_s c_p \left(T_{s,sup} - T_{s,ret}\right) \tag{B.8}$$

Interpretation. Secondary helium carries the process heat. The temperature window is fixed by the reformer and its preheat train, so the flow must adjust to the duty.

Derivation. Eq. A.8 on the secondary loop between IHX outlet and return.

Variables. $\dot{Q}_s$, heat carried by secondary helium, W.

Assumptions. SS, CP.

Source. [**P**] TEV-961 [36], Table 2 continued, PDF p. 18: 78.49 kg s^{-1}, 900 to 466 C.

Implementation. `units::ihx::cold_side`. The TEV-961 rows of the validation line use `props::helium::duty_w`, the same product

Validation. PUB, test `eq_b_08`. Computed here at the INL implied c_p of 5190: 176.8 MW, the Case A heat. The five ROT cases of TEV-961 and the TEV-953 helium duty: Validation results, 6 rows (T953-3 to T961-1-ROT925C): 5 pass, 1 fail. Failed: T961-1-ROT825C gives 149.4 against 149.1 MWt, limit 0.2236 MWt. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation B.9: Flow need

Tier R

$$\dot{m}_s = \frac{\dot{Q}_{proc}}{c_p \left(T_{s,sup} - T_{s,ret}\right)}, \qquad \dot{m}_p = \frac{\dot{Q}_R}{c_p \left(T_{ROT} - T_{s,ret} - \Delta T_{app}\right)} \tag{B.9}$$

Interpretation. Rearranging Eq. B.7 and Eq. B.8 gives each flow. A balanced exchanger runs both streams through the same temperature span, so the primary flow exceeds the secondary only by the ratio of the heats.

Derivation. Solve Eq. B.8 for $\dot{m}_s$. Solve Eq. B.7 at its limit with the whole reactor heat passing through the span. The second form is a sizing estimate, and Eq. B.5 shows how far it sits from an exact closure.

Variables. $\dot{Q}_{proc}$, process heat, W; $\dot{Q}_R$, reactor heat, W.

Assumptions. As Eq. B.7, Eq. B.8.

Source. [**C**] Calculated here.

Implementation. `units::circulator::secondary_flow_kg_s` and `units::circulator::primary_flow_kg_s`

Validation. GOLD, test `eq_b_09`. 53.6 and 79.2 kg s^{-1} at Case B.

Equation B.10: Compression

Tier R

$$\dot{W} = \frac{\dot{m} c_p T_1}{\eta_{is}} \left[\left(1 + \frac{\Delta p}{p} \right)^{(\kappa-1)/\kappa} - 1 \right], \qquad \frac{\kappa - 1}{\kappa} = 0.4 \tag{B.10}$$

Interpretation. Helium is a poor gas to pump. It is light, so raising its pressure a little takes a lot of work per kilogram, and the large flow turns that into megawatts.

Derivation. For an isentropic step of an ideal gas, $T_2/T_1 = (p_2/p_1)^{(\kappa-1)/\kappa}$. Eq. A.8 gives the ideal work $\dot{m} c_p (T_2 - T_1)$, and η_{is} scales it to the actual work. With $p_2/p_1 = 1 + \Delta p/p$ the statement follows. For $\Delta p/p \ll 1$ it reduces to $\dot{m} \Delta p / (\rho \eta_{is})$.

Variables. $\dot{W}$, shaft power, W; T_1, gas temperature at circulator inlet, K; Δp, loop pressure drop, Pa; η_{is}, isentropic efficiency, 0.75, 1.

Assumptions. IG, adiabatic.

Source. [**T**] Form. [**P**] $\eta_{is} = 0.75$, INL-21537 [21], Table 2, PDF p. 16.

Implementation. `units::circulator::shaft`

Validation. GOLD, test `eq_b_10`. 3.13 and 3.57 MW at Case B. Computed here: 58.0 kJ kg^{-1} on the secondary loop with 54.0 kg s^{-1}, and 45.1 kJ kg^{-1} on the primary loop with 79.2 kg s^{-1}. Target: the four budget rows of Eq. B.11. No measured shaft power exists, [**M13**].

Equation B.11: Pressure budget

Tier F

$$p = 7.0 \text{ MPa}, \qquad \Delta p_s = 200 \text{ kPa}, \qquad \Delta p_p = 150 \text{ kPa} \tag{B.11}$$

Interpretation. Without a loop design the drops are a budget. The budget is built up from component values that others have reported.

Justification. INL-11109 assumes 50 kPa for each heat exchanger and each pipe leg, and quotes MHR components at 30 to 70 kPa (PDF p. 11 to 12). The central values are project round figures on that scale. Other sets carried: low at 7.0 MPa with 150 and 100 kPa and $\eta_{is} = 0.80$; central at 5.15 MPa; high at 5.15 MPa with 300 and 200 kPa and $\eta_{is} = 0.65$. INL also labels the helium at 1005 psi and 945 psi across its preheaters on sheet A-55, which is 414 kPa and gives a sensitivity of $\dot{W}_s = 6.43$ MW from Eq. B.10.

Variables. Δp_s, secondary loop pressure drop, Pa; Δp_p, primary loop pressure drop, Pa.

Assumptions. A project budget. INL sources conflict on secondary pressure, at 9.1 and 7 MPa.

Source. [**A**] Built from INL-11109 [9], PDF p. 11 to 12, and JAERI [28]. Sensitivity from TEV-961 [36], A-55 and A-19 labels.

Implementation. `plant::anchored::Model::circ_set` holds the central, low, high and 5.15 MPa sets from the parameter file, and `plant::anchored::Model::circ_dp_central` and `plant::anchored::Model::circ_p_central_mpa` return the central drops and pressure

Validation. NONE, test `eq_b_11`. No benchmark exists. The project sensitivity adds up to 9.6 MW_{th} at Case A.

Equation B.12: Electric draw

Tier F

$$E_{circ} = \frac{\dot{W}_s + \dot{W}_p}{\eta_m}, \qquad \eta_m = 0.90 \tag{B.12}$$

Interpretation. The motors sit outside the gas, so their loss leaves as waste heat and only the shaft work reaches the helium.

Justification. An assumed drive efficiency. The electricity is charged to the plant in Eq. B.19, and the shaft work returns as heat in Eq. B.16, so a circulator costs less than its electrical rating suggests.

Variables. E_{circ}, circulator electricity, W; η_m, motor and drive efficiency, 1.

Assumptions. Motors outside the helium boundary.

Source. [**A**] Project assumption.

Implementation. `units::circulator::electric`

Validation. GOLD, test `eq_b_12`. 7.45 MW_e at Case B.

Equation B.13: Vessel loss

Tier E

$$\dot{Q}_{ves} = 0.81 \left(\frac{\dot{Q}_R}{30} \right)^{2/3} \text{ MW}, \qquad \text{high case: } 0.027\,\dot{Q}_R \tag{B.13}$$

Interpretation. Heat leaks through the vessel surface, and surface grows more slowly than power. Scaling from the HTTR by area therefore gives a smaller fractional loss for a larger reactor.

Fit range and origin. One anchor: the HTTR vessel cooling system removed 0.81 MW at 30 MW. The exponent assumes loss follows area with a fixed vessel shape, which is a project assumption. The high case keeps the HTTR fraction, $0.81/30 = 0.027$, and gives 4.82 MW at Case B.

Variables. $\dot{Q}_{ves}$, vessel heat loss, MW.

Assumptions. Area scaling [**A**].

Source. [**P**] IAEA-TECDOC-1382 [11], Table 3-21, PDF p. 164.

Implementation. `plant::anchored::Model::vessel_loss`, with the area law as the central mode and the proportional form as the high case

Validation. GOLD, test `eq_b_13`. 2.66 MW at Case B, 1.5% of $\dot{Q}_R$. The experimental anchor is a single test point.

Equation B.14: Duct loss

Tier E

$$\dot{Q}_{duct} = 0.45 \left(\frac{\dot{m}_s}{78.49} \right)^{1/2} \text{ MW at 90 m}, \qquad 3.12 \left(\frac{\dot{m}_s}{78.49} \right)^{1/2} \text{ MW at 500 m} \tag{B.14}$$

Interpretation. A duct loses heat through its surface, and at fixed pressure drop its diameter grows with the square root of flow. So the loss follows the square root of flow.

Fit range and origin. INL-11109 Table 3 gives 0.26 MW for 90 m and 1.82 MW for 500 m pipes at a loop flow of 26.7 kg s^{-1} (PDF p. 13). Multiplying by $\sqrt{78.49/26.67}$ gives 0.45 and 3.12 MW. The INL case is at 2 MPa, and a 7.1 MPa run on PDF p. 16 used 0.22 MW, so this scaling leans high at the project pressure.

Variables. $\dot{Q}_{duct}$, duct heat loss, MW.

Assumptions. Loss follows pipe diameter.

Source. [**P**] INL-11109 [9], Table 3, PDF p. 15, scaled [**C**].

Implementation. `units::circulator::duct_loss`

Validation. GOLD, test `eq_b_14`. This equation gives 0.37 MW at Case B against 0.31 MW in the screening reference, which scales with s. The difference is listed in Section M.

Equation B.15: IHX loss

Tier E

$$\dot{Q}_{ihx} = f_{ihx} \left(\dot{Q}_{proc} + \dot{Q}_{rec} \right), \qquad f_{ihx} = 0.008, \quad \text{range 0 to } 0.0458 \tag{B.15}$$

Interpretation. A small share of the heat through the exchanger never reaches the secondary side. The share comes from the gap between the two sides of a measured heat balance.

Fit range and origin. HTTR test data give $\dot{Q}_h = 9.48$ against $\dot{Q}_c = 9.41$ MW at 30 MW, a residual of 0.76%, and 4.96 against 4.90 MW at 15 MW, a residual of 1.28%. The residual is treated as a loss although instrument error may explain part of it. The value 0.0458 at the high end has no source.

Variables. f_{ihx}, IHX loss fraction, 1; $\dot{Q}_{ihx}$, IHX heat loss, MW.

Assumptions. The measured residual is a true loss.

Source. [**C**] From IAEA-TECDOC-1382 [11], Table 3-23, PDF p. 168.

Implementation. `units::ihx::loss`

Validation. GOLD, test `eq_b_15`. 0.97 MW at Case B.

Equation B.16: Available heat

Tier O

$$\dot{Q}_{av} = \dot{Q}_R + \dot{W}_s + \dot{W}_p - \dot{Q}_{ves} - \dot{Q}_{ihx} - \dot{Q}_{duct} \tag{B.16}$$

Interpretation. What the helium can deliver is the reactor heat plus the shaft work it picked up, less what leaked on the way.

Derivation. Eq. A.8 around both helium loops, with shaft work entering the gas by Eq. B.6.

Variables. $\dot{Q}_{av}$, heat available at the process interface, MW.

Assumptions. Shaft work returns to the helium.

Source. [**C**] Calculated here.

Implementation. Inside `plant::utility::reactor_required`: Eq. B.20 is this equation with the available heat set equal to its three uses

Validation. ID, test `eq_b_16`. Computed here: 181.3 MW at Case B, which equals the three uses in Eq. B.20.

Equation B.17: Conversion efficiency

Tier E

$$\eta_{pc} = 0.397, \qquad \text{alternative: } 0.475 = 0.66\left(1 - \frac{T_c}{T_h}\right) \tag{B.17}$$

Interpretation. Heat that goes to power becomes electricity at about this fraction. The alternative value treats the cycle as a fixed share of Carnot.

Fit range and origin. TEV-961 Table 2 continued prints 39.7% at 925 C ROT, with a note that it is interpolated from TEV-981. INL-21537 assumes 66% of Carnot (PDF p. 16) and prints a net 47.5% at 900 C (Table 4), which implies $T_c/T_h = 0.280$ computed here. TEV-981 is missing, so the net or gross status of 39.7% is unresolved. Taking either value changes the heat for power by a large factor, and the plant carries both.

Variables. η_{pc}, power conversion efficiency, 1; T_c, cycle cold temperature, K; T_h, cycle hot temperature, K.

Assumptions. Rankine cycle on excess heat.

Source. [**P**] TEV-961 [36], PDF p. 18, note 3, [**M2**]. INL-21537 [21], Table 4, PDF p. 20.

Implementation. `plant::anchored::Model::opts` sets the central η_{pc} from the parameter file. The alternative value is a parameter of its own

Validation. NONE, test `eq_b_17`. No benchmark exists. The net status of 39.7% is unresolved.

Equation B.18: Power branch

Tier O

$$E = \eta_{pc} \dot{Q}_{pc} \tag{B.18}$$

Interpretation. A fixed efficiency turns heat into electricity with no model of the cycle behind it.

Derivation. Definition of η_{pc}.

Variables. E, electricity generated, W; $\dot{Q}_{pc}$, heat into the power cycle, W.

Assumptions. Fixed efficiency.

Source. [**T**] Definition.

Implementation. Inside `plant::heatrec::extraction`, which divides the electricity that extraction does not meet by η_{pc}, Eq. B.19

Validation. ID, test `eq_b_18`. Round trip with Eq. B.19.

Equation B.19: Extraction

Tier O

$$w = \min\left(E,\, r_w \dot{Q}_{gap}\right), \qquad \dot{Q}_{br} = \dot{Q}_{gap} + w + \frac{E - w}{\eta_{pc}} \tag{B.19}$$

Interpretation. The reactor must supply the reboiler gap and the electricity from one branch of heat. Steam bound for the reboiler can first spin a turbine, so part of the electricity comes almost for free. The rest pays the full conversion penalty.

Derivation. Steam carrying $\dot{Q}_{gap}$ to the reboiler expands through a back-pressure stage and yields $r_w \dot{Q}_{gap}$ of work, limited by the demand E. That work costs heat one for one, because the condensing heat still reaches the reboiler. The remaining electricity $E-w$ needs $(E-w)/\eta_{pc}$ of heat from Eq. B.18. Summing gives $\dot{Q}_{br}$. At Case B, $w = 2.22$ MW.

Variables. w, work from the back-pressure stage, MW; $\dot{Q}_{br}$, heat to the power and steam branch, MW; r_w, work per unit of heat extracted, $\mathrm{MW_e\ MW_{th}^{-1}}$; $\dot{Q}_{gap}$, reboiler gap, MW.

Assumptions. Back-pressure co-generation.

Source. [**C**] Form calculated here. $r_w = 0.169$ from IEAGHG 2017-02 [1], Case 1A. 0.30 as high case [**A**].

Implementation. `plant::heatrec::extraction`, with the split exported by `plant::export::branch_split`

Validation. GOLD, test `eq_b_19`. 59.57 $\mathrm{MW_{th}}$ at Case B.

Equation B.20: Reactor allocation

Tier O

$$\dot{Q}_R = \dot{Q}_{proc} + \dot{Q}_{rec} + \dot{Q}_{ihx} + \dot{Q}_{duct} + \dot{Q}_{ves} + \dot{Q}_{br} - \dot{W}_s - \dot{W}_p \tag{B.20}$$

Interpretation. This is the answer to how big the reactor must be. Every use and every loss is added up, and the shaft work is taken off because it returns as heat.

Derivation. Set $\dot{Q}_{av}$ in Eq. B.16 equal to the sum of its uses, $\dot{Q}_{proc} + \dot{Q}_{rec} + \dot{Q}_{br}$, and solve for $\dot{Q}_R$. The electricity inside $\dot{Q}_{br}$ includes E_{circ}, so the balance is implicit through the flows and is solved by iteration.

Variables. $\dot{Q}_{rec}$, slip recycle heat, MW.

Assumptions. First law on the nuclear island.

Source. [**C**] Calculated here, from the screening balance.

Implementation. `plant::utility::reactor_required`

Validation. GOLD, test `eq_b_20`. 178.49 $\mathrm{MW_{th}}$ at Case B and 260.74 $\mathrm{MW_{th}}$ at Case A.

Equation B.21: Margin

Tier F

$$\dot{Q}_{spare} = \dot{Q}_{rated} - \dot{Q}_R \tag{B.21}$$

Interpretation. The selected reactor is larger than the plant needs. This line shows how much larger, and no income is booked against the difference.

Justification. A project decision: Case B at 210 $\mathrm{MW_{th}}$, with the margin kept idle. Its capital is charged in full in Section L.

Variables. $\dot{Q}_{spare}$, unused rating, $\mathrm{MW_{th}}$.

Assumptions. No revenue on the margin.

Source. [**C**] Calculated here.

Implementation. `plant::utility::margin`

Validation. GOLD, test `eq_b_21`. 31.5 $\mathrm{MW_{th}}$ at Case B.

Equation B.22: INL inlet fit

Tier E

$$T_{RI} = 0.8205\,T_{ROT} - 242.68 \qquad \text{(both in C)} \tag{B.22}$$

Interpretation. INL's rule of thumb from operating HTGRs ties inlet to outlet temperature. It is a check here, because this plant's inlet is set by the secondary return.

Fit range and origin. INL-21537 fits operating and past reactors and warns that the correlation serves that report only. In practice inlet temperature depends on more than the outlet.

Variables. T_{ROT}, reactor outlet temperature, K, or C in this fit.

Assumptions. Used as a check only.

Source. [**P**] INL-21537 [21], Table 2, PDF p. 16.

Implementation. Not implemented in the engine at this build. The fit is evaluated in this document only. Planned name `units::htgr::rit_check`.

Validation. PUB, test `eq_b_22`. Computed here at 925 C: 516.3 C, against 499.7 C from Eq. B.6, a gap of 16.6 K. The first sits above the 260 to 500 C range INL quotes and the second sits at its edge. Both are well above the 395 C normal limit of the HTTR vessel, TECDOC-1236 [12], Table 3-3.

Equation B.23: INL excess heat

Tier R

$$\dot{Q}_{ex} = \dot{m}_s c_p \left(T_{s,ret} - T_{s,ideal}\right) \tag{B.23}$$

Interpretation. INL asks how much hotter the helium returns than the process needs. That difference is heat the reactor could send to power instead.

Derivation. Eq. B.8 between the calculated and the ideal return. TEV-961 text quotes "71 C" and "3 C" at odds with its Table 2, so the table is used, and Section M records the conflict.

Variables. $\dot{Q}_{ex}$, excess heat for power, W; $T_{s,ideal}$, ideal helium return temperature, 461 C, K.

Assumptions. As Eq. B.8.

Source. [**P**] TEV-961 [36], Table 2 continued, PDF p. 18.

Implementation. `vv::gas_bench::tev961` evaluates it with `props::helium::duty_w` for the five TEV-961 cases of the validation line

Validation. PUB, test `eq_b_23`. Computed here: 2.04 $\mathrm{MW_t}$ at $c_p = 5190$ against the printed 2.0, so the reactor equivalent is 176.8 plus 2.0, the printed 178.8 $\mathrm{MW_t}$. The five ROT cases of TEV-961: Validation results, 5 rows (T961-2-ROT725C to T961-2-ROT925C): 5 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Worked example: reactor heat at Case B

Insert the Case B terms into Eq. B.20, in MW: process heat 120.80, slip heat 0.88, IHX loss 0.97, duct loss 0.31, vessel loss 2.66 and the branch 59.57. Their sum, less the shaft works 3.13 and 3.57, gives 178.49 MW_{th}. The branch is the term to notice: reboiler gap and electricity together take 59.57 MW_{th}, about half the process heat, because most of that electricity pays the full conversion penalty at $\eta_{pc} = 0.397$. Against the 210 MW_{th} rating, 31.5 MW_{th} stays idle.

Link to the sources

INL reports 178.8 MW_t for its Case 6 with 16.5 MW_e imported from the grid, and the circulators and losses stay outside that figure. This section adds them and the MEA reboiler gap, and it supplies all electricity from the reactor, so the required reactor at INL's own scale rises to 260.74 MW_{th}. Ahn and Lee print a higher process heat of 143.4 MW_{th} on a comparable duty, and the assumptions table keeps that case beside the central one.

5 C. Intermediate heat exchanger

The IHX passes reactor heat to a clean secondary loop. Its job here is hard for one reason: the secondary helium comes back at 466 C, so both ends of the exchanger run on a small temperature difference. This section takes the exchanger from energy balance to size, then checks the size against the only IHX that has operated, the HTTR unit.

Equation C.1: Hot side

Tier R

$$\dot{Q}_h = \dot{m}_p c_p \left(T_{h,in} - T_{h,out}\right) \tag{C.1}$$

Interpretation. Heat leaving the primary helium, read from its flow and temperature drop.

Derivation. Eq. A.8 on the hot channels with no shaft work and constant c_p.

Variables. $\dot{Q}_h$, heat lost by the hot stream, W; $T_{h,in}$, hot stream inlet, K.

Assumptions. SS, CP.

Source. [**T**] Consequence of Eq. A.8.

Implementation. `units::ihx::hot_side`

Validation. PUB, test `eq_c_01`. Computed here on HTTR data in TECDOC-1382 [11], Table 3-23: 9.48 MW at the 30 MW set and 4.96 MW at the 15 MW set.

Equation C.2: Cold side

Tier R

$$\dot{Q}_c = \dot{m}_s c_p \left(T_{c,out} - T_{c,in}\right) \tag{C.2}$$

Interpretation. Heat reaching the secondary helium, read in the same way.

Derivation. Eq. A.8 on the cold channels.

Variables. $\dot{Q}_c$, heat gained by the cold stream, W; $T_{c,in}$, cold stream inlet, K; $T_{c,out}$, cold stream outlet, K.

Assumptions. SS, CP.

Source. [**T**] Consequence of Eq. A.8.

Implementation. `units::ihx::cold_side`

Validation. PUB, test `eq_c_02`. Computed here on the same data: 9.41 and 4.90 MW. The two sides close within 1.28%.

Equation C.3: Closure

Tier R

$$\dot{Q}_h = \dot{Q}_c + \dot{Q}_{ihx} \tag{C.3}$$

Interpretation. The two sides differ only by what the exchanger loses to its surroundings.

Derivation. Eq. A.8 on the whole exchanger, with the loss from Eq. B.15.

Variables. $\dot{Q}_{ihx}$, IHX heat loss, W.

Assumptions. Loss from Eq. B.15.

Source. [**T**] Consequence of Eq. A.8.

Implementation. `units::ihx::closure`

Validation. ID, test `eq_c_03`. Closure inside the limit of Eq. A.15. The measured HTTR imbalance: Validation results, 2 rows (IHX-HTTR-15-CLOSURE to IHX-HTTR-30-CLOSURE): 2 reported without a limit. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence. Reported: IHX-HTTR-15-CLOSURE gives 0.01267 against 0; IHX-HTTR-30-CLOSURE gives 0.007509 against 0.

Equation C.4: Resistance network

Tier R

$$\frac{1}{UA} = \frac{1}{h_{fi} A_i} + \frac{R_{f,i}}{A_i} + \frac{\ln\left(d_o/d_i\right)}{2\pi k_w L N_t} + \frac{R_{f,o}}{A_o} + \frac{1}{h_{fo} A_o} \tag{C.4}$$

Interpretation. Heat crosses five barriers in series: two gas films, two fouling layers and the tube wall. As with resistors in series, the largest one controls.

Derivation. Steady conduction through a cylinder gives a resistance $\ln(d_o/d_i)/(2\pi k_w L)$ per tube. Convection gives $1/(hA)$ and fouling gives R_f/A. The same heat crosses each one, so resistances add.

Variables. UA, overall conductance, W K^{-1}; h_{fi}, film coefficient, tube side, W m^{-2} K^{-1}; h_{fo}, film coefficient, shell side, W m^{-2} K^{-1}; A_i, inner surface area, m^2; A_o, outer surface area, m^2; $R_{f,i}$, inner fouling resistance, m^2 K W^{-1}; $R_{f,o}$, outer fouling resistance, m^2 K W^{-1}; d_i, tube inner diameter, m; d_o, tube outer diameter, m; k_w, wall conductivity, W m^{-1} K^{-1}; L, tube length, m; N_t, number of tubes, 1.

Assumptions. Radial conduction.

Source. [**T**] Incropera [13].

Implementation. Not implemented in the engine at this build: no film coefficients are published for a helium IHX at this duty, so the engine sizes the exchanger on the conductance anchor of Eq. C.14. Planned name `units::ihx::ua_network`.

Validation. ID, test `eq_c_04`. Unit test on the series sum.

Equation C.5: Overall coefficient

Tier R

$$U_o = \frac{UA}{A_o} \tag{C.5}$$

Interpretation. The conductance spread over the outer area, which lets exchangers of different size be compared.

Derivation. Definition, on the outer area basis.

Variables. U_o, overall coefficient on outer area, W m^{-2} K^{-1}.

Assumptions. Outer area basis.

Source. [**T**] Incropera [13].

Implementation. Not implemented in the engine at this build. The HTTR values in the validation line are computed in this document from the measured set. Planned name `units::ihx::u_outer`.

Validation. EXP, test `eq_c_05`. Computed here from measured HTTR temperatures and flows, on an area of 212.9 m^2 derived from 96 tubes of 31.8 mm over 22.2 m: 428 W m^{-2} K^{-1} at the 30 MW set and 375 at the 15 MW set.

Equation C.6: Nusselt correlations

Tier E

$$Nu_{out} = 0.057\, C_0 Re^{0.8} Pr^{0.4} \ (800 < Re \le 7000), \quad 0.33\, C_0 Re^{0.6} Pr^{0.3} \ (7000 < Re)$$

$$Nu_{in} = 0.0228\, Re^{5/6} Pr^{0.4} \left(\frac{d}{D}\right)^{1/12} \left\{ 1 + \frac{0.0732}{\left[Re\,(d_{in}/D_h)^{2.5} \right]^{1/6}} \right\} \quad \text{(turbulent)} \tag{C.6}$$

Interpretation. These are the helical coil correlations behind the HTTR thermal model. They would give the film coefficients of Eq. C.4 if the shell factor were known.

Fit range and origin. Tochio and Nakagawa Eq 5, 6 and 8. Eq 6 defines $C_0 = \eta_f \eta_e \eta_\theta \eta_s E$, and the paper gives no number for it, so the shell side cannot be evaluated from the paper. The nomenclature defines D_h and d_{in} but gives no separate meaning for d/D in the turbulent form, and Eq 9 uses D for a Dean number, so the group is printed as published. The fit range is helical HTTR geometry. Design therefore uses the measured conductance in Eq. C.14.

Variables. Nu_{out}, shell-side Nusselt number, 1; Nu_{in}, tube-side Nusselt number, 1; C_0, shell-side correction factor, value not printed, 1; Re, Reynolds number, 1; Pr, Prandtl number, 1; d_{in}, tube inner diameter in the source, m; D_h, helix diameter in the source, m.

Assumptions. Helical HTTR geometry.

Source. **[P]** Tochio and Nakagawa [31], Eq 5, 6, 8, PDF p. 7. **[M15]**: C_0 is not printed.

Implementation. Not implemented in the engine at this build: the form needs C_0, which no source prints, so the engine carries no Nusselt form for the IHX. Planned name `units::ihx::nu_tochio`.

Validation. NONE, test `eq_c_06`. No benchmark can run without C_0. Design uses the conductance anchor Eq. C.14.

Equation C.7: Log-mean temperature difference

Tier R

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\Delta T_1 / \Delta T_2\right)} \tag{C.7}$$

Interpretation. The driving difference changes along the exchanger, so an average is needed. The right average is logarithmic because the difference itself decays exponentially with area.

Derivation. Along a counterflow exchanger, $d\dot{Q} = U \Delta T \, dA$, and each stream changes by $d\dot{Q}/C$, so $d(\Delta T) = -d\dot{Q}\,(1/C_h - 1/C_c)$. Substituting the first relation gives $d(\Delta T)/\Delta T = -U\, dA\,(1/C_h - 1/C_c)$. Integrating from end 1 to end 2 gives $\ln(\Delta T_2/\Delta T_1) = -UA(1/C_h - 1/C_c)$. The first relation integrated over the whole duty gives $\Delta T_2 - \Delta T_1 = -\dot{Q}(1/C_h - 1/C_c)$. Dividing removes the bracket and leaves $\dot{Q} = UA\,\Delta T_{lm}$.

Variables. ΔT_{lm}, log-mean temperature difference, K; ΔT_1, terminal difference at end 1, K; ΔT_2, terminal difference at end 2, K; U, overall heat transfer coefficient, W m^{-2} K^{-1}; A, heat transfer area, m^2; C_h, hot stream capacity rate, W K^{-1}; C_c, cold stream capacity rate, W K^{-1}.

Assumptions. Counterflow, constant U.

Source. **[T]** Incropera [13].

Implementation. `units::ihx::lmtd`

Validation. ID, PUB, test `eq_c_07`. Computed here for the HTTR 30 MW set, with end differences of 77 and 135 K: 103.3 K.

Equation C.8: Balanced limit

Tier R

$$\Delta T_{lm} \to \Delta T \quad \text{as} \quad \Delta T_1 \to \Delta T_2 = \Delta T \tag{C.8}$$

Interpretation. With equal capacity rates the two temperature lines run parallel, so the difference is the same everywhere and needs no averaging.

Derivation. Put $\Delta T_1 = \Delta T(1+\delta)$ and $\Delta T_2 = \Delta T$. Then $\ln(1+\delta) \approx \delta$ for small δ, and Eq. C.7 returns ΔT. The code uses this form when the two ends agree to within rounding, because Eq. C.7 is then zero over zero.

Variables. ΔT, local temperature difference, K.

Assumptions. Equal capacity rates.

Source. [**T**] Incropera [13].

Implementation. `units::ihx::lmtd`

Validation. ID, test `eq_c_08`. Limit test.

Equation C.9: Effectiveness

Tier R

$$\varepsilon_{HX} = \frac{\dot{Q}}{C_{min}\,(T_{h,in} - T_{c,in})} \tag{C.9}$$

Interpretation. The share of the thermodynamic maximum that the exchanger achieves. A value of one would need infinite area.

Derivation. The stream with the smaller capacity rate sees the larger temperature change, and no stream can change by more than the inlet difference. So $C_{min}(T_{h,in} - T_{c,in})$ is the largest possible duty.

Variables. ε_{HX}, exchanger effectiveness, 1; C_{min}, smaller capacity rate, W K^{-1}.

Assumptions. No loss in the definition.

Source. [**T**] Incropera [13]. The same definition is INL-11109 [9], Eq 1, PDF p. 12.

Implementation. `units::ihx::effectiveness`

Validation. PUB, test `eq_c_09`. Computed here: 0.867 for the HTTR 30 MW set and 0.946 for the balanced NUS-H2 exchanger, far tighter than the measured value.

Equation C.10: Number of transfer units

Tier R

$$NTU = \frac{UA}{C_{min}}, \qquad C^* = \frac{C_{min}}{C_{max}} \tag{C.10}$$

Interpretation. NTU says how large the exchanger is against the flow it serves. It measures size with the flow rate taken out.

Derivation. Definitions. For a balanced exchanger with $\dot{Q} = UA\,\Delta T$, it follows that NTU equals the stream temperature change over the approach.

Variables. NTU, number of transfer units, 1; C^*, capacity ratio, 1; C_{max}, larger capacity rate, W K^{-1}.

Assumptions. None.

Source. [**T**] Incropera [13].

Implementation. `units::ihx::ntu`

Validation. PUB, test `eq_c_10`. Computed here: 4.9 for HTTR, and 434 K over 25 K gives 17.4 at the NUS-H2 balanced point.

Equation C.11: Counterflow effectiveness

Tier R

$$\varepsilon_{HX} = \frac{1 - e^{-NTU(1-C^*)}}{1 - C^* e^{-NTU(1-C^*)}} \tag{C.11}$$

Interpretation. Given size and flows, this predicts the outlet temperatures with no iteration. The exchanger model runs on it.

Derivation. Take the cold stream as C_{min}. The difference at the end where the hot stream enters is $\Delta T_{hot} = (T_{h,in} - T_{c,in})(1 - \varepsilon_{HX})$, and at the other end it is $\Delta T_{cold} = (T_{h,in} - T_{c,in})(1 - C^* \varepsilon_{HX})$. The integration in Eq. C.7 gives $\Delta T_{hot}/\Delta T_{cold} = e^{-NTU(1-C^*)}$. Solving for ε_{HX} gives the statement. Swapping the roles of the streams gives the same result.

Variables. $\dot{Q}$, exchanger duty, W.

Assumptions. Constant U.

Source. [**T**] Incropera [13].

Implementation. `units::ihx::eps_ntu`, called by `units::ihx::rate`

Validation. EXP, test `eq_c_11`. Computed here for HTTR at 30 MW, with 828 and 248 C inlets, 4.1 and 3.6 kg s^{-1} and UA = 91 kW K^{-1}: 752 and 385 C, against measured 751 and 383 C, inside 5 K. At 15 MW with UA = 80 kW K^{-1}: 421 and 237 C against 420 and 235 C. Each UA was derived from the same data set, so this tests the algebra and the balance, and it is no blind prediction. HTTR-SR at 905 C has no row and stays a target. The engine rows on the HTTR sets and the INL IHX table: Validation results, 16 rows (IHX-HTTR-15-TP to IHX-INL-TH-CFG1_950C_7MPA): 12 pass, 4 reported without a limit. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence. Reported: IHX-INL-Q-CFG1_900C_2MPA gives 54 against 54 MW; IHX-INL-Q-CFG1_900C_7MPA gives 58.74 against 58.6 MW; IHX-INL-Q-CFG1_950C_2MPA gives 53.53 against 53.6 MW; IHX-INL-Q-CFG1_950C_7MPA gives 60.64 against 60.5 MW.

Equation C.12: Balanced effectiveness

Tier R

$$\varepsilon_{HX} = \frac{NTU}{1 + NTU} \tag{C.12}$$

Interpretation. At equal capacity rates, each extra unit of size pays back less. Going from effectiveness 0.87 to 0.95 needs about three times the NTU.

Derivation. Take $C^* \to 1$ in Eq. C.11. Both numerator and denominator vanish. Expanding the exponential to first order in $(1 - C^*)$ gives the statement.

Variables. C_h, hot stream capacity rate, W K^{-1}.

Assumptions. $C^* = 1$.

Source. [**T**] Incropera [13].

Implementation. `units::ihx::eps_ntu`

Validation. ID, test `eq_c_12`. Limit test against Eq. C.11.

Equation C.13: Area

Tier R

$$A = \frac{\dot{Q}}{U \Delta T_{lm}} \tag{C.13}$$

Interpretation. Small temperature differences cost area. Halving the approach doubles the surface.

Derivation. Rearrange the result of Eq. C.7.

Variables. A, heat transfer area, m^2.

Assumptions. Uniform U.

Source. [**T**] Incropera [13].

Implementation. `units::ihx::area`, called by `units::ihx::size`

Validation. PUB, test `eq_c_13`. Computed here for Case B, balanced at 25 K: UA =4.83 MW K^{-1}, which is 12,080 m^2 at 400 W m^{-2} K^{-1}, or 6,358 m^2 at 760 W m^{-2} K^{-1}.

Equation C.14: Module anchor

Tier E

$$UA_{mod} = \frac{UA}{N_{mod}}, \qquad UA_{HTTR} = 85 \text{ kW K}^{-1} \tag{C.14}$$

Interpretation. The only measured conductance for a large helical helium exchanger is the HTTR value. Dividing the plant duty over modules, and reporting the ratio to HTTR, shows the scale of the claim.

Fit range and origin. Tochio and Nakagawa plot UA at about 85 kW K^{-1} at full power in Fig 8, and Fig 10 shows it falling to about 65 kW K^{-1} at a lower LMTD. This is read from the figures. The module count of 2 to 4 is a project position, because no source supports a single unit at this conductance.

Variables. UA_{mod}, conductance per module, W K^{-1}; UA_{HTTR}, measured HTTR conductance, W K^{-1}; N_{mod}, number of modules, 1.

Assumptions. 2 to 4 modules. The ratio to HTTR is reported.

Source. [**P**] Tochio and Nakagawa [31], Figs 8, 10, PDF p. 7 to 8.

Implementation. `units::ihx::modules`

Validation. EXP, test `eq_c_14`. Computed here: Case B needs 57 times HTTR overall, so no single unit is claimed. With four modules, each needs 1.21 MW K^{-1}, or 14 times HTTR.

Equation C.15: Unbalanced option

Tier O

$$\dot{m}_p = \frac{\dot{Q}}{c_p\,(T_{h,in} - T_{h,out})}, \qquad \Delta T_{cold} = T_{h,out} - T_{c,in}, \qquad UA = \frac{\dot{Q}}{\Delta T_{lm}\,(\Delta T_{hot}, \Delta T_{cold})} \tag{C.15}$$

Interpretation. Run more flow on the primary side and the cold end opens up, so the exchanger needs far less conductance. The price is a hotter reactor inlet.

Derivation. Hold the 25 K hot end. Choose $T_{h,out}$. Then Eq. C.1 gives the primary flow, the cold end follows from the secondary return, and Eq. C.7 and Eq. C.13 give the conductance. This is a design option, and the central plant stays balanced.

Variables. ΔT_{cold}, cold-end temperature difference, K; ΔT_{hot}, hot-end temperature difference, K.

Assumptions. 25 K hot end fixed.

Source. [**C**] Calculated here.

Implementation. `units::ihx::size`, which takes the four terminal temperatures and returns the capacity rates, LMTD and UA. `plant::utility::level4` calls it with the primary leaving at the secondary return plus the approach

Validation. PUB, test `eq_c_15`. Computed here at $T_{h,out}$ = 600 C with an LMTD of 64.9 K. Case A: 104.7 kg s^{-1} and 2.72 MW K^{-1}. Case B: 71.6 kg s^{-1} and 1.86 MW K^{-1}, or 22 times HTTR.

Equation C.16: Tube friction

Tier E

$$\Delta p = f_D \frac{L}{D} \frac{\rho u^2}{2}, \qquad f_D = 0.3164\, Re^{-1/4} \tag{C.16}$$

Interpretation. Friction in a smooth tube at moderate Reynolds number. It is enough for a pressure estimate, and it leaves out coil effects.

Fit range and origin. Blasius, as printed in Sato Eq 10 with a coefficient of 0.3164. Yin Eq 14 prints the same law without the exponent on Reynolds number, which is a misprint recorded in Section M.

Variables. Δp, pressure drop, Pa.

Assumptions. Straight smooth tube.

Source. **[P]** Sato [26], Eq 10, PDF p. 4. **[M14]**: no measured IHX drop.

Implementation. `units::circulator::dp_blasius`, summed over the loop by `plant::utility::loop_drop`

Validation. PUB, test `eq_c_16`. Budget targets: HTR-10 design drops of 15 and 70 kPa, TECDOC-1236 [12] Table 5-5, PDF p. 89. INL assumes 50 kPa.

Equation C.17: Temperature cascade

Tier F

$$T_{s,sup} = T_{ROT} - \Delta T_{app} - \Delta T_{duct} \tag{C.17}$$

Interpretation. Each step from core to reformer spends temperature. This line shows how little is left to spend.

Justification. INL assumes a 25 K approach and no duct drop. At 925 C those give 900 C, and 29 K above the 871 C reformer exit. A 5 K duct drop leaves 24 K. A 50 K approach leaves 4 K, and with a 5 K drop the helium falls below the reformer exit, so a fired stage would return.

Variables. ΔT_{duct}, duct temperature drop, K.

Assumptions. 25 K approach, duct 0 K.

Source. **[P]** TEV-961 [36], printed p. 10. INL-21537 [21], Table 2, PDF p. 16.

Implementation. `units::ihx::cascade`

Validation. PUB, test `eq_c_17`. The figures above are arithmetic on published assumptions. Measured HTTR hot ends sit at 77 to 82 K.

Equation C.18: Constraints

Tier F

$$T_{s,sup} - T_{ref,out} \geq 10 \text{ K}, \qquad T_{tube} \leq 955 \text{ °C}, \qquad \Delta p_{tube} \leq 0.3 \text{ MPa} \tag{C.18}$$

Interpretation. Limits the design must respect. The first is an INL approach rule, and the other two are HTTR tube limits taken by analogy.

Justification. INL-21537 Table 2 sets a 10 C minimum process approach. Tochio Table 2 gives the allowable tube temperature of 955 C and differential pressure of 0.3 MPa for Hastelloy XR. They apply to NUS-H2 only by analogy, because the tube and service life differ.

Variables. T_{tube}, IHX tube temperature, K, stated in C; Δp_{tube}, differential pressure across the tube, Pa.

Assumptions. HTTR limits apply by analogy.

Source. **[P]** INL-21537 [21], Table 2, PDF p. 16. Tochio and Nakagawa [31], Table 2, PDF p. 5.

Implementation. `units::ihx::check_limits`

Validation. ID, test `eq_c_18`. A test sets each limit and reads the flag. Reported: a 50 K approach leaves a 4 K pinch and fails the first limit.

Equation C.19: Pressure sign

Tier F

$$p_s \geq p_p + 0.1 \text{ MPa} \quad \text{(HTTR)} \tag{C.19}$$

Interpretation. HTTR keeps the secondary helium above the primary, so a tube leak pushes clean gas into the reactor circuit and contains radioactivity. The project leaves its own sign to the safety case.

Justification. IAEA-TECDOC-1382 printed p. 8 states the HTTR rule and its reason. INL-11109 makes the same argument for a 7.1 MPa loop on PDF p. 16. No safety case exists for NUS-H2, so the balance holds both loops at 7.0 MPa and marks the sign as open.

Variables. p_s, secondary helium pressure, Pa; p_p, primary helium pressure, Pa.

Assumptions. An open item.

Source. [**P**] IAEA-TECDOC-1382 [11], printed p. 8, PDF p. 14.

Implementation. Not implemented in the engine at this build: the balance holds both loops at 7.0 MPa and checks no pressure sign. Planned as a further flag of `units::ihx::check_limits`.

Validation. NONE, test `eq_c_19`. No benchmark. The item stays open for the safety case.

Worked example: sizing the IHX at Case B

Take the balanced design with 120.80 MW and a 25 K approach at both ends. Eq. C.8 gives $\Delta T_{lm} = 25$ K, so Eq. C.13 gives UA =4.83 MW K^{-1}. Each stream changes by 434 K, so $NTU = 17.4$ by Eq. C.10 and $\varepsilon_{HX} = 0.946$ by Eq. C.12. At the HTTR scale of 400 W m^{-2} K^{-1}, the surface is 12,080 m^2. The measured HTTR conductance is 85 kW K^{-1}, so Case B needs 57 HTTR units in conductance. Four modules bring that to 14 per module, and the unbalanced option at a 600 C primary outlet brings the total to 22 at the cost of a hotter reactor inlet.

Link to the sources

Tochio and Nakagawa report a stable UA of about 85 kW K^{-1} over a 50-day high-temperature run, with no sign of degradation. That is the strongest evidence for this class of machine. It also marks the distance to this design: the balanced approach of 25 K sits a factor of three below the 77 to 82 K measured hot ends, and INL adopted 25 K as an assumption. The IHX therefore stays a labelled risk, and the 50 K sensitivity is carried through the cascade of Eq. C.17.

6 D. Steam methane reformer

The reformer is where nuclear heat becomes chemical energy. Two descriptions sit side by side here. The plant ledger uses equilibrium at the INL exit condition, Eq. D.11 at 871 C, with the INL heat basis. The Xu and Froment tube model then tests that assumption. It runs outside its fitted range, so its cards carry tier O, and no tube-model result enters the headline ledger.

Equation D.1: Reaction I

Tier R

$$\mathrm{CH_4 + H_2O \rightleftharpoons CO + 3\,H_2}, \qquad \Delta_r h^\circ_{298} = +206.1 \text{ kJ mol}^{-1} \tag{D.1}$$

Interpretation. Methane and steam make synthesis gas. The reaction takes in a large amount of heat and raises the number of moles, so it wants high temperature and low pressure.

Derivation. Stoichiometry from Xu and Froment Part I Table 2. The heat comes from Table 7. Table 2 lists $-\Delta H$ and Table 7 lists $+\Delta H$, so the sign must be read with care. Here it is positive for an endothermic reaction.

Variables. $\Delta_r h^\circ_{298}$, standard enthalpy of reaction at 298 K, J mol^{-1}.

Assumptions. None.

Source. [**P**] Xu and Froment Part I [37], Table 2, PDF p. 3, and Table 7, PDF p. 8.

Implementation. `props::equilibrium::Basis::amounts` and `units::smr_tube::Tube::flows` carry it as the extent of reaction I

Validation. ID, test `eq_d_01`. Element balance by Eq. A.5.

Equation D.2: Reaction II, shift

Tier R

$$\mathrm{CO} + \mathrm{H_2O} \rightleftharpoons \mathrm{CO_2} + \mathrm{H_2}, \qquad \Delta_r h^\circ_{298} = -41.15 \text{ kJ mol}^{-1} \tag{D.2}$$

Interpretation. Carbon monoxide takes oxygen from steam and frees more hydrogen. The heat released is small, so the reaction moves mainly with temperature through its equilibrium constant.

Derivation. As Eq. D.1. Section E applies this reaction in the shift beds.

Variables. $\Delta_r h^\circ_{298}$, standard enthalpy of reaction at 298 K, J mol^{-1}.

Assumptions. None.

Source. [**P**] Xu and Froment Part I [37], Table 2, PDF p. 3, and Table 7, PDF p. 8.

Implementation. `props::equilibrium::Basis::amounts` and `units::smr_tube::Tube::flows` carry it as the extent of reaction II

Validation. ID, test `eq_d_02`. Element balance by Eq. A.5.

Equation D.3: Reaction III

Tier R

$$\mathrm{CH_4} + 2\,\mathrm{H_2O} \rightleftharpoons \mathrm{CO_2} + 4\,\mathrm{H_2}, \qquad \Delta_r h^\circ_{298} = +164.9 \text{ kJ mol}^{-1} \tag{D.3}$$

Interpretation. The direct route to CO_2. It is the sum of the first two reactions, so it adds no thermodynamic freedom, though Xu and Froment give it its own rate.

Derivation. Add Eq. D.1 and Eq. D.2. The heats add to 164.95, which rounds to the printed value.

Variables. $\Delta_r h^\circ_{298}$, standard enthalpy of reaction at 298 K, J mol^{-1}.

Assumptions. None.

Source. [**P**] Xu and Froment Part I [37], Table 2, PDF p. 3, and Table 7, PDF p. 8.

Implementation. `units::xf::Kinetics::rates`, as the third rate with K_3 of Eq. D.9. The equilibrium solve needs only the two independent extents of Eq. A.6

Validation. ID, test `eq_d_03`. Sum of Eq. D.1 and Eq. D.2.

Equation D.4: Heavier gas

Tier O

$$\mathrm{C}_n\mathrm{H}_m + n\,\mathrm{H_2O} \rightarrow n\,\mathrm{CO} + \left(n + \frac{m}{2}\right)\mathrm{H_2} \tag{D.4}$$

Interpretation. Ethane and heavier gases reform much faster than methane, so they are taken to completion at the tube inlet and only methane is tracked after that.

Derivation. EVA measurements quoted in TECDOC-1236 show 30 to 40% of the ethane gone after 1 m of catalyst and all of it by 4 m. A heavier fraction therefore reaches the reaction zone only briefly. Natural gas in TEV-953 holds 93.571% methane, 3.749% ethane, 0.920% propane, 0.260% butane, 0.040% pentane and 0.010% hexane by mole, with 0.250% CO_2. Counting carbon gives 1.05379 mol of carbon per mole of gas.

Variables. n, carbon atoms in the hydrocarbon, 1; m, hydrogen atoms in the hydrocarbon, 1.

Assumptions. Rapid conversion.

Source. [**A**] Support from TECDOC-1236 [12], PDF p. 70. [**P**] Composition from TEV-953 [35], A-5, PDF p. 39.

Implementation. `props::equilibrium::basis` for the equilibrium unit and `units::smr_tube::pre_convert` for the tube, each at the inlet

Validation. PUB, test `eq_d_04`. Computed here: 1.05379 mol C per mol of gas from the printed composition, equal to the screening constant.

Equation D.5: Feed ratio

Tier F

$$S/C = \frac{F_{H_2O}}{F_C} = 3.0 \tag{D.5}$$

Interpretation. Excess steam keeps carbon off the catalyst and pushes conversion up. Each extra unit of steam costs heat to raise.

Justification. INL sets the steam-to-carbon ratio at 3.0 to avoid carbon formation. The Case 6 sheets are missing, so the same value is carried from the other cases and disclosed.

Variables. F_{H_2O}, steam molar feed, mol s^{-1}; F_C, carbon molar feed, mol s^{-1}.

Assumptions. The Case 6 value is not stated.

Source. [**P**] TEV-953 [35], PDF p. 18 and p. 64. [**M1**]: Case 6 sheets missing.

Implementation. `plant::flowsheet::S_TO_C`, applied to every carbon atom of feed and recycle in `plant::flowsheet::pass`

Validation. ID, test `eq_d_05`. Round trip with the mole table of Eq. D.6.

Equation D.6: Mole table from extents

Tier R

$$\begin{aligned} n_{CH_4} &= 1 - x_1, & n_{CO} &= x_1 - x_2, & n_{CO_2} &= \xi_{CO_2} + x_2, \\ n_{H_2} &= \xi_{H_2} + 3x_1 + x_2, & n_{H_2O} &= S_C - x_1 - x_2, & n_{tot} &= \Sigma_0 + 2x_1 \end{aligned} \tag{D.6}$$

Interpretation. Two numbers fix the whole composition once the feed is known: how far reaction I has gone and how far the shift has gone. Everything else follows from counting atoms.

Derivation. Take one mole of methane fed with S_C moles of steam, ξ_{CO_2} of CO_2 and ξ_{H_2} of H_2. Reaction I at extent x_1 removes x_1 of CH_4 and H_2O and adds x_1 of CO and $3x_1$ of H_2. Reaction II at extent x_2 moves x_2 from CO and H_2O to CO_2 and H_2. Reaction III needs no separate extent by Eq. A.6. Reaction I adds two moles per event and reaction II adds none, so the total is $\Sigma_0 + 2x_1$ with $\Sigma_0 = 1 + S_C + \xi_{CO_2} + \xi_{H_2}$ plus any inert.

Variables. n_i, moles of i per mole of CH_4 fed, 1; x_1, extent of reaction I, which is total methane conversion, 1; x_2, extent of reaction II, 1; ξ_{CO_2}, CO_2 fed per mole of CH_4, 1; ξ_{H_2}, H_2 fed per mole of CH_4, 1; S_C, steam fed per mole of CH_4, 1; Σ_0, total moles fed per mole of CH_4, 1.

Assumptions. x_1 and x_2 are the extents of reactions I and II.

Source. [**T**] Textbook result. The same split of conversions is in Xu and Froment Part II [38], PDF p. 3 and Notation.

Implementation. `props::equilibrium::Basis::amounts` and `props::equilibrium::Basis::total`, on the feed flows in mol s^{-1}, which carry this table scaled by the CH_4 fed

Validation. ID, test `eq_d_06`. Element balance by Eq. A.5.

Equation D.7: Equilibrium constant of reaction I

Tier E

$$K_1(T) = 1.198 \times 10^{13} \exp\left(-\frac{26830}{T}\right) \text{ bar}^2 \tag{D.7}$$

Interpretation. The reforming equilibrium climbs steeply with temperature. This is why a reformer needs to run hot and why 871 C matters so much.

Fit range and origin. Zhang et al. Table 1 prints $1.198 \times 10^{11} \exp(-26830/T)$ in MPa^2. One MPa^2 is 100 bar^2, which gives the bar form. Neither Xu and Froment paper prints $K(T)$, and the Hou and Hughes origin is not in the library. Zhang Eq 11 prints 10^{23}, which is the same constant in Pa^2. Zhang Eq 6 and 7 carry a stray $\times 100\%$. Hoseinzade and Adams use an atm^2 basis, which differs by 2.2%. Section M lists these.

Variables. K_1, equilibrium constant of reaction I, bar^2.

Assumptions. IG, standard state of 1 bar.

Source. [**P**] Zhang et al. [41], Table 1, PDF p. 5, converted from MPa^2 [**C**]. [**M3**]: origin not in the library.

Implementation. `props::equilibrium::k1_zhang_bar2`

Validation. PUB, test `eq_d_07`. Computed here against K_1 built from the NIST Shomate data over 680 to 1144 K: largest deviation 5.3%, and 3.3% at 1144 K. Validation results, 2 rows (PROPS-K-NIST-JANAF to PROPS-K-ZHANG-NIST): 2 pass. The results mark 1 of these rows as using the same data as the model, and 1 as independent.

Equation D.8: Equilibrium constant of reaction II

Tier E

$$K_2(T) = 1.767 \times 10^{-2} \exp\left(\frac{4400}{T}\right) \tag{D.8}$$

Interpretation. Shift weakens as temperature rises, so the reformer exit still holds carbon monoxide and downstream shift beds run cooler to finish the job.

Fit range and origin. Zhang et al. Table 1. Zhang Eq 12 misprints this constant as 1.176×10^{-2} with a negative exponent, and Hoseinzade and Adams Eq 52 prints an exponent of -4.063 where $\ln(1.767 \times 10^{-2}) = -4.036$. Table 1 is used.

Variables. K_2, equilibrium constant of reaction II, 1.

Assumptions. IG.

Source. [**P**] Zhang et al. [41], Table 1, PDF p. 5.

Implementation. `props::equilibrium::k2_zhang`

Validation. PUB, test `eq_d_08`. Computed here against NIST-based K_2 on the same range: largest deviation 3.6%, and 3.6% at 1144 K. Validation results, 2 rows (PROPS-K-ZHANG-NIST to PROPS-K2-ZHANG-LTS): 1 pass, 1 reported without a limit. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence. Reported: PROPS-K2-ZHANG-LTS gives 0.8321 against 1 ratio.

Equation D.9: Equilibrium constant of reaction III

Tier R

$$K_3 = K_1 K_2 \tag{D.9}$$

Interpretation. Reaction III is reaction I followed by reaction II, so its equilibrium constant is the product.

Derivation. Add the Gibbs energies of the two reactions and take exponentials. This is the consequence of Eq. A.6.

Variables. K_3, equilibrium constant of reaction III, bar^2.

Assumptions. Eq. A.6.

Source. [**P**] Hoseinzade and Adams [10], Eq 51 to 53, PDF p. 7.

Implementation. Inside `units::xf::Kinetics::rates`, which forms K_3 from the pair it is given

Validation. ID, test `eq_d_09`. Direct identity in the test.

Equation D.10: Reaction quotients

Tier R

$$Q_1 = \frac{p_{H_2}^3 p_{CO}}{p_{CH_4} p_{H_2O}}, \qquad Q_2 = \frac{p_{H_2} p_{CO_2}}{p_{CO} p_{H_2O}}, \qquad Q_3 = Q_1 Q_2 \tag{D.10}$$

Interpretation. A quotient measures where the mixture stands now. Compared with K, it shows which way the reaction will run.

Derivation. Mass action with fugacity equal to partial pressure and a 1 bar standard state. Pressure in bar makes Q_1 directly comparable with Eq. D.7.

Variables. Q_1, quotient of reaction I, bar^2; Q_2, quotient of reaction II, 1; Q_3, quotient of reaction III, bar^2.

Assumptions. IG, standard state 1 bar.

Source. [**T**] Textbook result.

Implementation. `props::equilibrium::Basis::residual`, on the logarithms

Validation. ID, test `eq_d_10`. $Q_3 = Q_1 Q_2$ in the test.

Equation D.11: Equilibrium

Tier R

$$Q_1\left(x_1, x_2, p\right) = K_1(T), \qquad Q_2\left(x_1, x_2\right) = K_2(T) \qquad \text{at the exit } T,\ p \tag{D.11}$$

Interpretation. Two equations and two unknowns. Solving them gives the composition the catalyst would reach given enough time, which is the upper limit of conversion at the exit condition.

Derivation. Substitute Eq. D.6 into Eq. D.10 with $p_i = p y_i$. In logarithms, $3\ln n_{H_2} + \ln n_{CO} - \ln n_{CH_4} - \ln n_{H_2O} + 2\ln(p/n_{tot}) = \ln K_1$ and $\ln n_{H_2} + \ln n_{CO_2} - \ln n_{CO} - \ln n_{H_2O} = \ln K_2$. Solve by damped Newton on (x_1, x_2), keeping each mole number positive.

Variables. n_{tot}, total moles per mole of CH_4 fed, 1.

Assumptions. Isobaric, IG.

Source. [**T**] Textbook result. INL uses Gibbs minimisation, TEV-953 [35], PDF p. 18.

Implementation. `props::equilibrium::reform`, called by `units::smr_eq::solve`

Validation. PUB, test `eq_d_11`. Computed here against TECDOC-1236 Tables 2.1 and 2.2, PDF p. 13, at $S/C = 3.98$ back-calculated from the printed gas: 66.2, 79.5 and 81.4% against printed 66.1, 79.8 and 81.3%, within 0.5 points. Computed here against INL raw syngas, TEV-953 A-40, PDF p. 74: H_2 48.2, CO 9.1, CO_2 5.3, H_2O 33.9 and CH_4 3.5 mol% against printed 47.8, 9.2, 5.3, 34.1 and 3.5, within 0.5 mol%. The engine rows on the printed feeds: Validation results, 52 rows (IA-1-800C-20BAR to T961-3-N2): 50 pass, 2 reported without a limit. Reported: PROPS-TE1236-825C-41.9BAR gives 71.92 against 72 %; PROPS-TE1236-850C-41.9BAR gives 77.42 against 77.6 %.

Equation D.12: Thermodynamic consistency

Tier R

$$\frac{d\ln K}{d(1/T)} = -\frac{\Delta_r h}{R} \tag{D.12}$$

Interpretation. The slope of $\ln K$ against $1/T$ is the heat of reaction. An exponential fit therefore carries an implied heat, which can be checked against Hess.

Derivation. From $\ln K = -\Delta_r g / RT$ and the Gibbs-Helmholtz relation. A pure exponential in $1/T$ has a constant slope, so the fit implies a constant $\Delta_r h$ across its range.

Variables. K, equilibrium constant, mixed.

Assumptions. Constant $\Delta_r h$ across the fit.

Source. [**T**] Textbook result.

Implementation. `props::equilibrium::k1_nist_bar2` and `props::equilibrium::k2_nist` build K from Gibbs energies, so they meet this relation by construction

Validation. PUB, test `eq_d_12`. Computed here: slopes of 26830 and 4400 K give 223.1 and 36.6 kJ mol^{-1}, against 224.0 and 37.3 from XF1 Table 7 at 948 K.

Equation D.13: Rate coefficients

Tier E

$$k_i = A(k_i) \exp\left(-\frac{E_{a,i}}{RT}\right), \qquad (D.13)$$

i	1	2	3
$A(k_i)$	4.225×10^{15}	1.955×10^{6}	1.020×10^{15}
$E_{a,i}$ (kJ mol^{-1})	240.1	67.13	243.9

Interpretation. Reforming has a high activation energy, so its rate rises fast with temperature. Shift has a low one, which is why it stays near equilibrium throughout the tube.

Fit range and origin. Xu and Froment Tables 5 and 6, read from the page image, because the text layer corrupts numbers. Fit range: 773 to 848 K and 3 to 15 bar for reforming, 573 to 673 K for reverse shift and methanation. At 1144 K this is an extrapolation of about 300 K. Tables 5 and 6 hold reference activity, an aged catalyst, and the paper multiplies all k by 2.246 for fresh catalyst. Hoseinzade and Adams Eq 48 to 50 include that factor without saying so. Units follow XF: $A(k_1)$ and $A(k_3)$ in kmol bar$^{0.5}$ kg^{-1} h^{-1}, and $A(k_2)$ in kmol kg^{-1} h^{-1} bar^{-1}. XF1 gives R in kJ/kmol while E is in kJ/mol, so the code holds R and E on the same mole basis.

Variables. k_i, rate coefficient of reaction i, XF units, see working; $A(k_i)$, pre-exponential factor of k_i, XF units; $E_{a,i}$, activation energy of reaction i, J mol^{-1}.

Assumptions. Reference activity. Fit range 773 to 848 K and 3 to 15 bar.

Source. [**P**] Xu and Froment Part I [37], Tables 5, 6, PDF p. 7, read as an image.

Implementation. Inside `units::xf::Kinetics::rates`, on the constants of `units::xf::TABLE6`

Validation. ID, test `eq_d_13`. Computed here: Table 6 evaluated at the Table 5 reference temperatures, over the Table 5 values, gives ratios from 0.996 to 1.015 across all seven constants, with two reference temperatures, 648 and 823 K. Validation results, 10 rows (XF-1-KCH4 to XF-2-TABLES): 10 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation D.14: Adsorption constants

Tier E

$$K_j = A(K_j) \exp\left(-\frac{\Delta H_j}{RT}\right) \qquad (D.14)$$

j	CO	H_2	CH_4	H_2O
$A(K_j)$	8.23×10^{-5}	6.12×10^{-9}	6.65×10^{-4}	1.77×10^{5}
ΔH_j (kJ mol^{-1})	-70.65	-82.90	-38.28	$+88.68$

Interpretation. Each adsorbed species occupies part of the nickel surface. Adsorption grows weaker as the catalyst gets hotter, except for water, whose fitted constant behaves the other way.

Fit range and origin. XF1 Tables 5 and 6. K_{CO}, K_{H_2} and K_{CH_4} are in bar^{-1} and K_{H_2O} is dimensionless. The positive value for water is a dissociative constant, and the paper notes it does not obey van't Hoff. The text on the same page prints 6.13×10^{-9} where Table 6 has 6.12×10^{-9}, and the table is used. The lower limit printed for K_{H_2O} in Table 5, 0.0317, is a misprint with no effect on the model. Hoseinzade and Adams Eq 55 prints 1.77×10^{3}, wrongly dividing a dimensionless constant by 100, so it must not be copied.

Variables. K_j, adsorption constant of species j, bar^{-1}, or 1 for H_2O; $A(K_j)$, pre-exponential factor of K_j, as K_j; ΔH_j, adsorption enthalpy of species j, J mol^{-1}.

Assumptions. Reference fit. Two reference temperatures in Table 5.

Source. [**P**] Xu and Froment Part I [37], Table 6, PDF p. 7, with Table 5 on the same page.

Implementation. Inside `units::xf::Kinetics::rates`, on the constants of `units::xf::TABLE6`

Validation. ID, test `eq_d_14`. Covered by the ratio test on Eq. D.13.

Equation D.15: Adsorption denominator

Tier E

$$DEN = 1 + K_{CO}p_{CO} + K_{H_2}p_{H_2} + K_{CH_4}p_{CH_4} + \frac{K_{H_2O}p_{H_2O}}{p_{H_2}} \tag{D.15}$$

Interpretation. DEN counts the competition for surface sites. The larger it is, the less room the reaction has. It is squared in each rate.

Fit range and origin. Langmuir-Hinshelwood form from the mechanism of Xu and Froment. The water term is a ratio to hydrogen because it describes the oxygen coverage set by the dissociation of steam.

Variables. DEN, adsorption denominator, 1; p_{CO}, partial pressure of CO, in bar in this form, bar.

Assumptions. Partial pressures in bar.

Source. [**P**] Xu and Froment Part I [37], Eq 3, PDF p. 5, printed with the rate group.

Implementation. Inside `units::xf::Kinetics::rates`

Validation. ID, test `eq_d_15`. Computed here at 1000 K and 30 bar with y of 0.15, 0.45, 0.25 and 0.03 for CH_4, H_2O, H_2 and CO: $DEN^2 = 82.7$.

Equation D.16: Rate of reaction I

Tier E

$$r_1 = \frac{k_1}{p_{H_2}^{2.5}} \left(p_{CH_4}p_{H_2O} - \frac{p_{H_2}^3 p_{CO}}{K_1} \right) \Big/ DEN^2 \tag{D.16}$$

Interpretation. The bracket is the distance from equilibrium, and it closes to zero there. The denominator holds the competition for sites. The leading hydrogen term means hydrogen slows its own formation.

Fit range and origin. Mechanistic rate of Xu and Froment. Note the singularity: as $p_{H_2} \to 0$ the rate diverges, so a feed needs some hydrogen or heavier hydrocarbon. INL adds hydrogen to 2 mol% in its gas feed, TEV-953 PDF p. 18.

Variables. r_1, rate of reaction I, kmol kg^{-1} h^{-1}.

Assumptions. Diverges as $p_{H_2} \to 0$, so the feed must hold hydrogen.

Source. [**P**] Xu and Froment Part I [37], Eq 3, PDF p. 5.

Implementation. `units::xf::Kinetics::rates`

Validation. ID, test `eq_d_16`. The rate is zero at the equilibrium of Eq. D.11. The rates against Xu and Froment's own experiments: Validation results, 1 row (X-1): 1 not run.

Equation D.17: Rate of reaction II

Tier E

$$r_2 = \frac{k_2}{p_{H_2}} \left(p_{CO}p_{H_2O} - \frac{p_{H_2}p_{CO_2}}{K_2} \right) \Big/ DEN^2 \tag{D.17}$$

Interpretation. The same structure for shift. Its rate coefficient is large, so the reaction runs near equilibrium and can change sign along the tube.

Fit range and origin. As Eq. D.16. Xu and Froment find the intrinsic r_2 changes sign at about 4 m in their commercial tube (Part II, PDF p. 4).

Variables. r_2, rate of reaction II, kmol kg^{-1} h^{-1}; p_{CO_2}, partial pressure of CO_2, in bar in this form, bar.

Assumptions. As Eq. D.16.

Source. [**P**] Xu and Froment Part I [37], Eq 3, PDF p. 5.

Implementation. `units::xf::Kinetics::rates`

Validation. ID, test `eq_d_17`. The rate is zero at equilibrium.

Equation D.18: Rate of reaction III

Tier E

$$r_3 = \frac{k_3}{p_{H_2}^{3.5}} \left(p_{CH_4} p_{H_2O}^2 - \frac{p_{H_2}^4 p_{CO_2}}{K_3} \right) \Big/ DEN^2 \tag{D.18}$$

Interpretation. The direct route to CO_2, with its own rate although Eq. D.9 removes its thermodynamic independence.

Fit range and origin. As Eq. D.16, with the steeper hydrogen power of the mechanism.

Variables. r_3, rate of reaction III, kmol kg^{-1} h^{-1}; p_{H_2}, partial pressure of H_2, in bar in this form, bar.

Assumptions. As Eq. D.16.

Source. [**P**] Xu and Froment Part I [37], Eq 3, PDF p. 5.

Implementation. `units::xf::Kinetics::rates`

Validation. ID, test `eq_d_18`. The rate is zero at equilibrium.

Equation D.19: Species rates

Tier R

$$r_{CO} = r_1 - r_2, \qquad r_{CO_2} = r_2 + r_3, \qquad r_{CH_4} = r_1 + r_3 \tag{D.19}$$

Interpretation. Each species rate is the sum over reactions that touch it. r_{CH_4} is the rate of disappearance, so it is positive for reforming.

Derivation. Eq. A.4 applied to the three reactions. Two of these rates are independent, as Xu and Froment state.

Variables. r_{CO}, rate of formation of CO, kmol kg^{-1} h^{-1}; r_{CO_2}, rate of formation of CO_2, kmol kg^{-1} h^{-1}; r_{CH_4}, rate of disappearance of CH_4, kmol kg^{-1} h^{-1}.

Assumptions. None.

Source. [**P**] Xu and Froment Part I [37], Eq 4, PDF p. 5.

Implementation. `units::xf::Rates::ch4` and `units::xf::Rates::co2`. The CO rate follows from the carbon balance

Validation. ID, test `eq_d_19`. Carbon balance on the three rates.

Equation D.20: Effectiveness factor

Tier O

$$\eta_j^{cat} = \frac{\int_{V_p} \rho_s r_j(\mathbf{p})\, dV}{\rho_s r_j(\mathbf{p}^s)\, V_p} \tag{D.20}$$

Interpretation. Reaction runs faster at the pellet surface than deep inside it, because reactants take time to diffuse in. This number is the rate the whole pellet achieves over the rate if all of it saw surface conditions.

Derivation. Xu and Froment Part II Eq 7 gives the observed rate and Eq 9 gives this ratio. They publish no table, only one simulated profile for a fired tube and ring catalyst. Read by eye from Fig 6, η_1 falls from about 0.025 to 0.014 along that tube. Those values do not transfer to a helium reformer, and the plant does not use them.

Variables. ρ_s, catalyst solid density, kg m^{-3}; $\mathbf{p}$, partial pressures inside the pellet, bar; $\mathbf{p}^s$, partial pressures at the pellet surface, bar; V_p, pellet volume, m^3.

Assumptions. XF values do not transfer.

Source. [**P**] Xu and Froment Part II [38], Eq 7 and Eq 9, PDF p. 3; Fig 6, PDF p. 5, by eye. [**M16**]: none for helium service.

Implementation. Not implemented in the engine at this build: the effectiveness is lumped into the activity of Eq. D.21. Planned name `units::xf::eta_cat`.

Validation. NONE, test `eq_d_20`. No benchmark exists at helium service.

Equation D.21: Lumped activity

Tier O

$$\eta_1^{cat} = \eta_2^{cat} = \eta_3^{cat} = a_{cat} \tag{D.21}$$

Interpretation. Since no effectiveness data exist for this service, one activity number stands in for all of it. It is a fitted parameter, and the fit is stated openly.

Derivation. Fit a_{cat} to JAERI test A-2 alone. Hold back A-3 and A-4, and judge the model on them. Exit composition barely separates models, because the mock-up reaches equilibrium near its hot end, so the axial profile and heat balance carry the weight.

Variables. a_{cat}, lumped catalyst activity, 1.

Assumptions. One parameter. Fit disclosed.

Source. [**C**] Calculated here, on JAERI [28] A-2, PDF p. 99.

Implementation. `units::xf::Kinetics::with_activity`, with the value fitted by `units::jaeri::fit_activity_on_a2`

Validation. EXP, test `eq_d_21`. A-2 is the calibration run, and A-3 to A-5 are the independent check. The fit stops at the upper limit of its activity search. Validation results, 8 rows (JA-A-2-X to JA-A-5-X-UNCAL): 2 pass, 6 fail. Failed: JA-A-2-X gives 0.511 against 0.5551, limit 0.03; JA-A-2-X-UNCAL gives 0.5106 against 0.5551, limit 0.03; JA-A-3-X gives 0.5157 against 0.567, limit 0.03; JA-A-3-X-UNCAL gives 0.5154 against 0.567, limit 0.03; and 2 more. JA-A-2-X is a calibration row, so it tests the fit only.

Equation D.22: Methane continuity

Tier O

$$\frac{dx_{CH_4}}{dz} = \frac{\Omega \rho_B}{F^0_{CH_4}} \left(\eta_1^{cat} r_1 + \eta_3^{cat} r_3 \right) \tag{D.22}$$

Interpretation. Conversion rises along the tube at a rate set by catalyst loading, local rate and throughput.

Derivation. Eq. A.13 with $F_{CH_4} = F^0_{CH_4}(1 - x_{CH_4})$ and $A_c = \Omega$ gives $dF_{CH_4}/dz = -\Omega\rho_B(\eta_1^{cat} r_1 + \eta_3^{cat} r_3)$, then divide by $-F^0_{CH_4}$. XF2 Eq 10a writes one factor $\eta_{CH_4} r_{CH_4}$, and this form splits it by reaction with Eq. D.19.

Variables. x_{CH_4}, total methane conversion, 1; z, tube coordinate, m; Ω, tube cross section, m^2; $F^0_{CH_4}$, methane feed rate per tube, kmol h^{-1}.

Assumptions. SS, 1D, PF.

Source. [**P**] Xu and Froment Part II [38], Eq 10a, PDF p. 3.

Implementation. `units::smr_tube::Tube::eval` returns the rate terms and `units::smr_tube::Tube::residual` holds the discrete balance on the extent of CH_4 consumed

Validation. EXP, test `eq_d_22`. Conversion of 0.555, 0.567 and 0.525 for JAERI A-2, A-3 and A-4, calculated from the printed gas analyses. Validation results, 4 rows (JA-A-2-X to JA-A-5-X): 1 pass, 3 fail. Failed: JA-A-2-X gives 0.511 against 0.5551, limit 0.03; JA-A-3-X gives 0.5157 against 0.567, limit 0.03; JA-A-5-X gives 0.5055 against 0.5476, limit 0.03. JA-A-2-X is a calibration row, so it tests the fit only.

Equation D.23: Carbon dioxide continuity

Tier O

$$\frac{dx_{CO_2}}{dz} = \frac{\Omega \rho_B}{F^0_{CH_4}} \left(\eta_2^{cat} r_2 + \eta_3^{cat} r_3 \right) \tag{D.23}$$

Interpretation. The second state variable. With x_{CH_4} it fixes the composition at each point, as Eq. D.6 requires.

Derivation. Eq. A.13 with $F_{CO_2} = F^0_{CH_4}(\xi_{CO_2} + x_{CO_2})$ and Eq. D.19.

Variables. x_{CO_2}, conversion of methane into CO_2, 1.

Assumptions. As Eq. D.22.

Source. [**P**] Xu and Froment Part II [38], Eq 10b, PDF p. 3.

Implementation. `units::smr_tube::Tube::eval` returns the rate terms and `units::smr_tube::Tube::residual` holds the discrete balance on the extent of CO_2 formed

Validation. EXP, test `eq_d_23`. Target: dry gas analyses of JAERI A-2 to A-5.

Equation D.24: Partial pressures

Tier R

$$p_i = p\, y_i = p\, \frac{n_i}{n_{tot}} \tag{D.24}$$

Interpretation. The bridge from the mole table to the rate equations.

Derivation. Eq. A.14 with n_i from Eq. D.6.

Variables. p, total pressure, bar, in rates.

Assumptions. IG.

Source. [**T**] Textbook result.

Implementation. Inside `units::smr_tube::Tube::eval`, which forms the partial pressures for the rate law

Validation. ID, test `eq_d_24`. Partial pressures sum to p.

Equation D.25: Tube energy balance

Tier O

$$c_p \rho_g u_s \frac{dT}{dz} = \rho_B \sum_j \left(-\Delta_r h_j \right) \eta_j^{cat} r_j + \frac{4U}{d_i} \left(T_w - T \right) \tag{D.25}$$

Interpretation. Gas temperature is a contest between the heat coming through the wall and the heat the reaction swallows. Near the inlet reaction can win, and the gas may even cool.

Derivation. Eq. A.13 divided by $A_c = \pi d_i^2/4$ with $\dot{q}' = \pi d_i U (T_w - T)$. XF2 Eq 10c prints the wall term as $-4(U/d_{ti})(T - T_r)$, which has the same sign. The notation list of that paper leaves T_r undefined and gives T_i and T_w for the inner and outer skin. Since their U in Eq 11 includes wall conduction, it is read here as the outer skin T_w. In this plant T_w is replaced by helium temperature through Eq. D.27 and Eq. D.28.

Variables. ρ_g, process gas density, kg m^{-3}; u_s, superficial velocity, m s^{-1}; T_w, tube wall temperature, K.

Assumptions. 1D, pseudo-homogeneous in the gas.

Source. [**P**] Xu and Froment Part II [38], Eq 10c, PDF p. 4, sign read from the image.

Implementation. `units::smr_tube::Tube::eval` and `units::smr_tube::Tube::residual`, on enthalpy flows of the formation basis, so the reaction heat sits inside the enthalpy as in Eq. A.10

Validation. EXP, test `eq_d_25`. JAERI stations 1 to 5, whose elevations have not been confirmed. Validation results, 37 rows (JA-A-2-Qproc to JA-A-5-S1-process): 37 reported without a limit. 12 of these rows are calibration rows, so they test the fit only. Simulated less published: −7.837 to 8.472 kW over 6 rows; −38.32 to 38.25 C over 31 rows.

Equation D.26: Tube momentum

Tier E

$$-\frac{dp}{dz} = \frac{G}{\rho_g D_p}\,\frac{1-\varepsilon_b}{\varepsilon_b^3}\left[\frac{150(1-\varepsilon_b)\mu}{D_p} + 1.75G\right] \tag{D.26}$$

Interpretation. Eq. A.12 applied to the process gas in the catalyst bed.

Fit range and origin. See Eq. A.12. The void fraction and bed density of the mock-up are not printed in the library, so they are assumed.

Variables. ε_b, bed porosity, 1.

Assumptions. Void and bed density assumed [**A**].

Source. [**P**] Hoseinzade and Adams [10], Eq 26, PDF p. 5. Xu and Froment Part II [38], Eq 10d, PDF p. 4.

Implementation. Inside `units::smr_tube::Tube::eval`, on the superficial velocity

Validation. EXP, test `eq_d_26`. Process-side drop of 0.121 MPa measured on the mock-up, Yin et al. [39], Table 2, PDF p. 5, with 0.118 MPa from their model. Validation results, 1 row (JA-A-2-dP): 1 reported without a limit. JA-A-2-dP is a calibration row, so it tests the fit only. Reported: JA-A-2-dP gives 58.41 against 120 kPa.

Equation D.27: Wall coefficient

Tier O

$$\frac{1}{U} = \frac{1}{h_{fi}} + R_f + \frac{t_w}{k_w} + \frac{A_i/A_o}{h_{fo}} \tag{D.27}$$

Interpretation. Resistances in series again, on the inner area. On the mock-up the process side is about 70% of the total, JAERI PDF p. 24, so helium side improvements would help little.

Derivation. Eq. C.4 per unit of inner area, with the thin wall form of the conduction term.

Variables. R_f, fouling resistance, m^2 K W^{-1}; t_w, wall thickness, m.

Assumptions. Thin wall.

Source. [**T**] Form. [**P**] Fouling of 0.000172 m^2 K W^{-1} from JAERI [28], Table 3.3, PDF p. 28.

Implementation. `units::tube_ht::FinnedTube::u_outer`, on the finned outer area with a fin efficiency, JAERI A.1.3

Validation. EXP, test `eq_d_27`. Target: 142.0 W m^{-2} K^{-1} calculated and 150.9 measured, JAERI Table 3.3.

Equation D.28: Helium side

Tier O

$$\dot{m}_{He} c_p \frac{dT_{He}}{dz} = \pm \pi d_i N_t U \left(T_{He} - T\right) \tag{D.28}$$

Interpretation. Helium cools as it hands heat to the tubes. It runs against the process gas, so the hottest helium meets the gas leaving the catalyst.

Derivation. Eq. A.8 on a slice of the shell flow. The sign follows the chosen z direction. U from Eq. D.27 is on the inner area, so the perimeter is πd_i.

Variables. $\dot{m}_{He}$, helium flow on the shell side, kg s^{-1}; T_{He}, helium temperature, K.

Assumptions. Counterflow.

Source. [**T**] Form. [**P**] Same balance in Hoseinzade and Adams [10], Eq 7, PDF p. 4.

Implementation. The helium row of `units::smr_tube::Tube::residual`

Validation. EXP, test `eq_d_28`. Sato [26] at 632.4 C measured against 634.5 C calculated, PDF p. 9, has no row and stays a target. The JAERI helium stations: Validation results, 16 rows (JA-A-2-S1-helium to JA-A-5-S5-helium): 16 reported without a limit. 5 of these rows are calibration rows, so they test the fit only. Simulated less published: −21.82 to 21.98 C over 16 rows.

Equation D.29: Finned annulus

Tier E

$$Nu = 0.039\, Re^{0.87} Pr^{0.4} \left(\frac{S_f}{D_e}\right)^{0.4} \left(\frac{H_f}{D_e}\right)^{-0.19} \tag{D.29}$$

Interpretation. Fins on the tube stir the helium and raise its film coefficient well above a plain annulus.

Fit range and origin. Knudsen correlation for an annular passage with transverse fins, JAERI Appendix A-1, with $Re = D_e G_{max}/\mu$ and $D_e = D_2 - D_f$. Sato Eq 9 prints the same form.

Variables. Nu, Nusselt number, 1; S_f, fin spacing, m; H_f, fin height, m; D_e, equivalent diameter of the annulus, m.

Assumptions. Knudsen fit, finned geometry.

Source. [**P**] JAERI [28], Appendix A-1, PDF p. 93. Sato [26], Eq 9, PDF p. 4.

Implementation. `units::tube_ht::alpha_o_knudsen`

Validation. EXP, test `eq_d_29`. Target: about 2325 W m^{-2} K^{-1}, JAERI Table 3.3.

Equation D.30: Plain annulus

Tier E

$$Nu = 0.205\, Re^{0.603} Pr^{0.33} \tag{D.30}$$

Interpretation. The EVA plant's fit for helium in a plain gap around a reformer tube, with radiation folded in.

Fit range and origin. TECDOC-1236 gives $Nu = c\, Re^m Pr^n$ with $c = 0.205$, $m = 0.603$ and $n = 0.33$, including radiation between guide tube and catalyst tube.

Variables. Re, Reynolds number, 1; Pr, Prandtl number, 1.

Assumptions. Includes radiation.

Source. [**P**] TECDOC-1236 [12], PDF p. 68, EVA-I.

Implementation. Not implemented in the engine at this build: both tubes the engine runs carry fins, so it uses Eq. D.29. Planned name `units::smr_tube::nu_plain`.

Validation. EXP, test `eq_d_30`. Applies only inside the EVA range.

Equation D.31: Process side

Tier E

$$h_w = 0.4\,\frac{k_g}{D_p}\left(2.58\,Re^{1/3}Pr^{1/3} + 0.094\,Re^{0.8}Pr^{0.4}\right) \tag{D.31}$$

Interpretation. Heat transfer from the tube wall into a packed bed. This is the controlling resistance of the reformer.

Fit range and origin. Hoseinzade and Adams Eq 31. Mixture properties are needed, and maker values in JAERI F-2 and G-1 with Wilke mixing supply them.

Variables. h_w, wall to bed film coefficient, W m^{-2} K^{-1}; k_g, gas conductivity, W m^{-1} K^{-1}.

Assumptions. Mixture properties **[M22]**.

Source. **[P]** Hoseinzade and Adams [10], Eq 31, PDF p. 6.

Implementation. Not implemented as written: the engine takes the bed coefficient from the Yagi-Kunii form of JAERI A.1.2 in `units::tube_ht::alpha_i`. Planned name `units::smr_tube::h_process`.

Validation. EXP, test `eq_d_31`. Target: about 540 W m^{-2} K^{-1}, JAERI Table 3.3.

Equation D.32: Radiation

Tier E

$$q = \frac{\sigma\varepsilon_{r,a}\varepsilon_{r,b}\pi d_o\left(T_a^4 - T_b^4\right)}{A_{sh}} \tag{D.32}$$

Interpretation. Radiation counts for less here than in a fired furnace, because helium is much cooler than flue gas.

Fit range and origin. Hoseinzade and Adams Eq 9, as a heat duty per unit volume of shell gas between two grey surfaces.

Variables. q, radiative heat per unit shell volume, W m^{-3}; σ, Stefan-Boltzmann constant, W m^{-2} K^{-4}; $\varepsilon_{r,a}$, emissivity of surface a, 1; $\varepsilon_{r,b}$, emissivity of surface b, 1; T_a, temperature of surface a, K; T_b, temperature of surface b, K; A_{sh}, shell cross section, m^2.

Assumptions. Grey surfaces.

Source. **[P]** Hoseinzade and Adams [10], Eq 9, PDF p. 4.

Implementation. Not implemented as written: radiation enters the engine only through the bed terms of `units::tube_ht::alpha_i`. Planned name `units::smr_tube::radiation`.

Validation. PUB, test `eq_d_32`. Target: 5.2% of shell heat transfer, Hoseinzade and Adams PDF p. 11.

Equation D.33: Performance

Tier R

$$X_{CH_4} = x_1\Big|_{exit}, \qquad Y_{H_2} = \frac{F_{H_2,out}}{F_{CH_4,in}} \tag{D.33}$$

Interpretation. Two figures of merit: how much methane was used, and how much hydrogen came out per mole fed.

Derivation. Definitions. For pure methane, Eq. D.6 gives $Y_{H_2} = 3x_1 + x_2$ before shift.

Variables. X_{CH_4}, methane conversion, 1; Y_{H_2}, hydrogen yield, 1.

Assumptions. None.

Source. **[T]** Definitions.

Implementation. `props::equilibrium::reform` returns the conversion, and `units::smr_eq::solve` passes it on. The yield is computed in this document

Validation. PUB, test `eq_d_33`. INL prints 78.1% from Gibbs minimisation, TEV-953 [35], PDF p. 18. Computed here for pure methane at 871 C, 30.3 bar and $S/C = 3$: 80.3%, with $Y_{H_2} = 2.70$. The INL

feed is natural gas with recycle, so the two figures are not on one basis. Validation results, 3 rows (IA-4 to T953-2): 2 pass, 1 fail. Failed: T953-2 gives 80.33 against 78.1 %, limit 1 %.

Equation D.34: Tube duty

Tier O

$$\dot{Q}_{tube} = \sum_{out} F_i h_i - \sum_{in} F_i h_i \tag{D.34}$$

Interpretation. The heat the reformer tube must receive, read from enthalpies. This is a check on the INL heat basis.

Derivation. Eq. A.8 on the tube with no work. Sheet A-55 for INL Case 5 gives 404.2 MMBtu h^{-1} to the reforming reactor out of 568.4 MMBtu h^{-1} from helium, which is 118.5 of 166.6 MW, with the remaining 48.1 MW in six preheaters. The labels sum to the total exactly. So process heat includes preheat, and the tube sees only part of it.

Variables. $\dot{Q}_{tube}$, reformer tube duty, W.

Assumptions. A check on the INL heat basis. Case 5 anchors, scaled on tube carbon.

Source. [**C**] Calculated here. Case 5 split from TEV-961 [36], A-55, PDF p. 79.

Implementation. `units::smr_eq::solve`, as the enthalpy rise of the stream on the formation basis

Validation. PUB, test `eq_d_34`. Computed here on scoped Case 5 anchors at Case A, from 718 F to 1600 F: 156.5 MW on the INL Selexol carbon basis and 167.1 MW on the recycle basis, against 176.8 MW. The gap is printed and left in place. The ledger keeps the INL heat basis, Eq. J.14, and the difference is listed in Section M. The reformer enthalpy rise on Case 5: Validation results, 1 row (T961-4): 1 pass.

Equation D.35: Tube count

Tier O

$$N_t = \frac{\dot{Q}_{tube}}{\pi d_o L \, q''}, \qquad q'' = 40 \text{ to } 60 \text{ kW m}^{-2} \tag{D.35}$$

Interpretation. Divide the duty by what one tube can pass. The average flux sits below that of a fired reformer, so a helium reformer needs many tubes.

Derivation. A flux of 40 to 60 kW m^{-2} is the average TECDOC-1236 gives for HTGR service. The geometry is that of Hoseinzade and Adams Table 6, with a 12 cm bore, a 1 cm wall and 14 m length.

Variables. q'', average heat flux on outer area, W m^{-2}.

Assumptions. Flux from HTGR studies.

Source. [**P**] TECDOC-1236 [12], PDF p. 9. Hoseinzade and Adams [10], Table 6, PDF p. 11.

Implementation. Not implemented in the engine at this build: the tube count of the engine is the one that brings the exit to a set temperature, `units::tube_case_b::tubes_for_exit`. Planned name `units::smr_tube::tube_count`.

Validation. PUB, test `eq_d_35`. Computed here: 72.9 MW over 199 tubes gives 59.5 kW m^{-2}, inside the range. At Case B with 120.8 MW: 392 tubes at 50 kW m^{-2}, and 327 to 490 across the range. This uses the full process heat as an upper bound on the tube duty. The engine tube model on the assumed tube of Hoseinzade and Adams, with helium supplied at 900 C, brings the exit to 782.1 C at 199 tubes, 88.9 K below the 871 C basis, and to 846.3 C at 5000 tubes, still 24.7 K below it.

Equation D.36: Hot end

Tier F

$$\Delta T_{hot} = T_{s,sup} - T_{ref,out} = 29 \text{ K} \tag{D.36}$$

Interpretation. Only 29 K separates the helium from the gas leaving the catalyst. No tube has run at this approach, and conversion depends on it.

Justification. INL's 871 C exit and 900 C supply give this number, from a Gibbs result with no tube model. The JAERI mock-up measured 102 to 125 K against a design of 100 K. The project keeps 871 C and carries sensitivities at 825 and 800 C, so the risk stays visible.

Variables. ΔT_{hot}, reformer hot-end approach, K.

Assumptions. INL Gibbs result, no tube model.

Source. [**P**] TEV-961 [36], PDF p. 14 and p. 18. Measured values from JAERI [28], Table 3.3, PDF p. 28.

Implementation. `units::ihx::check_limits` returns this difference, with the exit of `plant::flowsheet::T_REFORMER_OUT_C`

Validation. PUB, test `eq_d_36`. Computed here: equilibrium conversion of 80.3, 70.2 and 64.3% at 871, 825 and 800 C. Those lower values test the assumption, and this document draws no forecast of plant conversion from them. The flowsheet runs at the three exits give L5 at 25 y of S$189.7, 204.8 and 217.0/t. On the assumed tube of Eq. D.35, no count up to 5000 tubes reaches 871 C, so that exit rests on the INL Gibbs result alone.

Worked example: equilibrium at the reformer exit

Take one mole of pure methane at 871 C (1144.15 K), 30.3 bar and $S/C = 3$. Eq. D.7 and Eq. D.8 give $K_1 = 784.1$ bar^2 and $K_2 = 0.8268$. Solving Eq. D.11 gives a methane conversion x_1 of 80.3%, with $x_2 = 0.295$ and $n_{tot} = 5.606$. The wet composition is H_2 48.2, H_2O 33.9, CO 9.1, CO_2 5.3 and CH_4 3.5 mol%, close to the INL raw syngas printed in TEV-953. Hydrogen before shift is $3x_1 + x_2 = 2.70$ mol per mole of methane. Dropping the exit to 825 C takes conversion to 70.2%, and 800 C takes it to 64.3%, which is why the hot-end approach in Eq. D.36 is the first risk to read in this section.

Link to the sources

Xu and Froment built their rates on 3 to 15 bar and 773 to 848 K, with a catalyst crushed to remove diffusion. Applying them at 1144 K and 30 bar is an extrapolation, and the JAERI mock-up shows that exit composition cannot test it, since the rig runs near equilibrium at its hot end. The tube model therefore checks the heat path, and the plant result stands on the equilibrium of Eq. D.11, which reproduces TECDOC-1236 and INL within the margins stated on that card. Zhang et al. Table 1 is the source of $K(T)$ because the NIST data agree with it inside 5.3% across the range.

7 E. Water-gas shift

Shift turns the carbon monoxide left by the reformer into CO_2 and more hydrogen. INL runs two adiabatic beds in series in all its reforming cases, a high-temperature bed followed by a low-temperature bed, and the plant keeps that layout. The Case 6 sheets are missing, so operating points come from TEV-961 Case 5 as a proxy. Both beds are written as an approach to equilibrium. The Chinchen rate data for iron-chromium catalyst size the first bed as a check, and the library holds no rate law for the copper-zinc catalyst that a bed at this temperature would use.

In this section k_{HTS} replaces the plain k of the source, because k is thermal conductivity elsewhere, and X_{CO} replaces Chinchen's C. Numbers tagged [**C**] are hand arithmetic on printed source values, shown so that a reader can redo them. Implementation lines name the engine items that carry the card's doc tag, or say that the engine does not implement the card. The engine test run of this version is stated in the Reference Frame. A benchmark that a card states without a results outcome stays a target.

Equation E.1: CO conversion

Tier R

$$X_{CO} = \frac{F_{CO,in} - F_{CO,out}}{F_{CO,in}} \tag{E.1}$$

Interpretation. The share of incoming carbon monoxide that a bed consumes. It can be read across one bed or across the whole train.

Derivation. This is a definition. Shift keeps the total moles constant, so on wet mole fractions it reduces to $X_{CO} = 1 - y_{CO,out}/y_{CO,in}$, which is the form that a printed stream table allows.

Variables. X_{CO}, CO conversion, 1; $F_{CO,in}$, CO molar flow into the bed, mol s^{-1}; $F_{CO,out}$, CO molar flow out of the bed, mol s^{-1}.

Assumptions. None.

Source. [**T**] Definition. Stream values from IEAGHG 2017-02 [1], PDF p. 44.

Implementation. Not implemented in the engine at this build: the shift beds return the extent, and the conversion is computed in this document. Planned name `units::wgs::conversion`.

Validation. PUB, test `eq_e_01`. Target: 68% across the IEAGHG high-temperature bed. Streams 5 and 6 print y_{CO} falling from 0.1156 to 0.0366 at constant molar flow, which gives 68.3% [**C**].

Equation E.2: Species from one extent

Tier R

$$F_i = F_{i,in} + \nu_i \xi, \qquad \nu_{CO} = \nu_{H_2O} = -1, \quad \nu_{CO_2} = \nu_{H_2} = +1 \tag{E.2}$$

Interpretation. One number fixes the whole outlet composition. Each mole of shift removes a mole of CO and of steam and adds a mole of CO_2 and of hydrogen.

Derivation. Eq. A.3 at steady state, integrated over the bed with Eq. D.2 as the only reaction. Methane and nitrogen carry $\nu_i = 0$, so they pass through unchanged. The stoichiometric coefficients sum to zero, so the total molar flow stays constant across a bed.

Variables. ξ, extent of shift across a bed, mol s^{-1}.

Assumptions. SS. One reaction, with no methanation.

Source. [**T**] Consequence of Eq. A.3 and Eq. D.2.

Implementation. `units::wgs::shifted`

Validation. ID, test `eq_e_02`. Element balance by Eq. A.5 and constant total moles.

Equation E.3: Equilibrium limit

Tier R

$$Q_2\left(T_{out}\right) \leq K_2\left(T_{out}\right) \tag{E.3}$$

Interpretation. A bed can approach equilibrium at its exit temperature and can never pass it. Shift releases heat, so an adiabatic bed lowers its own limit as it warms.

Derivation. By the second law the reaction runs forward only while the quotient of Eq. D.10 stays below K_2. Eq. D.8 falls as temperature rises, while the quotient rises with ξ, so the two curves cross at one extent for a given feed.

Variables. T_{out}, gas temperature at the bed exit, K.

Assumptions. IG.

Source. [**T**] Second law. K_2 from Eq. D.8, Zhang et al. [41], Table 1, PDF p. 5.

Implementation. `units::wgs::extent_at`

Validation. ID, test `eq_e_03`. The solver never returns Q_2 above K_2 at the exit.

Equation E.4: Approach to equilibrium

Tier O

$$Q_2\left(T_{out}\right) = K_2\left(T_{out} + \Delta T_{eq}\right), \qquad \Delta T_{eq} = 10 \text{ K} \tag{E.4}$$

Interpretation. The exit gas sits where the equilibrium curve runs 10 K hotter than the gas itself. One number stands in for finite catalyst activity on both beds.

Derivation. Three printed outlets fix the value. Solving $K_2(T) = Q_2$ with Eq. D.8 gives the equilibrium temperature of each stream **[C]**. Case 5 high-temperature bed: raw gas from sheet A-50 and 3.01 mol% dry CO from A-53 give $Q_2 = 10.56$ against $K_2 = 11.43$ at 764 F, a ratio of 0.92 and a gap of 8.5 K. Case 5 low-temperature bed: 0.33 mol% dry CO gives $Q_2 = 122.8$ against $K_2 = 151.3$ at 415 F, a ratio of 0.81 and a gap of 11.4 K. IEAGHG high-temperature bed: $Q_2 = 9.78$ against $K_2 = 10.87$ at 412 C, a ratio of 0.90 and a gap of 11.5 K. The gaps run from 8.5 to 11.5 K, so 10 K is a round central value carried from two independent flowsheets.

Variables. ΔT_{eq}, equilibrium approach temperature, K.

Assumptions. Same approach on both beds.

Source. **[C]** Calculated here from TEV-961 [36], A-50 and A-53, PDF p. 74 and p. 77, and IEAGHG 2017-02 [1], PDF p. 44.

Implementation. `units::wgs::lts` applies the approach on the second bed. `units::wgs::hts` integrates Eq. E.7 on the first bed, and can size it to stop at the IEAGHG ratio of `units::wgs::ieaghg_hts_q_over_k`, in which case the third anchor sets that exit

Validation. PUB, test `eq_e_04`. Q_2/K_2 of 0.92 for the high-temperature bed, 0.81 for the low-temperature bed and 0.90 for the IEAGHG bed. The outlet CO rows of Case 5: Validation results, 3 rows (T961-5-CO to T961-6-CO-NIST): 2 pass, 1 reported without a limit. Reported: T961-6-CO-NIST gives 0.2866 against 0.33 mol%.

Equation E.5: Adiabatic bed

Tier R

$$\sum_i F_i h_i\Big|_{out} = \sum_i F_i h_i\Big|_{in} \tag{E.5}$$

Interpretation. No heat leaves the bed, so the reaction heat stays in the gas and appears as a temperature rise.

Derivation. Eq. A.8 with $\dot{Q} = 0$ and no shaft work. The enthalpies of Eq. A.9 sit on a formation basis, so the heat of reaction needs no separate term. INL prints a duty of 0.0 on both WGS-1 and WGS-2.

Variables. $\sum_i F_i h_i$, enthalpy flow of the stream, W.

Assumptions. AD, NKE, IG.

Source. **[T]** Consequence of Eq. A.8. INL duty from TEV-961 [36], A-56, PDF p. 80.

Implementation. `units::wgs::t_from_h`

Validation. PUB, test `eq_e_05`. 335 to 407 C on the high-temperature bed and 191 to 213 C on the low-temperature bed, Case 5, sheets A-53 and A-56. Hand check on the first bed: the NIST enthalpies with the printed outlet composition give 408 C **[C]**. The tools script holds the NIST water row from 500 K only, so no hand check was run on the second bed. Validation results, 2 rows (T961-5-T to T961-6-T): 2 pass.

Equation E.6: Heat release

Tier R

$$\dot{Q}_{WGS} = \xi\left(-\Delta_r h_2\right) \tag{E.6}$$

Interpretation. The heat set free by shift. The beds hold it as sensible heat, and the coolers of Section F take it out again.

Derivation. Eq. A.10 with one reaction. $\Delta_r h_2$ is negative, so the term is positive. Evaluate it at the mean bed temperature with Eq. A.9, because the 298 K value of Eq. D.2 sits slightly away from bed conditions.

Variables. $\dot{Q}_{WGS}$, heat released by shift, W.

Assumptions. None.

Source. [**T**] Consequence of Eq. A.10.

Implementation. `plant::heatrec::wgs_release_w`

Validation. ID, test `eq_e_06`. Equal to the sensible rise of Eq. E.5. This is a reported term of the plant balance: the Case B Level 3 flowsheet gives 8.07 MW across both beds. Hand value in the worked example.

Equation E.7: High-temperature rate

Tier E

$$\frac{dp_{CO}}{dt_r} = -k_{HTS}\,(p_{CO} - p_{CO,eq}) \tag{E.7}$$

Interpretation. The rate is proportional to how far CO sits above its equilibrium level. It closes to zero at equilibrium, as it must.

Fit range and origin. Fit form from semi-technical runs on ICI catalyst 15, in two pellet sizes, with 350 ml of catalyst at 310 to 450 C and 16 to 31 bar. The feed was 1.75 m^3/h of dry gas at 20 C and 1 bar, at a steam ratio of 1.0, and the gas was nominally 80% H_2, 10% CO and 10% CO_2. Chinchen prints the equation with no minus sign, so k there is a rate of disappearance. The sign is written out here.

Variables. k_{HTS}, pseudo-first-order shift constant, empty bed basis, s^{-1}; $p_{CO,eq}$, CO partial pressure at equilibrium, Pa; t_r, contact time on the empty bed, s.

Assumptions. Pseudo first order.

Source. [**P**] Chinchen et al. [4], PDF p. 6, printed p. 74. Feed from PDF p. 5.

Implementation. `units::wgs::hts_extent`, which integrates the rate along the bed volume

Validation. EXP, test `eq_e_07`. Chinchen Table 1, PDF p. 6. Table 1 is calibration data, so CH-1 tests the fit only. Validation results, 4 rows (CH-1 to W-2): 2 pass, 2 not run. The results mark 1 of these rows as using the same data as the model, and 2 as independent. CH-1 is a calibration row, so it tests the fit only.

Equation E.8: Integrated rate

Tier E

$$k_{HTS} = S_v \ln\left[\frac{1}{1 - X_{CO}/X_{CO,e}}\right] \tag{E.8}$$

Interpretation. This turns a measured conversion into a rate constant. A conversion close to equilibrium needs a long contact time, and the logarithm shows how fast that demand grows.

Fit range and origin. Separate Eq. E.7 and integrate from $t_r = 0$ to $t_r = 1/S_v$ at constant k_{HTS} and $p_{CO,eq}$, which gives $\ln[(p_{CO,0} - p_{CO,eq})/(p_{CO} - p_{CO,eq})] = k_{HTS}/S_v$. Conversion is linear in p_{CO}, so each difference becomes a difference in conversion, and the bracket follows. Chinchen defines both k and S_v on empty bed volume at reaction conditions.

Variables. S_v, space velocity at reaction conditions, empty bed, s^{-1}; $X_{CO,e}$, CO conversion at equilibrium, 1.

Assumptions. Plug flow, isothermal. Steam ratio 1.0 in the source against $S/DG = 0.50$ here, with no correction.

Source. [**P**] Chinchen et al. [4], PDF p. 6 for the equation and PDF p. 7 for the definitions.

Implementation. Not implemented as written: `units::wgs::hts_extent` integrates Eq. E.7 numerically, and the closed form serves as the hand check. Planned name `units::wgs::hts_volume`.

Validation. ID, test `eq_e_08`. Round trip with Eq. E.7 in the test.

Equation E.9: Rate constant table

Tier E

$k_{HTS}(T)$ from

T (C)	16 bar		21 bar		31 bar	
	S	L	S	L	S	L
310	0.019	0.018	0.018	0.018	0.017	0.017
330	·	0.041	0.046	0.044	0.043	0.043
350	0.103	0.090	0.114	0.091	0.092	0.097
370	·	0.156	0.216	0.139	0.190	0.154
400	0.461	0.334	·	·	0.404	0.256
425	·	·	·	·	0.586	0.358
450	1.200	0.829	·	·	0.918	0.565

(E.9)

Interpretation. The measured rate constant in s^{-1}. It rises between thirtyfold and sixtyfold across the range, and the larger pellet falls behind the smaller one above about 350 C, where the gas can no longer reach the inside of the pellet fast enough.

Fit range and origin. Table 1 read from the page image. S is the 5.4 by 3.6 mm pellet, type 15-5, and L is the 8.5 by 11.0 mm pellet, type 15-4. A dot marks a blank cell. Chinchen finds significant pore diffusion limits above 370 C for the small pellet and above 350 C for the large pellet at 31 bar, with apparent activation energy falling from about 125 to about 63 kJ mol^{-1}. The true value is 121.8 kJ mol^{-1}. Values are interpolated in $\ln k_{HTS}$ against $1/T$ between the 31 bar points, the pressure closest to the plant.

Variables. T, temperature, printed in C in the table, K.

Assumptions. Range 310 to 450 C and 16 to 31 bar. Catalyst age about 2 weeks on stream.

Source. **[P]** Chinchen et al. [4], Table 1, PDF p. 6. Diffusion limits from PDF p. 1 and p. 10, and true activation energy from PDF p. 12.

Implementation. `units::wgs::k_chinchen`, which holds both 31 bar columns and takes the pellet as an input

Validation. PUB, test `eq_e_09`. Table 1 itself. Validation results, 2 rows (CH-1 to CH-2): 2 pass. The results mark 1 of these rows as using the same data as the model, and 0 as independent. CH-1 is a calibration row, so it tests the fit only.

Equation E.10: Catalyst volume

Tier O

$$V_{cat} = \frac{\dot{V}}{k_{HTS}} \ln\left[\frac{1}{1 - X_{CO}/X_{CO,e}}\right] \tag{E.10}$$

Interpretation. A rough size for the first bed. It answers one question only: whether the assumed approach to equilibrium needs a plausible amount of catalyst.

Derivation. Put $S_v = \dot{V}/V_{cat}$ in Eq. E.8. At Case 5 the gas flow is 6.0 m^3 s^{-1} at the inlet and 7.0 m^3 s^{-1} at the outlet by the ideal gas law **[C]**. With 7 m^3 s^{-1}, k_{HTS} of 0.1 to 0.2 s^{-1} and $X_{CO}/X_{CO,e} = 0.92$, the volume is 88 to 177 m^3. Confidence is low, because the steam ratio differs by a factor of two from Chinchen's runs and most of the bed runs above the temperature where pore diffusion sets in.

Variables. V_{cat}, catalyst volume, empty bed basis, m^3; $\dot{V}$, gas volume flow at reaction conditions, m^3 s^{-1}.

Assumptions. Low confidence. Isothermal bed at a representative k_{HTS}.

Source. **[C]** Calculated here on Eq. E.8 with TEV-961 [36] Case 5 flows.

Implementation. `units::wgs::hts`, which returns the bed volume when the bed is sized to an exit ratio

Validation. PUB, test `eq_e_10`. Check only. No published bed volume exists for this plant, so the project band of 90 to 180 m^3 at Case 5 is the reference. The Case 5 rows at Chinchen's own space velocity: Validation results, 2 rows (T961-5-GHSV-CO to T961-5-GHSV-T): 2 fail. Failed: T961-5-GHSV-CO gives 7.063 against 3.01 mol%, limit 0.3 mol%; T961-5-GHSV-T gives 377.5 against 406.7 C, limit 10 C.

Equation E.11: Temperature profile

Tier O

$$\left(\sum_i F_i c_{p,i}\right) \frac{dT}{d\xi} = -\Delta_r h_2 \tag{E.11}$$

Interpretation. Each mole of shift heats the gas by the same small step. Plotted against extent, the bed temperature is close to a straight line.

Derivation. Differentiate Eq. E.5 with respect to ξ. The term $\sum_i \nu_i h_i$ is $\Delta_r h_2$ by Eq. A.10, and the term $\sum_i F_i c_{p,i}\, dT$ is the sensible change.

Variables. $dT/d\xi$, temperature rise per unit extent, K mol^{-1} s.

Assumptions. AD.

Source. [**T**] Consequence of Eq. E.5.

Implementation. `units::wgs::hts_extent` with `units::wgs::t_from_h`, which hold the enthalpy along the bed in place of the differential form

Validation. PUB, test `eq_e_11`. Compared through the end points of Eq. E.5.

Equation E.12: Bed pressure drop

Tier F

$$\Delta p_{bed} = 17 \text{ psi} = 1.17 \text{ bar} \quad \text{across each bed} \tag{E.12}$$

Interpretation. The pressure lost across one shift bed, taken directly from the INL sheets.

Justification. Case 5 labels give 436 to 419 psi across WGS-1 and 413 to 396 psi across WGS-2. The Ergun form Eq. A.12 is kept as a check only, because the bed geometry is unpublished.

Variables. Δp_{bed}, pressure drop across one shift bed, Pa.

Assumptions. INL Case 5 labels.

Source. [**P**] TEV-961 [36], A-55 and A-56, PDF p. 79 and p. 80.

Implementation. `units::case5::train`, which sets each bed outlet at its printed pressure

Validation. ID, test `eq_e_12`. The stream pressures close across the train.

Equation E.13: Steam to dry gas ratio

Tier F

$$S/DG = \frac{F_{H_2O}}{\sum_{dry} F_i} \tag{E.13}$$

Interpretation. The steam carried by each mole of dry gas. It pushes shift forward and protects the catalyst, and it falls from one bed to the next because shift consumes steam.

Justification. INL prints 0.50 with a minimum of 0.4 at the high-temperature inlet, and 0.37 with a minimum of 0.3 at the low-temperature inlet. TEV-953 says the 0.30 limit keeps good catalyst activity while inhibiting Fischer-Tropsch reactions.

Variables. S/DG, steam to dry gas mole ratio, 1.

Assumptions. INL values, 0.50 and 0.37, with minimum 0.4 and 0.3.

Source. **[P]** TEV-961 [36], A-53, PDF p. 77. TEV-953 [35], PDF p. 18 and p. 19.

Implementation. Not implemented in the engine at this build. The values are computed in this document. Planned name `units::wgs::check_window`.

Validation. PUB, test `eq_e_13`. Target: 0.50 and 0.37 as printed. Case 5 raw gas gives 33.4/66.6 = 0.50 at the first inlet, and the extent of the worked example gives 0.368 at the second **[C]**.

Equation E.14: Temperature window

Tier F

$$T_{HTS,in} \geq 600 \text{ F}, \qquad T_{LTS,in} \geq \max\left(375 \text{ F},\ T_{dp} + 27 \text{ F}\right) \tag{E.14}$$

Interpretation. Each bed needs its inlet warm enough to stay active, and the second bed must also stay clear of liquid water.

Justification. In SI the limits are 588.7 K and 463.7 K, with a dew margin of 15 K. Case 5 runs 635 F and 375 F against a printed dew point of 331 F, a margin of 44 F. Eq. F.3 gives the dew point.

Variables. $T_{HTS,in}$, high-temperature bed inlet, K; $T_{LTS,in}$, low-temperature bed inlet, K; T_{dp}, dew point, K.

Assumptions. INL limits carried unchanged.

Source. **[P]** TEV-961 [36], A-53, PDF p. 77, which prints the limits. Operating values also in TEV-953 [35], PDF p. 18 and p. 19.

Implementation. Not implemented in the engine at this build: a bed inlet is refused only when it is two-phase or short of CO or steam. Planned name `units::wgs::check_window`.

Validation. ID, test `eq_e_14`. The window check trips on inlets set just outside each limit.

Equation E.15: Low-temperature bed

Tier O

$$Q_2\left(T_{LTS,out}\right) = K_2\left(T_{LTS,out} + \Delta T_{eq}\right), \qquad \text{with Eq. E.5 across the bed} \tag{E.15}$$

Interpretation. The second bed has no kinetics at all. Its exit is set by the equilibrium approach and the energy balance.

Derivation. Solve Eq. E.2, Eq. E.4 and Eq. E.5 together for ξ and $T_{LTS,out}$. The library holds no copper-zinc rate law, so a rate form cannot be written here. TECDOC-1236 states only that copper-zinc catalyst allows 200 to 240 C.

Variables. $T_{LTS,out}$, low-temperature bed exit, K.

Assumptions. Equilibrium approach, anchored on INL operating points.

Source. **[C]** INL operating points, TEV-961 [36], A-53, PDF p. 77. **[M4]**: no copper-zinc rate law. TECDOC-1236 [12], PDF p. 36.

Implementation. `units::wgs::lts`

Validation. PUB, test `eq_e_15`. 0.33 mol% dry CO and 213 C, sheet A-53. Validation results, 4 rows (T961-6-CO to W-1): 2 pass, 1 reported without a limit, 1 not run. Reported: T961-6-CO-NIST gives 0.2866 against 0.33 mol%.

Equation E.16: Power law, listed and unused

Tier E

$$r = k_0 \exp\left(-\frac{E_{a,K}}{RT}\right) p_{CO}^{1.77}\, p_{H_2O}^{0.23}\, p_{CO_2}^{-0.17}\, p_{H_2}^{-0.12} \left(1 - \frac{Q_2}{K_2}\right) \tag{E.16}$$

Interpretation. A published rate law for iron-chromium catalyst on biomass gas. It is printed for completeness and takes no part in the plant.

Fit range and origin. Kraussler Eq 2 with $k_0 = 117.8$ mol g^{-1} Pa$^{-1.71}$ s^{-1} from Table 1. Table 3 raises $E_{a,K}$ from 101.9 to 126.6 kJ mol^{-1}, changing that parameter alone on one operating point. The rig ran at ambient pressure on real gas holding H_2S and tar. The orders sum to 1.71, so taking the law to 28 bar would multiply the rate by about 300, and Kraussler attributes the CO order of 1.77 to its low pressure, against about 1.0 elsewhere. That is why the law does not extrapolate to this plant. Kraussler's K_{MAL} and K_g are Q_2 and K_2 here.

Variables. r, shift rate per unit mass of catalyst, mol g^{-1} s^{-1}; k_0, pre-exponential factor, mol g^{-1} Pa$^{-1.71}$ s^{-1}; $E_{a,K}$, activation energy of this law, J mol^{-1}.

Assumptions. Ambient pressure, biomass gas.

Source. **[P]** Kraussler et al. [16], Eq 2 and Table 1, PDF p. 2, and Table 3, PDF p. 8.

Implementation. none. The equation is listed only.

Validation. NONE, test `eq_e_16`. No plant benchmark. The validation matrix keeps one ambient comparison, KR-1, on Kraussler's own rig as a weak test of the code. Validation results, 4 rows (KR-1-OP1 to KR-1-OP4): 4 not run.

Worked example: Case 5 shift, carried to Case B

Take 100 mol of Case 5 raw syngas from sheet A-50: H_2 48.8, CO 8.7, CO_2 4.8, H_2O 33.4, CH_4 3.6 and N_2 0.7. The dry gas is 66.6 mol, so $S/DG = 0.50$ by Eq. E.13.

First bed. A-53 prints 3.01 mol% dry CO at the exit. Each mole of shift removes one mole of CO and adds one mole of dry gas, so $(8.7 - \xi)/(66.6 + \xi) = 0.0301$ and $\xi = 6.50$ mol. By Eq. E.2 the gas becomes H_2 55.30, CO 2.20, CO_2 11.30 and H_2O 26.90, with $X_{CO} = 74.7\%$ by Eq. E.1. The quotient of Eq. D.10 is 10.56, close to the K_2 of 11.43 at 764 F.

Second bed. A-53 prints 0.33 mol% dry CO, so $\xi = 1.95$ mol. The gas becomes H_2 57.25, CO 0.25, CO_2 13.25 and H_2O 24.95, with 88.7% conversion across this bed and 97.2% across both. On a dry basis the gas holds 17.66 mol% CO_2, 76.3 mol% H_2, 4.80 mol% CH_4 and 0.33 mol% CO, which matches the 17.7 mol% printed at the Selexol inlet. The second bed adds 3.5% more hydrogen and 17% more CO_2 for capture, measured on the first bed outlet.

Heat. The mean molar mass is 12.32 g mol^{-1}, so 349,724 lb h^{-1} is 3,576 mol s^{-1}, which sits within 0.3% of the 3,568 mol s^{-1} from the printed 258 MMSCFD. Eq. E.6 then gives 232.4 mol s^{-1} at 38.5 kJ mol^{-1} on the first bed and 69.8 mol s^{-1} at 39.8 kJ mol^{-1} on the second, or 8.9 and 2.8 MW. Scaling by 1,927/1,970 for captured CO_2 and by $s = 0.6833$ for Case B gives 6.0 and 1.9 MW, a total of 7.8 MW. All of this is hand arithmetic on the Case 5 proxy.

Link to the sources

INL and IEAGHG reach almost the same outlet from different feeds, and the printed streams of both fall within 2 K of a 10 K approach to equilibrium. That agreement is the evidence behind Eq. E.4. Chinchen supports the first bed at the right pressure and a steam ratio twice the plant value. No library source supports the rate of the second bed, so Eq. E.15 stands on operating points alone, which is recorded as **[M4]**.

8 F. Syngas cooling and separation

After the second shift bed the gas falls from 213 C to 40 C and most of its water condenses. The INL sheets give the full train for Case 5, which stands in for the missing Case 6. Two results matter most for the plant balance: the heat that can reach the reboiler and the dry feed to the absorber. The condensate is tracked alongside them.

Inside this section T_{in} and T_{out} replace plain numbered temperatures, and the flash uses β_V with F_f, V_f and L_f, because V and L already name a control volume and a tube length. Numbers tagged **[C]**

are hand arithmetic on printed values. Implementation lines name the engine items that carry the card's doc tag, or say that the engine does not implement the card. The engine test run of this version is stated in the Reference Frame. A benchmark that a card states without a results outcome stays a target. The Case 5 duties are printed in MMBtu h^{-1}, and 1 MMBtu h^{-1} is 0.29307 MW.

Equation F.1: Sensible duty

Tier R

$$\dot{Q} = \sum_i F_i \left[h_i \left(T_{in} \right) - h_i \left(T_{out} \right) \right] \tag{F.1}$$

Interpretation. The heat a cooler removes while no water condenses. It reads straight from the enthalpies at the two ends.

Derivation. Eq. A.8 on the gas side with no work and no change of composition. The enthalpies of Eq. A.9 carry the temperature dependence, so no mean heat capacity is needed.

Variables. T_{in}, cooler inlet temperature, K; T_{out}, cooler outlet temperature, K.

Assumptions. SS, IG. No phase change above the dew point.

Source. **[T]** Consequence of Eq. A.8.

Implementation. `units::cooler::cool`, which also holds any condensed water at the split of Eq. F.7

Validation. PUB, test `eq_f_01`. TEV-961 [36], A-55 and A-56, PDF p. 79 and p. 80: HPS-GEN1 236.3 MMBtu h^{-1}, 69.3 MW, from 1600 to 635 F. WGS-EX1 66.7, 19.5 MW, from 764 to 480 F. WGS-EX2 24.2, 7.1 MW, from 480 to 375 F. Validation results, 3 rows (T961-7-HPS-GEN1 to T961-7-WGS-EX2): 3 pass.

Equation F.2: Cooling train

Tier R

$$\dot{Q}_{train} = \dot{Q}_{HPS} + \dot{Q}_{EX1} + \dot{Q}_{EX2} + \dot{Q}_{EX3} + \dot{Q}_{EX4} + \dot{Q}_{KO} \tag{F.2}$$

Interpretation. The whole train is six exchangers in series, with the two shift beds between them. Summing them gives the total heat taken out of the gas between the reformer and the absorber.

Derivation. Case 5 prints 236.3, 66.7, 24.2, 65.7, 117.1 and 19.7 MMBtu h^{-1} from HPS-GEN1 through KO-DRM-1. The sum is 529.7 MMBtu h^{-1}, or 155.2 MW **[C]**. WGS-EX3 runs from 415 to 300 F, WGS-EX4 from 300 to 158 F and the knock-out drum from 158 to 104 F. INL gives neither exchanger types nor areas.

Variables. $\dot{Q}_{train}$, total duty of the cooling train, W.

Assumptions. Case 5 proxy.

Source. **[P]** TEV-961 [36], A-55 and A-56, PDF p. 79 and p. 80. **[M1]**: Case 6 sheets missing.

Implementation. `units::case5::train` runs the six coolers in flow order

Validation. PUB, test `eq_f_02`. WGS-EX3 and WGS-EX4 on the first pass, with any misfit reported and left in place. Validation results, 2 rows (T961-7-WGS-EX3 to T961-7-WGS-EX4): 2 pass.

Equation F.3: Dew point

Tier R

$$y_{H_2O}\, p = p_{sat} \left(T_{dp} \right) \tag{F.3}$$

Interpretation. Water starts to condense when its partial pressure reaches the saturation pressure. Above that temperature Eq. F.1 holds. Below it the latent heat dominates.

Derivation. Raoult's law for a nearly pure liquid water phase, with $\gamma = 1$ and $x_w = 1$. The ideal gas sets the partial pressure.

Variables. y_{H_2O}, water mole fraction in the gas, 1; p_{sat}, water saturation pressure, Pa.

Assumptions. Raoult, $\gamma = 1$.

Source. [**T**] Textbook result.

Implementation. The dew test inside `units::flash::split`, at the set temperature. The engine solves for no T_{dp}

Validation. PUB, test `eq_f_03`. Target: 331 F printed on A-53, which is 166 C. Hand estimate at the second bed inlet: $y_{H_2O} = 0.269$ at 413 psi gives 7.66 bar of water, which saturates near 169 C [**C**], about 3 K above the printed value.

Equation F.4: Saturation pressure

Tier E

$$p_{sat}\,(T) = \text{IAPWS-IF97 saturation line} \tag{F.4}$$

Interpretation. Water properties come from the international steam tables through a maintained library.

Fit range and origin. The library holds no water saturation table and no Antoine coefficients. The screening Antoine fit is carried no further, because its constants are rated from 99 to 374 C and it gives 6.10 kPa at 35 C, against 5.63 kPa on the project input sheet. The maintained source is IF97 through `seuif97`, under an MIT licence.

Variables. $p_{sat}(T)$, saturation pressure at T, Pa.

Assumptions. Screening Antoine not carried.

Source. [**M9**]: no water saturation data in the library, IF97 used in its place.

Implementation. `props::steam::psat_pa`

Validation. PUB, test `eq_f_04`. The IF97 verification table. Validation results, 1 row (PROPS-IF97-VECTORS): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation F.5: Water left in the gas

Tier R

$$y_w p = x_w \gamma_w p_{sat}\,(T)\,, \qquad x_w \approx 1 \tag{F.5}$$

Interpretation. Gas leaving a cold knock-out drum still carries water, set by the drum temperature and pressure.

Derivation. The same balance as Eq. F.3, written for y_w at a known temperature. Dissolved gases are dilute, so x_w stays near one.

Variables. y_w, water mole fraction in the gas, 1; x_w, water mole fraction in the liquid, 1; γ_w, water activity coefficient, 1.

Assumptions. Dilute gases.

Source. [**T**] Textbook result. Stream value from IEAGHG 2017-02 [1], PDF p. 44.

Implementation. `units::flash::split`, with $\gamma_w = 1$

Validation. PUB, test `eq_f_05`. Target: 0.24 mol% at 35 C and 25.8 bar, IEAGHG stream 7. With 5.63 kPa from the project input sheet, this equation gives 0.22 mol% [**C**]. The printed stream implies 6.19 kPa, which is 10% higher.

Equation F.6: Dissolved gas

Tier E

$$x_{CO_2} = \frac{y_{CO_2}\, p}{H_x}, \qquad H_x = H^w_{CO_2}\, c_w \tag{F.6}$$

Interpretation. Some CO_2 dissolves in the condensate. Henry's law gives the amount from its partial pressure.

Fit range and origin. $H^w_{CO_2}$ is the water value of Eq. G.22 on a concentration basis, so the molar density of water c_w turns it into the mole fraction constant H_x. Aronu prints no unit beside its Eq 17. Dugas Eq 5.9 prints the same constants with Pa m^3 mol^{-1}. Both trace to Versteeg and van Swaaij (1988), a paper the library lacks. The local Versteeg file is the kinetics paper.

Variables. x_i, liquid mole fraction of i, 1; H_x, Henry constant, mole fraction basis, Pa; $H^w_{CO_2}$, Henry constant of CO_2 in water, Pa m^3 mol^{-1}; c_w, molar density of liquid water, mol m^{-3}.

Assumptions. Dilute CO_2 in pure water. No ionisation.

Source. [**P**] Aronu et al. [2], Chem. Eng. Sci. 66 (2011) 6393, Eq 17, PDF p. 4, with the unit from Dugas [6], Eq 5.9, PDF p. 129. [**M8**]: second hand.

Implementation. `units::flash::split` with the constant of `units::flash::henry_x_pa`, which holds every other gas insoluble

Validation. PUB, test `eq_f_06`. Target: 288 kg h^{-1} of CO_2 at Case 5, a project calculation. Hand check at 40 C and 4.74 bar of CO_2: 115 mol m^{-3} in 57.4 t h^{-1} of condensate, about 290 kg h^{-1} [**C**].

Equation F.7: Flash equation

Tier R

$$\sum_i \frac{z_i\,(K_i - 1)}{1 + \beta_V\,(K_i - 1)} = 0 \tag{F.7}$$

Interpretation. One equation with one unknown, the fraction of the feed that stays as vapour. It has a single root between the physical limits, so it solves reliably.

Derivation. Combine Eq. F.8 with $y_i = K_i x_i$ to get $x_i = z_i/[1 + \beta_V(K_i - 1)]$. Subtract $\sum_i x_i = 1$ from $\sum_i y_i = 1$ to reach this form, due to Rachford and Rice.

Variables. z_i, feed mole fraction of i, 1; K_i, vapour to liquid ratio of i, 1; β_V, vapour fraction of the feed, 1.

Assumptions. Vapour and liquid leave in equilibrium.

Source. [**T**] Rachford-Rice, textbook result.

Implementation. `units::flash::split`

Validation. ID, test `eq_f_07`. Residual of the root, and $\sum_i x_i = \sum_i y_i = 1$.

Equation F.8: Flash mass balance

Tier R

$$F_f = V_f + L_f, \qquad F_f z_i = V_f y_i + L_f x_i \tag{F.8}$$

Interpretation. What enters the drum leaves as gas or liquid. The same holds for each species.

Derivation. Eq. A.2 and Eq. A.3 on the drum at steady state, with no reaction.

Variables. F_f, flash feed, mol s^{-1}; V_f, flash vapour, mol s^{-1}; L_f, flash liquid, mol s^{-1}.

Assumptions. SS.

Source. [**T**] Consequence of Eq. A.2 and Eq. A.3.

Implementation. `units::flash::flash`

Validation. PUB, test `eq_f_08`. Target: 349,724 = 223,172 + 126,553 lb h^{-1}, sheet A-56. The printed sum is 349,725, so rounding sets the limit of this check.

Equation F.9: Flash energy balance

Tier R

$$\dot{Q} = V_f h_V + L_f h_L - F_f h_F \tag{F.9}$$

Interpretation. The heat the drum must lose to reach its temperature. Most of it is the latent heat of the water that condenses inside.

Derivation. Eq. A.8 on the drum with no shaft work. A negative value means heat leaves.

Variables. h_V, molar enthalpy of the vapour, J mol^{-1}; h_L, molar enthalpy of the liquid, J mol^{-1}; h_F, molar enthalpy of the feed, J mol^{-1}.

Assumptions. SS.

Source. [**T**] Consequence of Eq. A.8.

Implementation. `units::cooler::cool`. The drum of `units::flash::flash` carries no heat term, so the cooler ahead of it holds the duty

Validation. PUB, test `eq_f_09`. KO-DRM-1 19.7 MMBtu h^{-1}, 5.8 MW, sheet A-56. Validation results, 1 row (T961-7-KO-DRM-1): 1 pass.

Equation F.10: Condensate split

Tier R

$$L_{cond} = L_{ret} + L_{treat} \tag{F.10}$$

Interpretation. The condensate has two destinations. A small part returns as reformer water, and the rest goes to treatment.

Derivation. A mass balance at the splitter. Case 5 prints 126,553 lb h^{-1} from the drum, 12,945 lb h^{-1} to the reformer and 113,607 lb h^{-1} to biotreatment and reverse osmosis. The vent line reads 0. IEAGHG sends process condensate to boiler feed water through the deaerator, but gives no condensate chemistry, so that route is noted and left unassessed.

Variables. L_{cond}, total condensate, kg s^{-1}; L_{ret}, condensate returned to the reformer, kg s^{-1}; L_{treat}, condensate sent to treatment, kg s^{-1}.

Assumptions. Zero vent.

Source. [**P**] TEV-961 [36], A-54, A-56 and A-59, PDF p. 78, p. 80 and p. 83. Route from IEAGHG 2017-02 [1], PDF p. 37 and p. 38.

Implementation. Not implemented in the engine at this build: the drum condensate leaves as one stream. Planned name `units::flash::condensate`.

Validation. PUB, test `eq_f_10`. 57.4 t h^{-1}, and 4.36 t per t of hydrogen on 29,000 lb h^{-1} of product [**C**]. The drum condensate in total: Validation results, 1 row (T961-8-LIQ): 1 pass.

Equation F.11: Heat above a temperature limit

Tier O

$$\dot{Q}\left(T_{min}\right) = \int_{T_{min}}^{149\,\mathrm{C}} \dot{C}\, dT + \dot{m}_{cond}\Delta h_{vap} \tag{F.11}$$

Interpretation. The part of WGS-EX4 that is hot enough to drive a reboiler. Below the dew point most of the heat is latent, so a small temperature step carries a large duty.

Derivation. Screening form: dry gas of 9,644 kmol h^{-1}, heat capacities of 31 and 35 kJ kmol^{-1} K^{-1} for dry gas and vapour, a fixed latent heat of 2,170 kJ kg^{-1}, and vapour held at saturation by Eq. F.5 at a gas pressure of 27 bar. With n_v the water vapour flow at each end, condensation is $\dot{m}_{cond} = M_w[n_v(149\,\mathrm{C}) - n_v(T_{min})]$. At 130 C the curve gives 11.8 MW, against 11.5 MW from a reading of the WGS-EX4 line between 300 and 266 F. The engine uses IF97, so its value will differ slightly from the Antoine figure.

Variables. T_{min}, lowest useful gas temperature, K; $\dot{C}$, gas capacity rate, W K^{-1}; $\dot{m}_{cond}$, condensate formed over the interval, kg s^{-1}; Δh_{vap}, latent heat of water, J kg^{-1}.

Assumptions. Case 5 proxy.

Source. [**C**] Screening model, on TEV-961 [36], A-56, PDF p. 80.

Implementation. `plant::heatrec::syngas_above_w`, which takes enthalpies and the flash split from `units::cooler::heat_above` in place of fixed heat capacities

Validation. GOLD, test `eq_f_11`. 11.5 to 11.8 MW at 130 C, before scaling by s and 1,927/1,970. Port check only, so it says nothing about the physics. Validation results, 1 row (SCREENING_EX4_ABOVE_130_MW): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation F.12: CO_2 lost in condensate

Tier O

$$\dot{m}_{CO_2,cond} = \frac{\dot{m}_{cond}}{M_w}\, x_{CO_2}\, M_{CO_2} \tag{F.12}$$

Interpretation. Dissolved CO_2 leaves with the condensate and never reaches the absorber. The amount is small and real, so it is stated openly and kept out of the ledger.

Derivation. Eq. F.6 applied to the Case 5 drum at 40 C and 4.74 bar of CO_2. This gives 288 kg h^{-1}, about 0.4% of the 75 t h^{-1} of CO_2 in the shifted gas, or roughly 2,100 t y^{-1} at 85% availability [**C**]. INL prints zero vent and shows no gas stripped from this line.

Variables. $\dot{m}_{CO_2,cond}$, CO_2 leaving in condensate, kg s^{-1}; M_w, molar mass of water, kg mol^{-1}; M_{CO_2}, molar mass of CO_2, kg mol^{-1}.

Assumptions. About 0.4% of CO_2. Outside the ledger.

Source. [**C**] Calculated here with Eq. F.6.

Implementation. `units::flash::split`, as the dissolved CO_2 flow

Validation. NONE, test `eq_f_12`. Disclosed term. No published value exists for it.

Equation F.13: Absorber feed

Tier R

$$y_{i,dry} = \frac{F_i}{\sum_{dry} F_j}, \qquad i = CO_2,\ CO,\ CH_4,\ H_2,\ N_2 \tag{F.13}$$

Interpretation. The dry composition reaching the absorber. It fixes the CO_2 partial pressure that drives the column.

Derivation. Apply Eq. F.8 at the drum and strip water from the sum. CO_2 at 17.7 mol% and 26.8 bar gives 4.74 bar of CO_2, the bottom driving force of Section H. A-56 prints 104 F and 389 psi at this point.

Variables. $y_{i,dry}$, dry mole fraction of i, 1.

Assumptions. Water free basis. Case 5 proxy.

Source. [**P**] TEV-961 [36], A-53, PDF p. 77, for 17.7 mol% CO_2 and 0.33 mol% CO, and A-56, PDF p. 80, for temperature and pressure. [**C**] CH_4 and H_2 by the balance of the Section E example.

Implementation. `units::gas::Stream::dry_fraction`

Validation. PUB, test `eq_f_13`. 17.7, 0.33, 4.8 and 76.3 mol% dry for CO_2, CO, CH_4 and H_2. The dry CO_2 row: Validation results, 1 row (T961-8-CO2): 1 pass.

Worked example: heat and water at Case B

Case 5 sets the shape of the train, and Case B follows by the factor 1,927/1,970 for captured CO_2 and the factor $s = 0.6833$ on product, which together give 0.668.

Reboiler heat. WGS-EX4 passes the 130 C limit of the plant with 11.5 MW above it by the reading used in Eq. F.11. Scaling gives 11.25 MW at Case A and 7.69 MW at Case B. That heat is one layer

of the plant steam credit of 37.9 MW against a reboiler of 51.0 MW, which leaves the gap of 13.1 MW that the reactor must carry. The temperature limit matters more than the duty: all of WGS-EX3 runs above 130 C and INL already spends it on LP steam, while most of WGS-EX4 runs too cold to be useful.

Water. The drum condenses 57.4 t h^{-1} at Case 5, which scales to 38.4 t h^{-1} at Case B. Of this, 3.9 t h^{-1} returns to the reformer and 34.4 t h^{-1} goes to treatment by Eq. F.10. Dissolved CO_2 by Eq. F.12 scales to about 190 kg h^{-1}. The dry composition of Eq. F.13 stays the same, so the absorber sees 4.74 bar of CO_2 at either scale.

Link to the sources

Sheets A-55 and A-56 are the only complete cooling train in the library. IEAGHG shows the same layout down to the condensate separator, and its flowsheet with one shift bed leaves about 4.6 mol% CO in the same stream, which is why the second shift bed matters for this project. Water saturation and gas solubility rest on outside data, recorded as **[M9]** and **[M8]**.

9 G. MEA thermodynamics

The solvent model sets how much CO_2 each kilogram of solvent can hold and how hard it is to release. The selected model is the extended UNIQUAC of Aronu et al. [2], Chem. Eng. Sci. 66 (2011) 6393, with corrigendum 78 (2012) 246, cited below as Aronu. It carries tier R inside 15 to 60 mass% MEA and 40 to 120 C. The e-NRTL model of Zhang et al. [40], Fluid Phase Equilib. 311 (2011) 67, is left out because it needs Aspen Databank values that no local source prints, so its figures stay as read-offs for comparison.

Aronu prints the parameters and the top level of the model, and leaves the activity expressions to Thomsen (1997) and Thomsen and Rasmussen (1999), which the library lacks. The expressions in Eq. G.14, Eq. G.15 and Eq. G.17 therefore come from general knowledge of that framework, recorded as **[M7]**. Their support is indirect: with them, a prototype reproduces Aronu's published model column at 104 points. That prototype is a reading aid that ran on the printed tables. The Rust port in the `mea` crate has its own tests, and the run recorded for this version is stated in the Reference Frame. The benchmark rows of the solvent model are in the validation results file. A benchmark that a card states without a results outcome stays a target.

Symbols: $K_{x,j}$ is the mole fraction equilibrium constant, because K_j names adsorption in Section D. H_x is the Henry constant on a mole fraction basis, in Pa. Z_i is ion charge, z_c the coordination number, and A_{DH}, b_{DH} and σ_{DH} the Debye-Hückel terms. Implementation lines name the engine items that carry the card's doc tag, or say that the engine does not implement the card.

Equation G.1: Reactions

Tier R

$$
\begin{aligned}
&2\,H_2O \rightleftharpoons H_3O^+ + OH^- \\
&2\,H_2O + CO_2 \rightleftharpoons H_3O^+ + HCO_3^- \\
&H_2O + HCO_3^- \rightleftharpoons H_3O^+ + CO_3^{2-} \\
&H_2O + MEAH^+ \rightleftharpoons H_3O^+ + MEA \\
&H_2O + MEACOO^- \rightleftharpoons MEA + HCO_3^-
\end{aligned}
\tag{G.1}
$$

Interpretation. Dissolved CO_2 reacts on into bicarbonate and carbamate while MEA picks up protons. That chemistry is where the capacity of the solvent comes from.

Derivation. Aronu Eqs 4 to 8, with RNH_2 written as MEA. The carbamate is explicit, so nine species follow: H_2O, MEA, CO_2, H_3O^+, $MEAH^+$, OH^-, HCO_3^-, CO_3^{2-} and $MEACOO^-$. Aronu's text says

four reactions while listing five. Five is right, because nine species less four conservation laws leaves five independent reactions by Eq. A.6.

Variables. MEA, monoethanolamine, C_2H_7NO, species.

Assumptions. Carbamate explicit.

Source. [**P**] Aronu, Eqs 4 to 8, PDF p. 3.

Implementation. `mea::aronu::SPECIES`

Validation. ID, test `eq_g_01`. Element and charge balance on all five reactions by Eq. A.6.

Equation G.2: Equilibrium constants

Tier E

$$\ln K_{x,j} \text{ or } \ln H_x = C_1 + \frac{C_2}{T} + C_3 \ln T + C_4 T$$

	C_1	C_2	C_3	C_4	T (C)
H_2O	132.899	−13445.90	−22.4773	0	0 to 225
CO_2	231.465	−12092.10	−36.7816	0	0 to 225
HCO_3^-	216.049	−12431.70	−35.4819	0	0 to 225
MEA	−4.9074	−6166.12	0	−0.00098482	0 to 50
$MEACOO^-$	2.8898	−3635.09	0	0	25 to 120
H_x	170.7126	−8477.711	−21.95743	0.005781	0 to 100

(G.2)

Interpretation. Each reaction has its own temperature curve. The strong fall of the MEA and carbamate constants with cooling is what makes the solvent absorb at 40 C and release at 120 C.

Fit range and origin. Aronu Eq 16 and Table 2, read from the page image, because the text layer drops minus signs. Aronu states that the constants are on a mole fraction basis, so they are dimensionless, while H_x carries pascal. K_{MEA} is rated for 0 to 50 C and H_x for 0 to 100 C, yet both run to 120 C here. That is an extrapolation inside the Aronu regression itself.

Variables. $K_{x,j}$, mole fraction equilibrium constant of reaction j, 1; H_x, Henry constant of CO_2, mole fraction basis, Pa; C_1, C_2, C_3, C_4, coefficients of the fit, 1, K, 1, K^{-1}.

Assumptions. K_{MEA} rated 0 to 50 C, used to 120 C.

Source. [**P**] Aronu, Eq 16 and Table 2, PDF p. 4, read as an image.

Implementation. `mea::aronu::table2`, evaluated for each constant by `mea::aronu::Thermo::new`

Validation. PUB, test `eq_g_02`. Target: molal pK_a of 9.50 against 9.51 in Versteeg and van Swaaij [32], Chem. Eng. Sci. 43 (1988) 573, Table 10, PDF p. 12. Hand check at 298.15 K: $\ln K_{x,MEA} = -25.882$, so $pK_x = 11.24$, and removing $\log_{10}(1/M_w) = 1.74$ gives 9.50 [**C**].

Equation G.3: Chemical potential

Tier R

$$\mu_i = \mu_i^\circ(T) + RT \ln(\gamma_i x_i) \tag{G.3}$$

Interpretation. Each species has a reference value, and its chemical potential moves with its activity. Everything else in this section follows from this line.

Derivation. Definition of the activity coefficient. μ_i° belongs to the standard state of Eq. G.6.

Variables. μ_i, chemical potential of i, J mol^{-1}; μ_i°, standard chemical potential of i, J mol^{-1}; γ_i, activity coefficient of i, 1.

Assumptions. None.

Source. [**T**] Textbook result.

Implementation. Theory, no code. Parent of Eq. G.4.

Validation. ID, test `eq_g_03`. Covered by Eq. G.5.

Equation G.4: Equilibrium constant from standard potentials

Tier R

$$RT \ln K_{x,j} = -\sum_i \nu_{ij} \mu_i^\circ \tag{G.4}$$

Interpretation. An equilibrium constant is a statement about standard chemical potentials. It carries no information about the mixture.

Derivation. Aronu Eq 12. At equilibrium $\sum_i \nu_{ij} \mu_i = 0$, and substituting Eq. G.3 separates the standard terms from the activities.

Variables. ν_{ij}, stoichiometric coefficient of i in reaction j, 1.

Assumptions. None.

Source. [**P**] Aronu, Eq 12, PDF p. 4.

Implementation. Theory, no code. Parent of Eq. G.5.

Validation. ID, test `eq_g_04`. Covered by Eq. G.5.

Equation G.5: Equilibrium condition

Tier R

$$\ln K_{x,j} = \sum_i \nu_{ij} \ln (\gamma_i x_i) \tag{G.5}$$

Interpretation. Five nonlinear equations, one per reaction. With the four balances that follow they fix all nine mole fractions.

Derivation. Combine Eq. G.3 and Eq. G.4. The system is solved in the logarithms of the mole fractions with a Newton method, because the species span many orders of magnitude.

Variables. x_i, liquid mole fraction of i, 1.

Assumptions. Liquid phase in chemical equilibrium.

Source. [**T**] Consequence of Eq. G.4.

Implementation. `mea::speciate::Problem::residual`

Validation. ID, test `eq_g_05`. Scaled residual below 10^{-9}.

Equation G.6: Standard states

Tier R

$$\gamma_i \to 1 \text{ as } x_i \to 1 \;\; (\mathrm{H_2O},\ \mathrm{MEA}), \qquad \gamma_i^* = \frac{\gamma_i}{\gamma_i^\infty} \;\; \text{(solutes)} \tag{G.6}$$

Interpretation. Water and MEA are measured against the pure liquid. The seven solutes are measured against infinite dilution in water, where they behave ideally by definition.

Derivation. Aronu states that the reference for water and amine is the pure component, and that CO_2 takes Henry's law at infinite dilution in water. The model writes γ_i^∞ from the combinatorial and residual terms evaluated in pure water.

Variables. γ_i^*, unsymmetric activity coefficient of i, 1; γ_i^∞, activity coefficient of i at infinite dilution in water, 1.

Assumptions. None.

Source. [**P**] Aronu, Eq 11 and text, PDF p. 4.

Implementation. `mea::aronu::Thermo::ln_gamma`

Validation. ID, test `eq_g_06`. Limits at pure water and infinite dilution.

Equation G.7: Hidden convention on the MEA constant

Tier O

$$\ln K_{x,MEA,eff} = \ln K_{x,MEA} + \ln \gamma^{\infty}_{MEA} \tag{G.7}$$

Interpretation. Aronu's MEA constant is an aqueous pK_a, while the paper's stated reference for MEA is the pure liquid. This one line reconciles the two, and without it the model misses the published column by about 60%.

Derivation. Table 2 takes K_{MEA} from Bates and Pinching, where MEA is referred to infinite dilution in water, so its activity there is $\gamma^{*}_{MEA} x_{MEA}$. On the stated pure-liquid basis the activity is $\gamma_{MEA} x_{MEA}$. Then Eq. G.6 gives the factor γ^{∞}_{MEA} on K. Aronu does not state this. A project calculation found it from a ratio that was independent of loading and depended on temperature. Two independent values support it: $\ln \gamma^{\infty}_{MEA}$ at 313.15 K is -1.279 from Aronu's UNIQUAC and -1.320 from the NRTL of Zhang et al. Table 8, and Cownden and Lucquiaud [5], Carbon Capture Sci. Technol. 19 (2026) 100625, Table 1, regressed a nearby constant on a solvent reference. K_{MEACOO} stays as printed.

Variables. $K_{x,MEA,eff}$, MEA constant on the pure liquid basis, 1.

Assumptions. Not stated by Aronu.

Source. [**C**] Calculated here. Support from Cownden and Lucquiaud Table 1 and Zhang et al. Table 8.

Implementation. `mea::aronu::Thermo::ln_k_mea_effective`

Validation. PUB, test `eq_g_07`. Prototype figures at 30 mass%, 17 model column points: AARD 58.8% without this line and 0.11% with it. The engine test named for this card recomputes both on the Rust port and holds each to its recorded value.

Equation G.8: MEA balance

Tier R

$$n_{MEA} + n_{MEAH^+} + n_{MEACOO^-} = n^0 \tag{G.8}$$

Interpretation. Every MEA molecule is free, protonated or bound as carbamate.

Derivation. Eq. A.5 on nitrogen, the only element MEA alone carries here.

Variables. n^0, total MEA charged, mol.

Assumptions. No degradation.

Source. [**T**] Consequence of Eq. A.5.

Implementation. `mea::speciate::Problem::residual` carries it in the Newton system, and `mea::speciate::Speciation::balance_errors` reports it at the solution

Validation. ID, test `eq_g_08`. Balance error at the solution.

Equation G.9: Carbon balance

Tier R

$$n_{CO_2} + n_{HCO_3^-} + n_{CO_3^{2-}} + n_{MEACOO^-} = \alpha n^0 \tag{G.9}$$

Interpretation. Every absorbed CO_2 sits in one of four forms. Their sum is the loading times the MEA present.

Derivation. Eq. A.5 on the carbon that came from CO_2, which excludes the two carbon atoms of the MEA backbone.

Variables. α, CO_2 loading, mol mol^{-1}.

Assumptions. None.

Source. [**T**] Consequence of Eq. A.5.

Implementation. `mea::speciate::Problem::residual` carries it in the Newton system, and `mea::speciate::Speciation::balance_errors` reports it at the solution

Validation. ID, test `eq_g_09`. Balance error at the solution.

Equation G.10: Hydrogen balance

Tier R

$$\begin{aligned} 2n_{H_2O} + 3n_{H_3O^+} + n_{OH^-} + n_{HCO_3^-} + 7n_{MEA} + 8n_{MEAH^+} + 6n_{MEACOO^-} \\ = 2n^0_{H_2O} + 7n^0 \end{aligned} \tag{G.10}$$

Interpretation. Hydrogen is conserved as it moves among the nine species. This closes the system once charge is also balanced.

Derivation. Eq. A.5 on hydrogen, with MEA as C_2H_7NO, $MEAH^+$ as $C_2H_8NO^+$ and carbamate as $C_3H_6NO_3^-$. Bicarbonate holds one hydrogen, and carbonate and CO_2 hold none.

Variables. $n^0_{H_2O}$, apparent water charged, mol.

Assumptions. MEA is C_2H_7NO and carbamate is $C_3H_6NO_3^-$.

Source. [**T**] Consequence of Eq. A.5.

Implementation. `mea::speciate::Problem::residual` carries it in the Newton system, and `mea::speciate::Speciation::balance_errors` reports it at the solution

Validation. ID, test `eq_g_10`. Balance error at the solution.

Equation G.11: Electroneutrality

Tier R

$$n_{H_3O^+} + n_{MEAH^+} = n_{OH^-} + n_{HCO_3^-} + 2n_{CO_3^{2-}} + n_{MEACOO^-} \tag{G.11}$$

Interpretation. The liquid carries no net charge. In practice $MEAH^+$ balances carbamate and bicarbonate, and the other ions stay small.

Derivation. Charge is conserved in each reaction of Eq. G.1, and the solvent starts neutral.

Variables. n_i, moles of species i, mol.

Assumptions. None.

Source. [**T**] Textbook result.

Implementation. `mea::speciate::Problem::residual` carries it in the Newton system, and `mea::speciate::Speciation::balance_errors` reports it at the solution

Validation. ID, test `eq_g_11`. Balance error at the solution.

Equation G.12: Loading

Tier R

$$\alpha = \frac{\text{total } CO_2}{\text{total MEA}} \tag{G.12}$$

Interpretation. The usual measure of how full the solvent is. Lean solvent runs at 0.25 and rich solvent at 0.50 here.

Derivation. Definition, and the right side of Eq. G.9 over n^0. Total counts all bound forms, which is what a titration measures.

Variables. α_{lean}, lean loading, mol mol^{-1}; α_{rich}, rich loading, mol mol^{-1}.

Assumptions. None.

Source. [**T**] Definition.

Implementation. `mea::speciate::speciate`, which takes the loading as an input

Validation. ID, test `eq_g_12`. Round trip with Eq. G.9.

Equation G.13: Activity coefficient

Tier R

$$\ln \gamma_i = \ln \gamma_i^C + \ln \gamma_i^R + \ln \gamma_i^{DH} \tag{G.13}$$

Interpretation. Non-ideality splits into one term for each physical cause. Size and shape give the first, short-range contact the second and long-range ion forces the last.

Derivation. Aronu Eq 13 splits the excess Gibbs energy into combinatorial, residual and Debye-Hückel parts. Differentiating with respect to n_i gives this sum. Aronu prints only that top level.

Variables. γ_i^C, combinatorial activity coefficient, 1; γ_i^R, residual activity coefficient, 1; γ_i^{DH}, Debye-Hückel activity coefficient, 1.

Assumptions. Thomsen form of extended UNIQUAC.

Source. [**P**] Aronu, Eq 13, PDF p. 4. [**M7**]: the expressions of each term are not in the library.

Implementation. `mea::aronu::Thermo::ln_gamma`

Validation. PUB, test `eq_g_13`. The 104 published model points. Validation results, 1 row (AR-1): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation G.14: Combinatorial term

Tier R

$$\ln \gamma_i^C = \ln \frac{\phi_i}{x_i} + 1 - \frac{\phi_i}{x_i} - \frac{z_c}{2} q_i \left[\ln \frac{\phi_i}{\theta_i} + 1 - \frac{\phi_i}{\theta_i} \right], \quad \phi_i = \frac{x_i r_i}{\sum_k x_k r_k}, \quad \theta_i = \frac{x_i q_i}{\sum_k x_k q_k} \tag{G.14}$$

	H_2O	MEA	CO_2	H_3O^+	$MEAH^+$	OH^-	HCO_3^-	CO_3^{2-}	$MEACOO^-$
r_i	0.9200	4.2800	5.7410	0.13779	1.0241	9.3973	9.1571	9.7452	1.0741
q_i	1.4000	4.2800	6.0806	10^{-15}	2.5150	8.8171	6.3461	6.4614	0.1106

Interpretation. This is the part of non-ideality due to size and shape alone, with no energy in it. The r and q values act as a molecule's volume and surface area.

Derivation. Standard UNIQUAC combinatorial form, with $z_c = 10$ from the Thomsen framework. Table 7 prints r and q for all nine species, read from the image.

Variables. ϕ_i, volume fraction of i, 1; θ_i, surface fraction of i, 1; r_i, UNIQUAC volume parameter, 1; q_i, UNIQUAC surface parameter, 1; z_c, coordination number, 10, 1.

Assumptions. r and q from Table 7.

Source. [**M7**]: expression from general knowledge. [**P**] Aronu, Table 7, PDF p. 7.

Implementation. Inside `mea::aronu::Thermo::ln_gamma`

Validation. PUB, test `eq_g_14`. The 104 published model points. Validation results, 1 row (AR-1): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation G.15: Residual term

Tier R

$$\ln \gamma_i^R = q_i \left[1 - \ln \sum_k \theta_k \psi_{ki} - \sum_k \frac{\theta_k \psi_{ik}}{\sum_l \theta_l \psi_{lk}} \right], \qquad \psi_{ki} = \exp\left(-\frac{u_{ki} - u_{ii}}{T} \right) \tag{G.15}$$

Interpretation. Short-range energy between neighbours. A large positive u means two species avoid each other.

Derivation. Standard UNIQUAC residual form. The expression for ψ_{ki} is printed as Aronu Eq 14, and the rest is from the Thomsen framework.

Variables. ψ_{ki}, interaction factor between k and i, 1; u_{ki}, interaction energy between k and i, K.

Assumptions. Published parameters used with an expression from general knowledge.

Source. [**M7**]: expression from general knowledge. ψ_{ki} from Aronu, Eq 14, PDF p. 4.

Implementation. Inside `mea::aronu::Thermo::ln_gamma`

Validation. PUB, test `eq_g_15`. The 104 published model points. Validation results, 1 row (AR-1): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation G.16: Interaction parameters

Tier E

$$u_{ki} = u^0_{ki} + u^T_{ki}\,(T - 298.15)\,, \qquad u_{ki} = u_{ik} \tag{G.16}$$

Interpretation. Each pair of species has one energy that drifts linearly with temperature. The values in the two tables below are Aronu's regression.

Fit range and origin. Aronu Eq 15 with Tables 8 and 9, read from the page image because the text layer drops minus signs. The tables are symmetric, so only the lower triangle is printed. Pairs held at 10^9 never meet in practice, so those pairs drop out of the model.

Variables. u^0_{ki}, interaction energy at 298.15 K, K; u^T_{ki}, temperature slope of the interaction energy, 1.

Assumptions. Tables 8 and 9, signs from the image.

Source. [**P**] Aronu, Eqs 14 and 15, PDF p. 4, with Tables 8 and 9, PDF p. 7.

Implementation. `mea::aronu::u_ij`

Validation. PUB, test `eq_g_16`. The 104 published model points. Validation results, 1 row (AR-1): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

u^0_{ki}	H_2O	MEA	CO_2	H_3O^+	$MEAH^+$	OH^-	HCO_3^-	CO_3^{2-}	$MEACOO^-$
H_2O	0								
MEA	173.9645	414.6924							
CO_2	−151.4573	87.5583	40.5176						
H_3O^+	10^4	10^9	10^9	0					
$MEAH^+$	−20.7732	310.1293	30.8035	10^9	0				
OH^-	600.4952	10^9	2500	10^9	10^9	1562.8810			
HCO_3^-	517.0278	655.0881	597.9726	10^9	732.7007	2500	743.6159		
CO_3^{2-}	361.3877	10^9	2500	10^9	10^9	1588.0250	719.159	1458.3440	
$MEACOO^-$	2758.3852	10^9	10^9	10^9	10^9	10^9	10^9	10^9	1500

u^T_{ki}	H_2O	MEA	CO_2	H_3O^+	$MEAH^+$	OH^-	HCO_3^-	CO_3^{2-}	$MEACOO^-$
H_2O	0								
MEA	0.8027	0.6647							
CO_2	6.0908	4.6660	13.6290						
H_3O^+	0	0	0	0					
$MEAH^+$	−1.9174	0.1213	7.3541	0	0				
OH^-	8.5455	0	0	0	0	5.6169			
HCO_3^-	6.9504	15.2488	5.8077	0	2.8863	0	17.1148		
CO_3^{2-}	3.3516	0	0	0	0	2.7496	2.6115	−1.3448	
$MEACOO^-$	16.9192	0	0	0	0	0	0	0	0

Equation G.17: Debye-Hückel term

Tier R

$$\begin{gathered}
\ln\gamma_i^{DH} = -Z_i^2 \frac{A_{DH}\sqrt{I}}{1 + b_{DH}\sqrt{I}}, \qquad \ln\gamma_w^{DH} = \frac{2}{3} M_w A_{DH} I^{3/2} \sigma_{DH}\left(b_{DH}\sqrt{I}\right) \\
\sigma_{DH}(x) = \frac{3}{x^3}\left[1 + x - \frac{1}{1+x} - 2\ln(1+x)\right] \\
A_{DH} = 1.131 + 1.335 \times 10^{-3} t_C + 1.164 \times 10^{-5} t_C^2, \qquad b_{DH} = 1.5
\end{gathered} \tag{G.17}$$

Interpretation. Ions feel each other over long distances, and the effect grows with charge squared. Water takes a small correction from the same term.

Derivation. Extended Debye-Hückel form of the Thomsen framework, with molality on water alone. Neither the expressions, nor $A_{DH}(T)$, nor b_{DH} appear in any local source. They are stated from general knowledge and supported only by the test on Eq. G.13. Scaling A_{DH} by 1.3 or setting $b_{DH} = 1.0$ moves the 30 mass% AARD to 11.0% and 10.2%, against 0.11% with these values, so the published column does discriminate between them.

Variables. Z_i, ion charge of i, 1; A_{DH}, Debye-Hückel constant, $\mathrm{kg}^{1/2}\ \mathrm{mol}^{-1/2}$; b_{DH}, Debye-Hückel size parameter, 1.5, $\mathrm{kg}^{1/2}\ \mathrm{mol}^{-1/2}$; σ_{DH}, Debye-Hückel water function, 1; t_C, temperature in Celsius, here only, C; γ_w^{DH}, Debye-Hückel coefficient of water, 1.

Assumptions. Molality on water.

Source. [**M7**]: general knowledge, supported by the test on Eq. G.13.

Implementation. `mea::aronu::a_dh` and `mea::aronu::Thermo::ln_gamma`

Validation. PUB, test `eq_g_17`. The 104 published model points. Validation results, 1 row (AR-1): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation G.18: Ionic strength

Tier R

$$I = \frac{1}{2}\sum_i m_i Z_i^2 \tag{G.18}$$

Interpretation. A single measure of how many ions are present, weighted by charge squared.

Derivation. Definition, with molality taken on water: $m_i = x_i/(x_w M_w)$.

Variables. I, ionic strength, mol kg^{-1}; m_i, molality of i, mol kg^{-1}.

Assumptions. None.

Source. [**T**] Definition.

Implementation. Inside `mea::aronu::Thermo::ln_gamma`

Validation. ID, test `eq_g_18`. Direct evaluation on a charge-balanced mixture.

Equation G.19: CO_2 vapour-liquid equilibrium

Tier R

$$\varphi_{CO_2} y_{CO_2} p = \gamma^*_{CO_2} x_{CO_2} H_x \exp\left[\frac{v^\infty_{CO_2}\left(p - p^s_{H_2O}\right)}{RT}\right] \tag{G.19}$$

Interpretation. The pressure of CO_2 above the solvent. It is the single most important output of this section, because it sets the driving force in both columns.

Derivation. Aronu Eq 10. Equate the fugacity of CO_2 in each phase, with the liquid referred to Henry's law at infinite dilution in water at the saturation pressure of water. The exponential is the Poynting correction from that pressure to the system pressure. It is 1.029 at 21 bar and 40 C, which shifts loading by only 0.002 to 0.003. Below, $p^*_{CO_2}$ means this pressure at the liquid composition, with $\varphi_{CO_2} = 1$ unless stated.

Variables. φ_i, fugacity coefficient of i, 1; $v^\infty_{CO_2}$, partial molar volume of CO_2 at infinite dilution, $\mathrm{m}^3\ \mathrm{mol}^{-1}$; $p^s_{H_2O}$, saturation pressure of water, Pa; $p^*_{CO_2}$, equilibrium partial pressure of CO_2, Pa.

Assumptions. $v^\infty_{CO_2} = 3.5 \times 10^{-5}\ \mathrm{m}^3\ \mathrm{mol}^{-1}$ [**A**], as Aronu prints no value.

Source. [**P**] Aronu, Eq 10, PDF p. 3 to p. 4.

Implementation. `units::solvent::Solvent::pstar_gas_pa`, with the absorber fugacity coefficient `units::solvent::PHI_CO2_ABSORBER`. The equilibrium pressure itself is `mea::pstar::pstar_kpa`, with the gas-side form `mea::pstar::pco2_gas_kpa`.

Validation. PUB, test `eq_g_19`. The 104 model points, with the measured data on AR-2 and AR-3. Validation results, 4 rows (AR-1 to AR-3): 4 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation G.20: Water and MEA vapour-liquid equilibrium

Tier R

$$\varphi_i y_i p = \gamma_i x_i p_i^s \varphi_i^s \exp\left[\frac{v_i\,(p - p_i^s)}{RT}\right] \tag{G.20}$$

Interpretation. Modified Raoult's law for the solvent itself. This gives the water that leaves with the treated gas and the MEA lost as vapour.

Derivation. Aronu Eq 11, from the pure-liquid reference of Eq. G.6. Aronu takes p_i^s from DIPPR, which is not local. Water uses IF97 by Eq. F.4. For MEA the library holds only 164 Pa at 40 C and 666 Pa at 60 C, from Dugas, so MEA vapour pressure rests on two points.

Variables. p_i^s, saturation pressure of i, Pa; φ_i^s, fugacity coefficient of saturated i, 1; v_i, partial molar volume of i, $\mathrm{m^3\ mol^{-1}}$.

Assumptions. DIPPR not local, IF97 for water. MEA two points only.

Source. [**P**] Aronu, Eq 11, PDF p. 4. Dugas [6], PDF p. 125, for the MEA points.

Implementation. `mea::pstar::water_partial_pressure_pa`, for water only. MEA volatility is left out

Validation. EXP, test `eq_g_20`. Aronu total pressure AARD, quoted as 11.7%. Validation results, 1 row (AR-4): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation G.21: Vapour fugacity

Tier R

$$\begin{gathered} Z^3 - Z^2 + \left(A_S - B_S - B_S^2\right) Z - A_S B_S = 0 \\ \ln \varphi_i = \frac{b_i}{b_m}\,(Z-1) - \ln\left(Z - B_S\right) - \frac{A_S}{B_S}\left(\frac{2\sum_j y_j a_{ij}}{a_m} - \frac{b_i}{b_m}\right)\ln\left(1 + \frac{B_S}{Z}\right) \end{gathered} \tag{G.21}$$

Interpretation. A correction for non-ideal gas at pressure. In the plant it matters only slightly, and it is kept so that the published model column can be matched.

Derivation. Soave-Redlich-Kwong, as Aronu states. Here $A_S = a_m p/(RT)^2$ and $B_S = b_m p/(RT)$, with van der Waals mixing and $a_{ij} = \sqrt{a_i a_j}$. Critical constants are textbook values. The code holds CO_2 and water only. No binary parameter for H_2 exists in any local source, so it would be taken as zero in a hydrogen mixture.

Variables. A_S, SRK attraction group, 1; B_S, SRK volume group, 1; a_m, SRK mixture attraction, $\mathrm{Pa\ m^6\ mol^{-2}}$; b_m, SRK mixture covolume, $\mathrm{m^3\ mol^{-1}}$.

Assumptions. H_2 k_{ij} absent, taken as 0.

Source. [**P**] Aronu, PDF p. 3, for the choice of SRK. [**T**] Form and constants.

Implementation. `mea::pstar::srk_phi_co2` and `mea::pstar::srk_co2_h2o`

Validation. ID, test `eq_g_21`. Limit $\varphi_i \to 1$ at low pressure. Used only for the model column comparison.

Equation G.22: Henry constant in water

Tier E

$$H^{w}_{CO_2} = 2.82 \times 10^{6} \exp\left(-\frac{2044}{T}\right) \tag{G.22}$$

Interpretation. How easily CO_2 stays in physical solution in water. A larger value means lower solubility, and it grows on heating.

Fit range and origin. Aronu Eq 17 prints the constants with no unit. Dugas Eq 5.9 prints the same constants in Pa m^3 mol^{-1}. Both cite Versteeg and van Swaaij (1988) in J. Chem. Eng. Data 33, which the library lacks, so the values are second hand.

Variables. $H^{w}_{CO_2}$, Henry constant of CO_2 in water, Pa m^3 mol^{-1}.

Assumptions. Second hand.

Source. [**P**] Aronu, Eq 17, PDF p. 4. Dugas [6], Eq 5.9, PDF p. 129. [**M8**]: original not local.

Implementation. `mea::aronu::henry_water_co2_pa_m3_mol`

Validation. PUB, test `eq_g_22`. The Table 2 form. Hand check at 298.15 K: $H_x = 1.61 \times 10^8$ Pa from Eq. G.2 against $H^{w}_{CO_2} c_w = 1.64 \times 10^8$ Pa [**C**], within 2%. The Versteeg Henry constants: Validation results, 1 row (VER-X): 1 not run.

Equation G.23: N_2O analogy

Tier E

$$H_{app} = H_{N_2O} \frac{H^{w}_{CO_2}}{H^{w}_{N_2O}}, \qquad H^{w}_{N_2O} = 8.55 \times 10^{6} \exp\left(-\frac{2284}{T}\right), \qquad H_{app} = \gamma^{*}_{CO_2} H^{w}_{CO_2} \tag{G.23}$$

Interpretation. CO_2 reacts in amine, so its physical solubility cannot be measured directly. N_2O has similar size and shape and does not react, so it stands in.

Fit range and origin. Aronu Eqs 18 and 19. Dugas Eq 5.10 prints the same N_2O constants with the unit. A basis warning applies: Eq 19 sets a mole fraction activity coefficient equal to a ratio of concentration-based constants. Total molar density is 43.4 kmol m^{-3} at 40 C and loading 0.5, against 55.1 for water, a factor of 1.27. So $\gamma^{*}_{CO_2}$ stays inside the equilibrium model and does not pass to rate calculations.

Variables. H_{app}, apparent Henry constant of CO_2 in the solvent, Pa m^3 mol^{-1}; H_{N_2O}, Henry constant of N_2O in the solvent, Pa m^3 mol^{-1}; $H^{w}_{N_2O}$, Henry constant of N_2O in water, Pa m^3 mol^{-1}.

Assumptions. Mixed basis in the source, as stated above.

Source. [**P**] Aronu, Eqs 18 and 19, PDF p. 4. Dugas [6], Eq 5.10, PDF p. 129.

Implementation. `mea::pstar::henry_apparent_pa_m3_mol`

Validation. EXP, test `eq_g_23`. Target: Hartono et al. [8], J. Chem. Eng. Data 59 (2014) 1808, Table 11, PDF p. 6, $n = 15$, AARD 3.4% prototype figure, against 2.7% quoted by Aronu.

Equation G.24: Heat of absorption

Tier O

$$\Delta h_{abs} = -R \left(\frac{\partial \ln p^{*}_{CO_2}}{\partial \left(1/T\right)} \right)_{p,x} \tag{G.24}$$

Interpretation. The slope of the equilibrium curve gives the heat released when CO_2 is absorbed. It drops above a loading of about 0.5, where absorption turns more physical.

Derivation. Aronu Eq 22, the Gibbs-Helmholtz relation applied to Eq. G.19, which makes Δh_{abs} a positive number for exothermic absorption. The code takes a central difference 2 K either side at fixed apparent composition. Aronu warns that this estimate is not necessarily accurate, and the calorimetric data of Kim and Svendsen exist locally only as figures. The prototype gives 84.7 and 61.5 kJ mol^{-1} at 40 C and loadings 0.30 and 0.50, as a reading aid only.

Variables. Δh_{abs}, heat of absorption of CO_2, positive when exothermic, J mol^{-1}.

Assumptions. Aronu warns about accuracy.

Source. [**C**] Calculated here with Aronu, Eq 22, PDF p. 11.

Implementation. `mea::enthalpy::dh_abs_j_mol`

Validation. NONE, test `eq_g_24`. No numeric benchmark exists. Comparison is against figures only, Figs 12 to 14 of Zhang et al.

Equation G.25: Solvent heat capacity

Tier F

$$c_{p,s} = 3.6 \text{ kJ kg}^{-1} \text{ K}^{-1} \tag{G.25}$$

Interpretation. A single value for solvent heat capacity, used in the column and exchanger energy balances.

Justification. No local source tabulates this quantity. Plaza Fig 3.13, for 7 m MEA at loadings 0.14, 0.36 and 0.54 from 40 to 120 C, spans an axis of 3.2 to 3.8 kJ kg^{-1} K^{-1}. Read by eye, the points run from about 3.25 at high loading and low temperature to about 3.7 at low loading and high temperature, so 3.6 sits in the upper part of that range for loaded solvent.

Variables. $c_{p,s}$, solvent mass heat capacity, J kg^{-1} K^{-1}.

Assumptions. Constant.

Source. [**A**] Assumed. Range from Plaza [24], Fig 3.13, PDF p. 79, by eye. [**M21**]: no tabulated value.

Implementation. `units::solvent::CP_SOLVENT_J_KG_K`

Validation. NONE, test `eq_g_25`. No benchmark exists.

Equation G.26: Density

Tier E

$$\begin{gathered} V_{sln} = x_{MEA}V_{MEA} + x_{H_2O}V_{H_2O} + x_{CO_2}V_{CO_2} + x_{MEA}x_{H_2O}V^* \\ V_{MEA} = \frac{M_{MEA}}{a_\rho T^2 + b_\rho T + c_\rho}, \qquad \rho_{sln} = \frac{\sum_i x_i M_i}{V_{sln}} \end{gathered} \tag{G.26}$$

Interpretation. Solvent density from a molar volume with one mixing correction. Dissolved CO_2 adds mass and almost no volume, so density rises with loading.

Fit range and origin. Weiland et al. as printed by Plaza Eqs 3.16 and 3.17 and by Dugas Eqs 5.17 to 5.20. Here x_i are apparent mole fractions on MEA, water and CO_2, with V in mL mol^{-1}, $a_\rho = -5.35162 \times 10^{-7}$, $b_\rho = -4.51417 \times 10^{-4}$, $c_\rho = 1.19451$, $V^* = -1.8218$ and $V_{CO_2} = 0.04747$. Both theses print V_{CO_2} with this value, and reading it as 47.47 raises AARD from 0.51% to 4.84%. Eq 3.17 prints $M_{MEA} = 61.09$ g mol^{-1}, while the Plaza subroutine holds 61.08, which the code uses. Water molar volume comes from IF97. Hartono's own correlation stays a benchmark, because its unloaded base cannot be rebuilt from that paper.

Variables. V_{sln}, molar volume of the solution, m^3 mol^{-1}; V_{MEA}, molar volume of MEA, m^3 mol^{-1}; V_{H_2O}, molar volume of water, m^3 mol^{-1}; V_{CO_2}, dissolved volume of CO_2, m^3 mol^{-1}; V^*, interaction volume of MEA and water, m^3 mol^{-1}; ρ_{sln}, solution density, kg m^{-3}; a_ρ, b_ρ, c_ρ, MEA density coefficients, g mL^{-1} with T in K.

Assumptions. Apparent mole fractions. Checked from 20 to 80 C only.

Source. [**P**] Plaza [24], Eq 3.16 and Eq 3.17, PDF p. 71, and subroutine, PDF p. 295. Dugas [6], Eqs 5.17 to 5.20 and Table 5.2, PDF p. 133.

Implementation. `mea::density::density_kg_m3`

Validation. EXP, test `eq_g_26`. Hartono et al. Table 3, PDF p. 3, 30 mass%, 20 to 80 C, loading to 0.50. Validation results, 1 row (HA-1): 1 pass.

Equation G.27: Viscosity

Tier E

$$\mu_{sln} = \mu_w \exp\left\{\frac{(21.186\,\Omega_w + 2373)\,[\alpha\,(0.01015\,\Omega_w + 0.0093\,T - 2.2589) + 1]\,\Omega_w}{T^2}\right\} \tag{G.27}$$

Interpretation. MEA thickens water, and loading thickens it further. Viscosity matters because it slows diffusion, and that reduces the liquid film coefficient.

Fit range and origin. Weiland correlation. Dugas Eq 5.16 with Table 5.1, Plaza Eq 3.18 and Khan Eq 7 agree once the zero terms of Table 5.1 are dropped. Ω_w is mass percent MEA on a CO_2-free basis, so 30 here. Water viscosity comes from the IAPWS release.

Variables. μ_{sln}, solution viscosity, Pa s; μ_w, water viscosity, Pa s; Ω_w, MEA mass percent, 1.

Assumptions. Loading to 0.50.

Source. **[P]** Dugas [6], Eq 5.16 and Table 5.1, PDF p. 132. Khan et al. [15], Chem. Eng. Res. Des. 89 (2011) 1600, Eq 7, PDF p. 3. Plaza [24], Eq 3.18, PDF p. 73 to p. 74.

Implementation. `mea::viscosity::viscosity_pa_s`

Validation. EXP, test `eq_g_27`. Hartono et al. Table 7, PDF p. 5. Validation results, 1 row (HA-2): 1 pass.

Equation G.28: Diffusivity

Tier E

$$D_w = 2.35 \times 10^{-6} \exp\left(-\frac{2119}{T}\right), \qquad D_l = D_w \left(\frac{\mu_w}{\mu_{sln}}\right)^{0.8} \tag{G.28}$$

Interpretation. CO_2 diffuses more slowly in thick solvent. The exponent says the slowdown follows viscosity closely, though a little less than in direct proportion.

Fit range and origin. Khan Eqs 6a and 6b, from Versteeg and van Swaaij. Dugas Eq 5.15 prints the same water value with the unit $m^2\ s^{-1}$. Dugas Eq 5.14 uses an exponent of 0.72 in a different kinetic set, and these sets must not be mixed.

Variables. D_w, CO_2 diffusivity in water, $m^2\ s^{-1}$; D_l, CO_2 diffusivity in solution, $m^2\ s^{-1}$.

Assumptions. Exponent 0.8 goes with the Khan set.

Source. **[P]** Khan et al., Eqs 6a and 6b, PDF p. 3. Dugas [6], Eq 5.15, PDF p. 132.

Implementation. `mea::kinetics::d_co2_solution_m2_s`

Validation. PUB, test `eq_g_28`. Target: Dugas Tables 4.2 and B.1.

Equation G.29: Free MEA

Tier O

$$f_{free} = \frac{n_{MEA}}{n^0} \tag{G.29}$$

Interpretation. Only free MEA reacts with CO_2. As the solvent fills, this fraction collapses, so the reaction slows long before thermodynamics runs out.

Derivation. Read from the solution of Eq. G.5 to Eq. G.11 at the local temperature and loading. The prototype gives 0.4% of total MEA at 40 C and 1.2% at 60 C at equilibrium rich loading. That is the kinetic reason for the cap in Eq. G.32. Dugas Table 5.4 shows the same trend for 7 m MEA at 20 C: its calculated k'_g falls from 2.5×10^{-6} at loading 0.25 to 1.7×10^{-7} mol s^{-1} Pa^{-1} m^{-2} at 0.50.

Variables. f_{free}, free fraction of MEA, 1.

Assumptions. Kinetic limit near equilibrium loading.

Source. [**C**] Calculated here. Trend from Dugas [6], Table 5.4, PDF p. 192.

Implementation. `mea::speciate::speciate`

Validation. PUB, test `eq_g_29`. Target: 0.4% at 40 C and equilibrium loading, prototype figure.

Equation G.30: Empirical comparator

Tier E

$$\ln p^*_{Xu} = 44.2 - \frac{116{,}000}{RT} - 29.7\,\alpha + 11{,}600\,\frac{\alpha}{T} + 17.3\,\alpha^2 \tag{G.30}$$

Interpretation. A simple fit for equilibrium pressure, in Pa. It is useful as a second opinion and fails at the rich end.

Fit range and origin. Dugas Eq 5.26, from Xu in Rochelle et al., valid for MEA from 40 to 160 C, with T in K and R in J mol^{-1} K^{-1}. It runs low above a loading of about 0.45.

Variables. p^*_{Xu}, equilibrium CO_2 pressure by the Xu fit, Pa.

Assumptions. Fails above loading about 0.45.

Source. [**P**] Dugas [6], Eq 5.26, PDF p. 136.

Implementation. Not implemented in the engine at this build. The comparison in the validation line is computed in this document. Planned name `mea::xu_check`.

Validation. EXP, test `eq_g_30`. Target: ratio to data of 0.45 at loading 0.496 and 40 C, a project calculation.

Equation G.31: Plant point

Tier O

$$p^*_{CO_2}(T, \alpha^*) = p_{CO_2}, \qquad \alpha^*\,(378\text{ kPa},\ 40\text{ C}) = 0.741 \tag{G.31}$$

Interpretation. The loading the solvent would reach at rest with the feed gas. It sits far above the 0.50 cap, so equilibrium leaves room and rate sets the limit.

Derivation. Invert Eq. G.19 for loading. Aronu's 30 mass% data stop at 11.8 kPa and loading 0.524 at 40 C, so this point extrapolates inside the source's own regression set. The 378 kPa point comes from the project input sheet at 21 bar and 18 mol%. The design absorber at 26.8 bar and 17.7 mol% sees 474 kPa, which the engine will evaluate. That value has not been computed here.

Variables. α^*, equilibrium loading, mol mol^{-1}.

Assumptions. Extrapolates 30 mass% data.

Source. [**C**] Calculated here with Eq. G.19.

Implementation. `mea::pstar::alpha_at`

Validation. EXP, test `eq_g_31`. Target read-offs from figures, both printed because they disagree: Jou, about 0.74 through Zhang et al. Fig 9, and Wong et al. [34], RSC Adv. 6 (2016) 10816, Fig 3, about 0.63.

Equation G.32: Range guards

Tier F

$$313.15\text{ K} \le T \le 393.15\text{ K}, \qquad \alpha \le 0.50 \text{ in columns}, \qquad p_{CO_2} \le 10\text{ bar} \tag{G.32}$$

Interpretation. The range where the model is trusted. Outside it the column code must stop or flag the result.

Justification. The temperature limits are Aronu's data range. The loading cap holds because all kinetic and physical property fits in the library end near 0.50. The pressure limit comes from Wong Table 4, where the prototype moves from a mean of +0.05 in loading at or below 10 bar to −0.15 above it.

The direct solver accepts a wider window, from 298.15 to 403.15 K and loading 0.001 to 1.4, so these limits belong to the column layer.

Variables. T, temperature, K.

Assumptions. Rich loading 0.55 stays an upper case only.

Source. [**M11**]: no kinetics or properties above loading 0.50. Limits from Aronu, PDF p. 3 to p. 5.

Implementation. `mea::speciate::check_range`

Validation. ID, test `eq_g_32`. Each guard trips on a value just outside it.

Equation G.33: Table for column calls

Tier O

$$\ln p^{*}\left(T,\alpha\right)-2\ln\alpha=\sum_{r=1}^{4}\sum_{s=1}^{4}w_{r}\left(T\right)w_{s}\left(\alpha\right)\Lambda_{rs} \tag{G.33}$$

Interpretation. A column calls p^{*} thousands of times, so the solved model is stored on a grid and read back by interpolation.

Derivation. The grid holds 16 temperatures from 30 to 130 C and 40 loadings from 0.02 to 0.75, for one MEA mass fraction, each node from a direct solve. Interpolation is cubic Lagrange in each direction. At low loading p^{*} varies as α^{2}, so the table stores the smooth remainder $\ln p^{*}-2\ln\alpha$ and restores the term afterwards. Every node still equals a direct solve.

Variables. w_r, four-point Lagrange weight, 1; Λ_{rs}, stored value of $\ln p^{*}-2\ln\alpha$, 1.

Assumptions. Design strength of 30 mass%. Direct solve above that strength.

Source. [**C**] Calculated here.

Implementation. `mea::pstar::stencil`, inside the table built by `mea::pstar::pstar_table`

Validation. ID, test `eq_g_33`. Against Eq. G.19. Grid error below 1% is the stated target, and the crate holds a tighter 0.5% gate at 30 mass%.

Worked example: Case B lean and rich solvent at 40 C

Take 1 kg of CO_2-free solvent at 30 wt% MEA: 4.912 mol MEA and 38.86 mol water. At the lean loading of 0.25 this carries 1.228 mol CO_2, so the apparent mole fractions are 0.1092, 0.8636 and 0.0273. At 313.15 K, Eq. G.26 gives $V_{MEA}=61.04$ mL mol^{-1}, and water at 992.2 kg m^{-3} gives 18.16 mL mol^{-1}. Then $V_{sln}=22.17$ mL mol^{-1}, the mean molar mass is 23.43 g mol^{-1}, the density is 1,057 kg m^{-3} and the MEA concentration is 4.92 kmol m^{-3}. Eq. G.27 gives $\mu_{sln}/\mu_{w}=3.13$. Eq. G.28 starts from $D_w=2.71\times10^{-9}$ m^2 s^{-1} and gives $D_l=1.09\times10^{-9}$ m^2 s^{-1}.

At the rich loading of 0.50 the same steps give 1,110 kg m^{-3}, $\mu_{sln}/\mu_{w}=3.90$ and $D_l=9.1\times10^{-10}$ m^2 s^{-1}. Hartono Table 3 prints 1,059.8 kg m^{-3} at loading 0.29 and 1,098.7 at 0.50, so the Weiland form runs 1.1% high at the rich end. These are hand figures for a loading range that Eq. G.32 allows. The Case B plant moves 894 t h^{-1} of this solvent, on a CO_2-free basis, by Eq. H.28.

Link to the sources

The prototype reproduced Aronu's model column at 104 points only after the convention of Eq. G.7 was found, and it did so with activity expressions the library does not hold. That is strong indirect support for [**M7**], and a copy of Thomsen and Rasmussen remains the first item to request. Against measured data the same model gives an AARD of 21.9% on Aronu Table 4 and 8.4% on Dugas Table 4.4, which was outside Aronu's regression. Both figures describe the prototype. At high pressure and in hydrogen-rich gas, no local source holds equilibrium data for this solvent.

10 H. Rate-based MEA absorber

Rate-based here describes the method. The column is a one-dimensional film model with an approximate enhancement factor, so these cards carry tier O, and it runs in a regime no local source has measured: 30 wt% MEA on hydrogen-rich gas at 26.8 bar. The closest experiments use 20 wt% MEA on methane at similar CO_2 partial pressure. Each result from this section is therefore an estimate with a stated test behind it, and none enters the headline ledger as a measured value.

Symbols keep this section apart from the rest of the document. $\mathcal{E}$, $\mathcal{E}_1$ and $\mathcal{E}_\infty$ are enhancement factors and $\mathcal{M}$ is the Hatta group, because E and M name stored energy and mass in Section A. G_n and L_n are molar flows, because G is a mass velocity and L a tube length. a_p is packing area and a_e the effective area that Sonderby writes as a. h_{GL}, q_{GL} and S_{ch} replace plain h, q and S. Khan et al. [15], Chem. Eng. Res. Des. 89 (2011) 1600, is cited as Khan, and Isa et al. [14], Int. J. Greenh. Gas Control 56 (2017) 173, as Isa. The column model is in the engine, and each card names the function that implements it. Each validation line prints the outcome that the validation results file records. Gate M6 judged the column rows and failed, with 6 of 13 judged rows outside their limits (SA-1, DICE-SRD, KW-SRD, MO-K16-TOUT, VI-3-SS2, VI-3-SS3). Terms T02 and T06 to T08 therefore stay anchored in the Headline, and no modelled column value enters the ledger.

Equation H.1: Gas film

Tier O

$$N_{CO_2} = k_G \left(p_{CO_2} - p_{int}\right) \tag{H.1}$$

Interpretation. CO_2 crosses a thin gas film by diffusion, driven by the drop in partial pressure from the bulk gas to the interface.

Derivation. Whitman two-film theory, as Khan Fig 1 shows it. Each film resists in series and holds no material at steady state.

Variables. N_{CO_2}, CO_2 flux from gas to liquid, mol m^{-2} s^{-1}; k_G, gas film coefficient, mol m^{-2} s^{-1} Pa^{-1}; p_{int}, CO_2 partial pressure at the interface, Pa.

Assumptions. Two-film.

Source. [**P**] Khan, Fig 1 and text, PDF p. 2.

Implementation. `units::absorber::flux`

Validation. ID, test `eq_h_01`. Sign and zero flux at zero driving force.

Equation H.2: Liquid film

Tier O

$$N_{CO_2} = \mathcal{E}\, k_L^{\circ}\, \frac{p_{int} - p^*_{CO_2}}{H} \tag{H.2}$$

Interpretation. Reaction in the liquid film speeds absorption up. $\mathcal{E}$ measures that gain over plain physical absorption.

Derivation. Physical absorption gives $k_L^{\circ}(C_{int} - C_b)$. Sonderby's first assumption places all reaction inside the film and leaves the bulk at equilibrium, so $C_b = p^*_{CO_2}/H$, with $p^*_{CO_2}$ from Eq. G.33. Then Eq. H.3 gives this form.

Variables. $\mathcal{E}$, enhancement factor, 1; k_L°, physical liquid film coefficient, m s^{-1}; H, Henry constant of CO_2 on the rate side, Pa m^3 mol^{-1}.

Assumptions. All reaction in the film and the bulk at equilibrium.

Source. [**P**] Sonderby et al. [29], assumption 1, PDF p. 3.

Implementation. `units::absorber::flux`

Validation. ID, test `eq_h_02`. Flux changes sign when p_{int} passes $p^*_{CO_2}$.

Equation H.3: Interface equilibrium

Tier R

$$p_{int} = H\, C_{int} \tag{H.3}$$

Interpretation. The two sides of the interface are at equilibrium, even though the bulk phases are far from it.

Derivation. Henry's law at the interface, which has no resistance in two-film theory.

Variables. C_{int}, CO_2 concentration at the interface, mol m^{-3}.

Assumptions. Henry.

Source. [**T**] Textbook result.

Implementation. `units::absorber::flux`

Validation. ID, test `eq_h_03`. Direct identity.

Equation H.4: Overall coefficient

Tier O

$$N_{CO_2} = K_G \left(p_{CO_2} - p^*_{CO_2}\right), \qquad \frac{1}{K_G} = \frac{1}{k_G} + \frac{H}{\mathcal{E}\, k^\circ_L} \tag{H.4}$$

Interpretation. Two resistances in series. Whichever is larger controls, and in this column the liquid film is expected to dominate.

Derivation. Write Eq. H.1 and Eq. H.2 as $N_{CO_2}/k_G = p_{CO_2} - p_{int}$ and $N_{CO_2} H/(\mathcal{E} k^\circ_L) = p_{int} - p^*_{CO_2}$, and add them to remove p_{int}. Khan Eq 3 is the same form per unit volume, with $p^*_{CO_2}$ set to zero.

Variables. K_G, overall gas side coefficient, mol m^{-2} s^{-1} Pa^{-1}.

Assumptions. Two-film.

Source. [**P**] Sonderby et al. [29], Eqs 9 and 10, PDF p. 3. Khan, Eq 3, PDF p. 3.

Implementation. `units::absorber::k_overall`

Validation. EXP, test `eq_h_04`. Sonderby runs through the full column, on RSP-250 geometry standing in for Mellapak 250Y, which has no local data. Validation results, 5 rows (SO-R1-CAP to SO-R5-CAP): 5 reported without a limit. Simulated less published: −13.89 to −0.967 % over 5 rows.

Equation H.5: Rate constant

Tier E

$$\log_{10} k_2 = 10.99 - \frac{2152}{T} \tag{H.5}$$

Interpretation. The second-order rate constant of CO_2 with MEA. It doubles about every 14 K, so the temperature bulge inside the column speeds the reaction up.

Fit range and origin. Hikita's correlation through Khan Eq 2. Khan prints no unit. Taken in m^3 kmol^{-1} s^{-1} it gives 5,918 at 298.15 K, close to 5,993 from $4.4 \times 10^{11} \exp(-5400/T)$ in Part 5b [27], Eq 41. Rate constants are tied to the activity and diffusivity models they were fitted with, so this one stays with the Khan set and must not be mixed with the Plaza or Dugas sets.

Variables. k_2, second-order rate constant, m^3 kmol^{-1} s^{-1}.

Assumptions. Stays with the Khan set. Fit runs to loading about 0.50.

Source. [**P**] Khan, Eq 2, PDF p. 3.

Implementation. `units::absorber::k2`, which calls `mea::kinetics::k2_hikita`

Validation. PUB, test `eq_h_05`. Versteeg and van Swaaij [32], Table 10, PDF p. 12: Hikita rows 4.42, 5.87 and 7.72 m^3 mol^{-1} s^{-1} at 293, 298 and 303 K, the other laboratories of the same table, and the Brønsted relation. Validation results, 12 rows (VER-1-293K to VER-3-303K): 11 pass, 1 fail. Failed: VER-2-298K-PennyRitter1983 gives 5.868 against 4.89 m3/mol/s, limit 0.978 m3/mol/s. The results

mark 3 of these rows as using the same data as the model, and 9 as independent. Part 5b [27], Table 1, is a further source with no results row.

Equation H.6: Hatta group and first-order enhancement

Tier O

$$\mathcal{M} = \frac{D_l k_2 C_B}{\left(k_L^\circ\right)^2}, \qquad \mathcal{E}_1 = \frac{\sqrt{\mathcal{M}}}{\tanh\sqrt{\mathcal{M}}} \tag{H.6}$$

Interpretation. $\mathcal{M}$ compares reaction speed with diffusion through the film. When it is large, $\mathcal{E}_1$ tends to $\sqrt{\mathcal{M}}$ and absorption depends on reaction with no need for bulk mixing.

Derivation. Film theory with a pseudo-first-order reaction gives this result. Khan Eq 4 prints $\mathcal{E}_1$ in this form beside M^2 on the left of the group, and calls M the Hatta modulus. Those two lines conflict. The form here follows film theory and agrees with $E = \sqrt{M}$ in Sonderby Table 1. C_B is free MEA from Eq. G.29, in kmol m^{-3} with k_2 of Eq. H.5.

Variables. $\mathcal{M}$, Hatta group, the Hatta number squared, 1; $\mathcal{E}_1$, pseudo-first-order enhancement, 1; C_B, free MEA in the bulk liquid, kmol m^{-3}.

Assumptions. C_B is free MEA. Decision S5: the code reads Khan Eq 4 as $\mathcal{E}_1 = \mathrm{Ha}/\tanh \mathrm{Ha}$ with $\mathrm{Ha} = \sqrt{\mathcal{M}}$, which corrects the M^2 form Khan prints.

Source. [**P**] Khan, Eq 4, PDF p. 3, with the correction stated above. Sonderby et al. [29], Table 1, PDF p. 4.

Implementation. `units::absorber::enhancement`, which calls `mea::kinetics::hatta` and `mea::kinetics::e1_of_hatta`

Validation. ID, test `eq_h_06`. Limits at small and large $\mathcal{M}$.

Equation H.7: Instantaneous enhancement

Tier O

$$\mathcal{E}_\infty = 1 + \frac{C_B D_B}{b\, D_l C_{int}} \tag{H.7}$$

Interpretation. The ceiling on enhancement. Reaction can go no faster than MEA reaches the interface. At high CO_2 pressure the interface concentration is large, so this ceiling comes down.

Derivation. Khan Eq 4. With $b = 2$ from the carbamate reaction $CO_2 + 2RNH_2$, two MEA arrive for each CO_2. Khan gives no correlation for D_B, so the caller supplies it.

Variables. $\mathcal{E}_\infty$, enhancement for an instantaneous reaction, 1; D_B, MEA diffusivity in solution, m^2 s^{-1}; b, MEA consumed per CO_2, 2, 1.

Assumptions. $b = 2$.

Source. [**P**] Khan, Eq 4 and reaction R1, PDF p. 2 and p. 3.

Implementation. `units::absorber::enhancement`, which calls `mea::kinetics::e_infinity`

Validation. PUB, test `eq_h_07`. Reference values from a project calculation, with no results row: about 11 at 4.74 bar and about 355 at 0.13 bar, with the Hatta number near 100.

Equation H.8: Enhancement by Wellek

Tier O

$$\mathcal{E} = 1 + \frac{1}{\left[\left(\dfrac{1}{\mathcal{E}_\infty - 1}\right)^{1.35} + \left(\dfrac{1}{\mathcal{E}_1 - 1}\right)^{1.35}\right]^{1/1.35}} \tag{H.8}$$

Interpretation. A smooth blend of the two limits. The smaller of $\mathcal{E}_1$ and $\mathcal{E}_\infty$ controls, so at the column bottom the instantaneous ceiling is expected to set the rate.

Derivation. Khan Eq 4, an explicit fit to the second-order problem. Khan states it deviates by less than 3% from van Krevelen and Hoftijzer. Sonderby uses the implicit form instead and finds all forms close at flue gas conditions. Pseudo-first-order forms from flue gas work will overpredict rate near the instantaneous limit, which is why this form is used.

Variables. $\mathcal{E}$, enhancement factor, 1.

Assumptions. Within 3% of the van Krevelen fit.

Source. [**P**] Khan, Eq 4, PDF p. 3.

Implementation. `units::absorber::enhancement`, which calls `mea::kinetics::enhancement_wellek`

Validation. EXP, test `eq_h_08`. Khan [15], Table 1 runs. Validation results, 5 rows (KH-1-T13 to KH-2-I06): 5 not run. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence. The runs cannot be set up, because Khan prints no specific area or critical surface tension for these random packings and no local source holds them, so Onda's area correlation cannot run.

Equation H.9: Henry constant in the Khan set

Tier E

$$H = 10^{\,5.3 - 0.035\,C_{MEA} - 1140/T} \tag{H.9}$$

Interpretation. Rate-side solubility of CO_2 in the solvent. It belongs to the Khan set and stays with it, while equilibrium uses Eq. G.19.

Fit range and origin. Khan Eq 8, after Danckwerts. Khan prints no unit, so atm L mol^{-1} is assumed. At zero MEA and 298.15 K that reading gives 29.95, against 29.32 from Aronu Eq 17 in the same unit. Multiply by 101.325 for Pa m^3 mol^{-1}. The form has no loading term, while the measured N_2O Henry constant rises by about 70% from loading 0 to 0.5 at 25 C, so a misfit at high loading is expected. Hartono's own correlation cannot replace it, because its unloaded base is not reconstructible.

Variables. C_{MEA}, total MEA concentration, kmol m^{-3}.

Assumptions. Unit not stated, assumed atm L/mol.

Source. [**P**] Khan, Eq 8, PDF p. 4.

Implementation. `units::absorber::henry_khan`, which calls `mea::kinetics::henry_khan_pa_m3_mol`

Validation. EXP, test `eq_h_09`. Hartono et al. [8], Table 11, through the N_2O analogy of Eq. G.23. Validation results, 1 row (HA-3): 1 fail. Failed: HA-3 gives 40.41 % AARD, limit 30 % AARD.

Equation H.10: Liquid film coefficient

Tier E

$$k_L^{\circ} = 2\left(\frac{D_l C_E U_L}{\pi S_{ch}}\right)^{1/2}, \qquad C_E = 0.9 \tag{H.10}$$

Interpretation. Penetration theory: liquid renews itself at each corrugation, and shorter contact gives faster transfer.

Fit range and origin. Rocha form as printed in Isa Table 4, Eq 10, with $C_E = 0.9$ from the nomenclature. S_{ch} for the assumed Mellapak 250Y is not local **[M17]**, and geometry may not be taken from memory. Until the vendor sheet or Rocha is in hand, the working case runs Onda on a random packing, Khan Eq 5b: $k_L^{\circ}(\rho_L/\mu_L g)^{1/3} = 0.0051(L_m/a_w\mu_L)^{2/3}(\mu_L/\rho_L D_l)^{-0.5}(a_p d_p)^{0.4}$, with L_m a mass velocity. Isa Eq 16 prints this form differently, and Khan's form is used.

Variables. C_E, surface renewal correction, 0.9, 1; U_L, superficial liquid velocity, m s^{-1}; S_{ch}, corrugation channel side, m; L_m, liquid mass velocity, kg m^{-2} s^{-1}; a_w, wetted area per unit volume, Onda, m^2 m^{-3}; a_p, packing area per unit volume, m^2 m^{-3}; d_p, nominal packing size, m; ρ_L, liquid density, kg m^{-3}; μ_L, liquid viscosity, Pa s.

Assumptions. S_{ch} for the packing is missing.

Source. [**P**] Isa, Table 4, Eq 10, PDF p. 6, with C_E from PDF p. 2. Khan, Eq 5b, PDF p. 3. [**M17**]: Rocha and packing data not local.

Implementation. `units::absorber::kl_rocha`

Validation. EXP, test `eq_h_10`. Only through Eq. H.4, on the Sonderby column runs. Validation results, 5 rows (SO-R1-CAP to SO-R5-CAP): 5 reported without a limit. Simulated less published: −13.89 to −0.967 % over 5 rows.

Equation H.11: Gas film coefficient

Tier E

$$k_{G,v} = 0.054\,\frac{D_G}{S_{ch}}\left[\frac{(U_L + U_G)\,\rho_G S_{ch}}{\mu_G}\right]^{0.8}\left(\frac{\mu_G}{D_G \rho_G}\right)^{0.33}, \qquad k_G = \frac{k_{G,v}}{RT} \tag{H.11}$$

Interpretation. Turbulent gas flow through corrugated channels. Hydrogen makes this gas light and very diffusive, so the gas film should offer little resistance.

Fit range and origin. Rocha form as printed in Isa Table 4, Eq 11, with velocities taken relative to the falling liquid. The result is in m s^{-1}, and dividing by RT gives the unit of Eq. H.1. The working Onda case is Khan Eq 5a: $k_G RT/(a_p D_G) = C_1 (G_m/a_p\mu_G)^{0.7}(\mu_G/\rho_G D_G)^{1/3}(a_p d_p)^{-2.0}$. Khan's C_1 depends on packing size and has no printed value, while Isa uses 5.23.

Variables. $k_{G,v}$, gas film coefficient as a velocity, m s^{-1}; D_G, gas diffusivity of CO_2, m^2 s^{-1}; U_G, superficial gas velocity, m s^{-1}; ρ_G, gas density, kg m^{-3}; μ_G, gas viscosity, Pa s; G_m, gas mass velocity, kg m^{-2} s^{-1}; C_1, Onda packing constant, 1.

Assumptions. D_G by the Fuller method [**M20**].

Source. [**P**] Isa, Table 4, Eq 11, PDF p. 6. Khan, Eq 5a, PDF p. 3.

Implementation. `units::absorber::kg_rocha`

Validation. EXP, test `eq_h_11`. Only through Eq. H.4, on the Sonderby column runs. Validation results, 5 rows (SO-R1-CAP to SO-R5-CAP): 5 reported without a limit. Simulated less published: −13.89 to −0.967 % over 5 rows.

Equation H.12: Effective area, central

Tier E

$$\frac{a_e}{a_p} = \begin{cases} C_{h,BS}\left(\dfrac{U_L\rho_L}{a_p\mu_L}\right)^{0.15}\left(\dfrac{U_L^2 a_p}{g}\right)^{0.1}, & Re_L < 5 \\ 0.85\,C_{h,BS}\left(\dfrac{U_L\rho_L}{a_p\mu_L}\right)^{0.25}\left(\dfrac{U_L^2 a_p}{g}\right)^{0.1}, & Re_L \geq 5 \end{cases} \qquad C_{h,BS} = 0.554 \tag{H.12}$$

Interpretation. Only part of the packing surface takes part in mass transfer. This correlation estimates that share from the liquid flow alone.

Fit range and origin. Billet and Schultes in Isa Table 5, Eqs 17 and 18, the only local form that is stated to 30 bar and needs no surface tension. The two branches meet at $Re_L = 5$ to within 0.2%, and both are printed as the source gives them. The plant sits on the second branch. With a liquid load of 40 to 100 m^3 m^{-2} h^{-1}, the solvent of the Section G example at 40 C, water viscosity of 0.653 mPa s from the IAPWS release and the 420 m^2 m^{-3} of Isa's Flexipac 1Y, Re_L runs from about 14 to 34 on lean solvent and from 12 to 29 on rich solvent [**C**]. $C_{h,BS}$ is the packing constant in Isa's nomenclature, for a column that holds Flexipac 1Y, so carrying it to another packing is an assumption. The engine plan names the Wang form as its working area, so the choice is confirmed when the column first runs.

Variables. a_e, effective area per unit volume, m^2 m^{-3}; $C_{h,BS}$, Billet and Schultes packing constant, 1; Re_L, liquid Reynolds number, $U_L\rho_L/a_p\mu_L$, 1.

Assumptions. Stated to 30 bar. Needs no surface tension.

Source. [**P**] Isa, Table 5, Eqs 17 and 18, PDF p. 6, with $C_{h,BS}$ from PDF p. 2 and the packing area from PDF p. 9.

Implementation. `units::absorber::area_bs`

Validation. EXP, test `eq_h_12`. Isa [14], Table 8, PDF p. 9. Validation results, 1 row (I-2): 1 not run. Fig 10 is a raster image and Table 8 prints no operating condition for its single area value.

Equation H.13: Effective area, alternatives

Tier E

$$\left.\frac{a_e}{a_p}\right|_R = F_{SE}\frac{29.12\,(We_L Fr_L)^{0.15}\,S_{ch}^{0.359}}{Re_L^{0.2}\varepsilon_p^{0.6}\,(1-0.93\cos\gamma_c)\,(\sin\theta)^{0.3}}$$
$$\left.\frac{a_e}{a_p}\right|_W = \left.\frac{a_e}{a_p}\right|_R 0.17\left(\frac{\rho_L}{\rho_G}\right)^{0.16} e^{-0.15/U_G}, \qquad \left.\frac{a_e}{a_p}\right|_{Gu} = \left.\frac{a_e}{a_p}\right|_R \frac{1.2}{1+0.2\,e^{15U_L/U_G}} \tag{H.13}$$

Interpretation. Other published forms, kept to show how far the answer moves with the choice of correlation. In Isa's high-pressure column it moved a great deal.

Fit range and origin. Rocha Eq 19, Wang Eq 20 and Gualito Eq 23 of Isa Table 5, with stated ranges of 0.33 to 20.4, 6.9 to 27.6 and 0.05 to 27.2 bar. Sonderby sets $F_{SE} = 0.25$ for a small column, against 0.35 in the original. Rocha and Onda need the surface tension of loaded solvent [**M19**], and Onda also appears as Khan Eq 5c. Isa Table 7 shows predicted heights from 0.00 m to 5.2 million m across correlation pairs for a 1.505 m column.

Variables. F_{SE}, surface enhancement factor, 1; We_L, liquid Weber number, 1; Fr_L, liquid Froude number, 1; ε_p, packing void fraction, 1; γ_c, liquid contact angle, rad.

Assumptions. Surface tension is missing.

Source. [**P**] Isa, Table 5, Eqs 19, 20 and 23, PDF p. 6, and Table 7, PDF p. 9. Sonderby et al. [29], PDF p. 4. [**M19**]: surface tension.

Implementation. Not implemented in the engine at this build. The column runs Eq. H.12 only, and this form stays a listed alternative. Planned name `units::absorber::area_alt`.

Validation. NONE, test `eq_h_13`. Sensitivity only, with no benchmark of its own.

Equation H.14: Gas flow along the column

Tier O

$$\frac{dG_n}{dz} = -\left(N_{CO_2} + N_{H_2O}\right) a_e A_c \tag{H.14}$$

Interpretation. The gas loses moles as it rises, because CO_2 and water leave it. With 18 mol% CO_2 that loss is large enough to matter.

Derivation. Sonderby Eq 1, a mole balance on a slice of height dz with z measured up from the gas inlet. Khan holds gas flow constant, which suits flue gas and fails here, so the variable form is used.

Variables. G_n, total gas molar flow, mol s^{-1}; N_{H_2O}, water flux from gas to liquid, mol m^{-2} s^{-1}.

Assumptions. IN for H_2, CH_4 and CO. Variable gas flow.

Source. [**P**] Sonderby et al. [29], Eq 1, PDF p. 3.

Implementation. `units::absorber::rhs`

Validation. EXP, test `eq_h_14`. Sonderby Tables 3 to 6 through the full column. Validation results, 27 rows (SO-R1-CAP to SO-R5-V9): 27 reported without a limit. Simulated less published: −13.89 to −0.967 % over 5 rows; −0.07697 to 0.011 mol/mol over 22 rows.

Equation H.15: Gas CO_2 fraction

Tier O

$$\frac{dy_{CO_2}}{dz} = \frac{N_{CO_2} a_e A_c \left(y_{CO_2} - 1\right) + N_{H_2O} y_{CO_2} a_e A_c}{G_n} \tag{H.15}$$

Interpretation. CO_2 fraction falls as CO_2 leaves. The second term is a small correction, because losing water concentrates what remains.

Derivation. A balance on CO_2 gives $d(G_n y_{CO_2})/dz = -N_{CO_2} a_e A_c$. Expand the product and substitute Eq. H.14.

Variables. y_{CO_2}, CO_2 mole fraction in the gas, 1.

Assumptions. As Eq. H.14.

Source. [**P**] Sonderby et al. [29], Eq 2, PDF p. 3.

Implementation. `units::absorber::rhs`

Validation. EXP, test `eq_h_15`. Sonderby Tables 3 to 6, through the full column of Eq. H.14.

Equation H.16: Gas water fraction

Tier O

$$\frac{dy_{H_2O}}{dz} = \frac{N_{H_2O} a_e A_c \left(y_{H_2O} - 1\right) + N_{CO_2} y_{H_2O} a_e A_c}{G_n} \tag{H.16}$$

Interpretation. The same balance for water. Water can move either way, depending on the local temperature of the two phases.

Derivation. As Eq. H.15, starting from $d(G_n y_{H_2O})/dz = -N_{H_2O} a_e A_c$.

Variables. y_{H_2O}, water mole fraction in the gas, 1.

Assumptions. As Eq. H.14.

Source. [**P**] Sonderby et al. [29], Eq 3, PDF p. 3.

Implementation. `units::absorber::rhs`

Validation. EXP, test `eq_h_16`. Sonderby Tables 3 to 6, through the full column of Eq. H.14.

Equation H.17: Liquid flow along the column

Tier O

$$\frac{dL_n}{dz} = -N_{H_2O} a_e A_c \tag{H.17}$$

Interpretation. The liquid is counted on a CO_2-free basis, so only water changes its flow.

Derivation. Liquid falls toward smaller z, so a balance across dz gives $L_n(z) = L_n(z+dz) + N_{H_2O} a_e A_c \, dz$. Sonderby Eq 4 follows.

Variables. L_n, liquid molar flow, CO_2-free basis, mol s^{-1}.

Assumptions. CO_2-free basis.

Source. [**P**] Sonderby et al. [29], Eq 4, PDF p. 3.

Implementation. `units::absorber::rhs`

Validation. EXP, test `eq_h_17`. Sonderby Tables 3 to 6, through the full column of Eq. H.14.

Equation H.18: Liquid composition

Tier O

$$\frac{dX_{CO_2}}{dz} = \frac{\left(N_{H_2O} X_{CO_2} - N_{CO_2}\right) a_e A_c}{L_n}, \qquad \frac{dX_{H_2O}}{dz} = \frac{N_{H_2O} \left(X_{H_2O} - 1\right) a_e A_c}{L_n} \tag{H.18}$$

Interpretation. Moving down the column, the liquid gains CO_2, so X_{CO_2} falls as z rises. Loading follows directly from X_{CO_2}.

Derivation. Balances on CO_2 and water in the liquid give $d(L_n X_i)/dz = -N_i a_e A_c$. Expand and use Eq. H.17. Sonderby Eq 6 prints its left side as X_{H_2O}/dz, with the leading d missing, and the form here is the balance it means.

Variables. X_{CO_2}, CO_2 in the liquid per mole of CO_2-free liquid, 1; X_{H_2O}, water fraction of the CO_2-free liquid, 1.

Assumptions. CO_2-free basis.

Source. [**P**] Sonderby et al. [29], Eqs 5 and 6, PDF p. 3.

Implementation. `units::absorber::rhs`

Validation. EXP, test `eq_h_18`. Sonderby Tables 3 to 6, through the full column of Eq. H.14.

Equation H.19: Gas temperature

Tier O

$$\frac{dT_G}{dz} = -\frac{q_{GL} a_e A_c}{G_n c_{p,G}} \tag{H.19}$$

Interpretation. The rising gas takes up heat from warmer liquid or gives it to cooler liquid. The sign depends on where in the column it sits.

Derivation. Sonderby Eq 7, an energy balance on the gas with heat crossing the interface. The sensible heat carried by the transferred species is charged to the liquid in Eq. H.20.

Variables. T_G, gas temperature, K; q_{GL}, gas to liquid heat flux, W m^{-2}; $c_{p,G}$, gas molar heat capacity, J mol^{-1} K^{-1}.

Assumptions. Adiabatic column, assumption 5.

Source. [**P**] Sonderby et al. [29], Eq 7, PDF p. 3, read as an image.

Implementation. `units::absorber::rhs`

Validation. EXP, test `eq_h_19`. Sonderby temperature profiles. Validation results, 55 rows (SO-R1-T-RMS to SO-R5-TI9): 55 reported without a limit. Simulated less published: 3.598 to 35.13 K over 5 rows; −24.83 to 42.58 C over 50 rows.

Equation H.20: Liquid temperature

Tier O

$$\frac{dT_L}{dz} = \frac{\left(N_{CO_2} c_{p,CO_2} + N_{H_2O} c_{p,H_2O}\right) a_e A_c \left(T_L - T_G\right)}{L_n c_{p,L}} - \frac{q_{GL} a_e A_c}{L_n c_{p,L}} + \frac{\left(N_{CO_2} \Delta h_{CO_2} + N_{H_2O} \Delta h_{H_2O}\right) a_e A_c}{L_n c_{p,L}} \tag{H.20}$$

Interpretation. Absorption releases heat into the liquid, and this drives the temperature bulge seen in absorbers. The first term brings transferred gas to liquid temperature, and the second is exchange with the gas.

Derivation. Liquid falls toward smaller z, so the balance across dz gives $L_n c_{p,L}[T_L(z) - T_L(z+dz)]$ on one side and the three heat inputs on the other. Signs follow the page image of Sonderby Eq 8. The middle term is negative because heat from a hotter gas warms the falling liquid. The last term carries enthalpy changes that are negative for exothermic absorption and condensation, so it also warms the falling liquid, and $\Delta h_{CO_2} = -\Delta h_{abs}$ from Eq. G.24. Δh_{H_2O} is the condensation enthalpy of water.

Variables. T_L, liquid temperature, K; $c_{p,L}$, liquid molar heat capacity, J mol^{-1} K^{-1}; Δh_{CO_2}, enthalpy change of CO_2 absorption, J mol^{-1}; Δh_{H_2O}, enthalpy change of water condensation, J mol^{-1}.

Assumptions. Signs from the image. Interface at the liquid temperature, assumption 2.

Source. [**P**] Sonderby et al. [29], Eq 8 and assumption 2, PDF p. 3, read as an image.

Implementation. `units::absorber::rhs`

Validation. EXP, test `eq_h_20`. The temperature bulge in the Sonderby profiles, through the rows of Eq. H.19.

Equation H.21: Heat flux

Tier O

$$q_{GL} = h_{GL}\left(T_G - T_L\right), \qquad h_{GL} = k_{G,v}\, c_G\, c_{p,G} \left(\frac{Sc}{Pr}\right)^{2/3} \tag{H.21}$$

Interpretation. Heat and mass cross the gas film by the same mechanism, so the mass transfer coefficient also gives the heat transfer coefficient.

Derivation. Sonderby Eq 11 gives the flux. Chilton and Colburn set the heat and mass transfer j factors equal, $h_{GL} Pr^{2/3}/(c_G c_{p,G} u) = k_{G,v} Sc^{2/3}/u$, which gives h_{GL}. The same area serves heat and mass, as assumption 4 states.

Variables. h_{GL}, gas side heat transfer coefficient, W m^{-2} K^{-1}; c_G, gas molar density, mol m^{-3}; Sc, gas Schmidt number, 1.

Assumptions. Chilton-Colburn analogy. Same area for heat and mass.

Source. [**P**] Sonderby et al. [29], Eq 11 and assumption 4, PDF p. 3. [**T**] Analogy.

Implementation. `units::absorber::heat_flux`

Validation. ID, test `eq_h_21`. Direct identity at equal Schmidt and Prandtl numbers.

Equation H.22: Water flux

Tier O

$$N_{H_2O} = k_{G,H_2O}\left(p_{H_2O} - p^*_{H_2O}\right) \tag{H.22}$$

Interpretation. Water moves between the phases with almost no liquid side resistance, so the gas film alone sets its rate.

Derivation. Sonderby's third assumption neglects liquid side resistance for the volatile solvent, so the interface sits at $p^*_{H_2O}$ from Eq. G.20. The coefficient follows from Eq. H.11 with the diffusivity of water.

Variables. k_{G,H_2O}, gas film coefficient of water, mol m^{-2} s^{-1} Pa^{-1}; p_{H_2O}, water partial pressure in the bulk gas, Pa; $p^*_{H_2O}$, water pressure at equilibrium with the solvent, Pa.

Assumptions. Liquid side resistance small.

Source. [**P**] Sonderby et al. [29], assumption 3, PDF p. 3.

Implementation. `units::absorber::water_flux`

Validation. ID, test `eq_h_22`. Sign and zero flux at equilibrium.

Equation H.23: Column pressure drop

Tier F

$$\frac{\Delta p}{Z_c} = \left(\frac{\Delta p}{Z_c}\right)_{ref} \left(\frac{F_s}{F_{s,ref}}\right)^2, \qquad F_s = U_G\sqrt{\rho_G} \tag{H.23}$$

Interpretation. Pressure loss grows with the kinetic energy of the gas. Without a local packing correlation, this simple scaling stands in.

Justification. Friction in the turbulent range scales with $\rho_G U_G^2$. The reference point comes from Sonderby Table 5, which prints small differences across the column, 5.8 mbar in run R1, with some runs changing sign, so it is weak even at 1 bar. INL prints no pressure loss across its Selexol block, 389 psi in and out.

Variables. Z_c, packed height, m; F_s, gas load factor, $\mathrm{Pa}^{1/2}$.

Assumptions. Scaling from Sonderby Table 5.

Source. [**M18**]: no packed-column correlation in the library. [**A**] Assumed. INL from TEV-961 [36], A-56, PDF p. 80.

Implementation. Not implemented in the engine at this build: the absorber solves at the one pressure of its specification and carries no pressure drop. Planned name `units::absorber::dp`.

Validation. NONE, test `eq_h_23`. No benchmark exists at pressure.

Equation H.24: Grid

Tier O

$$\left.\frac{dG_n}{dz}\right|_k \approx \frac{G_{n,k} - G_{n,k-1}}{\Delta z}, \qquad \left.\frac{dL_n}{dz}\right|_k \approx \frac{L_{n,k+1} - L_{n,k}}{\Delta z}, \qquad \Delta z = \frac{Z_c}{N_z} \tag{H.24}$$

Interpretation. The column is cut into slices. Each phase takes its value from upstream, which means from below for gas and from above for liquid.

Derivation. Finite volume form of Eq. H.14 to Eq. H.20, written for one variable of each phase. Upwinding keeps the scheme stable at this order. Khan used forward differences in a spreadsheet, solving each phase in turn. Here all slices are solved together, because the temperature bulge couples both directions.

Variables. N_z, number of slices, 1; Δz, slice height, m.

Assumptions. $N_z = 50$ and 100 compared.

Source. [**T**] Textbook result. Khan, section 2.5, PDF p. 4, for the forward difference scheme.

Implementation. `units::absorber::grid`

Validation. ID, test `eq_h_24`. Engine test: capture changes by less than 0.002, as a fraction, when the grid doubles.

Equation H.25: Boundary conditions

Tier O

$$z = 0: \; G_n, \; y_{CO_2}, \; y_{H_2O}, \; T_G \text{ given}; \qquad z = Z_c: \; L_n, \; X_{CO_2}, \; X_{H_2O}, \; T_L \text{ given} \tag{H.25}$$

Interpretation. Gas enters at the bottom and liquid at the top, so half the conditions sit at each end. This makes the column a two-point problem.

Derivation. Sonderby states the problem as a boundary value problem on Eqs 1 to 8, with inlet flows, composition, pressure and temperature defined. The eight unknowns of Eq. H.14 to Eq. H.20 match the eight conditions here.

Variables. Z_c, packed height, m.

Assumptions. Two-point problem.

Source. [**P**] Sonderby et al. [29], PDF p. 4.

Implementation. `units::absorber::rhs`, where the end values enter the residual

Validation. ID, test `eq_h_25`. End values equal the inputs.

Equation H.26: Solution method

Tier O

$$\mathbf{J}\left(\mathbf{w}^{(m)}\right)\Delta\mathbf{w} = -\mathbf{f}\left(\mathbf{w}^{(m)}\right), \qquad \mathbf{w}^{(m+1)} = \mathbf{w}^{(m)} + \lambda\Delta\mathbf{w}, \qquad \|\mathbf{f}\|_\infty < 10^{-8} \tag{H.26}$$

Interpretation. All slices are solved at once by repeated linearisation. The matrix is banded, so each step stays fast.

Derivation. Newton's method on the system from Eq. H.24 and Eq. H.25. The damping factor λ reduces a step that would push a composition negative. The start is an isothermal guess.

Variables. **w**, all unknowns on the grid, mixed; **f**, residual vector, mixed; **J**, Jacobian of the residual, mixed; λ, damping factor, 1.

Assumptions. Start from an isothermal guess.

Source. [**T**] Textbook result.

Implementation. `units::absorber::solve`, with the damped Newton of `units::newton::solve`

Validation. ID, test `eq_h_26`. A repeat run gives identical output.

Equation H.27: Per-pass capture

Tier R

$$\eta_{pp} = \frac{y_{in} n_{in} - y_{out} n_{out}}{y_{in} n_{in}} \tag{H.27}$$

Interpretation. The share of CO_2 removed in one pass through the absorber. Plant capture is a different number, and Section J relates the two.

Derivation. Definition, on the gas side and with total flows in and out, as both Vinjarapu and Kwak write it.

Variables. η_{pp}, per-pass capture, 1; y_{in}, CO_2 fraction in the feed gas, 1; y_{out}, CO_2 fraction in the treated gas, 1; n_{in}, gas flow in, mol s^{-1}; n_{out}, gas flow out, mol s^{-1}.

Assumptions. None.

Source. [**P**] Vinjarapu et al. [33], Appl. Energy 355 (2024) 122193, Eq 20, PDF p. 11. Kwak et al. [17], Eq 2, PDF p. 3.

Implementation. `units::absorber::capture`

Validation. EXP, test `eq_h_27`. Salvinder et al. [25], Table 2, PDF p. 5, 85.0% at 10.5 bar, and Tan et al. [30], Section 3.2, PDF p. 4, 76, 95 and 100% at 0.1, 1 and 5 MPa with 50% CO_2 in the feed. Both use 20 wt% MEA on natural gas. Tan defines removal on inlet and outlet concentrations alone. With half the feed absorbed that basis reads low: 95% on concentrations is 97.4% by this equation if only CO_2 transfers [**C**]. Validation results, 5 rows (SA-1 to TAN-TREND): 1 pass, 1 fail, 3 reported without a limit. Failed: SA-1 gives 72.58 against 85 %, limit 5 %. Reported: TAN-P0.1 gives 19.82 against 76 %; TAN-P1 gives 25.96 against 95 %; TAN-P5 gives 47.93 against 100 %. The trend row judges the sign of the pressure trend only, so it is not counted as evidence on magnitude.

Equation H.28: Solvent flow

Tier R

$$\dot{m}_{solv} = \frac{\dot{n}_{CO_2} M_{MEA}}{w_{MEA} \left(\alpha_{rich} - \alpha_{lean}\right)} \tag{H.28}$$

Interpretation. Each kilogram of solvent carries a fixed swing of CO_2, so the solvent flow follows from the swing and the mass fraction alone.

Derivation. A CO_2 balance over the column: $\dot{n}_{CO_2} = \dot{n}_{MEA}(\alpha_{rich} - \alpha_{lean})$, with $\dot{n}_{MEA} = w_{MEA}\dot{m}_{solv}/M_{MEA}$. The result is on a CO_2-free basis, so the lean stream itself is heavier by the CO_2 it carries.

Variables. $\dot{m}_{solv}$, lean solvent, CO_2-free basis, kg s^{-1}; $\dot{n}_{CO_2}$, CO_2 absorbed, mol s^{-1}; M_{MEA}, molar mass of MEA, kg mol^{-1}; w_{MEA}, MEA mass fraction, CO_2-free basis, 1.

Assumptions. $\alpha_{rich} \leq 0.50$.

Source. [**C**] Calculated here.

Implementation. `units::absorber::solvent_rate`

Validation. GOLD, test `eq_h_28`. 894 t h^{-1} at Case B and 1,309 t h^{-1} at Case A on the anchored port, the screening values, which is a port check. Hand comparison with DICE [22], Fig 5-2, PDF p. 192: 3,517 lbmol h^{-1} of CO_2, or 70.2 t h^{-1}, with loadings 0.16 and 0.40 at 30 wt% gives 1,354 t h^{-1} on this basis. The lean stream prints 1,400 t h^{-1}, and its 47 t h^{-1} of CO_2 closes the difference **[C]**.

Equation H.29: Column diameter

Tier O

$$D_c = \sqrt{\frac{4\dot{V}_L}{\pi\, \ell_L}}, \qquad \ell_L = 40 \text{ to } 100 \text{ m}^3 \text{ m}^{-2} \text{ h}^{-1} \tag{H.29}$$

Interpretation. Liquid sets this diameter. Syngas is so light that the liquid to gas ratio is about five times that of a flue gas absorber, so gas capacity is the easier limit.

Derivation. Divide the liquid flow by an allowed load. L/G runs 12 to 15 kg kg^{-1} here against 2.4 to 3.0 in flue gas pilots **[C]**. The load range is a project judgement, with no vendor rating behind it.

Variables. D_c, column diameter, m; $\dot{V}_L$, liquid volume flow, m^3 s^{-1}; ℓ_L, liquid load, m s^{-1}.

Assumptions. L/G 12 to 15 kg/kg.

Source. **[C]** Calculated here.

Implementation. `units::absorber::diameter`

Validation. PUB, test `eq_h_29`. A project calculation gives 4 to 6.5 m at about 1,400 t h^{-1} of solvent. Validation results, 5 rows (ABS-CASEB-DIAMETER to ABS-CASEB-SPEC): 1 pass, 4 reported without a limit. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation H.30: Co-absorption

Tier F

$$\dot{n}_{H_2,abs} = \dot{n}_{CH_4,abs} = \dot{n}_{CO,abs} = 0 \tag{H.30}$$

Interpretation. Hydrogen and methane also dissolve a little and leave with the solvent. No data exist for this solvent, so the central case sets them to zero and states the reference value beside it.

Justification. No local source gives H_2, CH_4 or CO solubility in loaded MEA. The nearest reference is activated MDEA in IEAGHG Case 1A. Flash gas stream 16 and CO_2 stream 17 hold 2.80 kmol h^{-1} of H_2 and 0.28 of CH_4, against 4,989 and 201 in feed stream 7, or 0.056% and 0.137% **[C]**. Yield loss is small on that reference. The effect on CO_2 purity is larger, and it is disclosed.

Variables. $\dot{n}_{H_2,abs}$, H_2 absorbed, mol s^{-1}; $\dot{n}_{CH_4,abs}$, CH_4 absorbed, mol s^{-1}; $\dot{n}_{CO,abs}$, CO absorbed, mol s^{-1}.

Assumptions. Zero central. Reference aMDEA, 0.056% H_2 and 0.137% CH_4.

Source. **[C]** Calculated here from IEAGHG 2017-02 [1], streams 7, 16 and 17, PDF p. 75 and p. 76. **[M10]**: no solubility data for MEA.

Implementation. `units::absorber::coabsorb`

Validation. NONE, test `eq_h_30`. Disclosed term, with no benchmark for MEA.

Equation H.31: Pressure index, listed and unused

Tier E

$$f = \frac{L_v C^0_{MEA}}{a_p} \int_{\alpha_1}^{\alpha_2} \frac{d\alpha}{H\sqrt{D_l k_{MEA} C_{MEA}}\,(p - p^*)} \tag{H.31}$$

Interpretation. A fitted correction for high pressure. It is printed for completeness and takes no part in the plant.

Fit range and origin. Isa Eq 42, with $k_{MEA} = 4.61 \times 10^9 \exp(-4412/T)$ from Eq 43. Table 10 gives f from 9.92 to 2.92 for the Tsai area and from 77.47 to 22.82 for the Billet area between 20 and 60 bar. It was fitted to 20 wt% MEA with no regeneration, on CO_2 and CH_4 at 1:1, in one column. Moving it to 30 wt% on hydrogen would have no basis, so the plant leaves it out.

Variables. f, pressure modification index, 1; L_v, liquid flow in Isa's balance, m^3 s^{-1}; C^0_{MEA}, MEA concentration fed, mol m^{-3}; k_{MEA}, reaction rate constant of Isa Eq 43, m^3 mol^{-1} s^{-1}.

Assumptions. Fitted to 20 wt% on a CH_4 carrier.

Source. [**P**] Isa, Eqs 42 and 43, PDF p. 7, and Table 10, PDF p. 10.

Implementation. none. The equation is listed only.

Validation. NONE, test `eq_h_31`. Listed for reference. Table 10 sits in the benchmark fixture and no test uses it.

Worked example: Case B solvent, diameter and driving force

Solvent. Case B captures 48.3 t h^{-1} of CO_2, or 305 mol s^{-1}. With $w_{MEA} = 0.30$ and a swing of 0.25, Eq. H.28 gives 248.4 kg s^{-1}, which is the 894 t h^{-1} of the ledger. The lean stream also carries its own loading of 0.25, equal to the swing, so it adds another 48.3 t h^{-1} and weighs 942 t h^{-1}.

Diameter. At the 1,057 kg m^{-3} of the Section G example that is 892 m^3 h^{-1}. Eq. H.29 then gives 3.4 m at a load of 100 and 5.3 m at a load of 40 m^3 m^{-2} h^{-1}. These sizes are for one column and use no gas side check.

Driving force. The feed holds 17.7 mol% CO_2 at 26.8 bar, so $p_{CO_2} = 4.74$ bar at the bottom. With 4.9 kmol m^{-3} of MEA at 40 C, Eq. H.9 gives 30.8 atm L mol^{-1}, or 3,117 Pa m^3 mol^{-1}, so C_{int} reaches 152 mol m^{-3} if the gas film is taken as negligible. Eq. H.5 gives $k_2 = 13{,}100$ m^3 kmol^{-1} s^{-1} at this temperature. Eq. H.7 then gives $\mathcal{E}_\infty = 1 + 16\, D_B/D_l$ with total MEA as an upper bound on C_B. Khan gives no D_B, and a ratio near 0.6 recovers the target of about 11. Free MEA at the rich end is far smaller than the total, so the real ceiling there is lower still, which is the plain reason the rich loading stays capped at 0.50.

Link to the sources

Sonderby supplies the column structure, tested at 30 wt% and 1 bar, and Khan the kinetic set with a test at up to 7.24 bar. The high-pressure work of Isa and Salvinder uses 20 wt% MEA on natural gas, and its findings warn against trust in any one correlation pair. So the absorber result carries a named gap at every step, and the 90% per-pass capture stays an assumption of the ledger until the column runs and its result is reported, whether it meets that figure or falls short.

11 I. MEA stripper

The stripper takes the loaded solvent back to its lean state and releases CO_2 at low pressure. Heat is the whole cost of this step, so the question here is how much reboiler duty each tonne of CO_2 needs and what grade of heat can supply it. The model is reduced order: at 120 C reactions are fast, so the column is treated as equilibrium limited. The specific reboiler duty (SRD) of 3.8 GJ t^{-1} stays the ledger input. The stripper and regeneration train are in the engine, and gate M6 judged their benchmark rows and failed, so the input stays anchored. Eq. I.13 is an independent bottom-up check, and it never replaces the input without a benchmark on syngas, which does not exist in the library [**M12**]. All worked numbers are at Case B unless the card says otherwise.

Equation I.1: CO_2 balance

Tier R

$$\dot{n}_{CO_2} = \dot{n}_{MEA}\,(\alpha_{rich} - \alpha_{lean}) \tag{I.1}$$

Interpretation. The CO_2 released equals the loading the solvent gives up on one pass. The cyclic capacity $\alpha_{rich} - \alpha_{lean}$ therefore sets how much solvent must circulate.

Derivation. Eq. A.3 for CO_2 on the column at steady state, with MEA conserved: $\dot{n}_{MEA}\alpha_{rich} = \dot{n}_{MEA}\alpha_{lean} + \dot{n}_{CO_2}$.

Variables. $\dot{n}_{CO_2}$, CO_2 released, mol s^{-1}; $\dot{n}_{MEA}$, MEA circulation, mol s^{-1}.

Assumptions. SS. No solvent degradation.

Source. [**T**] Consequence of Eq. A.3.

Implementation. `units::stripper::co2_balance`

Validation. ID, test `eq_i_01`. Unit identity. Computed here: 1097.90 kmol h^{-1} over a capacity of 0.25 needs 4,392 kmol h^{-1} of MEA.

Equation I.2: Desorption driving force

Tier O

$$N_{CO_2} = K_G \left(p^* - p_{CO_2}\right) \tag{I.2}$$

Interpretation. Absorption run in reverse. CO_2 leaves the liquid while the equilibrium pressure over the solvent exceeds the pressure in the gas.

Derivation. Eq. H.4 with the sign reversed, so that a positive flux leaves the liquid. The form is kept only to scope the claim made in Eq. I.3. By Eq. H.5 the second-order rate constant rises by a factor of 25 between 40 and 120 C, which is why this document stops short of integrating a rate model here.

Variables. N_{CO_2}, CO_2 flux, positive out of the liquid, mol m^{-2} s^{-1}; K_G, overall gas-phase coefficient, mol m^{-2} s^{-1} Pa^{-1}; p^*, equilibrium CO_2 pressure, Pa; p_{CO_2}, CO_2 partial pressure in the gas, Pa.

Assumptions. Equilibrium limited. This equation is not integrated.

Source. [**T**] Reverse of Eq. H.4.

Implementation. `units::stripper::desorb`

Validation. Scoped, test `eq_i_02`. No separate benchmark. The stage model in Eq. I.3 carries the test.

Equation I.3: Equilibrium stage

Tier O

$$y_{CO_2,k}\, p = p^*_{CO_2}\left(T_k, \alpha_k\right), \qquad k = 1, \ldots, N \tag{I.3}$$

Interpretation. Each of N stages brings vapour and liquid to equilibrium. This suits a hot, fast system, where the column is limited by how much steam reaches each height and hardly at all by reaction rate.

Derivation. Set the flux of Eq. I.2 to its limit of zero driving force on each stage, then link stages by Eq. I.1 and an enthalpy balance. Equilibrium pressures come from Section G. The kinetic expressions in Plaza are quoted for reference and are not used.

Variables. $y_{CO_2,k}$, CO_2 mole fraction on stage k, 1; $p^*_{CO_2}$, equilibrium CO_2 pressure over the solvent, Pa; T_k, stage temperature, K; α_k, stage loading, mol mol^{-1}; N, number of equilibrium stages, 1.

Assumptions. Fast reaction at 120 C. Stage efficiency of one.

Source. [**P**] Plaza [24], Eq 3.46, 3.47, PDF p. 98, kinetics quoted and unused. Cownden and Lucquiaud [5], PDF p. 11, compare regenerator temperature profiles with pilot data.

Implementation. `units::stripper::stages`

Validation. EXP, test `eq_i_03`. NCCC runs K14, K15 and K16 (Morgan [19], SI Table S4, PDF p. 5, with the rich and pressure inputs in SI Table S2), and the lean loadings of Vinjarapu et al. [33], Table 5, at the printed reboiler duty. Validation results, 9 rows (MO-K14-LEAN to VI-3-SS3): 3 pass, 3 fail, 3 reported without a limit. Failed: MO-K16-TOUT gives 121.3 against 122.9 C, limit 1.4 C; VI-3-SS2 gives 0.1596 against 0.23 mol/mol, limit 0.03 mol/mol; VI-3-SS3 gives 0.1968 against 0.25 mol/mol,

limit 0.03 mol/mol. Reported: MO-K14-LEAN gives 0.253 against 0.224 mol/mol; MO-K15-LEAN gives 0.2566 against 0.224 mol/mol; MO-K16-LEAN gives 0.1392 against 0.124 mol/mol.

Equation I.4: Rich-end pinch

Tier O

$$\frac{\dot{n}_{H_2O}}{\dot{n}_{CO_2}} = \frac{p - p^*_{CO_2}}{p^*_{CO_2}} \tag{I.4}$$

Interpretation. Water leaving the top of the column carries heat that the condenser throws away. The higher the equilibrium pressure over the rich feed, the less water each mole of CO_2 takes with it.

Derivation. The overhead vapour leaves close to equilibrium with the rich feed, so $y_{CO_2} = p^*_{CO_2}/p$. Water makes up the rest, so $y_{H_2O} = 1 - y_{CO_2}$, and the mole ratio is the ratio of the two.

Variables. $\dot{n}_{H_2O}$, water in the overhead, mol s^{-1}; p, column pressure, Pa.

Assumptions. Vapour leaves near equilibrium with the feed. MEA and inert gases neglected.

Source. [**C**] This document.

Implementation. `units::stripper::steam_ratio`

Validation. PUB, test `eq_i_04`. Computed here from DICE [22], Fig 5-2, stream 23, PDF p. 192: 3,183 and 3,519 lbmol h^{-1} give 0.905. At 27.35 psia this implies $p^*_{CO_2} = 0.99$ bar at the top. The stripper model on the DICE rich stream: Validation results, 1 row (DICE-RATIO): 1 reported without a limit. Reported: DICE-RATIO gives 2.505 against 0.9045 mol/mol.

Equation I.5: Lean-end bubble point

Tier R

$$p^*_{CO_2}(T_{reb}, \alpha_{lean}) + p^*_{H_2O}(T_{reb}, \alpha_{lean}) + p^*_{MEA}(T_{reb}, \alpha_{lean}) = p_{strip} \tag{I.5}$$

Interpretation. The reboiler sits at the boiling point of the lean solvent. Choose the pressure and the lean loading, and the temperature follows.

Derivation. At the bottom of the column the liquid boils, so the partial pressures over it sum to the total pressure. All three terms come from Section G.

Variables. T_{reb}, reboiler temperature, K; $p^*_{H_2O}$, equilibrium water pressure over the solvent, Pa; p^*_{MEA}, equilibrium MEA pressure over the solvent, Pa; p_{strip}, stripper pressure, 2.0 bar, Pa.

Assumptions. Liquid at its bubble point. Pressure 2.0 bar.

Source. [**A**] Project solvent specification, 121.9 C at 2.0 bar.

Implementation. `units::stripper::t_reboiler`

Validation. PUB, test `eq_i_05`. DICE [22], Fig 5-2, PDF p. 192, gives lean solvent at 253.8 F, which is 123.2 C, at 28.21 psia, with loading 0.160. Vinjarapu et al. [33], Fig 12, PDF p. 13, runs 117 to 119.8 C near 1.9 bar, read from the figure. Validation results, 3 rows (DICE-TREB to KW-TREB): 3 reported without a limit. Reported: DICE-TREB gives 122.2 against 123.2 C; DICE-TTOP gives 111.7 against 96.56 C; KW-TREB gives 114.5 against 113.8 C.

Equation I.6: Reboiler energy balance

Tier R

$$\dot{Q}_{reb} = \sum_{out} \dot{n}\, h - \sum_{in} \dot{n}\, h \tag{I.6}$$

Interpretation. The reboiler supplies whatever the streams leaving the stripper carry above the streams entering it.

Derivation. Eq. A.8 around the column and reboiler with no shaft work.

Variables. $\dot{n}$, molar flow of a stream, mol s^{-1}; h, molar enthalpy, J mol^{-1}.

Assumptions. SS. Adiabatic column.

Source. [**T**] Consequence of Eq. A.8.

Implementation. `units::stripper::reboiler`

Validation. ID, test `eq_i_06`. Unit identity on the column balance.

Equation I.7: Condenser

Tier R

$$\dot{Q}_{cond} = \dot{Q}_{H_2O} + \dot{Q}_{steam} + \dot{Q}_{CO_2} \tag{I.7}$$

Interpretation. The condenser removes latent and sensible heat from the wet overhead. This duty is the heat lost at the top of the column, and it is the size of the stripping term in Eq. I.13.

Derivation. Enthalpy balance on the overhead between the column top and the separator: steam cooled and condensed, condensate cooled and CO_2 cooled.

Variables. $\dot{Q}_{cond}$, condenser duty, W; $\dot{Q}_{H_2O}$, cooling of condensed water, W; $\dot{Q}_{steam}$, cooling and condensing of steam, W; $\dot{Q}_{CO_2}$, cooling of CO_2, W.

Assumptions. Vent saturated at separator temperature.

Source. [**P**] Vinjarapu et al. [33], Appl. Energy 355 (2024) 122193, Eq 24 to 27, PDF p. 13.

Implementation. `units::stripper::condenser`

Validation. PUB, test `eq_i_07`. 64.4 MMBtu h^{-1} for the DICE stripper condenser [22], Table 5-3, PDF p. 197. Computed here, that is 0.97 GJ t^{-1}. Validation results, 1 row (DICE-QCOND): 1 reported without a limit. Reported: DICE-QCOND gives 52.36 against 18.76 MW.

Equation I.8: Reflux

Tier R

$$\dot{n}_{ref} = \dot{n}_{H_2O} - \dot{n}_{CO_2} \frac{p_{sat}\left(T_{cond}\right)}{p - p_{sat}\left(T_{cond}\right)} \tag{I.8}$$

Interpretation. Almost all the water in the overhead returns to the column. Only the small amount that stays saturated in the CO_2 vent leaves.

Derivation. Water balance on the separator, with the vent saturated: $y_{H_2O} = p_{sat}/p$, so the vent carries $y_{H_2O}/(1 - y_{H_2O})$ moles of water per mole of CO_2.

Variables. $\dot{n}_{ref}$, condensate returned, mol s^{-1}; T_{cond}, separator temperature, K; p_{sat}, water saturation pressure, Pa.

Assumptions. Vent saturated.

Source. [**P**] Saturated vent as in Vinjarapu et al. [33], Appl. Energy 355 (2024) 122193, Eq 23, PDF p. 12.

Implementation. `units::stripper::reflux`

Validation. ID, test `eq_i_08`. Water balance identity.

Equation I.9: Circulation

Tier R

$$\dot{m}_{solv} = \frac{\dot{n}_{CO_2} M_{MEA}}{w\left(\alpha_{rich} - \alpha_{lean}\right)} \tag{I.9}$$

Interpretation. The solvent rate follows directly from the capacity: half the capacity means twice the pumping and twice the sensible heat.

Derivation. Eq. I.1 with $\dot{n}_{MEA} = w\dot{m}_{solv}/M_{MEA}$. The same relation is Eq. H.28.

Variables. M_{MEA}, molar mass of MEA, 61.08, g mol^{-1}; w, MEA mass fraction, 0.30, kg kg^{-1}.

Assumptions. Lean solvent basis.

Source. [**C**] Restated from Eq. H.28.

Implementation. `units::absorber::solvent_rate`

Validation. GOLD, test `eq_i_09`. 894 t h^{-1} at Case B.

Equation I.10: Rich-lean exchanger

Tier R

$$\dot{m}_r c_p \left(T_{r,out} - T_{r,in}\right) = \dot{m}_l c_p \left(T_{l,in} - T_{l,out}\right), \qquad T_{r,out} = T_{l,in} - \Delta T_{hx} \tag{I.10}$$

Interpretation. Hot lean solvent preheats the cold rich solvent. Each kelvin of approach left at the hot end must be paid for in the reboiler, so this exchanger sets the sensible term of Eq. I.13.

Derivation. Eq. A.8 on each side with equal heat capacity. The approach is fixed at 10 K as a design choice.

Variables. $\dot{m}_r$, rich solvent flow, kg s^{-1}; $\dot{m}_l$, lean solvent flow, kg s^{-1}; $T_{r,out}$, rich solvent leaving the exchanger, K; $T_{l,in}$, lean solvent entering the exchanger, K; ΔT_{hx}, hot-end approach, 10 K, K.

Assumptions. CP, with equal heat capacity on both sides. [**M21**]: the solvent value of 3.6 kJ kg^{-1} K^{-1} is a figure axis range.

Source. [**C**] This document.

Implementation. `units::regen::rich_lean`

Validation. PUB, test `eq_i_10`. Computed here from DICE [22]: rich solvent at 138.1 F and 3,242,115 lb h^{-1} (Fig 5-2, PDF p. 192), exchanger duty 274 MMBtu h^{-1} (PDF p. 215) and lean solvent at 253.8 F give an approach of 9.7 K.

Equation I.11: Heating steam

Tier R

$$\dot{m}_{st} = \frac{\dot{Q}_{reb}}{\Delta h_{cond}} \tag{I.11}$$

Interpretation. Steam gives up its latent heat in the reboiler, so the steam flow is the duty over that latent heat.

Derivation. Eq. A.8 on the steam side, with saturated steam in and saturated condensate out. The code takes Δh_{cond} from IAPWS-IF97.

Variables. $\dot{m}_{st}$, heating steam flow, kg s^{-1}; Δh_{cond}, latent heat of the heating steam, J kg^{-1}.

Assumptions. Saturated steam, full condensation.

Source. [**T**] Consequence of Eq. A.8.

Implementation. `units::regen::steam`

Validation. PUB, test `eq_i_11`. Computed here. DICE [22]: 242 MMBtu h^{-1} (PDF p. 215) on 263.3 klb h^{-1} (Table 5-3, PDF p. 197) is 919 Btu lb^{-1}. INL: 121.7 MMBtu h^{-1} on 134,540 lb h^{-1} (TEV-961 [36], A-57, PDF p. 81, read as an image) is 905 Btu lb^{-1}. At the INL value Case B needs 24.2 kg s^{-1}.

Equation I.12: Specific reboiler duty

Tier R

$$SRD = \frac{\dot{Q}_{reb}}{\dot{m}_{CO_2}} \tag{I.12}$$

Interpretation. The headline energy number for a solvent plant: heat per tonne captured.

Derivation. Definition. Vinjarapu writes it as $RD \cdot 3.6/m_{CO_2}$ with duty in kW and CO_2 in kg h^{-1}, which returns GJ t^{-1}.

Variables. SRD, specific reboiler duty, J kg^{-1}.

Assumptions. Reboiler duty basis, heat loss included.

Source. [**P**] Vinjarapu et al. [33], Appl. Energy 355 (2024) 122193, Eq 19, PDF p. 11.

Implementation. `units::regen::srd`

Validation. EXP, test `eq_i_12`. Computed here from Vinjarapu et al. [33], Table D.19, PDF p. 38, as the benchmark fixture holds it: the 16 steady states with capture of 88 to 92% have a median of 3.75 GJ t^{-1}, from 3.46 to 4.82. DICE gives 3.64 on its 70.2 t h^{-1}. Both are flue gas. Validation results, 6 rows (DICE-SRD to VI-2): 2 pass, 2 fail, 2 reported without a limit. Failed: DICE-SRD gives 5.293 against 3.638 GJ/t, limit 0.3638 GJ/t; KW-SRD gives 4.977 against 3.92 GJ/t, limit 0.392 GJ/t. Reported: KW-SRD-90C gives 4.982 against 3.92 GJ/t; KW-SRD-95C gives 4.972 against 3.92 GJ/t. Gate M6 judged these rows with the other column rows and failed, with 6 of 13 judged rows outside their limits (SA-1, DICE-SRD, KW-SRD, MO-K16-TOUT, VI-3-SS2, VI-3-SS3). The ledger therefore keeps 3.8 GJ t^{-1}.

Equation I.13: SRD split

Tier O

$$SRD = q_{des} + q_{sens} + q_{strip}, \qquad q_{des} = \frac{\Delta h_{abs}}{M_{CO_2}} \tag{I.13}$$

Interpretation. The first term breaks the chemical bond and is set by the solvent. The other two, warming the solvent over the last few kelvin and raising steam that the condenser later removes, respond to design choices.

Derivation. Split Eq. I.6 by where the enthalpy goes. The first term is the heat of absorption per unit mass of CO_2. The second is $\dot{m}_r c_p \Delta T_{hx} / \dot{m}_{CO_2}$ from Eq. I.10. The third is $\dot{Q}_{cond} / \dot{m}_{CO_2}$ from Eq. I.7.

Variables. q_{des}, desorption term, J kg^{-1}; q_{sens}, sensible term, J kg^{-1}; q_{strip}, stripping steam term, J kg^{-1}; Δh_{abs}, heat of absorption, J mol^{-1}; M_{CO_2}, molar mass of CO_2, 44.01, g mol^{-1}.

Assumptions. No separate heat loss term.

Source. [**C**] This document, on DICE [22] data.

Implementation. `units::regen::srd_split`

Validation. PUB, test `eq_i_13`. Computed here for DICE: $q_{sens} = 0.73$ and $q_{strip} = 0.97$, so a total of 3.64 leaves $q_{des} = 1.94$ GJ t^{-1}. The desorption term is back-calculated, so this split closes by construction and tests nothing about it. At Case B, $q_{sens} = 0.70$ and with the same two other terms the bottom-up total is 3.61 GJ t^{-1}. It stays a check on flue-gas terms.

Equation I.14: Carried SRD

Tier F

$$SRD = 3.8 \text{ GJ t}^{-1}, \qquad 3.4 \le SRD \le 4.2 \text{ GJ t}^{-1} \tag{I.14}$$

Interpretation. The value the ledger actually uses. It sits above the bottom-up figure and close to the measured pilot median.

Justification. The central value rounds the pilot median of Eq. I.12 up slightly, and allows for heat loss in a plant that has never run on syngas. The range spans good heat recovery at the low end and unsettled flue-gas campaigns at the high end.

Variables. SRD, carried specific reboiler duty, J kg^{-1}.

Assumptions. Flue-gas evidence extrapolated to syngas.

Source. [**A**] Project value. [**M12**]: no SRD on syngas in the library.

Implementation. `units::regen::srd_input`

Validation. NONE, test `eq_i_14`. No benchmark on syngas exists. Flue-gas values are in Eq. I.12.

Equation I.15: Stripping intensity

Tier R

$$I_{str} = \frac{\dot{Q}_{reb}}{\dot{m}_{rich}} \tag{I.15}$$

Interpretation. Heat per kilogram of solvent. Too little leaves the solvent rich. Too much makes excess steam. Pilots show a U-shaped curve of SRD against this number.

Derivation. Definition. Vinjarapu writes $RD \cdot 3600/m_{rich}$ with duty in kW and solvent in kg h^{-1}, which returns kJ kg^{-1}.

Variables. I_{str}, stripping intensity, J kg^{-1}; $\dot{m}_{rich}$, rich solvent flow, kg s^{-1}.

Assumptions. Total rich solvent basis.

Source. [**P**] Vinjarapu et al. [33], Appl. Energy 355 (2024) 122193, Eq 16, 17, PDF p. 11.

Implementation. `units::regen::intensity`

Validation. EXP, test `eq_i_15`. The pilot optimum is 200 to 210 kJ kg^{-1} at a minimum SRD of 3.65 GJ t^{-1} (same paper, Fig 10(a), PDF p. 11). Computed here for Case B: 195 kJ kg^{-1}, just below that band.

Equation I.16: Reboiler area

Tier R

$$\dot{Q}_{reb} = U_r A_r \Delta T_{LMTD} \tag{I.16}$$

Interpretation. Reboiler duty needs a surface and a driving difference. Hotter steam shrinks the reboiler, and it also costs more.

Derivation. Eq. C.7 applied to a kettle reboiler.

Variables. U_r, reboiler overall coefficient, W m^{-2} K^{-1}; A_r, reboiler area, m^2; ΔT_{LMTD}, reboiler log-mean difference, K.

Assumptions. Uniform coefficient.

Source. [**P**] Vinjarapu et al. [33], Appl. Energy 355 (2024) 122193, Eq 22, PDF p. 11.

Implementation. `units::regen::reboiler_area`

Validation. ID, test `eq_i_16`. Unit identity.

Equation I.17: Source temperature

Tier F

$$T_{source} \geq T_{reb} + \Delta T_{min} \tag{I.17}$$

Interpretation. Heat must arrive hotter than the reboiler. This line decides which plant streams count as reboiler heat, and it rules out most low-grade syngas heat.

Justification. Second law, with a minimum approach. DICE heats its kettle with 60 psia saturated steam, which is 145 C by the Antoine fit in the engine inputs. INL distributes LP steam at 307 F and 75 psia, which is 153 C. With a reboiler at 121.9 C those set the working band, and syngas counts only above 130 C.

Variables. T_{source}, heating medium temperature, K; ΔT_{min}, minimum approach, K.

Assumptions. Steam near 145 to 153 C.

Source. [**P**] DICE [22], PDF p. 186. TEV-961 [36], A-57, PDF p. 81, read as an image.

Implementation. `plant::heatrec::check_grade`

Validation. PUB, test `eq_i_17`. Grade rule only. The stack in Eq. J.17 counts syngas above 130 C and nothing colder.

Equation I.18: Interheating, listed and not carried

Tier F

$$SRD_{tot} = SRD_{reb} + q_{IH} \tag{I.18}$$

Interpretation. A stripper interheater moves duty out of the reboiler to a cooler source. The reboiler figure falls and the total heat barely changes.

Justification. Vinjarapu report 2.71 GJ t^{-1} on the reboiler with 28% interheating, against 3.71 in the base case. Adding the interheater back at 53 kJ kg^{-1} on 264 kg h^{-1} of solvent and 13.1 kg h^{-1} of CO_2 gives 1.07, for a total of 3.78 GJ t^{-1}. So 2.71 is a reboiler figure, and it is never quoted here as a saving.

Variables. SRD_{tot}, total stripping heat per unit CO_2, J kg^{-1}; SRD_{reb}, reboiler share, J kg^{-1}; q_{IH}, interheater share, J kg^{-1}.

Assumptions. Not in the design.

Source. [**P**] Vinjarapu et al., Energy 320 (2025) 134973, Table 4, PDF p. 17. [**C**] for the sum.

Implementation. none. No code path.

Validation. NONE, test `eq_i_18`. Not carried, because total heat is unchanged and the plant has no surplus cool source above the band in Eq. I.17.

Worked example: regeneration at Case B

Start from 1097.90 kmol h^{-1} captured, which is 48.3 t h^{-1}. Eq. I.1 with a capacity of 0.25 gives 4,392 kmol h^{-1} of MEA, and Eq. I.9 turns that into 894 t h^{-1} of solvent at 30 wt%. Eq. I.12 at 3.8 GJ t^{-1} gives a reboiler duty of 51.0 MW. Dividing by the rich solvent flow gives 195 kJ kg^{-1} by Eq. I.15, which sits just under the pilot optimum. The plant steam stack of Eq. J.17 covers 37.9 MW of this, and the other 13.1 MW falls to the reactor.

Link to the sources

The flue-gas pilots agree on a narrow band of SRD, and the DICE design shows how that heat divides. What does not transfer cleanly is the absorber side: syngas at 26.8 bar reaches its rich loading with little pinch, yet the stripper still works at 2 bar, so the desorption term is unchanged. That is why the carried value stays near flue-gas practice, and why [**M12**] remains open.

12 J. Integrated plant

This section joins the unit models into one plant and asks how large the reactor must be. The order of work matters: scale the INL reference first, then close the carbon balance, then the heat and electricity balances, and only then report. INL Case 5 stands in for Case 6 on every syngas duty because the Case 6 sheets are not in the library [**M1**], scaled by 1,927 over 1,970 where used.

The reported terms map to cards as follows: reactor heat Eq. J.26; IHX duty and loss Eq. C.1 and Eq. B.15; process heat Eq. J.14; shift heat Eq. E.6; syngas recovery Eq. F.2; regeneration Eq. J.16; steam generation Eq. J.17; electricity Eq. B.18; helium Eq. B.12; pumping and compression Eq. J.20 and Eq. J.22; cooling Eq. J.31; unmet heat Eq. J.18; unmet electricity Eq. J.29. Values here are the Headline preset of the final results: the anchored port with term T05, syngas heat above 130 C, modelled, because gate M4 passed for it. Gates M6 and M7 failed, so the column and flowsheet terms stay anchored and their modelled values appear only in the drift table of the results. The comparison with the screening values (GOLD) runs on the Anchored preset, so a Headline value differs from its screening value by the T05 term alone.

Equation J.1: Scale

Tier F

$$s = \frac{\dot{m}_{H_2,B}}{\dot{m}_{H_2,A}} = \frac{8988}{13154} \tag{J.1}$$

Interpretation. Every extensive INL quantity is carried to the SOXAL-scale plant by this one ratio. Intensive conditions stay at their INL values.

Justification. Case B is 8,988 kg h^{-1} by Eq. J.2. Case A is 29,000 lb h^{-1}, which is 13,154 kg h^{-1}, and the INL product holds 0.1 mol% CH_4, or 0.79% by mass. On a volume basis the ratio would be 0.6891. The mass basis is used throughout, so the choice is stated once here.

Variables. $\dot{m}_{H_2,A}$, Case A hydrogen product, kg s^{-1}; $\dot{m}_{H_2,B}$, Case B hydrogen product, kg s^{-1}.

Assumptions. Linear scaling of the INL flowsheet.

Source. [**P**] TEV-953 [35], PDF p. 16. [**C**] for the ratio.

Implementation. `plant::anchored::Model::scale`

Validation. GOLD, test `eq_j_01`. $s = 0.6833$.

Equation J.2: Hydrogen mass rate

Tier R

$$\dot{m} = \rho_N \dot{V}_N, \qquad \rho_N = 0.08988 \text{ kg Nm}^{-3} \tag{J.2}$$

Interpretation. Plant capacity is quoted in normal cubic metres, and everything downstream needs mass.

Derivation. Definition of normal density. IEAGHG uses a value that is 0.07% higher, 8.994 t h^{-1} at the same volume.

Variables. ρ_N, hydrogen normal density, kg m^{-3}; $\dot{V}_N$, normal volumetric flow, 100,000, Nm^3 h^{-1}.

Assumptions. 100,000 Nm^3 h^{-1}, the SOXAL-scale train.

Source. [**P**] Air Liquide releases of 15 January 2008 and 30 November 2011 for capacity, not read at source. IEAGHG 2017-02 [1], Table 2, PDF p. 16, for its own density.

Implementation. `plant::anchored::Model::case_summary`, key h2_kg_h

Validation. GOLD, test `eq_j_02`. 8,988 kg h^{-1}.

Equation J.3: Annual hydrogen

Tier R

$$M_{H_2} = \dot{m}\, t_h, \qquad t_h = 8760 f_{av} \tag{J.3}$$

Interpretation. Hydrogen runs only while the reactor runs, so availability sets the annual output directly.

Derivation. Time integral at constant rate. No output is counted in outage hours.

Variables. t_h, annual operating hours, h y^{-1}.

Assumptions. 85% availability. Backup firing stays a sensitivity, Eq. K.20.

Source. [**C**] This document.

Implementation. `plant::anchored::Model::case_summary`, key h2_t_y

Validation. GOLD, test `eq_j_03`. 66,925 t y^{-1} on 7,446 h y^{-1}. Validation results, 1 row (HEADLINE_H2_T_Y): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation J.4: Gas unit conversion

Tier R

$$1 \text{ MMSCFD} = \frac{10^6}{379.49 \cdot 24} \text{ lbmol h}^{-1} \tag{J.4}$$

Interpretation. INL reports gas in standard volume per day. The balances need moles per second.

Derivation. One lbmol of ideal gas occupies 379.49 scf at 60 F and 14.696 psia, which gives 13.83 mol s^{-1} per MMSCFD.

Variables. $\dot{n}_{NG}$, natural gas molar flow, mol s^{-1}.

Assumptions. IG at 60 F and 14.696 psia.

Source. [**P**] TEV-961 [36], Table 2 note 1, PDF p. 18, for the 60 F standard. [**C**] for the molar volume.

Implementation. `plant::anchored::Model::case_summary`, key ng_mmscfd

Validation. GOLD, test `eq_j_04`. 23.23 MMSCFD at Case B, from 34.0 MMSCFD at Case A. Validation results, 2 rows (FS_T13_NG_FOR_CASE_B_H2 to HEADLINE_NG_MMSCFD): 2 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation J.5: Feed carbon

Tier R

$$F_C = 1.05379\, \dot{n}_{NG} \tag{J.5}$$

Interpretation. Natural gas carries slightly more than one carbon atom per molecule because of its ethane and heavier gases.

Derivation. Sum of carbon number times mole fraction over the feed composition: 93.571 CH_4, 3.749 C_2, 0.920 C_3, 0.260 C_4, 0.040 C_5, 0.010 C_6 and 0.250 CO_2, all in mol%.

Variables. F_C, feed carbon, mol s^{-1}.

Assumptions. TEV-953 gas, unchanged for the Singapore supply.

Source. [**P**] TEV-953 [35], A-5, PDF p. 39. [**C**] for the sum.

Implementation. `plant::flowsheet::natural_gas`

Validation. PUB, test `eq_j_05`. Computed here: 1.05379 from the composition. Case B is 1219.23 kmol h^{-1}, against 2,077.6 st d^{-1} of CO_2 equivalent printed for Case A. The Level 3 flowsheet against the anchored ledger: Validation results, 1 row (FS_CARBON_VS_ANCHOR_FEED): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation J.6: Carbon split after shift

Tier O

$$\phi_c = \frac{F_{CO_2,abs}}{T_C} = 0.775, \qquad 0.76 \le \phi_c \le 0.79 \tag{J.6}$$

Interpretation. Only CO_2 can be captured by the solvent. This ratio says what share of the carbon leaving the shift reactors has reached that form.

Derivation. Case 5 reconstruction: 3,754 kmol h^{-1} CO_2 with 1,020 CH_4 and 70 CO gives 0.775. Case 6 sheets are missing, so the range covers the proxy.

Variables. ϕ_c, CO_2 share of carbon after shift, 1; $F_{CO_2,abs}$, CO_2 at the absorber inlet, mol s^{-1}; T_C, tube carbon, mol s^{-1}.

Assumptions. Case 5 proxy [**M1**].

Source. [**C**] Reconstruction from TEV-961 [36], A-53 and A-56.

Implementation. `plant::anchored::Model::carbon`, which takes ϕ_c as an argument

Validation. GOLD, test `eq_j_06`. $\phi_c = 0.775$. The Level 3 flowsheet against the anchored ledger: Validation results, 1 row (FS_CARBON_VS_ANCHOR_CO2_ABS): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation J.7: INL recycled fraction

Tier O

$$T_6 = \frac{F_{cap,6}}{\eta_{Sel}\phi_c}, \qquad G_6 = T_6 - F_{cap,6} - P, \qquad f_r = \frac{T_6 - F_C}{G_6} \tag{J.7}$$

Interpretation. INL recycles the methane-rich cut of the tail gas. Working back from its published capture gives the share of tail-gas carbon that returns to the tubes.

Derivation. Capture at Case 6 is $F_{cap,6} = \eta_{Sel}\phi_c T_6$. Carbon reaching the tail gas is what the tubes hold less capture less product slip. Carbon in the tubes exceeds fresh feed by exactly the recycle, so $T_6 - F_C = f_r G_6$. At Case A, 1,927 st d^{-1} gives $F_{cap,6}$ = 1,655.1, T_6 = 2,157.2 and G_6 = 495.6 kmol h^{-1}.

Variables. T_6, tube carbon of INL Case 6, mol s^{-1}; $F_{cap,6}$, carbon captured in INL Case 6, mol s^{-1}; G_6, tail-gas carbon of INL Case 6, mol s^{-1}; P, carbon in the product, mol s^{-1}; η_{Sel}, Selexol capture, 0.99, 1; f_r, recycled fraction of tail-gas carbon, 1.

Assumptions. $\eta_{Sel} = 0.99$.

Source. [**P**] TEV-961 [36], Table 2, PDF p. 17, and A-53, PDF p. 77.

Implementation. `plant::flowsheet::recycle_fraction`

Validation. GOLD, test `eq_j_07`. $f_r = 0.752$.

Equation J.8: Tube carbon

Tier O

$$T_C = \frac{F_C - f_r P}{1 - f_r\,(1 - \eta_{pp}\phi_c)} \tag{J.8}$$

Interpretation. MEA lets 10% of the CO_2 slip, so more carbon returns through the recycle than INL sent back. The tubes then carry more carbon than INL's Selexol plant at the same feed.

Derivation. Start from $T_C = F_C + f_r G$. Tail gas is $G = T_C - C - P$ and capture is $C = \eta_{pp}\phi_c T_C$, so $G = T_C(1 - \eta_{pp}\phi_c) - P$. Substituting and collecting T_C gives the statement.

Variables. G, tail-gas carbon, mol s^{-1}; C, carbon captured, mol s^{-1}; η_{pp}, per-pass capture, 1.

Assumptions. Slip passes the cryogenic unit with the INL split, which INL did not model.

Source. [**C**] This document.

Implementation. `plant::anchored::Model::carbon`

Validation. GOLD, test `eq_j_08`. 1574.06 kmol C h^{-1} at Case B. The Level 3 flowsheet against the anchored ledger: Validation results, 1 row (FS_CARBON_VS_ANCHOR_TUBE): 1 fail. Failed: FS_CARBON_VS_ANCHOR_TUBE gives 1,497 against 1,574 kmol C/h, limit 2% relative. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation J.9: Captured carbon

Tier O

$$C = \eta_{pp}\phi_c T_C \tag{J.9}$$

Interpretation. Per-pass capture acts only on the CO_2 share of the tube carbon.

Derivation. Definition of η_{pp} on absorber inlet CO_2, with Eq. J.6.

Variables. C, carbon captured, mol s^{-1}.

Assumptions. Per-pass capture 0.90.

Source. [**C**] This document.

Implementation. `plant::anchored::Model::carbon`

Validation. GOLD, test `eq_j_09`. 1097.90 kmol h^{-1} at Case B. The Level 3 flowsheet against the anchored ledger: Validation results, 1 row (FS_CARBON_VS_ANCHOR_CAPTURED): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation J.10: Tail gas

Tier O

$$G = T_C - C - P, \qquad \dot{n}_{rec} = f_r G, \qquad \dot{n}_{burn} = (1 - f_r)\, G \tag{J.10}$$

Interpretation. Carbon that escapes capture and product ends up in the PSA reject. The cryogenic unit sends the methane-rich share back and burns the nitrogen-rich share.

Derivation. Carbon balance around capture and PSA. The route of the reject through the cryogenic unit is INL's.

Variables. $\dot{n}_{rec}$, recycled carbon, mol s^{-1}; $\dot{n}_{burn}$, carbon burned in the nitrogen-rich cut, mol s^{-1}.

Assumptions. Nitrogen-rich cut burned on site.

Source. [**P**] Route from TEV-961 [36], A-16, A-18, PDF p. 40, p. 42.

Implementation. `plant::carbon::CarbonLedger::recycle_share`

Validation. GOLD, test `eq_j_10`. 471.7, 354.82 and 116.90 kmol h^{-1} at Case B. The Level 3 flowsheet against the anchored ledger: Validation results, 3 rows (FS_CARBON_VS_ANCHOR_BURNED to FS_CARBON_VS_ANCHOR_TAIL_GAS): 3 fail. Failed: FS_CARBON_VS_ANCHOR_BURNED gives 93.92 against 116.9 kmol C/h, limit 2% relative; FS_CARBON_VS_ANCHOR_RECYCLED gives 285.1 against 354.8 kmol C/h, limit 2% relative; FS_CARBON_VS_ANCHOR_TAIL_GAS gives 379 against 471.7 kmol C/h, limit 2% relative. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence. These misses are why gate M7 failed and terms T04, T09 and T13 stay anchored, with the carbon lines TUBE, TAIL_GAS, RECYCLED, BURNED outside the 2% limit against the anchored ledger.

Equation J.11: Carbon closure

Tier R

$$F_C = C + (1 - f_r)\, G + P \tag{J.11}$$

Interpretation. Feed carbon must equal stored carbon plus stack carbon plus product carbon. The recycle is internal and cancels, so a slip anywhere in the flowsheet shows up here.

Derivation. Sum Eq. J.10 into Eq. J.8. The recycle cancels because it is internal.

Variables. P, carbon in the product, 0.1 mol% CH_4, mol s^{-1}.

Assumptions. 0.1 mol% CH_4 in product.

Source. [**P**] TEV-953 [35], A-40, PDF p. 74.

Implementation. `plant::carbon::close_carbon` and `plant::carbon::CarbonLedger::closure`

Validation. ID, test `eq_j_11`. Limit 0.01 kmol h^{-1}. Case B: 1097.90 + 116.90 + 4.42 against 1219.23. Validation results, 8 rows (CARBON_CLOSURE_ALL_RUNS to FS_ENVELOPE_ELEMENTS_FS_B_REC_F_T825): 8 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation J.12: PSA block

Tier X

$$F_{H_2,prod} = \xi_{PSA} F_{H_2,in}, \qquad \xi_{PSA} = 0.88 \tag{J.12}$$

Interpretation. The PSA is a specified split. It is an external block, outside the simulated plant, so nothing here predicts its performance.

Boundary. The boundary is fixed by project decision. The PSA is not simulated, and its 88% recovery and 99.9% purity are INL's specified inputs. The reject returns as a stream that crosses the boundary, so its carbon stays in every balance.

Variables. $F_{H_2,prod}$, hydrogen in the product, mol s^{-1}; $F_{H_2,in}$, hydrogen to the PSA, mol s^{-1}; ξ_{PSA}, PSA hydrogen recovery, 1.

Assumptions. Specified recovery and purity.

Source. [**P**] TEV-953 [35], PDF p. 19, p. 21. TEV-961 [36], A-53, PDF p. 77.

Implementation. `plant::flowsheet::psa_boundary`

Validation. NONE, test `eq_j_12`. No benchmark. The product is never to be described as simulated.

Equation J.13: Hydrogen stream ledger

Tier R

$$F_{H_2,in} = \frac{F_{H_2,prod}}{\xi_{PSA}}, \qquad F_{H_2,rej} = (1 - \xi_{PSA})\, F_{H_2,in} \tag{J.13}$$

Interpretation. Reformer syngas, H_2-rich gas after capture, product after PSA, recovery and annual hydrogen are separate rows, because they answer separate questions.

Derivation. Eq. J.12 rearranged and a hydrogen balance on the PSA.

Variables. $F_{H_2,rej}$, hydrogen in the PSA reject, mol s^{-1}.

Assumptions. No hydrogen lost in capture [**M10**].

Source. [**C**] This document.

Implementation. `plant::flowsheet::h2_ledger`

Validation. ID, test `eq_j_13`. Case B: 10,214 kg h^{-1} to the PSA, 8,988 to product, 1,226 to reject. Validation results, 1 row (FS_H2_RETURN_CLOSURE): 1 pass. FS_H2_RETURN_CLOSURE is a calibration row, so it tests the fit only.

Equation J.14: Process heat

Tier O

$$\dot{Q}_{proc} = 176.8\, s \tag{J.14}$$

Interpretation. The heat the helium must deliver to the process at Case B. It covers preheat as well as reforming, and preheat is about 29% of the fired duty in Case 5.

Derivation. INL's 925 C process heat demand scaled by Eq. J.1. The species-enthalpy duty Eq. D.34 sits below this basis, and the ledger keeps the INL figure.

Variables. $\dot{Q}_{proc}$, process heat, W.

Assumptions. Linear in s. Includes preheat.

Source. [**P**] TEV-961 [36], Table 2, PDF p. 17.

Implementation. `plant::anchored::Model::balance`

Validation. GOLD, test `eq_j_14`. 120.80 MW. Ahn and Lee [18], Table 6, PDF p. 6, print 143.4 MW_{th} on a different flowsheet, see Eq. J.28. Validation results, 1 row (HEADLINE_PROCESS_HEAT): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation J.15: Slip heat

Tier O

$$\dot{Q}_{rec} = \Delta F_{slip} c_{p,CO_2} \Delta T \tag{J.15}$$

Interpretation. CO_2 that MEA lets through and the recycle returns must be reheated in the reformer. INL's Selexol plant had far less of it.

Derivation. Sensible heat on the extra CO_2 over the INL case. At Case A, $\Delta F_{slip} = 161.8$ kmol h^{-1}. Reverse shift is not counted.

Variables. ΔF_{slip}, extra CO_2 slip over INL, mol s^{-1}; c_{p,CO_2}, CO_2 heat capacity, 50, J mol^{-1} K^{-1}; ΔT, temperature rise, 571, K.

Assumptions. Sensible only. Reverse water-gas shift excluded.

Source. [**A**] Project values.

Implementation. `plant::anchored::Model::balance`

Validation. GOLD, test `eq_j_15`. 0.88 MW at Case B.

Equation J.16: Reboiler duty

Tier R

$$\dot{Q}_{reb} = \dot{m}_{CO_2} SRD \tag{J.16}$$

Interpretation. Regeneration heat is capture rate times the specific duty.

Derivation. Eq. I.12 rearranged. 48.3 t h^{-1} at 3.8 GJ t^{-1}, divided by 3.6, gives megawatts.

Variables. $\dot{m}_{CO_2}$, CO_2 captured, kg s^{-1}.

Assumptions. Carried SRD of Eq. I.14.

Source. [**C**] From Eq. I.12.

Implementation. `units::regen::q_reb` on the column model, and `plant::flowsheet::AnchoredMea::reboiler_w` on the anchored solvent loop

Validation. GOLD, test `eq_j_16`. 51.0 MW at Case B.

Equation J.17: Heat stack for the reboiler

Tier O

$$\dot{Q}_{stack} = \dot{Q}_{Sel} + \dot{Q}_{exp} + \dot{Q}_{syn} \tag{J.17}$$

Interpretation. Three plant streams can serve the reboiler before the reactor is asked. They are spent in a fixed order: freed Selexol steam, then export steam, then syngas above 130 C.

Derivation. Removing Selexol frees its LP steam, 65.0 MMBtu h^{-1} in Case 5. Export steam is the 83.7 MMBtu h^{-1} of Case 6. Syngas above 130 C gives 11.5 MW in Case 5. Case 5 duties are scaled by 1,927 over 1,970, and all three by s. Results match the screening order whenever the whole stack is used, which holds at all central cases.

Variables. $\dot{Q}_{stack}$, plant steam credit, W; $\dot{Q}_{Sel}$, freed Selexol steam, W; $\dot{Q}_{exp}$, export steam, W; $\dot{Q}_{syn}$, syngas heat above 130 C, W.

Assumptions. Case 5 proxy [**M1**]. Zero-liquid-discharge steam stays with its evaporator.

Source. [**P**] TEV-961 [36], A-56, PDF p. 80, and Table 2, PDF p. 17. [**C**] for the syngas cut.

Implementation. `plant::heatrec::allocate`

Validation. GOLD, test `eq_j_17`. 12.73, 16.76 and 8.39 MW, total 37.9 MW at Case B. The syngas term is T05 modelled, which the Headline uses because gate M4 passed for it; its anchored value is 7.69 MW.

Equation J.18: Reboiler gap

Tier O

$$\dot{Q}_{gap} = \max\left(0, \dot{Q}_{reb} - \dot{Q}_{stack}\right) \tag{J.18}$$

Interpretation. Whatever the stack cannot supply falls to the reactor. It is the unmet thermal demand if the reactor cannot cover it.

Derivation. Heat balance on the reboiler, with the stack spent first.

Variables. $\dot{Q}_{gap}$, reboiler gap, W.

Assumptions. Stack fully usable at grade, Eq. J.30.

Source. [**C**] This document.

Implementation. `plant::heatrec::gap`

Validation. GOLD, test `eq_j_18`. 13.12 MW at Case B on the Headline. The screening value differs by the T05 term, see the section opening.

Equation J.19: Scaled electricity rows

Tier O

$$E_i = E_{i,6}\, s \tag{J.19}$$

Interpretation. Rows that do not depend on the capture step follow the plant size.

Derivation. Table 2 rows at 925 C: reforming 1.2, steam 0.3, cooling 0.1, water 1.5 $\mathrm{MW_e}$. The Selexol row is removed. PSA stays at 17.3 kW from Case 5.

Variables. E_i, electric row i, W; $E_{i,6}$, row i in INL Case 6, W.

Assumptions. Linear in s.

Source. [**P**] TEV-961 [36], Table 2, PDF p. 17, and A-56, PDF p. 80.

Implementation. `plant::anchored::Model::balance`, with the rows held in `plant::anchored::ERows`

Validation. GOLD, test `eq_j_19`. 0.82, 0.012, 0.20, 0.068 and 1.02 $\mathrm{MW_e}$ at Case B, for reforming, PSA, steam, cooling and water.

Equation J.20: Lean pump

Tier O

$$E_{pump} = \frac{\dot{m}_{solv}\,(p_{abs} - p_{strip})}{\rho\,\eta_{pump}\eta_m} \tag{J.20}$$

Interpretation. Solvent must go back up from 2 to 26.8 bar. This load is new against a flue-gas plant and has no letdown recovery here.

Derivation. Hydraulic work per unit volume is the pressure rise, divided by pump and motor efficiency.

Variables. E_{pump}, lean pump electricity, W; η_{pump}, pump efficiency, 0.75, 1; η_m, motor efficiency, 0.90, 1.

Assumptions. Solvent density 1,050 kg $\mathrm{m^{-3}}$. No letdown turbine.

Source. [**A**] Project values.

Implementation. `plant::flowsheet::AnchoredMea::pump_w`

Validation. GOLD, test `eq_j_20`. 0.87 $\mathrm{MW_e}$ at Case B.

Equation J.21: Cooling fans

Tier O

$$E_{fans} = k_{cool}\dot{Q}_{reb} \tag{J.21}$$

Interpretation. Reboiler heat comes back out in the lean cooler and condenser, and moving it to the air takes electricity.

Derivation. Assumed ratio of 0.015 $\mathrm{MW_e\ MW_{th}^{-1}}$. DICE gives an order check, with cooling tower fans at 977 kW and circulating water pumps at 2,407 kW. No published ratio exists.

Variables. E_{fans}, cooling electricity, W; k_{cool}, electric load per unit reboiler heat, 1.

Assumptions. Ratio assumed.

Source. [**A**] Project value. Order check on DICE [22], PDF p. 18.

Implementation. `plant::flowsheet::AnchoredMea::fans_w`

Validation. GOLD, test `eq_j_21`. 0.77 $\mathrm{MW_e}$ at Case B.

Equation J.22: CO_2 compression

Tier O

$$E_{CO_2} = 6.4\,\frac{C}{C_6}\ \mathrm{MW_e} \tag{J.22}$$

Interpretation. Compression to 2,015 psia scales with the CO_2 handled. INL tuned the power to vendor data.

Derivation. INL Case 6 row scaled on captured carbon, C/C_6. A ship route needs liquefaction, whose power is not in the library [**M28**], so the pipeline pressure is kept as a proxy and the tariff carries transport.

Variables. E_{CO_2}, CO_2 compression electricity, W; C_6, carbon captured in INL Case 6, mol $\mathrm{s^{-1}}$.

Assumptions. Pipeline pressure kept for a ship route.

Source. [**P**] TEV-961 [36], Table 2, PDF p. 17, and A-53, PDF p. 77.

Implementation. `plant::anchored::Model::balance` for the anchored row of term T09, and `plant::flowsheet::co2_compression_isothermal_w` for its modelled cross-check

Validation. GOLD, test `eq_j_22`. 4.25 $\mathrm{MW_e}$ at Case B. Computed here: DICE at 7,838 kW on 70.2 t $\mathrm{h^{-1}}$ ([22], PDF p. 18) is 27% higher per tonne. This stays an open risk.

Equation J.23: Recycle row

Tier O

$$E_{pur} = 5.7\,s\left(1 + \frac{\Delta F_{slip}}{1473}\right) \tag{J.23}$$

Interpretation. INL's Purge Recovery row runs the fuel-gas recycle and the nitrogen removal equipment. More CO_2 in the reject means more gas to handle, so the row grows with the slip.

Derivation. Table 2 row restored by project decision, scaled by s, then raised in proportion to the extra flow over the Case 5 reject of 1,473 kmol $\mathrm{h^{-1}}$. The split of the row is unpublished [**M30**].

Variables. E_{pur}, recycle and nitrogen removal electricity, W.

Assumptions. Load proportional to reject flow.

Source. [**P**] TEV-961 [36], Table 2, PDF p. 17. [**A**] for the slip term.

Implementation. `plant::anchored::Model::balance`

Validation. GOLD, test `eq_j_23`. 4.32 $\mathrm{MW_e}$ at Case B, 3.89 without the slip term.

Equation J.24: Total electricity

Tier R

$$E_{tot} = E_{proc} + E_{circ} \tag{J.24}$$

Interpretation. Everything the plant draws, process rows plus helium circulators.

Derivation. Sum of Eq. J.19 to Eq. J.23 and Eq. B.12.

Variables. E_{circ}, circulator electricity, W.

Assumptions. No grid import at Case B.

Source. [**C**] This document.

Implementation. `plant::utility::e_total`

Validation. GOLD, test `eq_j_24`. 12.33 + 7.45 = 19.78 $\mathrm{MW_e}$.

Equation J.25: Coupling

Tier O

$$\dot{m}_p = \dot{m}_p\left(\dot{Q}_R\right), \quad \dot{W}_p = \dot{W}_p\left(\dot{m}_p\right), \quad \dot{Q}_{ves} = \dot{Q}_{ves}\left(\dot{Q}_R\right) \tag{J.25}$$

Interpretation. Reactor power sets primary flow, primary flow sets circulator power, and circulator power changes the heat the reactor must supply. The balance therefore needs an iteration.

Derivation. Eq. B.9 and Eq. B.13 depend on $\dot{Q}_R$. Eq. B.10 feeds Eq. B.20.

Variables. $\dot{m}_p$, primary helium flow, kg $\mathrm{s^{-1}}$; $\dot{W}_p$, primary circulator shaft power, W.

Assumptions. Algebraic coupling only.

Source. [**C**] This document.

Implementation. `plant::utility::reactor_required`

Validation. ID, test `eq_j_25`. Structure test on the residual.

Equation J.26: Fixed point for the reactor heat

Tier O

$$\dot{Q}_R^{(k+1)} = \dot{Q}_{proc} + \dot{Q}_{rec} + \dot{Q}_{ihx} + \dot{Q}_{duct} + \dot{Q}_{ves}^{(k)} + \dot{Q}_{br}^{(k)} - \dot{W}_s - \dot{W}_p^{(k)} \tag{J.26}$$

Interpretation. The required reactor rating is the number that reproduces itself through the whole balance. It is the plant's main design output.

Derivation. Iterate Eq. B.20 from 220 s to a change below 10^{-9} MW. The loop gain is small because circulator work and vessel loss are small parts of the total, and convergence takes well under 200 steps.

Variables. $\dot{Q}_{br}$, heat to the power branch and reboiler gap, W.

Assumptions. Converges in under 200 steps.

Source. [**C**] This document.

Implementation. `plant::anchored::Model::balance`, which iterates with `numerics::fixed_point`

Validation. GOLD, test `eq_j_26`. 178.49 $\mathrm{MW_{th}}$ at Case B on the Headline and 260.74 at Case A. The Anchored preset gives 179.04 at Case B, which is the value the screening comparison uses. Validation results, 2 rows (FS_TEAR_THREE_STARTS to HEADLINE_REACTOR_NEED): 2 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation J.27: Minimum capacity with corners

Tier O

$$\dot{Q}_{R,c} = \dot{Q}_R\left(\text{SRD, circulator set, loss set}\right)_c \tag{J.27}$$

Interpretation. The same balance run at sets of inputs that all lean one way. The corners bound the requirement and carry no probability.

Derivation. Solve Eq. J.26 at each named set.

Variables. $\dot{Q}_{R,c}$, reactor heat at corner c, W.

Assumptions. Best, low, central and high sets.

Source. [**C**] This document.

Implementation. `plant::drift::high_corner` for the high corner. The central value is the rating that `plant::anchored::Model::balance` solves.

Validation. GOLD, test `eq_j_27`. Best 155.8, central 178.5, high 214.1 $\mathrm{MW_{th}}$ at Case B. The low set, 163.0 in the screening table, has no results row at this build. Only the high corner exceeds 210 $\mathrm{MW_{th}}$.

Equation J.28: Heat basis sensitivity

Tier O

$$\dot{Q}_{proc} \rightarrow 143.4\ \mathrm{MW_{th}} \tag{J.28}$$

Interpretation. A recent study puts process heat well above the scaled INL figure at this H_2 rate. Trying it shows how much of the 31 MWth margin is at risk.

Derivation. Replace Eq. J.14 and repeat Eq. J.26. Ahn and Lee use S/C 2.7, 875 C and MDEA, so the value is a sensitivity on a different flowsheet.

Variables. $\dot{Q}_{proc}$, alternative process heat, W.

Assumptions. Ahn S/C 2.7, 875 C and MDEA differ from this plant.

Source. [**P**] Ahn and Lee [18], Table 1, PDF p. 4, and Table 6, PDF p. 6.

Implementation. `plant::sens::sweeps`, sweeps HEAT_BASIS and HEAT_BASIS_HE_HELD

Validation. PUB, test `eq_j_28`. Sensitivity results: 203.5 $\mathrm{MW_{th}}$ with helium flow scaled, 202.4 with helium flow held. An earlier hand estimate of about 201.6 sits below both, and that gap is not reconciled.

Equation J.29: Fixed rating and import

Tier O

$$E_{imp} = E_{nuc} - \max\left(0, \dot{Q}_{rated} - \dot{Q}_{heat}\right)\eta_{pc} \tag{J.29}$$

Interpretation. If the rating is fixed, heat duties are served first and any electricity the leftover heat cannot make is imported. This is the unmet electrical demand.

Derivation. Heat first: $\dot{Q}_{heat}$ is Eq. J.26 with the power branch removed. What remains of the rating runs through η_{pc}.

Variables. E_{imp}, imported electricity, W; E_{nuc}, electricity asked of the reactor, W; $\dot{Q}_{heat}$, heat duties at fixed rating, W; $\dot{Q}_{rated}$, selected reactor rating, W.

Assumptions. Heat served first.

Source. [**C**] This document.

Implementation. `plant::utility::import`

Validation. GOLD, test `eq_j_29`. 0.0 $\mathrm{MW_e}$ at Case B, 22.1 $\mathrm{MW_e}$ at Case A with MEA.

Equation J.30: Pinch on the stack

Tier F

$$T_{stack,j} \geq T_{source,min} \quad \forall j \tag{J.30}$$

Interpretation. Each stream counted in the stack must meet the grade of Eq. I.17. The hot water loop is excluded because it is far too cold for a reboiler.

Justification. INL's hot water loop supplies at 200 F, which is 93 C, against a reboiler near 122 C. LP steam at 307 F qualifies.

Variables. $T_{stack,j}$, temperature of stack stream j, K; $T_{source,min}$, lowest usable source, K.

Assumptions. 93 C loop too cold.

Source. [**P**] TEV-961 [36], A-57, PDF p. 81, read as an image.

Implementation. `plant::heatrec::check_grade`

Validation. ID, test `eq_j_30`. Rule test: no hot water row enters the stack.

Equation J.31: Cooling duty

Tier O

$$\dot{Q}_{cool} = \dot{Q}_{KO} + \dot{Q}_{HW} + \dot{Q}_{lean} + \dot{Q}_{cond} + (1 - \eta_{pc})\,\dot{Q}_{pc} \tag{J.31}$$

Interpretation. All heat not sold or used ends up in the cooling water. The sum sizes the cooling system and shows what Singapore must reject to the sea.

Derivation. Whole-plant energy balance. Syngas knock-out and hot water trim come from Case 5, the lean cooler and condenser from the solvent loop, and the last term from the power cycle.

Variables. $\dot{Q}_{cool}$, total cooling duty, W; $\dot{Q}_{KO}$, knock-out drum cooler, W; $\dot{Q}_{HW}$, hot water trim cooler, W; $\dot{Q}_{lean}$, lean solvent cooler, W; $\dot{Q}_{pc}$, heat into the power cycle, W.

Assumptions. Case 5 proxy [**M1**].

Source. [**C**] This document, on TEV-961 [36], A-55 to A-57.

Implementation. Not implemented in the engine at this build: the balance carries cooling as electricity rows only, Eq. J.19 and Eq. J.21. Planned name `plant::utility::cooling`.

Validation. PUB, test `eq_j_31`. Computed here for the two published rows before scale: 19.7 and 74.1 MMBtu h^{-1} are 5.8 and 21.7 MW. The engine carries no cooling duty, so no total is printed.

Equation J.32: Thermal efficiency

Tier O

$$\eta_{th} = \frac{\dot{m}_{H_2} LHV_{H_2}}{\dot{n}_{NG} LHV_{NG} + \dot{Q}_R} \tag{J.32}$$

Interpretation. Hydrogen energy out over all energy in, counting reactor heat at par with gas.

Derivation. First-law efficiency on an LHV basis. The gas HHV of 1,044 Btu scf^{-1} is converted with the 0.90 rule of thumb for gaseous fuels.

Variables. η_{th}, thermal efficiency, 1; LHV_{H_2}, hydrogen lower heating value, J kg^{-1}; LHV_{NG}, gas lower heating value, J mol^{-1}.

Assumptions. HHV to LHV ratio 0.90.

Source. [**P**] GHG Protocol stationary combustion guidance, PDF p. 46, for the ratio. TEV-961 [36], A-50, PDF p. 74, for HHV. [**C**] for the result.

Implementation. Not implemented in the engine at this build. The value in the validation line is computed in this document. Planned name `plant::utility::eta_th`.

Validation. PUB, test `eq_j_32`. Computed here: 0.67 at the required 178.49 MW_{th}. INL prints 82.5% on an HHV basis (Table 2, PDF p. 17). Its definition was not traced here, so the two are not compared.

Equation J.33: Capture pair

Tier R

$$\eta_{pp} = \frac{C}{\phi_c T_C}, \qquad \eta_{pl} = \frac{C}{F_C} \tag{J.33}$$

Interpretation. Two different percentages that are easy to confuse. Per-pass is what the absorber does to the gas it sees. Plant capture is what the whole site does to the carbon it buys.

Derivation. Definitions, printed together by project decision. The recycle returns most of the slip for another pass, so plant capture stays near the per-pass value even though only 0.775 of tube carbon reaches the absorber as CO_2.

Variables. η_{pl}, plant capture, 1.

Assumptions. Direct carbon basis.

Source. [**C**] This document.

Implementation. `plant::carbon::CarbonLedger::capture_pair`

Validation. GOLD, test `eq_j_33`. 0.90 and 0.9005. The results tables hold a second form, $C/\left[C + (1 - f_r)G\right] = 0.9038$, which leaves the product carbon out of the denominator. Both are printed so the basis stays visible.

Equation J.34: Yield

Tier R

$$Y_m = \frac{\dot{m}_{NG}}{\dot{m}_{H_2}}, \qquad Y_v = \frac{\dot{V}_{H_2}}{\dot{V}_{NG}} \tag{J.34}$$

Interpretation. How much hydrogen each unit of gas buys, in both of the forms the sources use.

Derivation. Ratios of Eq. J.2 and Eq. J.4. The mass ratio takes the molar mass of the feed composition, 17.18 g mol^{-1}.

Variables. Y_m, gas mass per unit hydrogen, kg kg^{-1}; Y_v, hydrogen volume per unit gas, 1; $\dot{m}_{NG}$, gas mass flow, kg s^{-1}; $\dot{V}_{H_2}$, hydrogen standard volume flow, m^3 s^{-1}.

Assumptions. Case 6 gas rate.

Source. [**P**] TEV-961 [36], Table 2, PDF p. 17, prints 3.82. [**C**] for the mass ratio.

Implementation. Not implemented in the engine at this build. The values in the validation line are computed in this document. Planned name `plant::utility::yield`.

Validation. PUB, test `eq_j_34`. Computed here: 130 over 34.0 is 3.82 scf scf^{-1}, and the mass ratio is 2.21 kg kg^{-1}.

Equation J.35: Sizing walk

Tier O

$$\dot{Q}_{R,0} \;\rightarrow\; \dot{Q}_{R,1} \;\rightarrow\; \cdots \;\rightarrow\; \dot{Q}_R \tag{J.35}$$

Interpretation. The walk explains why the requirement is well above the first estimate by adding one missing item at a time, so each difference has a name.

Derivation. Start from process heat plus electricity over η_{pc} without the purge row. Then restore the purge row, add losses, add circulators, swap Selexol for MEA with zero reboiler heat, and add the reboiler gap.

Variables. $\dot{Q}_{R,0}$, first estimate of reactor heat, W.

Assumptions. Case A order, repeated at Case B.

Source. [**C**] This document.

Implementation. `plant::anchored::Model::sizing_walk`

Validation. GOLD, test `eq_j_35`. Case B on the Anchored preset: 140.8, 150.6, 154.3, 165.9, 166.2, 179.0 MW_{th}. Case A: 206.0, 220.4, 225.3, 242.4, 242.8, 261.5 MW_{th}, read from the design decisions table because no results row holds them. Validation results, 7 rows (SIZING_WALK_STEP_0 to SIZING_WALK_STEP_6): 7 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation J.36: Rated against required

Tier O

$$\dot{Q}_{spare} = \dot{Q}_{rated} - \dot{Q}_R \tag{J.36}$$

Interpretation. The reactor is rated at 210 MWth and the plant needs less. The spare heat stays as margin with no revenue, yet the capital is still charged on the full rating.

Derivation. Eq. B.21 at Case B, then the ledger of Section L at each charge.

Variables. $\dot{Q}_{spare}$, rating less requirement, W.

Assumptions. No revenue on the spare heat.

Source. [**C**] This document.

Implementation. `plant::utility::margin`

Validation. GOLD, test `eq_j_36`. 31.5 MW_{th} spare. Charged at 210 and at 178.5 MW_{th}, BOAK, the full cost set gives 189.7 and 161.2 S$ t^{-1} at 25 y, 165.6 and 140.4 at 60 y. See Eq. L.26. The screening comparison runs on the Anchored preset: Validation results, 1 row (HEADLINE_SPARE): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Worked example: closing Case B

Carbon first. Eq. J.5 gives 1219.23 kmol h^{-1} of feed carbon. With $f_r = 0.752$, $\eta_{pp} = 0.90$ and $\phi_c = 0.775$, Eq. J.8 gives 1574.06 in the tubes, Eq. J.9 gives 1097.90 captured and Eq. J.10 leaves 116.90 in the stack. Eq. J.11 closes with 4.42 in the product.

Heat next. Process heat is 120.80 MW, with 0.88 of slip heat. The reboiler wants 51.0 MW, and the stack gives 37.9, so the gap is 13.12. Electricity totals 19.78 MW_e, of which 7.45 runs the circulators. Eq. J.26 then converges at 178.49 MW_{th}, leaving 31.5 MW_{th} of the 210 rating unused.

Link to the sources

INL's 925 C case reaches 178.8 MWt on heat and power (TEV-961, Table 2, PDF p. 18). That figure is close to the 178.5 here at a smaller plant only by coincidence: this design scales down by 0.6833 and then adds a reboiler gap, the restored purge row, losses and circulators that INL did not carry. The walk in Eq. J.35 shows each step.

13 K. Emissions

This section turns the carbon balance into tonnes per year and asks how much CO_2 the plant keeps out of the atmosphere. The headline basis is direct emissions to atmosphere against a same-size unabated SMR, a new-build at SOXAL scale, using the INL baseline. Per-pass and plant capture are shown together, and lifecycle stays a sensitivity. The PSA stays external, and its reject carbon is still counted, so the boundary cannot hide carbon. The compressor and offshore storage sit inside the dashed line of Figure 2. Eq. K.17 explains this, and the accounting of carbon stored outside Singapore remains open.

Equation K.1: Baseline intensity

Tier R

$$e_{base} = \frac{\dot{E}_{base,A}}{\dot{m}_{H_2,A}} \tag{K.1}$$

Interpretation. What an unabated SMR emits per kilogram of hydrogen. Abatement is measured against this number.

Derivation. INL's conventional plant without capture emits 3,205 st d^{-1} of CO_2 while making 29,000 lb h^{-1} of hydrogen. INL's ton is the short ton, at 907.185 kg. Dividing gives the statement.

Variables. e_{base}, baseline intensity, kg kg^{-1}; $\dot{E}_{base,A}$, CO_2 of the INL conventional plant, kg s^{-1}.

Assumptions. Short tons. Same gas and H_2 basis as INL.

Source. [**P**] TEV-953 [35], Table 2, PDF p. 30.

Implementation. `plant::anchored::Model::case_summary`, key baseline_kg_kg

Validation. GOLD, test `eq_k_01`. 9.21 kg kg^{-1}.

Equation K.2: Baseline bounds

Tier F

$$e_{base} \in \{9.35,\ 9.00\}\ \text{kg kg}^{-1} \tag{K.2}$$

Interpretation. Two other baselines bracket the INL number from above and below, so the result never rests on one plant.

Justification. 9.35 is the mean of Cho et al. for 33 US facilities, from the abstract only [**M25**]. IEAGHG gives 0.8091 kg Nm^{-3} of H_2 at 8.994 t h^{-1} per 100,000 Nm^3 h^{-1}, which is 8.996 kg kg^{-1}, rounded to 9.00.

Variables. e_{base}, alternative baseline intensity, kg kg^{-1}.

Assumptions. Cho abstract only.

Source. [**P**] IEAGHG 2017-02 [1], Table 2, PDF p. 16. Cho 2022 abstract.

Implementation. `plant::anchored::Model::case_summary`, keys base_cho and base_ieaghg

Validation. PUB, test `eq_k_02`. Published values printed. The IEAGHG ratio is computed here.

Equation K.3: Annual baseline

Tier R

$$E_{base} = e_{base} M_{H_2} \tag{K.3}$$

Interpretation. What the unabated plant would emit in a year making the same hydrogen in the same hours.

Derivation. Eq. K.1 times Eq. J.3.

Variables. E_{base}, annual baseline CO_2, t y^{-1}.

Assumptions. Same H_2, same hours.

Source. [**C**] This document.

Implementation. `plant::emissions::Model::emissions`, field baseline

Validation. GOLD, test `eq_k_03`. 616,361 t y^{-1}.

Equation K.4: Furnace CO_2 of the conventional plant

Tier R

$$\dot{E}_{furn} = M_{CO_2}\left(f_{fire}F_C^{conv} + \dot{n}_{rej}^{conv}\right) \tag{K.4}$$

Interpretation. A conventional reformer burns gas and PSA reject in its furnace, so its stack is the one point source that a helium-heated plant removes.

Derivation. Carbon balance on the furnace. INL fires 15.14% of the natural gas for heat in the conventional case. IEAGHG's fuel share computed from its Table 2 is 14.2%, the same order.

Variables. $\dot{E}_{furn}$, furnace CO_2, kg s^{-1}; f_{fire}, share of gas fired, 1; F_C^{conv}, feed carbon of the conventional plant, mol s^{-1}; $\dot{n}_{rej}^{conv}$, reject carbon burned, mol s^{-1}.

Assumptions. Conventional plant.

Source. [**P**] TEV-953 [35], A-7, PDF p. 41. IEAGHG 2017-02 [1], Table 2, PDF p. 16.

Implementation. Not implemented as a separate engine line, because the INL total of Eq. K.1 already holds the furnace. Planned name `plant::emissions::furnace`.

Validation. PUB, test `eq_k_04`. Shares compared here. The furnace is not a separate line in the results, because the INL total of Eq. K.1 already holds it.

Equation K.5: Process minimum

Tier R

$$e_{st} = \frac{M_{CO_2}}{4M_{H_2}} \tag{K.5}$$

Interpretation. Even a perfect plant makes this much CO_2 from methane, because the carbon has to go somewhere.

Derivation. Reaction III, $CH_4 + 2H_2O \rightarrow CO_2 + 4H_2$, run to completion. One mole of CO_2 per four of H_2.

Variables. e_{st}, stoichiometric CO_2 per unit H_2, kg kg^{-1}; M_{H_2}, molar mass of hydrogen in this ratio, g mol^{-1}.

Assumptions. Reaction III complete.

Source. [**T**] Stoichiometry.

Implementation. Not implemented in the engine at this build. The value is computed in this document. Planned name `plant::emissions::stoich`.

Validation. ID, test `eq_k_05`. Computed here: 5.46 kg kg^{-1}.

Equation K.6: Carbon bound

Tier R

$$e_{feed} = \frac{F_C M_{CO_2}}{\dot{m}_{H_2}} \tag{K.6}$$

Interpretation. Burn all the carbon fed to this plant and this is the most CO_2 it could make per kilogram of H_2. It sits above the stoichiometric floor because some gas stays unconverted and heavier gases add carbon.

Derivation. Carbon balance, all feed carbon to CO_2.

Variables. e_{feed}, feed carbon intensity, kg kg^{-1}.

Assumptions. All feed carbon counted.

Source. [**C**] This document.

Implementation. `plant::export::results_block`, which writes the feed carbon row

Validation. GOLD, test `eq_k_06`. Computed here: 5.97 kg kg^{-1}. A process share of 65% of the 9.35 baseline would be 6.08, which is 1.8% above this bound, one reason the split from the Cho abstract is dropped.

Equation K.7: Captured CO_2 Tier R

$$E_{cap} = C\, M_{CO_2} t_h \tag{K.7}$$

Interpretation. Tonnes sent to storage each year.

Derivation. Eq. J.9 in mass units over the operating hours.

Variables. E_{cap}, CO_2 captured, t y^{-1}.

Assumptions. No loss in compression or transport.

Source. [**C**] This document.

Implementation. `plant::emissions::Model::emissions`, field captured

Validation. GOLD, test `eq_k_07`. 359,777 t y^{-1}.

Equation K.8: Residual to atmosphere Tier R

$$E_{atm} = (1 - f_r)\, G\, M_{CO_2} t_h \tag{K.8}$$

Interpretation. The nitrogen-rich cut is burned, so its carbon is the plant's only direct stack CO_2.

Derivation. Eq. J.10 in mass units. Its carbon is the whole residual because the rest of the tail gas recycles.

Variables. E_{atm}, residual CO_2 to atmosphere, t y^{-1}.

Assumptions. Nitrogen-rich cut burned.

Source. [**C**] This document.

Implementation. `plant::emissions::Model::vent_condensate` for the vented-condensate case of decision S14. The central residual is the atmosphere line of `plant::emissions::Model::emissions`.

Validation. GOLD, test `eq_k_08`. 38,309 t y^{-1}. With the condensate CO_2 vented (decision S14), abatement falls by 1,731 t y^{-1} and the L5 cost at BOAK and 25 y rises by S$0.29/t.

Equation K.9: Electricity emissions Tier O

$$E_{el} = E_{imp} t_h EF_{grid} \tag{K.9}$$

Interpretation. Grid electricity would carry its own CO_2. At Case B the reactor supplies all power, so this line is zero.

Derivation. Energy times emission factor.

Variables. E_{el}, CO_2 from imported power, t y^{-1}; EF_{grid}, grid emission factor, 0.402, t MWh^{-1}.

Assumptions. Zero central. [**M27**]: the factor was not re-checked at source.

Source. [**C**] This document.

Implementation. Not implemented as written: the engine applies the grid factor only to the baseline power of the system view in Eq. K.18, and the import at the rating is 0.0 MW_e at Case B. Planned name `plant::emissions::electricity`.

Validation. GOLD, test `eq_k_09`. 0.0 at Case B.

Equation K.10: Operating hours

Tier R

$$t_h = 8760 f_{av} \tag{K.10}$$

Interpretation. The same hours as hydrogen output, so no tonne of abatement is counted outside production.

Derivation. Definition.

Variables. f_{av}, availability, 1.

Assumptions. 85% central.

Source. [**C**] This document.

Implementation. `plant::anchored::Model::case_summary`, key hours

Validation. ID, test `eq_k_10`. 7,446 h y^{-1}.

Equation K.11: Net abatement

Tier R

$$\mathcal{A} = E_{base} - E_{atm} \tag{K.11}$$

Interpretation. The headline climate number: CO_2 that would have reached the atmosphere and now does not.

Derivation. Direct emissions to atmosphere on the same hydrogen. Stored CO_2 is not an emission on this boundary.

Variables. $\mathcal{A}$, direct abatement, t y^{-1}.

Assumptions. Direct, atmosphere view.

Source. [**C**] This document.

Implementation. `plant::emissions::Model::emissions`, field a_atm

Validation. GOLD, test `eq_k_11`. 578,052 t y^{-1}. Validation results, 1 row (HEADLINE_ABATEMENT): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation K.12: Abatement on the bounds

Tier R

$$\mathcal{A}\left(e_{base}\right) = e_{base} M_{H_2} - E_{atm} \tag{K.12}$$

Interpretation. Only the baseline moves. The plant's own emissions stay the same, so a higher baseline simply credits more abatement.

Derivation. Replace Eq. K.1 by Eq. K.2 in Eq. K.3.

Variables. $\mathcal{A}$, abatement at a chosen baseline, t y^{-1}.

Assumptions. Baselines of Eq. K.2.

Source. [**C**] This document.

Implementation. `plant::anchored::Model::case_summary`, with the baseline of each bound

Validation. GOLD, test `eq_k_12`. 0.587 and 0.564 Mt y^{-1}.

Equation K.13: Carbon intensity

Tier R

$$CI = \frac{E_{atm}}{M_{H_2}} \tag{K.13}$$

Interpretation. The direct intensity of the product, the number to set against low-carbon hydrogen standards.

Derivation. Eq. K.8 over Eq. J.3.

Variables. CI, direct carbon intensity, kg kg^{-1}.

Assumptions. Direct only.

Source. [**C**] This document.

Implementation. `plant::anchored::Model::case_summary`, keys ci_kg_kg and reduction

Validation. GOLD, test `eq_k_13`. 0.572 kg kg^{-1}, 93.8% below the baseline.

Equation K.14: Capture pair with abatement

Tier R

$$\{\mathcal{A},\ \eta_{pp},\ \eta_{pl}\} \tag{K.14}$$

Interpretation. Abatement is never printed without saying which capture figure sits behind it.

Derivation. Reporting rule, set by project decision. See Eq. J.33.

Variables. η_{pp}, per-pass capture, 1.

Assumptions. Reporting rule.

Source. [**C**] This document.

Implementation. `plant::carbon::CarbonLedger::capture_pair`

Validation. ID, test `eq_k_14`. Both capture figures sit in the same row of the emissions results.

Equation K.15: Capture sensitivity

Tier O

$$\mathcal{A}\left(\eta_{pp}\right) = E_{base} - \left(1 - f_r\right) G\left(\eta_{pp}\right) M_{CO_2} t_h \tag{K.15}$$

Interpretation. Abatement is much less sensitive to per-pass capture than a simple scaling would suggest, because slip mostly returns through the recycle and gets another pass.

Derivation. Eq. J.8 and Eq. J.10 with η_{pp} varied, then Eq. K.11.

Variables. η_{pp}, per-pass capture, 0.85 to 0.95, 1.

Assumptions. 0.85 to 0.95. Results also hold 0.80.

Source. [**C**] This document.

Implementation. `plant::sens::sweeps`, sweep CAPTURE

Validation. GOLD, test `eq_k_15`. 0.571 and 0.584 Mt y^{-1} at 0.85 and 0.95, and 0.564 at 0.80.

Equation K.16: Availability sensitivity

Tier R

$$\mathcal{A} \propto f_{av} \tag{K.16}$$

Interpretation. Fewer running hours mean fewer tonnes abated. The target margin is wide enough to survive it.

Derivation. Eq. K.3 and Eq. K.8 are both linear in t_h, since baseline and plant emissions run on the same hours.

Variables. f_{av}, availability, 0.65 to 0.90, 1.

Assumptions. Baseline scales too.

Source. [**C**] This document.

Implementation. `plant::sens::sweeps`, sweep AVAILABILITY

Validation. GOLD, test `eq_k_16`. 0.442, 0.510, 0.578 and 0.612 Mt y^{-1} at 0.65, 0.75, 0.85 and 0.90.

Equation K.17: Tail-gas fates

Tier O

$$\mathcal{A}_{fate} = E_{base} - M_{CO_2} t_h \left(\dot{n}_{burn} + \dot{n}_{exp}\right) \tag{K.17}$$

Interpretation. Carbon in the PSA reject has three possible fates: recycle, burn on site or export. Moving it across a boundary never removes it. The atmosphere view counts it wherever it is finally burned.

Derivation. Eq. K.11 with exported carbon added. An export route would look clean at the site boundary, since $\dot{n}_{burn}$ is then zero, yet its carbon still reaches the air. That is why the headline counts the atmosphere. Compressor and offshore storage follow Figure 2 and sit inside the line. Whether carbon stored outside Singapore counts as Singapore abatement is an open question and is not settled here.

Variables. $\mathcal{A}_{fate}$, abatement by tail-gas fate, t y^{-1}; $\dot{n}_{exp}$, carbon leaving the boundary as fuel gas, mol s^{-1}.

Assumptions. Atmosphere view.

Source. [**C**] This document.

Implementation. `plant::sens::sweeps`, sweep TAIL_GAS, on the scenarios of `plant::anchored::Model::carbon`

Validation. GOLD, test `eq_k_17`. Case B: recycle 0.578, burn on site 0.467, export 0.467 Mt y^{-1}. Burn on site with gas held instead of hydrogen gives 0.357.

Equation K.18: System view

Tier O

$$\mathcal{A}_{sys} = \mathcal{A} - E_{lost\,steam} + E_{base\,power} \tag{K.18}$$

Interpretation. The plant diverts export steam that a neighbour must then raise by burning gas, while an unabated SMR would have bought grid power. This view counts both.

Derivation. Lost steam over a 90% boiler at 56.1 kg GJ^{-1} LHV gives $E_{lost\,steam}$. Baseline grid power of 6.3 MW_e scaled by s at 0.402 t MWh^{-1} gives $E_{base\,power}$.

Variables. $\mathcal{A}_{sys}$, system abatement, t y^{-1}; $E_{lost\,steam}$, CO_2 to replace lost steam, t y^{-1}; $E_{base\,power}$, CO_2 of the avoided baseline grid power, t y^{-1}.

Assumptions. Boiler 90%, 56.1 kg GJ^{-1} LHV.

Source. [**C**] This document. The emission factor is the IPCC default as carried in the project inputs. The EU ETS table of the GHG Protocol guidance, Table 14, PDF p. 50, was checked for its location and the factor was not read from it. [**M27**] for the grid factor.

Implementation. `plant::emissions::Model::system_adjust`

Validation. GOLD, test `eq_k_18`. 0.528 Mt y^{-1}.

Equation K.19: Lifecycle view

Tier O

$$\mathcal{A}_{LC} = \mathcal{A} + E_{up} - E_{nuc} - E_{TS} \tag{K.19}$$

Interpretation. A sensitivity only. Saved gas also saves its upstream emissions, while the nuclear chain and storage have small burdens of their own.

Derivation. Upstream gas at 11.5 kg GJ^{-1} on the gas saved, nuclear at 5.5 kg MWh^{-1} on electricity-equivalent output at the charged rating, and transport and storage at 2.5% of the tonnes stored.

Variables. $\mathcal{A}_{LC}$, lifecycle abatement, t y^{-1}; E_{up}, saved upstream gas emissions, t y^{-1}; E_{nuc}, nuclear lifecycle burden, t y^{-1}; E_{TS}, transport and storage burden, t y^{-1}.

Assumptions. Sensitivity only. The factors were not re-read at source.

Source. [**C**] This document. Factors as carried in the project inputs.

Implementation. `plant::emissions::Model::lifecycle_terms`

Validation. GOLD, test `eq_k_19`. 0.616 Mt y^{-1} with saved gas counted, 0.566 with burdens only.

Equation K.20: Backup firing

Tier O

$$e_{bk} = e_{site} + \frac{\dot{Q}_{proc} EF}{\eta_{bk} \dot{m}_{H_2}} \tag{K.20}$$

Interpretation. Central hydrogen runs only in the 85% availability hours. If the plant kept running through reactor outages, it would burn gas for process heat, and the intensity of those hours would rise sharply.

Derivation. Site intensity plus fired heat per kilogram of H_2 at boiler efficiency η_{bk} and emission factor EF. Computed here: $e_{site} = 0.572$ and the fired term is 3.19, so $e_{bk} = 3.77$ kg kg^{-1}, or 44,476 t y^{-1} on the extra H_2. The engine runs this card as its heater route: 44,476 t y^{-1} emitted and 64,293 t y^{-1} of added abatement. It also runs two conventional-train routes at INL's rates: 29,017 t y^{-1} emitted with capture, adding 79,753, and 108,770 unabated, adding 0. Over 1,314 h y^{-1} every route adds 11,810 t y^{-1} of H_2. Furnace and heater capital are unpriced.

Variables. e_{bk}, intensity in backup hours, kg kg^{-1}; e_{site}, site intensity, kg kg^{-1}; EF, combustion emission factor, kg J^{-1}; η_{bk}, backup firing efficiency, 0.85, 1.

Assumptions. Sensitivity only, by project decision. $\eta_{bk} = 0.85$ assumed.

Source. [**M29**]. The need for backup is stated in IAEA NP-T-4.3, PDF p. 23.

Implementation. `plant::emissions::Model::backup_heater`, with the conventional-train routes in `plant::emissions::Model::backup`

Validation. NONE, test `eq_k_20`. No benchmark. The hand route above is an independent check on the engine heater route.

Equation K.21: SMR plus CCS comparator

Tier O

$$\eta_{cmp} = \frac{\dot{E}_{cap}^{cmp}}{\dot{E}_{cap}^{cmp} + \dot{E}_{em}^{cmp}} \tag{K.21}$$

Interpretation. One labelled alternative at the same hydrogen rate, so the reader sees what plain capture on a fired plant achieves. Two published cases are in view, one partial and one near full capture.

Derivation. INL's conventional plant with capture stores 2,143 and emits 855 st d^{-1}, which gives 71.5%. IEAGHG Case 3, MEA on flue gas at 100,000 Nm^3 h^{-1}, captures 90.0% with 0.0888 kg Nm^{-3} emitted. The cost row is Eq. L.32.

Variables. η_{cmp}, comparator capture, 1; $\dot{E}_{cap}^{cmp}$, comparator CO_2 captured, kg s^{-1}; $\dot{E}_{em}^{cmp}$, comparator CO_2 emitted, kg s^{-1}.

Assumptions. New-build, same H_2, one labelled row.

Source. [**P**] TEV-953 [35], Table 2, PDF p. 30. IEAGHG 2017-02 [1], Table 2, PDF p. 16.

Implementation. `plant::comparator::Model::comparator`

Validation. PUB, test `eq_k_21`. Computed here: 71.5% for INL, and 0.536 Mt y^{-1} for IEAGHG Case 3 against its own baseline at this availability.

Equation K.22: Target test

Tier R

$$\mathcal{A} > 0.25 \text{ Mt y}^{-1} \tag{K.22}$$

Interpretation. The project statement's abatement requirement.

Derivation. Compare Eq. K.11 with the target.

Variables. $\mathcal{A}$, direct abatement, t y^{-1}.

Assumptions. Direct abatement.

Source. [**A**] Project statement.

Implementation. `plant::export::results_block`, which writes the row of abatement above the target

Validation. ID, test `eq_k_22`. 0.578 Mt y^{-1} at Case B, 328,052 t y^{-1} above the target. The lowest mass-balanced case is burn on site with gas held, at 0.357 Mt y^{-1}.

Worked example: abatement at Case B

Eq. K.1 gives 9.21 kg kg^{-1}, and 66,925 t y^{-1} of hydrogen makes the baseline 616,361 t y^{-1} by Eq. K.3. The plant stores 359,777 t y^{-1} and emits 38,309 t y^{-1} from the nitrogen-rich cut. The rest of the baseline never forms as CO_2 because reactor heat replaces the furnace and recycle cuts the gas bought. Eq. K.11 gives 578,052 t y^{-1}, or 0.578 Mt y^{-1}, with per-pass capture 0.90 and plant capture 0.9005.

Link to the sources

INL published a plant emitting 142 st d^{-1} at Case 6 (TEV-961, Table 2, PDF p. 17). Scaled, that is below the residual here, because MEA slip raises the burned cut above the Selexol case. The difference is the price of swapping a physical solvent for MEA, and it is the reason Eq. J.8 carries the recycle explicitly.

14 L. Economics

This section prices the plant by two independent methods that feed one result.

Method 1, annual incremental ledger. The ladder L0 to L5 lists annual cost differences against a same-size unabated SMR, with capital annualised. Eq. L.1 to Eq. L.33 build it. Method 1 answers the question "what does each item add?"

Method 2, discounted cash flow. Eq. L.34 to Eq. L.37 run year-by-year real cash flows at 8% for two plants of equal hydrogen output on a common horizon, the unabated SMR and NUS-H2. Method 2 answers the question "what does each plant cost, and what does the difference buy?"

The central ledger is Case B with recycle, $n_s = 1.0$ with 0.6 and 0.31 at equal prominence, bottom-up capture and T&S at S$60/t. Prices are in 2025 S$. The headline is the full cost set, which is L5 and the DCF, with the reactor charged at 210 MWth at BOAK and FOAK, at 25 y and 60 y. L3 prints beneath it as a named step. Carbon tax stays out of the headline. The S$100/t target is tested at the end of the section, and no claim is made unless a computed central case passes.

Method 1: annual incremental ledger

Equation L.1: Capital recovery factor

Tier R

$$CRF = \frac{i}{1 - (1+i)^{-t_\ell}} \tag{L.1}$$

Interpretation. The level annual payment that repays one dollar of capital over the life, with interest. It works like a mortgage payment.

Derivation. A unit payment at each year end has present value $\sum_{y=1}^{t_\ell}(1+i)^{-y} = [1-(1+i)^{-t_\ell}]/i$, a geometric sum. Setting that annuity times the payment equal to the capital and solving for the payment gives the factor.

Variables. CRF, capital recovery factor, y^{-1}; i, real discount rate, 1; t_ℓ, life or horizon, y; y, year index, y.

Assumptions. End-year payments.

Source. [**T**] Standard annuity result.

Implementation. `plant::anchored::crf`

Validation. PUB, test `eq_l_01`. Computed here at 8%: 0.09368, 0.08386 and 0.08080 at 25, 40 and 60 y. The results row SCREENING_CRF_25 records the first.

Equation L.2: Annualised capital

Tier R

$$C_{ann} = C\,CRF \tag{L.2}$$

Interpretation. Turns a lump of capital into a yearly cost that can sit beside fuel and labour.

Derivation. Definition from Eq. L.1.

Variables. C_{ann}, annualised capital, S$ y^{-1}; C, capital, S$.

Assumptions. Constant real payment.

Source. [**T**] Standard annuity result.

Implementation. `plant::economics::Model::ledger`

Validation. ID, test `eq_l_02`. Unit identity.

Equation L.3: Sinking fund factor

Tier R

$$SFF = \frac{i}{(1+i)^{t_\ell}-1} \tag{L.3}$$

Interpretation. The yearly deposit that grows to one dollar at the end of the life. Long lives make it tiny, which is why decommissioning weighs so little at 8%.

Derivation. Deposits at year ends grow to $[(1+i)^{t_\ell}-1]/i$ by the horizon, a geometric sum. Inverting gives the factor. It equals $CRF\,(1+i)^{-t_\ell}$.

Variables. SFF, sinking fund factor, y^{-1}.

Assumptions. End-year deposits at the discount rate.

Source. [**T**] Standard annuity result.

Implementation. `plant::anchored::sff`

Validation. GOLD, test `eq_l_03`. Computed here: 0.01368 at 25 y and 0.00080 at 60 y.

Equation L.4: Interest during construction

Tier R

$$f_{IDC} = \frac{(1+i)^{t_c}-1}{i\,t_c} \tag{L.4}$$

Interpretation. Money spent during a long build sits idle until the plant runs. This factor says how much the overnight cost grows by the first day of operation.

Derivation. Equal spends at year ends reach $(C/t_c)[(1+i)^{t_c}-1]/i$ at completion, the future value of an annuity. Dividing by C gives the factor. It is applied at a non-integer t_c as a smooth extension, which is one source of the gap in Eq. L.37.

Variables. f_{IDC}, interest during construction factor, 1; t_c, construction time, 5.5, y.

Assumptions. Uniform spend, 5.5 y base case.

Source. [**P**] Construction time from INL/RPT-23-74378 (NZW09), Table ES-1, PDF p. 7. [**C**] for the factor.

Implementation. `plant::economics::Model::idc_factor`

Validation. GOLD, test `eq_l_04`. Computed here: 1.198 at 8% and 5.5 y.

Equation L.5: Overnight capital, no turbine

Tier R

$$c_{kWt} = c_{kWe}\,(1 - 0.09) \cdot 0.40 \tag{L.5}$$

Interpretation. INL quotes reactors per electrical kilowatt. A heat-only plant buys no turbine, so that share comes off, and 40% efficiency converts to thermal kilowatts.

Derivation. INL's worked example for a non-electric HTGR, steps 3 and 4. It uses low and high OCC of 4,000 and 7,000 US$$_{2019}$ kWe^{-1}. The project applies the same recipe to the 6,000 medium value of Table 1.

Variables. c_{kWt}, overnight capital cost, no turbine, US$$_{2019}$ kWt^{-1}; c_{kWe}, BOAK overnight capital cost, US$$_{2019}$ kWe^{-1}.

Assumptions. Turbine share 9%, efficiency 40%.

Source. [**P**] INL/RPT-23-72972 Rev 3, PDF p. 47 and Table 1, PDF p. 5. [**C**] for the central value.

Implementation. `plant::economics::Model::reactor_capex`, on the overnight cost per kWt of the registry

Validation. PUB, test `eq_l_05`. INL prints US$1,456 and US$2,548 per kWt. The recipe gives US$2,184 at the medium value.

Equation L.6: Reactor capital

Tier O

$$C_R = c_{kWt}\dot{Q}_{ref}F_{19}m_{stage}\left(\frac{\dot{Q}_{rated}}{\dot{Q}_{ref}}\right)^{n_s} \tag{L.6}$$

Interpretation. Cost per kilowatt at a reference size, re-based to 2025 S$, adjusted for build stage, then scaled to the rating. With $n_s = 1$ the reference size drops out and the cost is linear in rating.

Derivation. Reference cost is $c_{kWt}\dot{Q}_{ref}$. Scaling by a power law in size gives the bracket. INL's survey found no significant difference in specific cost between large reactors and small modular reactors, which is the basis for $n_s = 1$ as central.

Variables. C_R, reactor capital, S$; F_{19}, US$2019 to S$2025 factor, 1.6195, S$ US$$^{-1}$; m_{stage}, stage multiplier, 1; $\dot{Q}_{ref}$, reference size, 600, MW$_{\text{th}}$; n_s, size exponent, 1.

Assumptions. $\dot{Q}_{ref} = 600$ MWth assumed, with no effect at $n_s = 1$.

Source. [**P**] INL/RPT-23-72972 Rev 3, PDF p. 27, p. 45, p. 47. Size exponent central by project decision.

Implementation. `plant::economics::Model::reactor_capex`

Validation. GOLD, test `eq_l_06`. BOAK at 210 MW$_{\text{th}}$: S$742.8M, S$1,130.4M and S$1,531.5M at $n_s = 1.0$, 0.6 and 0.31.

Equation L.7: Stage multiplier

Tier O

$$m_{stage} = 1.0 \text{ (BOAK)}, \qquad m_{stage} = 1.6 \text{ (FOAK, 1.3 to 2.1)} \tag{L.7}$$

Interpretation. INL's cost table describes a build between first and Nth of a kind. A first plant in a new country sits nearer FOAK, so both are printed.

Derivation. Read directly from INL. BOAK is the base of Table 1, and FOAK is the premium range of Table 11. NZW09 gives 1.5 as a single value (Table ES-1, PDF p. 7).

Variables. m_{stage}, stage multiplier, 1.

Assumptions. BOAK sits between first and Nth of a kind.

Source. [**P**] INL/RPT-23-72972 Rev 3, Table 1, PDF p. 5, and Table 11, PDF p. 40.

Implementation. `plant::economics::Model::stage_mult`

Validation. PUB, test `eq_l_07`. Published values printed.

Equation L.8: Learning from BOAK

Tier O

$$m_{stage} = 0.9^{\log_2 10} = 0.7047 \tag{L.8}$$

Interpretation. A 10% cost fall per doubling, carried to the tenth unit from the BOAK cost.

Derivation. INL's multi-unit example applies $BOAK \times (1-0.1)^{\log_2 N}$ to second and third units. Taken to $N = 10$ this gives the multiplier.

Variables. N, unit number, 10, 1.

Assumptions. Printed as a route, never as a headline.

Source. [**P**] INL/RPT-23-72972 Rev 3, PDF p. 46.

Implementation. `plant::economics::Model::stage_mult`, stage 10OAK from BOAK

Validation. PUB, test `eq_l_08`. Computed here: US$1,539 per kWt on the central OCC.

Equation L.9: Learning from FOAK

Tier O

$$m_{stage} = 1.6 \cdot 0.7047 = 1.128 \tag{L.9}$$

Interpretation. The same learning, started from the FOAK cost as in INL's non-electric example. The order of the routes is an artefact: this tenth unit costs more than a BOAK plant.

Derivation. INL applies $FOAK \times (1-LR)^{\log_2 N}$ to its FOAK range of US$1,892 to US$5,350 per kWt.

Variables. m_{stage}, tenth-unit multiplier from FOAK, 1.

Assumptions. Stage order on this route is an artefact.

Source. [**P**] INL/RPT-23-72972 Rev 3, PDF p. 47.

Implementation. `plant::economics::Model::stage_mult`, stage 10OAK from FOAK

Validation. PUB, test `eq_l_09`. Computed here: US$2,462 per kWt on the central OCC. On INL's range, 3,770 and 1,333, against the rounded US$4,000 and US$1,500 printed. The results rows SCREENING_OAK10_INL_HIGH and LOW record both.

Equation L.10: Size exponent

Tier E

$$n_s \in \{1.0,\ 0.6,\ 0.31\}, \qquad 0.31 = 1 + \frac{\ln(26553/18313)}{\ln(350/600)} \tag{L.10}$$

Interpretation. How fast specific cost rises as a reactor gets smaller. The value drives the result more than any other input, so the three are printed at equal prominence.

Fit range and origin. 1.0 is INL's recipe with no size penalty. 0.6 is the classic chemical-plant rule, assumed. 0.31 follows from two NGNP FOAK points: $c \propto \dot{Q}^{n_s-1}$ gives the fraction, with specific costs of US$18,313 and US$26,553 per kWe at 600 and 350 MWth. It is a weak adverse bound from single-unit FOAK estimates that lie well above the survey range.

Variables. n_s, size exponent, 1.

Assumptions. Equal prominence, by project decision. 0.31 rests on two points.

Source. [**P**] INL/RPT-23-72972 Rev 3, PDF p. 25 for the two costs. [**A**] for 0.6.

Implementation. `plant::economics::Model::reactor_capex`, argument n_s

Validation. GOLD, test `eq_l_10`. Computed here: 0.3107.

Equation L.11: Reactor O&M

Tier O

$$C_{OM} = c_{OM} F_{19} t_h \left[f_{fuel} \dot{Q} + (1 - f_{fuel}) \dot{Q}_{ref} \left(\frac{\dot{Q}}{\dot{Q}_{ref}} \right)^{n_s} \right] \tag{L.11}$$

Interpretation. Fuel scales with heat produced. Staff and the rest scale with plant size by the same exponent as capital.

Derivation. INL converts OPEX of US$15 to US$35 per MWh to US$6 to US$14 per MWt-h at 40% efficiency, so the medium US$25 is US$10. The HTGR fuel share is 44.38% of OPEX.

Variables. C_{OM}, reactor O&M and fuel, S$ y^{-1}; c_{OM}, O&M rate, 10, US$$_{2019}$ $MWth^{-1}$ h^{-1}; f_{fuel}, fuel share of OPEX, 0.4438, 1; $\dot{Q}$, charged reactor heat, MW_{th}.

Assumptions. Charged on the rating.

Source. [**P**] INL/RPT-23-72972 Rev 3, Table 9, PDF p. 37, and PDF p. 47.

Implementation. `plant::economics::Model::reactor_om`

Validation. GOLD, test `eq_l_11`. S$25.32M per year at 210 MW_{th}.

Equation L.12: Capture capital

Tier O

$$C_{cap} = K_{cap} \dot{m}_{CO_2} \tag{L.12}$$

Interpretation. Capital per tonne-per-hour captured, taken from a real difference between two costed plants.

Derivation. IEAGHG Case 1A less the Base Case: TCR of 263.91 less 222.89 is EUR 41.02M at 46.6 t h^{-1}. Re-based at 2.322, that is 2.04 S$M per t h^{-1}. The low value takes the 15% class bound, 1.74. The high value uses Case 3, MEA on flue gas, at 5.09. Case 1A uses aMDEA on syngas, so the central value is a proxy for MEA on syngas.

Variables. C_{cap}, capture capital, S$; K_{cap}, capture capital per unit rate, S$ t^{-1} h.

Assumptions. aMDEA-based central, flue-gas MEA high.

Source. [**P**] IEAGHG 2017-02 [1], Table 2, PDF p. 16, and Table 3, PDF p. 17. [**C**] for the ratio.

Implementation. `plant::economics::Model::cost_items`, line CAPTURE_CAP of `plant::economics::Model::ledger`

Validation. GOLD, test `eq_l_12`. S$98.8M, or S$9.25M per year on CRF at 25 y.

Equation L.13: Capture fixed O&M

Tier F

$$C_{FOM} = 0.027\, C_{cap} \tag{L.13}$$

Interpretation. Labour, maintenance and insurance for the capture plant as a share of its capital.

Justification. Central 2.7% per year, derived from the IEAGHG cases. The high case uses 6%, the maintenance share in Ahn and Lee.

Variables. C_{FOM}, capture fixed O&M, S$ y^{-1}.

Assumptions. 2.7% central, 6% high.

Source. [**A**] Project value. [**P**] Ahn and Lee [18], Table 3, PDF p. 5, for 6%.

Implementation. `plant::economics::Model::cost_items`, line CAPTURE_FOM of `plant::economics::Model::ledger`

Validation. GOLD, test `eq_l_13`. S$2.67M per year.

Equation L.14: Variable O&M

Tier F

$$C_{VOM} = 0 \text{ (central)}, \qquad C_{VOM} = 0.1 \text{ EUR M y}^{-1} \cdot F_{14} \text{ (sensitivity)} \tag{L.14}$$

Interpretation. MEA make-up and reclaiming are real costs that the ledger leaves unpriced at the centre. The sensitivity shows they are small.

Justification. IEAGHG fixes the cost of chemicals at EUR 100,000 per year for all cases. Re-based, that is 0.40 S$ t^{-1} of abatement.

Variables. C_{VOM}, capture variable O&M, S$ y^{-1}; F_{14}, EUR2014 to S$2025 factor, 2.322, S$ EUR^{-1}.

Assumptions. Disclosed sensitivity only.

Source. [**P**] IEAGHG 2017-02 [1], PDF p. 13.

Implementation. Not implemented in the engine at this build: the ledger carries no variable O&M line. Planned name `plant::economics::vom`.

Validation. NONE, test `eq_l_14`. Computed here only. The line is not in the results.

Equation L.15: Transport and storage

Tier F

$$C_{TS} = p_{TS} E_{cap} \tag{L.15}$$

Interpretation. Every tonne stored pays a tariff. Singapore has no store of its own, so this price is an assumption with a wide range.

Justification. Central S$60/t, from S$23.2/t low (IEAGHG EUR 10/t re-based) to S$112/t high (ERIA integrated transport figure read from a chart). Capture energy stays in the heat and power balance, so no all-in capture price is added.

Variables. C_{TS}, transport and storage cost, S$ y^{-1}; p_{TS}, transport and storage price, S$ t^{-1}.

Assumptions. Project value. No all-in capture price on top.

Source. [**A**] Project value. [**P**] IEAGHG 2017-02 [1], report basis, and ERIA 2024, Fig 2.13, as carried in the project inputs.

Implementation. `plant::economics::Model::cost_items`, line TS of `plant::economics::Model::ledger`

Validation. GOLD, test `eq_l_15`. S$21.59M per year.

Equation L.16: Gas saving

Tier R

$$C_{gas} = -\left(52.5\,s - 34.0\,s\right) HHV\, d\, p_{gas} \tag{L.16}$$

Interpretation. The main benefit of the plant: gas no longer burned or fed. It enters the ledger as a negative cost.

Derivation. The unabated SMR uses 52.5 MMSCFD at Case A and the recycle plant 34.0. Scale both, convert to energy at 1,044 Btu scf^{-1} and price over the operating days.

Variables. C_{gas}, gas cost difference, S\$ y^{-1}; HHV, gas higher heating value, J m^{-3}; d, operating days, 310.25, d y^{-1}; p_{gas}, gas price, 15, S\$ GJ^{-1}.

Assumptions. Same H_2. Gas priced on HHV.

Source. [**P**] TEV-953 [35], Table 2, PDF p. 30. TEV-961 [36], Table 2, PDF p. 17, and A-50, PDF p. 74.

Implementation. `plant::economics::Model::cost_items`, line GAS of `plant::economics::Model::ledger`

Validation. GOLD, test `eq_l_16`. S\$-64.80M per year on 12.6 MMSCFD saved.

Equation L.17: Grid electricity

Tier O

$$C_{el} = E_{imp} t_h p_{el} \tag{L.17}$$

Interpretation. The plant supplies itself at Case B, so this line is active only when the rating runs short.

Derivation. Energy times price, with Eq. J.29 for the import.

Variables. C_{el}, imported electricity cost, S\$ y^{-1}; p_{el}, electricity price, 0.20, S\$ kWh^{-1}.

Assumptions. [**M27**]: the price was not re-checked at source.

Source. [**A**] Project value.

Implementation. Not implemented as a ledger line at this build: the import at the rating is 0.0 MW_e at Case B, so the line would be zero. Planned name `plant::economics::electricity`.

Validation. GOLD, test `eq_l_17`. 0.0 MW_e imported at Case B, so the line is zero.

Equation L.18: IHX and secondary loop

Tier E

$$C_{ihx} = C_a \frac{\dot{Q}_{proc} + \dot{Q}_{rec}}{170\ \text{MW}} \tag{L.18}$$

Interpretation. The exchanger and secondary helium loop scale with duty from a single published loop cost.

Fit range and origin. US\$69M at about 2012 for 170 MW, escalated by 128.885 over 93.1835 and converted at 1.3065, gives C_a = S\$124.7M. At Case B, C_{ihx} = S\$89.2M, annualised at 25 y. The anchor is a generic estimate, which sets the fit range: one point.

Variables. C_{ihx}, IHX and loop capital, S\$; C_a, loop cost anchor, S\$.

Assumptions. Linear in duty. 25 y life.

Source. [**M23**]: IAEA-TECDOC-1682 Table 3.17, carried from earlier project files and not opened at source.

Implementation. `plant::economics::Model::cost_items`, line IHX of `plant::economics::Model::ledger`

Validation. GOLD, test `eq_l_18`. S\$8.36M per year.

Equation L.19: Licensing

Tier E

$$C_{lic} = C_{NRC}\, FX\, CRF \tag{L.19}$$

Interpretation. A first-licence country has work the INL estimate does not hold. A design certification fee stands in for it, as a proxy.

Fit range and origin. US$50M central, 0 to 137.5. INL's OCC already holds plant licensing at 0.56% and site permits at 0.23%, so the line overlaps by up to 0.79% of OCC. Setting up a regulator is not priced. The fit range is one fee.

Variables. C_{lic}, annualised licensing cost, S$ y^{-1}; C_{NRC}, design certification fee proxy, US$; FX, S$ per US$, 1.3065, S$ US$$^{-1}$.

Assumptions. Proxy. No deflator applied.

Source. [**M23**]: NRC blog summary, carried from earlier project files. [**P**] INL/RPT-23-72972 Rev 3, PDF p. 34, for the overlap.

Implementation. `plant::economics::Model::ledger`, line LICENCE

Validation. GOLD, test `eq_l_19`. S$6.12M per year at 25 y, S$5.28M at 60 y.

Equation L.20: Decommissioning

Tier E

$$C_{dec} = c_{dec}\left(0.40\,\dot{Q}\right) FX\, SFF \tag{L.20}$$

Interpretation. A fund built over the life pays for dismantling at the end. Discounting makes the line almost vanish at 60 y.

Fit range and origin. NZW09 gives US$750, 1,000 and 1,250 per kWe in 2022 dollars. Applied at 0.40 kWe kWt^{-1}, the central total is S$109.7M, then Eq. L.3. The data are US LWR history, which sets the fit range, and graphite and TRISO disposal are not priced.

Variables. C_{dec}, decommissioning fund payment, S$ y^{-1}; c_{dec}, decommissioning cost, US$$_{2022}$ kWe^{-1}.

Assumptions. LWR history. 3% fund rate in the high case.

Source. [**P**] INL/RPT-23-74378 (NZW09), Table ES-1, PDF p. 7.

Implementation. `plant::economics::Model::ledger`, line DECOM

Validation. GOLD, test `eq_l_20`. S$1.50M per year at 25 y, S$0.09M at 60 y.

Equation L.21: IDC line

Tier O

$$C_{IDC} = C_R\,(f_{IDC} - 1)\, CRF \tag{L.21}$$

Interpretation. The ledger shows interest during construction as its own rung, so the reader can see it.

Derivation. The excess of Eq. L.4 over one, annualised with Eq. L.1 at the reactor life. Applied to the reactor only.

Variables. C_{IDC}, annualised construction interest, S$ y^{-1}.

Assumptions. Rung L4.

Source. [**C**] This document.

Implementation. `plant::economics::Model::ledger`, line IDC, with `plant::economics::Model::idc_factor`

Validation. GOLD, test `eq_l_21`. S$13.75M per year at 25 y BOAK.

Equation L.22: Power island

Tier O

$$C_{pi} = c_{pi} E_{tot} F_{19} m_{stage} \tag{L.22}$$

Interpretation. Eq. L.5 took the turbine out, yet the plant does make its own electricity. This line puts back a turbine for the megawatts actually generated.

Derivation. 9% of US$6,000 per kWe gives US$540, applied to Eq. J.24, annualised at the reactor life. The stage multiplier applies at the central cost level.

Variables. C_{pi}, power island capital, S$; c_{pi}, turbine share, 540, US$$_{2019}$ kWe^{-1}.

Assumptions. Rung L4.

Source. [**C**] From Eq. L.5.

Implementation. `plant::economics::Model::ledger`, line POWER_ISLAND

Validation. GOLD, test `eq_l_22`. S$1.62M per year at 25 y BOAK, on capital of S$17.3M.

Equation L.23: Recycle train

Tier E

$$C_{rec} = C_{rec,A}\, s, \qquad C_{ann,rec} = C_{rec}\,(CRF_{25} + 0.027) \tag{L.23}$$

Interpretation. Recycling tail gas needs compressors and a cryogenic unit that no INL table prices. A compression cost per megawatt stands in for them.

Fit range and origin. IEAGHG Case 1A prices CO_2 compression at EUR 17.0M total plant cost. Over INL's 3.32 kW per st d^{-1} at 46.6 t h^{-1}, that is a cost per MW. Applied to 5.7 MWe, with TCR over TPC of 1.3 and re-based at 2.322, this gives S$71.5M at Case A and S$48.8M at Case B. The fit range is one proxy, and the high case uses IEAGHG's cryogenic block.

Variables. C_{rec}, recycle train capital, S$; $C_{rec,A}$, recycle train capital at Case A, S$; $C_{ann,rec}$, recycle train annual line, S$ y^{-1}.

Assumptions. No published cost [**M31**].

Source. [**C**] From IEAGHG 2017-02 [1], PDF p. 223 for the compression cost, as carried in the project inputs.

Implementation. `plant::economics::Model::cost_items`, line RECYCLE_TRAIN of `plant::economics::Model::ledger`

Validation. GOLD, test `eq_l_23`. S$5.89M per year.

Equation L.24: Lost steam and baseline power

Tier O

$$C_{st} = \left(\dot{Q}_{exp}^{conv} - \dot{Q}_{exp}^{kept}\right) t_h p_{gas}, \qquad C_{pw} = -E_{conv} t_h p_{el} \tag{L.24}$$

Interpretation. An unabated SMR sells export steam and buys grid power. NUS-H2 spends its steam in the reboiler and buys no power. Charging both differences is what moves L4 to L5.

Derivation. 189.2 MMBtu h^{-1} scaled by s is the steam of the unabated plant. At Case B all of NUS-H2's export steam goes to the reboiler, so none is kept. Steam is valued at the gas price. The baseline plant buys 6.3 MW$_e$ scaled by s.

Variables. C_{st}, lost steam line, S$ y^{-1}; C_{pw}, baseline power line, S$ y^{-1}; $\dot{Q}_{exp}^{conv}$, export steam of the unabated SMR, W; $\dot{Q}_{exp}^{kept}$, export steam the plant keeps, W; E_{conv}, electricity of the unabated SMR, W.

Assumptions. Steam valued at gas price. Rung L5.

Source. [**P**] TEV-953 [35], Table 2, PDF p. 30.

Implementation. `plant::economics::Model::ledger`, lines LOST_STEAM and BASE_POWER

Validation. GOLD, test `eq_l_24`. S$15.23M and S$-6.41M per year.

Equation L.25: Incremental cost ladder

Tier R

$$\Delta C_k = \sum_{j \in L_0 \cup \cdots \cup L_k} C_j, \qquad k = 0, \ldots, 5 \tag{L.25}$$

Interpretation. Each rung adds one group of real costs, so the reader sees exactly what moves the result.

Derivation. The rungs are fixed: L0 gas, reactor capital, capture capital, capture fixed O&M and T&S. L1 adds reactor O&M. L2 adds the IHX. L3 adds licensing and decommissioning. L4 adds IDC, the power island and the recycle train. L5 adds lost steam, baseline power and any export revenue.

Variables. ΔC_k, cumulative annual cost difference at rung k, S$ y^{-1}; C_j, ledger line j, S$ y^{-1}.

Assumptions. Annual differences against the same-size unabated SMR.

Source. [**C**] This document.

Implementation. `plant::economics::Ledger::ladder`

Validation. GOLD, test `eq_l_25`. S$79.60M per year at L3 and S$109.69M per year at L5, BOAK, 25 y. Validation results, 1 row (GOLD_LEDGER_GRID): 1 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation L.26: Abatement cost

Tier R

$$AC = \frac{\Delta C}{\mathcal{A}} \tag{L.26}$$

Interpretation. The extra cost per tonne kept out of the air, the measure set against the S$100/t target.

Derivation. Definition, annual cost over annual tonnes.

Variables. AC, abatement cost, S$ t^{-1}; ΔC, annual cost difference, S$ y^{-1}.

Assumptions. Direct abatement.

Source. [**C**] This document.

Implementation. `plant::economics::Ledger::l3` and `plant::economics::Ledger::l5`, over the abatement of `plant::emissions::Model::emissions`

Validation. GOLD, test `eq_l_26`. L3 at 210 MW_{th}: 137.7 and 117.2 at 25 and 60 y BOAK, 209.9 and 179.5 FOAK. L3 at 178.5 MW_{th}: 112.7 and 95.1 BOAK. L5 values are in Eq. L.27. The screening comparison of L3: Validation results, 6 rows (HEADLINE_L3_BOAK_25Y_210 to HEADLINE_L3_FOAK_60Y_210): 6 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Equation L.27: Size and stage table

Tier R

$$AC = AC\left(n_s, \text{stage}, t_\ell\right) \tag{L.27}$$

Interpretation. The side-by-side table of the choices that matter most, printed so that no single exponent looks like the answer.

Derivation. Evaluate Eq. L.26 over the grid in the table below, with the same abatement throughout.

Variables. n_s, size exponent, 1.

Assumptions. Equal prominence, by project decision.

Source. [**C**] This document.

Implementation. `plant::economics::Model::ledger`, run over the grid of stages, lives and n_s

Validation. GOLD, test `eq_l_27`. The table below, read from the results.

Stage	Life	L3, S$/t $n_s = 1.0$	0.6	0.31	L5, S$/t $n_s = 1.0$	0.6	0.31
BOAK	25 y	137.7	213.2	291.4	189.7	277.7	368.7
BOAK	60 y	117.2	184.1	253.4	165.6	243.2	323.6
FOAK	25 y	209.9	323.1	440.3	277.9	411.0	548.8
FOAK	60 y	179.5	278.9	381.8	241.7	358.2	478.8

Equation L.28: Levelised form of the ledger

Tier O

$$LAC = \frac{\mathrm{NPV}\,(\Delta C)}{\mathrm{NPV}\,(\mathcal{A})} \tag{L.28}$$

Interpretation. Cost and tonnes are both spread over time, so both are discounted. Method 1 is the special case in which every year looks the same.

Derivation. If ΔC and $\mathcal{A}$ are constant from year 1 to t_ℓ, each NPV is the constant times the same annuity sum, the sum cancels, and $LAC = AC$ of Eq. L.26. Method 1 reaches that form by annualising capital. Method 2 keeps the true timing, and Eq. L.36 evaluates it year by year.

Variables. LAC, levelised abatement cost, S$ t^{-1}; NPV, net present value, S$.

Assumptions. Tonnes discounted at the same rate as cost.

Source. [**C**] This document.

Implementation. `plant::dcf::Model::dcf`

Validation. ID, test `eq_l_28`. Identity at constant flows. The compared values are in Eq. L.37.

Equation L.29: One-at-a-time sensitivity

Tier O

$$AC_q = AC\,(q)\,, \qquad \text{other inputs central} \tag{L.29}$$

Interpretation. Change one input at a time and read the effect. Gas price, T&S and availability matter most after the reactor cost itself.

Derivation. Rerun the ledger at each value of input q. Reactor capital moves with INL's low and high overnight cost, and capture capital runs from the low factor to IEAGHG Case 3, MEA on flue gas.

Variables. q, swept input, various.

Assumptions. Central set otherwise. BOAK, $n_s = 1.0$, 210 MWth.

Source. [**C**] This document.

Implementation. `plant::sens::sweeps`

Validation. GOLD, test `eq_l_29`. The table below, read from the sensitivity results.

Change	L3, 25 y	L3, 60 y	L5, 25 y	L5, 60 y
Central	137.7	117.2	189.7	165.6
Gas S$10/GJ	175.1	154.6	218.3	194.2
Gas S$20/GJ	100.3	79.9	161.2	137.1
T&S S$23.2/t	114.8	94.3	166.8	142.7
T&S S$112/t	170.1	149.6	222.1	198.0
Availability 0.70	173.8	149.0	233.8	204.5
Availability 0.90	128.3	109.0	178.3	155.6
Baseline 9.35	135.5	115.4	186.7	163.0
Baseline 9.00	141.2	120.2	194.6	169.8
Process heat 143.4 MWth	140.4	119.9	192.6	168.5
Charged at 178.5 MWth	112.7	95.1	161.2	140.4
Reactor OCC US$1,456/kWt	97.6	82.6	141.7	124.2
Reactor OCC US$2,548/kWt	157.8	134.5	213.8	186.4
Capture capital S$1.74M per t/h	134.6	114.2	186.7	162.5
Capture capital S$5.09M per t/h	168.5	148.0	220.5	196.4
No recycle, tail gas burned	252.9	227.6	283.1	253.4

Equation L.30: Discount rate sweep

Tier O

$$AC\,(i) \text{ through } CRF\,(i)\,,\ SFF\,(i)\,,\ f_{IDC}\,(i) \tag{L.30}$$

Interpretation. A nuclear plant is capital heavy, so the rate moves its cost more than almost anything else.

Derivation. Only the capital lines depend on i. Rebuild them with Eq. L.1, Eq. L.3 and Eq. L.4 and keep the annual lines fixed.

Variables. i, real discount rate, 1.

Assumptions. 5, 8 and 10%.

Source. [**C**] This document.

Implementation. `plant::sens::sweeps`, sweep DISCOUNT

Validation. ID, test `eq_l_30`. The engine sweep gives L5 BOAK at 5, 8 and 10%: 136.4, 189.7 and 231.4 S$ t^{-1} at 25 y, 104.4, 165.6 and 212.3 at 60 y. Rebuilt here from the ledger lines with Eq. L.1, Eq. L.3 and Eq. L.4: 136.4, 189.7 and 231.4 at 25 y, 104.4, 165.6 and 212.3 at 60 y. The sequence rises with i as it should.

Equation L.31: Levers to the target

Tier O

$$AC\,(q^*) = 100 \text{ S\$ t}^{-1} \tag{L.31}$$

Interpretation. How far one input alone would have to move to reach the target. A lever that lands outside any credible range shows that the target cannot be met on that input alone.

Derivation. The ceiling is $100\mathcal{A}$ = S$57.8M per year. The ledger is linear in gas price, T&S and reactor OCC, so each lever solves directly from the excess over the ceiling.

Variables. q^*, value of an input that meets the target, various.

Assumptions. One lever at a time, stated per lever.

Source. [**C**] This document.

Implementation. `plant::sens::levers`, with the stacked runs `plant::sens::mature_rows`, `plant::sens::stepdown_rows` and `plant::sens::cofunding_rows`

Validation. ID, test `eq_l_31`. The results give, at BOAK and 25 y, L5: gas S$30.7/GJ, OCC US$824/kWt, T&S S$-84.2/t. At 60 y, L5: S$26.5/GJ, US$1,031/kWt, S$-45.5/t. At 25 y, L3: S$20.0/GJ, US$1,500/kWt, S$-0.6/t. A negative T&S price means that lever cannot close the gap

alone. The OCC lever keeps the same stage premium and sits below INL's low value of US$1,456 per kWt at L5. The stacked run MATURE_60Y, 10th of a kind on INL's BOAK route over 60 y with $n_s = 1.0$ and 210 MWth charged, gives S$128.2/t on L5 and S$128.5/t on the DCF at the central gas price of S$15/GJ. Gas at S$20.63/GJ brings L5 to S$96/t, and gas at S$20.68/GJ brings the DCF there. The printed price is the larger of the two solves rounded up to S$0.1/GJ, S$20.7/GJ, where L5 gives S$95.6/t and the DCF S$95.9/t. A higher gas price lowers this cost, because NUS-H2 burns less gas than the unabated plant. The step-down from the base, L5 with the DCF beside it in S$/t: BOAK at 25 y, 189.7 and 190.1; 10th of a kind at 25 y, 146.3 and 146.6; 10th of a kind at 60 y, 128.2 and 128.5; then gas at the printed price, 95.6 and 95.9. Public co-funding of the S$742.8M reactor capital brings BOAK to the same target at 25 y with 65.0% (S$483.0M) on L5 and 65.1% (S$483.6M) on the DCF. At 60 y the shares are 56.0% (S$416.0M) and 56.2% (S$417.1M). These are named scenarios beside the base result.

Equation L.32: SMR plus CCS comparator

Tier O

$$AC_{cmp} = F_{14} CAC_3 + (p_{TS} - p_{TS,3}) \frac{e_{cap,3}}{e_{av,3}} \tag{L.32}$$

Interpretation. What a fired SMR with MEA on its flue gas costs per tonne avoided on the same tariff, one labelled row for comparison.

Derivation. IEAGHG Case 3 prints a CO_2 avoidance cost of EUR 69.8/t with T&S at EUR 10/t. Re-based at 2.322 that is 162.1. Moving T&S to the project tariff multiplies the change by captured over avoided CO_2, 0.8004 over 0.7203 kg Nm^{-3}. Gas price and availability differ from this ledger, so the row is indicative. An earlier lifecycle comparator of this project sits on a different basis and is not carried.

Variables. AC_{cmp}, comparator cost, S$ t^{-1}; CAC_3, IEAGHG Case 3 avoidance cost, EUR t^{-1}; $p_{TS,3}$, T&S price in the IEAGHG case, 23.2, S$ t^{-1}; $e_{cap,3}$, CO_2 captured per unit H_2, kg Nm^{-3}; $e_{av,3}$, CO_2 avoided per unit H_2, kg Nm^{-3}.

Assumptions. One labelled comparator, by project decision.

Source. [**P**] IEAGHG 2017-02 [1], Table 2, PDF p. 16, and Table 5, PDF p. 18. [**C**] for the adjustment.

Implementation. `plant::comparator::Model::comparator`

Validation. PUB, test `eq_l_32`. Computed here: 162.1 as published, 203.0 at S$60/t and 260.8 at S$112/t, with abatement of 0.536 Mt y^{-1} at 85%.

Equation L.33: Guards

Tier F

$$F_{19} = 1.6195, \qquad F_{14} = 2.322, \qquad E_{cap\ energy} \text{ counted once} \tag{L.33}$$

Interpretation. Rules that keep the ledger honest. Capture energy is paid in heat and power, so it is never priced a second time. Every price is carried to one base year, and carbon tax stays outside the ledger.

Justification. F_{19} is the GDP deflator ratio 128.885 over 103.9745 times 1.3065. F_{14} is the corresponding EUR2014 factor from the project inputs. A GDP deflator understates nuclear escalation, so reactor prices err low. Bottom-up capture carries capital and fixed O&M only. Carbon tax credits sit outside every line.

Variables. F_{19}, US$2019 to S$2025 factor, S$ US$$^{-1}$.

Assumptions. GDP deflator.

Source. [**C**] Project inputs.

Implementation. The factors are registry entries read by `plant::anchored::Model::new`. No single guard function exists.

Validation. GOLD, test `eq_l_33`. Inputs read by the model. The results row PARAMS_COMPLETE records that every input is read.

Method 2: discounted cash flow

Year 0 is the end of construction, and flows fall at year ends, so operation runs from year 1 to the horizon. Both plants make the same hydrogen in the same hours. The horizon is the reactor life, 25 y or 60 y. The cards below follow `engine/plant/src/dcf.rs` line by line.

Line	Plant	Timing
SMR island capital	both	Year 0, replaced every 25 y, credit at horizon
SMR island fixed O&M, water and chemicals	both	Years 1 to t_ℓ
Gas; steam credit	both, own values	Years 1 to t_ℓ
Purchased power	SMR	Years 1 to t_ℓ
Reactor capital	NUS-H2	Uniform over 5.5 y to year 0, part year first
Capture capital; IHX; recycle train	NUS-H2	Year 0, replaced every 25 y, credit at horizon
Capture fixed O&M; T&S; reactor O&M; recycle fixed O&M	NUS-H2	Years 1 to t_ℓ
Capture all-in price; export revenue	NUS-H2	Years 1 to t_ℓ, zero in the central case
Licence; power island	NUS-H2	Year 0 once
Decommissioning	NUS-H2	Year t_ℓ, paid from cash flow, no sinking fund

Equation L.34: Discounted cash flow

Tier R

$$\mathrm{NPV}_x = \sum_j \sum_y c_{j,x}\,(y)\,(1+i)^{-y}$$

$$c_j^{rep}\,(y) = \begin{cases} C_j & y = 25m < t_\ell \\ -C_j\,u/25 & y = t_\ell \end{cases}, \qquad u = 25\left\lceil \frac{t_\ell}{25} \right\rceil - t_\ell \tag{L.34}$$

Interpretation. Each plant is a list of dated bills. Discounting brings every bill to year 0 and adds them. Items with a 25 y life are bought again on schedule, and the part of the last unit still unused at the horizon comes back as a straight-line credit.

Derivation. The sum is the definition of present value at real rate i. Reactor capital spreads uniformly over t_c, a part year first as the earliest bucket: $C_R \cdot 0.5/5.5$ at year -5 and $C_R/5.5$ at years -4 to 0. The SMR island enters at year 0 at Total Capital Requirement, because TCR already holds the island's own interest during construction, so spreading it would count that interest twice. IEAGHG's TCR is EUR 222.89M and its fixed O&M EUR 7,545,514 per year. Raw water and chemicals with catalysts, EUR 99,365 and EUR 420,000 per year, are stated at 95% availability and scale to 85%. Re-based at 2.322 these are S$517.6M, S$17.52M per year and S$1.08M per year. Its feedstock, fuel and electricity lines are left out because the gas, steam and power lines cover them. Both plants carry the island unchanged, on the assumption that reformer differences cancel. Replacement runs at $m = 0, 1, \ldots$ while $25m < t_\ell$, so at 60 y units are bought at years 0, 25 and 50, and 15 of the last 25 y are unused, which gives a credit of $0.6C_j$ at year 60. Nothing is bought at the horizon itself.

Variables. NPV_x, present cost of plant x, S$; $c_{j,x}$, cash flow of line j on plant x, S$ y^{-1}; c_j^{rep}, purchase or credit of a replaced item, S$; C_j, capital of item j, S$; u, unused years of the last unit at the horizon, y; m, replacement index, 1.

Assumptions. Real flows at 8%. Whole-year horizon. Costs from the same set as Method 1. Replaced lines: SMR island, capture, IHX, recycle train, each on a 25 y life. At the high cost level only, the IHX life is its rated hours over the operating hours, cut to whole years.

Source. **[P]** IEAGHG 2017-02 [1], Table 3, PDF p. 17, and Table 4, PDF p. 18. INL/RPT-23-74378 (NZW09), Table ES-1, PDF p. 7, for 5.5 y. **[C]** for the method.

Implementation. `plant::dcf::Model::dcf`, with `plant::dcf::uniform_spend` and `plant::dcf::with_replacement`

Validation. ID, test `eq_l_34`. The engine unit tests cover the spend sum, the replacement schedule on both plants, and the 95% to 85% scaling. Computed here from the printed equations, independent of the engine: BOAK present costs of S$2,585M and S$3,758M at 25 y, S$2,998M and S$4,185M at 60 y.

Equation L.35: Levelised cost of hydrogen

Tier R

$$LCOH_x = \frac{\mathrm{NPV}_x}{\mathrm{NPV}\left(M_{H_2}\right)} = \frac{\mathrm{NPV}_x}{M_{H_2} \sum_{y=1}^{t_\ell} (1+i)^{-y}} \tag{L.35}$$

Interpretation. The constant price per kilogram that would pay for the plant exactly. Kilograms are discounted like dollars, since a kilogram sold later is worth less to the plant today.

Derivation. Find the price π for which revenue has the present value of cost: $\sum_y \pi M_{H_2}(1+i)^{-y} = \mathrm{NPV}_x$. Solving gives the statement. The annuity sum is 10.675 at 25 y and 12.377 at 60 y.

Variables. $LCOH_x$, levelised cost of hydrogen of plant x, S$ kg^{-1}.

Assumptions. Same hydrogen on both plants.

Source. [**C**] This document.

Implementation. `plant::dcf::Model::dcf`, fields `lcoh_smr` and `lcoh_nush2`

Validation. ID, test `eq_l_35`. Results, BOAK, 25 and 60 y: SMR S$3.62 and S$3.62/kg, NUS-H2 S$5.26 and S$5.05/kg. FOAK NUS-H2: S$6.02 and S$5.71/kg.

Equation L.36: Levelised abatement cost

Tier R

$$LAC = \frac{\mathrm{NPV}_{NUS} - \mathrm{NPV}_{SMR}}{\mathcal{A} \sum_{y=1}^{t_\ell} (1+i)^{-y}} \tag{L.36}$$

Interpretation. The present cost difference between the two plants, spread over present tonnes abated. The SMR island appears on both sides and cancels, so only the new items and the gas, steam and power differences remain.

Derivation. Eq. L.28 with $\mathrm{NPV}(\Delta C) = \mathrm{NPV}_{NUS} - \mathrm{NPV}_{SMR}$ and constant annual tonnes. Dividing through by M_{H_2} and the annuity gives $LAC = (LCOH_{NUS} - LCOH_{SMR})\, M_{H_2}/\mathcal{A}$, a useful check.

Variables. NPV_{NUS}, present cost of NUS-H2, S$; NPV_{SMR}, present cost of the unabated SMR, S$.

Assumptions. Tonnes discounted at the same rate as cost.

Source. [**C**] This document.

Implementation. `plant::dcf::Model::dcf`, field `lac`

Validation. ID, test `eq_l_36`. Results: 190.1 and 166.0 at BOAK, 278.4 and 242.2 at FOAK, 25 and 60 y. Computed here from the printed equations, independent of the engine: 190.1, 166.0, 278.4 and 242.2, each within 0.01 S$ t^{-1} of the results. The LCOH identity gives 190.1 at BOAK, 25 y.

Equation L.37: Gap test between the methods

Tier R

$$\begin{aligned} g &= \frac{LAC - AC_{L5}}{AC_{L5}}, \qquad |g| \le 0.10 \\ LAC - AC_{L5} &= \frac{C_R\,(f_s - f_{IDC})\,CRF + \sum_j C_j\,(\Theta\,CRF - CRF_{25})}{\mathcal{A}} \end{aligned} \tag{L.37}$$

Interpretation. Both methods price the same items, so they must nearly agree. Any difference has to be a named timing effect. A gap above 10% must be explained by a named term in the documentation,

and it is never tuned away. The test checks the bookkeeping, and an input wrong in both methods would pass it.

Derivation. Annual lines match exactly, because they enter both methods unchanged. Decommissioning matches too: $D(1+i)^{-t_\ell}$ spread by CRF equals $D\,SFF$, by the note in Eq. L.3. Licence and power island at year 0 match by Eq. L.1. Two terms remain. Reactor timing: the discrete spend gives $f_s = [0.5(1+i)^5 + \sum_{k=0}^{4}(1+i)^k]/5.5 = 1.2002$, against the smooth Eq. L.4, 1.198. Replacement: the sum over j runs over capture, IHX and recycle train, with $\Theta = \sum_m (1+i)^{-25m} - (u/25)(1+i)^{-t_\ell}$, which is 1 at 25 y and 1.1614 at 60 y. The SMR island nets out of the difference.

Variables. g, relative gap between methods, 1; AC_{L5}, ledger abatement cost at L5, S$ t^{-1}; f_s, spend factor of the discrete schedule, 1; Θ, present worth of replacements per unit capital, 1; CRF_{25}, capital recovery factor at 25 y, y^{-1}.

Assumptions. Same inputs in both methods. Limit 10%.

Source. [C] This document, limit set by the project cost method.

Implementation. `plant::dcf::Model::dcf`, fields `gap_abs`, `gap_rel` and `gap_terms`

Validation. ID, test `eq_l_37`. Results on all twelve grid rows are recorded as LAC_DCF_VS_L5 in the validation results, and the largest $|g|$ is 0.2%. BOAK at $n_s = 1$: 0.31 S$ t^{-1} at 25 y (0.2%) and 0.33 at 60 y (0.2%). Computed here from the closed form: 0.309 at 25 y, all reactor timing. At 60 y, 0.267 reactor timing plus 0.066 replacement, total 0.333. The test uses the same inputs on both sides, so it says nothing about the inputs. Validation results, 12 rows (LAC_DCF_VS_L5_BOAK_25Y_N0P31 to LAC_DCF_VS_L5_FOAK_60Y_N1P00): 12 pass. The results mark these rows as using the same data as the model or as a check of the code, so they are not validation evidence.

Worked example: both methods at Case B

The ledger first. BOAK, $n_s = 1.0$, 210 MWth charged and 578,052 t y^{-1} abated:

Rung	Line	BOAK, S$M/y 25 y	BOAK, S$M/y 60 y	FOAK, S$M/y 25 y	FOAK, S$M/y 60 y
L0	Gas saved	-64.80	-64.80	-64.80	-64.80
L0	Reactor capital	69.58	60.01	111.33	96.02
L0	Capture capital	9.25	9.25	9.25	9.25
L0	Capture fixed O&M	2.67	2.67	2.67	2.67
L0	T&S	21.59	21.59	21.59	21.59
L1	Reactor O&M and fuel	25.32	25.32	25.32	25.32
L2	IHX and secondary loop	8.36	8.36	8.36	8.36
L3	Licensing	6.12	5.28	6.12	5.28
L3	Decommissioning	1.50	0.09	1.50	0.09
	L3, S$/t	**137.7**	**117.2**	**209.9**	**179.5**
L4	IDC on reactor	13.75	11.86	22.01	18.98
L4	Power island	1.62	1.40	2.59	2.24
L4	Recycle train	5.89	5.89	5.89	5.89
	L4, S$/t	174.5	150.4	262.7	226.4
L5	Lost steam export	15.23	15.23	15.23	15.23
L5	Baseline power avoided	-6.41	-6.41	-6.41	-6.41
	L5, S$/t	**189.7**	**165.6**	**277.9**	**241.7**

Then the cash flow at BOAK. The present value of each NUS-H2 group, computed here from Eq. L.34:

Group, S$M present value	25 y	60 y
SMR island, capital and O&M	716.1	831.3
Gas	1,271.2	1,473.9
Reactor capital at year 0	891.5	891.5
Capture capital and fixed O&M	127.2	147.7
T&S	230.4	267.2
Reactor O&M and fuel	270.3	313.4
IHX	89.2	103.7
Licence and power island	82.6	82.6
Decommissioning	16.0	1.1
Recycle train and fixed O&M	62.9	73.0
NUS-H2 total	**3,758**	**4,185**
Unabated SMR total	2,585	2,998

Dividing the difference by 10.675 times 578,052 t y^{-1} at 25 y gives the Eq. L.36 figure, 190.1 S$ t^{-1}.

Headline and target test

Rebuilt full cost at Case B, $n_s = 1.0$, reactor charged at 210 MWth:

Stage	Life	L5, S$/t	LAC, S$/t	g, %	$LCOH_{SMR}$, S$/kg	$LCOH_{NUS}$, S$/kg	L3, S$/t
BOAK	25 y	189.7	190.1	0.2	3.62	5.26	137.7
BOAK	60 y	165.6	166.0	0.2	3.62	5.05	117.2
FOAK	25 y	277.9	278.4	0.2	3.62	6.02	209.9
FOAK	60 y	241.7	242.2	0.2	3.62	5.71	179.5

The results count 0 of 8 central cases at or below S$100/t, on L5 and the DCF, at BOAK and FOAK, at 25 y and 60 y. The cost target is therefore missed on this basis, and no S$100/t claim is made. BOAK at 25 y exceeds the ceiling of S$57.8M per year by S$51.9M per year on the full cost set, and by S$21.8M per year at L3. Charging only the 178.5 MWth the process needs brings L3 to 112.7 and 95.1 at 25 and 60 y, and L5 to 161.2 and 140.4. The abatement target is met by a wide margin, Eq. K.22.

Link to the sources

INL's cost survey gives quartiles rather than a price for a 925 C reactor, and it warns against reading more precision into them than they hold. The result here inherits that spread: the size exponent and stage move the abatement cost by more than any plant design choice in Sections I to K. The DCF confirms that the ledger's arithmetic is sound, and what stays uncertain is the reactor price itself. Items carried on proxies, [**M23**] and [**M31**], are named on their cards.

15 M. Missing sources and source errata

This section lists what the derivations could not read at source, the printing errors found in the sources that were read, and the places where this document departs from the earlier screening model. A card that leans on an item here carries its tag, [**Mn**]. Each item stays open until its source is in the project library.

Missing source register

Id	Missing	Effect	What this document does instead
M1	TEV-961 Case 6 sheets, A-60 onward, and Appendix G	No Case 6 stream data	Case 5 used as a proxy, scaled by 1,927 over 1,970 where used, stated on every card
M2	TEV-981	Whether the 39.7% conversion efficiency is a net value is unknown	39.7% kept, 47.5% as a sensitivity; Case A spans 253 to 271 MW_{th}
M3	Hou and Hughes (2001), origin of $K(T)$	Neither Xu and Froment paper prints $K(T)$	Zhang et al. Table 1 [41], cross-checked against NIST data
M4	Copper-zinc low-temperature shift kinetics	No rate law for the second shift bed	A 10 K approach to equilibrium on INL and IEAGHG operating points
M5	Helium property correlation in the library		Petersen (1970) [23], read on the web and cited as an external source
M6	Species heat capacity and formation enthalpy		NIST SRD 69 Shomate form [20], coefficients read from the WebBook
M7	Thomsen (1997); Thomsen and Rasmussen (1999)	Activity expressions and the parameters $A(T)$, b and z come from general knowledge of the framework	Indirect support: a prototype reproduces Aronu's published model at 104 points, AARD 0.20%. The cards say this plainly
M8	Versteeg and van Swaaij (1988), J. Chem. Eng. Data 33, 29 to 34	Henry constants second hand	Aronu Eqs 17 and 18; Dugas Eqs 5.9 to 5.11 [6]
M9	Water saturation data		IAPWS-IF97 in its place
M10	H_2, CH_4 and CO solubility in loaded MEA	No co-absorption model	Zero in the central case, with an aMDEA reference; the purity effect is disclosed
M11	MEA kinetics and properties above a loading of 0.50		Rich loading capped at 0.50
M12	Specific reboiler duty measured on syngas		3.8 GJ t^{-1}, range 3.4 to 4.2, labelled assumed
M13	Measured circulator power		Eq. B.10 with a budget
M14	Measured IHX pressure drop		HTR-10 design and the INL 50 kPa; GTHTR300C values of 31 and 58 kPa stay unchecked. JAEA-Technology 2008-093 would close it
M15	Tochio and Nakagawa constant C_0	Shell-side Nusselt correlation cannot be used	UA anchor [31]
M16	Catalyst effectiveness in helium-heated service		One activity factor, fitted and disclosed
M17	Rocha (1993, 1996) and packing data	Specific area of Mellapak 250Y, C_L and C_G are not local	Mellapak 250Y assumed, with geometry to come from the vendor sheet or Rocha. Until then the working case runs Onda, Khan Eqs 5a and 5b, on a random packing
M18	Packed-column pressure-drop correlation		Eq. H.23 scaling, labelled assumed. Billet and Schultes (1999) would close it
M19	Surface tension of loaded MEA	The Rocha and Onda areas need it	The Billet and Schultes area, Eq. H.12, needs none
M20	Gas diffusivity method		Fuller method **[T]**, from Poling, Prausnitz and O'Connell
M21	Tabulated heat capacity of the loaded solvent		3.6 kJ kg^{-1} K^{-1}, from a figure axis range
M22	Syngas transport properties		Maker values in JAERI F-2 and G-1, PDF p. 185 and p. 187 [28], with Wilke mixing **[T]**
M23	IAEA-TECDOC-1682 Table 3.17, GA-A24265, an NRC blog summary	IHX and licensing proxies carry S$21.6 per tonne at Case A	Values carried from earlier project files, labelled as not opened at source

Id	Missing	Effect	What this document does instead
M24	TEV-954 and TEV-962	INL's own hydrogen cost analyses are not used	Section L rests on the cost sources named on its cards
M25	Cho et al. full text	The split of the 9.35 kg kg^{-1} mean is unavailable	Abstract only; the 35/65 split is dropped
M26	ETC exhibit	US$60 per tonne not checked at source	Not used in the central case
M27	S$0.20 per kWh and 0.402 t MWh^{-1}	Not checked at source	Sensitivity and system view only
M28	Liquefaction power for ship transport	The project flowsheet shows a ship, while INL compresses to 139 bar	Compression power scaled from INL kept as a proxy and flagged; the T&S tariff carries transport
M29	Basis for backup firing		Sensitivity with an assumed efficiency
M30	Split of the 5.7 MWe row		Worded as fuel-gas recycle and nitrogen removal equipment; the split is unpublished
M31	Recycle train cost		Proxy from the IEAGHG compression cost per MW

Source errata

Constants were read from page images wherever a text layer was in doubt. The rows below record what the pages print and how this document handles each case. XF1 and XF2 are Xu and Froment Parts I and II [37, 38]; HA is Hoseinzade and Adams [10].

Id	Where	What is printed	Handling
T1	XF1 Table 5, PDF p. 7	Lower limit of K_{H_2O} printed as 0.0317	The value 0.4152 and upper limit 0.5032 imply a symmetric lower limit of 0.3272 **[C]**. A misprint that touches only the quoted uncertainty
T2	HA Eq 55, PDF p. 7	$K_{H_2O} = 1.77 \times 10^3$ where XF1 has 1.77×10^5	K_{H_2O} is dimensionless (XF1 PDF p. 8), so the bar to kPa division must not apply. At a sample point the rate changes by a factor of 27 **[C]**. HA constants are not used
T3	HA Eqs 48 to 50, PDF p. 7	Rate constants carry an unstated factor	Each is 2.246 times the XF value **[C]**, the fresh-catalyst activity. Not copied
T4	HA Eq 52	Exponent of K_2 printed as -4.063	$\ln(1.767 \times 10^{-2}) = -4.036$ **[C]**, a likely transposition that lowers K_2 by 2.7%
T5	HA against Zhang et al., K_1	Pressure bases differ	HA's 1.193×10^{13} atm^2 is 1.225×10^{13} bar^2, against Zhang's 1.198×10^{13} bar^2 **[C]**, 2.2% in K_1
T6	Zhang et al. Eq 12, PDF p. 5	$1.176 \times 10^{-2} \exp(-4400/T)$ with $(1 - X_1)$ in the denominator	Table 1 gives $1.767 \times 10^{-2} \exp(+4400/T)$ and stoichiometry needs $(1 - X_2)$ **[C]**. Table 1 is used and the equation derived here
T7	Zhang et al. Eq 11 and Table 1	1.198×10^{23} in Eq 11 against 1.198×10^{11} in Table 1	Eq 11 is in Pa^2 and Table 1 in MPa^2. Eqs 6 and 7 also carry a stray $\times 100\%$
T8	XF1 Table 2, PDF p. 3, and Table 7, PDF p. 8	Table 2 lists $-\Delta H$ and Table 7 lists $+\Delta H$; reaction III reads -165.0 in one and 164.9 in the other	Sign convention, read with care

Id	Where	What is printed	Handling
T9	XF1 notation, PDF p. 8	R given in kJ/kmol with no kelvin; activation energies in kJ/mol while rates are in kmol	XF2 PDF p. 6 gives kJ/(kmol K). Kinetics are coded in the published units and converted at the function boundary
T10	XF1 Table 5	Two reference temperatures: 648 K for k, K_{CO} and K_{H_2}, and 823 K for K_{CH_4} and K_{H_2O} in a footnote	Each constant uses its own reference temperature
T11	XF1 PDF p. 7 text against Table 6	$A(K_{H_2})$ printed as 6.13×10^{-9} in the text and 6.12×10^{-9} in the table	Back-calculation gives 6.146×10^{-9} **[C]**. Table 6 is used
T12	XF2 PDF p. 3	Binary diffusivities attributed to Wilke and Chang (1955), whose title is on dilute solutions, a liquid-phase correlation	A gas-phase method is needed to reproduce the effectiveness factors
T13	XF2 notation, PDF p. 6	Gas conductivity in W/(m h K), viscosity in kg/(m s), other terms in hours	Unit mixing. Read from the text layer only
T14	XF1 text layer	The text gives -241.3 where the page image shows -247.3	Constants are taken from page images only
T15	Yin Eq 14, PDF p. 5	Blasius printed as $0.3164/Re$ with no exponent	Misprint. Sato Eq 10 prints $Re^{-1/4}$ and is used in Eq. C.16
T16	Yin Eq 15, PDF p. 5	Ergun with d_2 defined as the catalyst tube inner diameter, and $150/Re$ without $(1 - \varepsilon_b)$	Not used as printed. HA Eq 26 or Sato Eq 11 instead
T17	HA Eq 13, PDF p. 4	A cross-flow correlation applied to helium over tubes, then raised by 2.7 on the coefficient and 2.3 on area for fins (PDF p. 8)	JAERI's 2.7 is the finned outer to inner area ratio, with h_o/h_i near 4.8 (JAERI PDF p. 24). Not a helium annulus model
T18	Aronu et al., PDF p. 4	The text says four reactions	Five are listed and five are correct
T19	Part 5b, Eq 35	2985	2895 is the correct value
T20	Hartono et al., Eqs 3 to 5	The basis of the variables is unclear	Read as working with Φ^1 and mass fraction, which is an inference
T21	Isa et al., Eqs 16, 25 and 28	Forms differ from Khan's	Khan's forms are used
T22	Khan et al.	Units of the Henry constant are not stated	Read with care
T23	Wong, Tables 2 and 3	Unusable as printed	Not used
T24	TEV-961 text	"71 C" and "3 C" disagree with Table 2	Table 2 is used
T25	IAEA NP-T-4.3, Table 16	IHX temperatures that are impossible as printed	Not used
T26	IAEA-TECDOC-1236	IHX tube outer diameter printed as 3.2 mm	A misprint for 31.8 mm
T27	Tochio and Nakagawa, Fig 9	Temperatures cannot be resolved	Only UA and the hot-end gap are quoted
T28	IEAGHG 2017-02, stream 13	CH_4 of 0.0363 printed in the oxygen row	The balance confirms 201 kmol/h of CH_4 **[C]**; read as CH_4
T29	DICE Table 5-3	263.3	A steam flow in 1000 lb/h. The reboiler duty is 242 MMBtu/h, per E-104
T30	Salvinder et al., Table 1	"H_2O 0.8 mol, MEA 0.2 mol" against 20 wt% in the text	A mole fraction of 0.2 is 45.9 wt% **[C]**, so the label is likely a mass-fraction error
T31	Halim et al.	Text gives mol/(m^2 h kPa), figures give mol/(m^3 h kPa)	The figures are used
T32	Versteeg and van Swaaij	The local file is the kinetics paper	The Henry constant paper is separate, see M8

Differences from the screening model

The engine reproduces the earlier screening model wherever the two share a method. Reference tests against that model expect the five differences below, and a test that misses by more than the listed effect points to a porting error in the code.

1. **Duct loss.** Eq. B.14 scales the loss with $\sqrt{\dot{m}}$, where the screening model scales it with s: 0.37 against 0.31 MW at Case B.
2. **Reboiler steam order.** The screening model draws freed Selexol steam, then EX4 and ZLD steam, then export steam. This document follows the project order: freed Selexol steam, export steam, syngas above 130 C, then reactor steam. The two agree whenever the reboiler uses the whole stack, which holds in every central case. They part only when the reboiler needs less than the stack, for example at an SRD of 3.4 with ZLD steam freed, and then the export steam kept and the lost steam change.
3. **Water saturation.** Saturation in Eq. F.11 comes from IAPWS-IF97, so the result differs slightly from the Antoine value of the screening model.
4. **Size exponent.** The default here is $n_s = 1.0$, against 0.6 in the screening model.
5. **Reformer duty.** The species-enthalpy duty of Eq. D.34 sits below the INL heat basis. The ledger keeps Eq. J.14 and prints both.

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